

or for the $(n - 1)$ th stage,

$$A(X_0^A - X_{n-1}^A) = B(X_1^B - X_n^B)$$

Multiplying the equation by -1 , we obtain

$$A(X_{n-1}^A - X_0^A) = B(X_n^B - X_1^B) \quad (d)$$

If we rearrange Eq. (d) assuming that A , B , X_0^A , and X_1^B are constants (i.e., steady-state operation) and X^A and X^B are the variables as we go from stage to stage, we can write

$$AX_{n-1}^A = \frac{B}{A}X_n^B + \left(X_0^A - \frac{B}{A}X_1^B\right) \quad (e)$$

Equations (a)–(e) represent an unusual type of equation, one that gives the relationship between discrete points rather than continuous variables; it is called a *difference* equation. In Eq. (e) the locus of these points will fall upon a line with the slope B/A and an intercept $[X_0^A - (B/A)X_1^B]$, as shown in Fig. E2.22b.

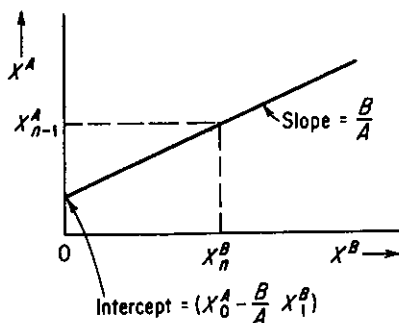
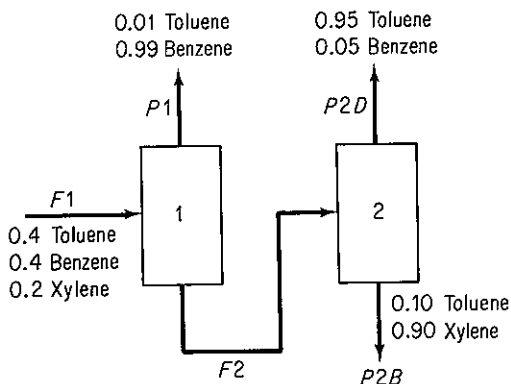


Figure E2.22b

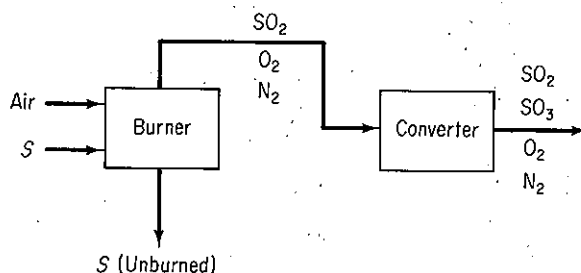
In the next section we look at material balances in which some material is recycled (fed back) from upstream in a process back to the start of the process.

Self-Assessment Test

1. A two-stage separations unit is shown in the figure. Given that the input stream F1 is 1000 lb/hr, calculate the value of F2 and the composition of F2.



2. A simplified process for the production of SO_3 to be used in the manufacture of sulfuric acid is illustrated in the figure. Sulfur is burned with 100% excess air in the burner, but for the reaction $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$, only 90% conversion of the S to SO_2 is achieved. In the converter, the conversion of SO_2 to SO_3 is 95% complete. Calculate the lb of air required per 100 lb of sulfur burned, and the concentration in mole fraction or percent of the exit gas from the burner and from the converter.



2.6 RECYCLE, BYPASS, AND PURGE CALCULATIONS

Your objectives in studying this section are to be able to:

1. Draw a flow diagram for problems involving recycle, bypass, and purge.
2. Apply the 10-step strategy to solve steady-state problems (with and without chemical reaction) involving recycle, and/or bypass, and/or purge streams.
3. Solve problems in which a modest number of interconnected units are involved by making appropriate balances.
4. Use the overall conversion and single-pass (once-through) conversion concepts to solve recycle problems involving reactors.
5. Explain the purpose of a recycle stream, a bypass stream, and a purge stream.

Recycle stream is a term denoting a process stream that conducts material exiting or downstream from a unit back to the inlet or upstream of the same unit. For example, in a reactor, unreacted material is separated from the reactor products and *fed back* and joins with a stream of reactants that enter the reactor. Examine Figure 2.16. In Figure 2.4 you can observe the recycle of C_6H_6 from the settler back to the evaporator. As another example, in planning long space missions, all the food and water will have to be provided from stores on board the spacecraft. Figure 2.17 shows the recycle of O_2 and water.

Many industrial processes employ recycle streams. In some drying operations, the humidity in the air is controlled by recirculating part of the wet air that leaves

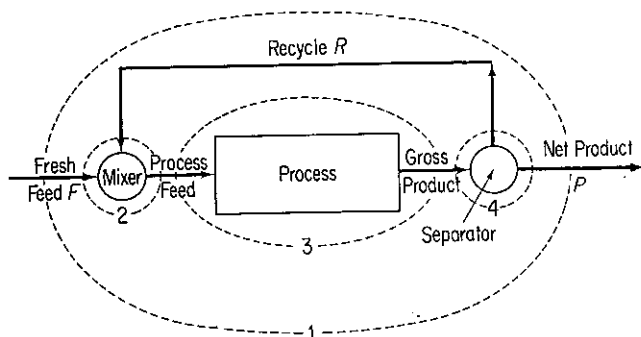


Figure 2.16 Process with recycle (the numbers designate possible system boundaries for the material balances—see the text).

the dryer. In chemical reactions, exit catalyst is returned to the reactor for reuse. Another example of the use of recycling is in fractionating columns where part of the distillate is refluxed through the column to maintain the quantity of liquid within the column.

Do not let recycle streams confuse you. The steps in the analysis and solution of material balance problems involving recycle are the same as described in Table 2.4. With a little practice in solving problems involving recycle, you should experience little difficulty in solving recycle problems in general. The essential point you should grasp with respect to recycle calculations in this chapter is that the processes such as shown in Fig. 2.4 or 2.16 are in the *steady state*.

No buildup or depletion of material takes place inside the process or in the recycle stream.

The values of F , P , and R in Fig. 2.16 are *constant*. Unsteady-state processes such as startup and shutdown are discussed in Chapter 6.

2.6-1 Recycle in Processes without Chemical Reaction

The strategy listed in Table 2.4 is the strategy to be used in solving recycle problems. You can make component and total material balances for each subsystem as discussed in Sec. 2.5, as well as component and total balances for the overall process. Not all of the equations so formulated will be independent, of course. Depending on the information available concerning the amount and composition of each stream, you can determine the amount and composition of the unknowns. If tie components are available, they often simplify the calculations.

Examine Fig. 2.16. Material balances can be written for several different systems, four of which are shown by dashed lines in Fig. 2.16:

- About the entire process including the recycle stream, as indicated by the dashed lines (marked 1 in Fig. 2.16)
- About the junction point at which the fresh feed is combined with the recycle stream (marked 2 in Fig. 2.16)
- About the process only (marked 3 in Fig. 2.16)
- About the junction point at which the gross product is separated into recycle and net product (marked 4 in Fig. 2.16)

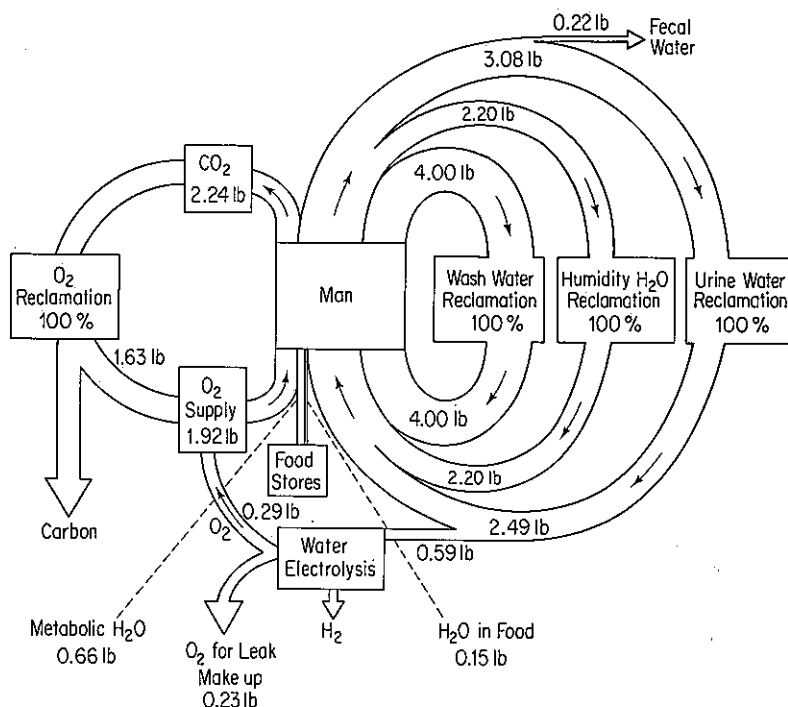


Figure 2.17 Water and oxygen recycle in a space vehicle.

In addition, balances can be made about combinations of subsystems, such as the process plus the separator (3 + 4). Only three of the four balances (a)–(d) are independent for one component. However, balance 1 will not include the recycle stream, so that the balance will not be directly useful in calculating a value for the recycle R . Balances 2 and 4 do include R . You could write a material balance for the combination of subsystems 2 and 3 or 3 and 4 and include the recycle stream.

EXAMPLE 2.23 Recycle

Examine the flow sheet in Fig. E2.23, a flowsheet that contains recycle streams. What is the maximum number of independent material balances that can be written for the system if each stream contains three components, ethanol, acetone, and methanol?

Solution

Three material balances (corresponding to three components) can be written each for units A, B, and C, for a total of 9. Any other material balance, such as one for the combined subsystem A and B or A and C, can be obtained by appropriate combination of the nine component material balances. However, all nine of the material balances may not be independent. How many would be independent would depend on the specific values of the flow streams and concentrations that are known and unknown.

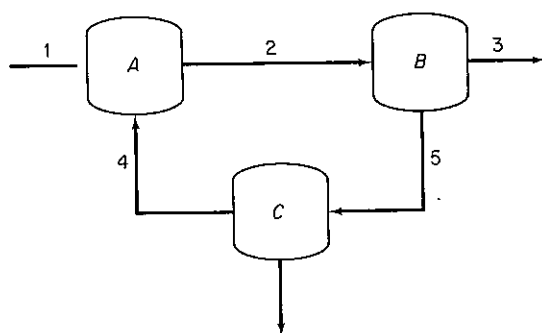


Figure E2.23

EXAMPLE 2.24 Recycle without Chemical Reaction

A distillation column separates 10,000 kg/hr of a 50% benzene–50% toluene mixture. The product D recovered from the condenser at the top of the column contains 95% benzene, and the bottoms W from the column contain 96% toluene. The vapor stream V entering the condenser from the top of the column is 8000 kg/hr. A portion of the product from the condenser is returned to the column as reflux, and the rest is withdrawn for use elsewhere. Assume that the compositions of the streams at the top of the column (V), the product withdrawn (D), and the reflux (R) are identical because the V stream is condensed completely. Find the ratio of the amount refluxed to the product withdrawn (D).

Solution

This is a steady-state problem without reaction occurring.

Steps 1, 2, and 3 See Fig. E2.24 for the known data, symbols, and other information.

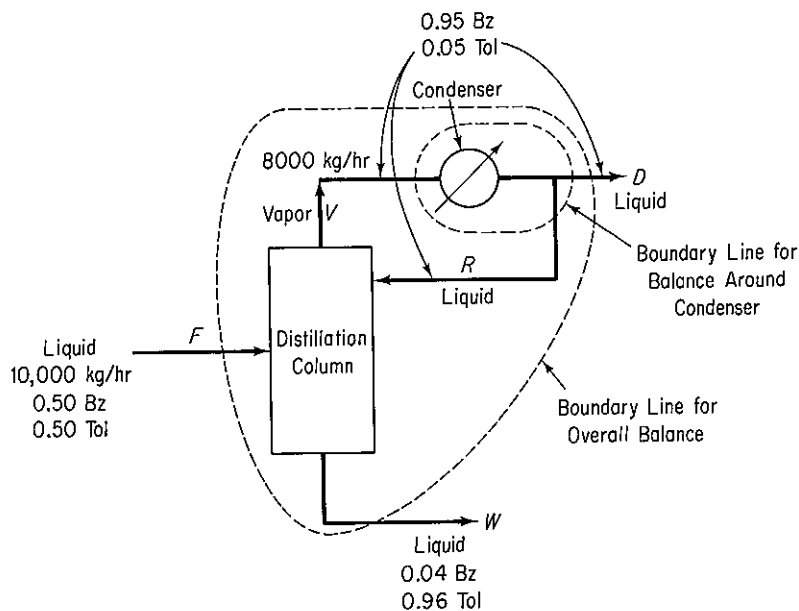


Figure E2.24

Step 4 Select a basis of 1 hr (equal to $F = 10,000$ kg).

Steps 5 and 6 All the compositions are known and three stream flows, D , W , and R , are unknown. No tie components are evident in this problem. Two component material balances can be made for the still and two for the condenser. Presumably three of these are independent; hence the problem has a unique solution. We can check as we proceed. A balance around either the distillation column or the condenser would involve the stream R . An overall balance would involve D and W but not R .

Steps 7, 8, and 9 What balances to select to solve for R is somewhat arbitrary. We will choose to use overall balances first to get D (and W), and then use a balance on the condenser to get R . Once D is obtained, R can be obtained by subtraction.

Overall Material balances:

Total material:

$$\begin{aligned} F &= D + W \\ 10,000 &= D + W \end{aligned} \tag{a}$$

Component (benzene):

$$\begin{aligned} F\omega_F &= D\omega_D + W\omega_w \\ 10,000(0.50) &= D(0.95) + W(0.04) \end{aligned} \tag{b}$$

Solving (a) and (b) together, we obtain

$$\begin{aligned} 5000 &= (0.95)(10,000 - W) + 0.04W \\ W &= 4950 \text{ kg/hr} \\ D &= 5050 \text{ kg/hr} \end{aligned}$$

Balance around the condenser:

Total material:

$$\begin{aligned} V &= R + D \\ 8000 &= R + 5050 \\ R &= 2950 \text{ kg/hr} \\ \frac{R}{D} &= \frac{2950}{5050} = 0.58 \end{aligned} \tag{c}$$

Would the benzene or toluene balances on the condenser yield additional information to that obtained from the total balance, Eq. (c)? Write the balances down and check to see if they are redundant with Eq. (c).

EXAMPLE 2.25 Recycle without Chemical Reaction

The manufacture of such products as penicillin, tetracycline, vitamins, and other pharmaceuticals, as well as photographic chemicals, dyes, and other fine organic compounds, usually requires separating the suspended solids from their mother liquor by centrifuging, and then drying the wet cake. A closed-loop system (see Fig. E2.25a) for centrifuge unloading, drying, conveying, and solvent recovery is comprised of equipment especially designed for handling materials requiring sterile and contamination-free conditions.

Given the experimental measurements on the pilot plant equipment outlined in Fig. E2.25a, what is the lb/hr of the recycle stream R ?

Overall mass balances:

$$V: \quad 0.20(98) = 0 \quad + 0.96P \quad (a)$$

$$H_2O: \quad 0.80(98) = (1.0)W + 0.04P \quad (b)$$

$$\text{Total:} \quad 98 \quad = \quad W \quad + \quad P \quad (c)$$

Observe that V is a tie component so that P can be calculated directly in Eq. (a): $P = 20.4$ lb, and W can be calculated from Eq. (c).

$$W = 98 - 20.4 = 77.6 \text{ lb}$$

Steps 7, 8, and 9 (Continued) To determine the recycle stream R , we need to make a balance that involves the stream R . Either (a) balances around the centrifuge or (b) balances around the filter will do. The latter are easier to formulate since the mixing of R and F does not have to be calculated.

Total balance on filter:

$$\begin{aligned} C &= R + P \\ C &= R + 20.4 \end{aligned} \quad (d)$$

Component V balance on filter:

$$\begin{aligned} C\omega_C &= R\omega_R + P\omega_P \\ 0.6C &= 0.286R + 0.96(20.4) \end{aligned} \quad (e)$$

Solving Eqs. (d) and (e), we obtain $R = 23.4$ lb/hr.

Step 10 Check the value of R using a material balance around the centrifuge.

2.6-2 Recycle in Processes with Chemical Reaction

Now let us turn to recycle problems in which a chemical reaction occurs. Recall from Sec. 1.9 that not all of the limiting reactant necessarily reacts in a process. Do you remember the concept of conversion as discussed in Sec. 1.9? Two bases for conversion are used in describing a process; examine Fig. 2.18.

(a) **Overall fraction conversion:**

$$\frac{\text{mass (moles) of reactant in fresh feed} - \text{mass (moles) of reactant in output of the overall process}}{\text{mass (moles) of reactant in fresh feed}}$$

(b) **Single-pass ("once-through") fraction conversion:**

$$\frac{\text{mass (moles) of reactant fed into the reactor} - \text{mass (moles) of reactant exiting the reactor}}{\text{mass (moles) of reactant fed into the reactor}}$$

When the fresh feed consists of more than one material, the conversion must be stated for a single component, usually the limiting reactant, the most expensive reactant, or some similar compound.

Note the distinction between *fresh feed* and *feed to the process*. The feed to the process itself is made up of two streams, the fresh feed and the recycled material.

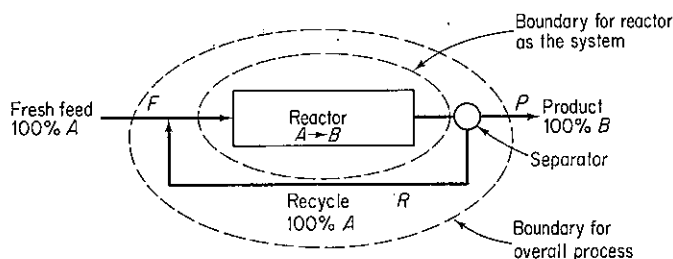


Figure 2.18 Recycle problem.

The gross product leaving the process is separated into two streams, the net product and the material to be recycled. In some cases the recycle stream may have the same composition as the gross product stream, while in other instances the composition may be entirely different depending on how the separation takes place and what happens in the process. Suppose that you are given the data that 30% of the A is converted to B on a single pass through the reactor, as illustrated in Fig. 2.18, and are asked to calculate the value of R , the recycle on the basis of 100 moles of fresh feed, F . We will make a balance for A with the reactor as the system.

Recall from Eq. (2.1) that for a specific chemical compound the steady-state material balance for a reactor is (the accumulation term in zero)

$$\left\{ \begin{array}{c} \text{input} \\ \text{through} \\ \text{system} \\ \text{boundary} \end{array} \right\} - \left\{ \begin{array}{c} \text{output} \\ \text{through} \\ \text{system} \\ \text{boundary} \end{array} \right\} + \left\{ \begin{array}{c} \text{generation} \\ \text{within the} \\ \text{system} \end{array} \right\} - \left\{ \begin{array}{c} \text{consumption} \\ \text{within the} \\ \text{system} \end{array} \right\} = 0 \quad (2.12)$$

Reactants are consumed and products are generated. If a reaction takes place within the system, you must be given (or look up) information about the reaction stoichiometry and extent of reaction. Or, perhaps the question is to calculate the extent of conversion given some of the process data. In any case, the fraction of feed converted to products is always an essential additional piece of information that helps determine values of the terms in Eq. (2.12).

Let us examine how to apply Eq. (2.12) for a recycle reactor such as shown in Fig. 2.18, in which A is converted to B . How much A exits the reactor itself? The unconverted A is 70% of the A that enters the reactor. No A occurs in the P stream. The system is the reactor and the basis is 100 moles of fresh feed. The A balance is

$$\frac{\text{Input of } A}{[(1.0)(100) + (1.0)R]} - \frac{\text{Output of } A}{[(1.0)R + 0(P)]} - \frac{\text{Consumption of } A}{[0.30(100 + R)]} = 0$$

or

$$\frac{0.70(100 + R)}{\text{moles of } A \text{ unconverted}} = \frac{R}{\text{moles of } A \text{ leaving the reactor}}$$

$$R = 233 \text{ moles}$$

Note that in Fig. 2.18 all the A was recycled for simplicity of illustration of the principle, but such may not be the case in general. Nevertheless, Eq. (2.12) still applies.

The single-pass (mole) balance on A provides the crucial information to evaluate R . Will an A balance, or total balance, for the overall process enable you to solve for R ? Try one and see why not. What is the overall fraction conversion of A for the entire process? Does that information help you solve for R ?

EXAMPLE 2.26 Recycle with a Reaction Occurring

Immobilized glucose isomerase is used as a catalyst in producing fructose from glucose in a fixed-bed reactor (water is the solvent). For the system shown in Fig. E2.26a, what percent conversion of glucose results on one pass through the reactor when the exit stream/recycle ratio in moles is equal to 8.33? The reaction is

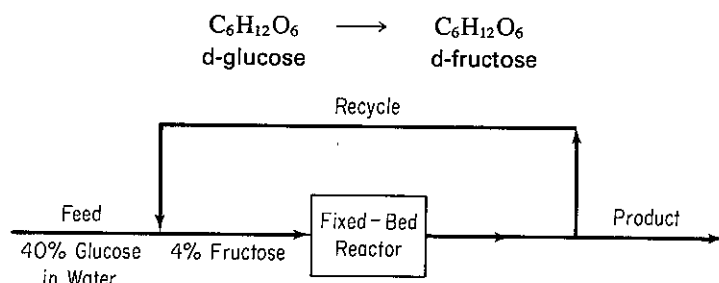


Figure E2.26a

Solution

We have a steady-state process with a reaction occurring.

Steps 1, 2, and 3 Figure E2.26b includes all the known and unknown values of the variables using appropriate notation (W stands for water, G for glucose, and F for fructose in the second position of the mass fraction subscripts). Note that the recycle stream and product stream have the same composition and consequently the same mass fraction symbols are used in the diagram for each stream.

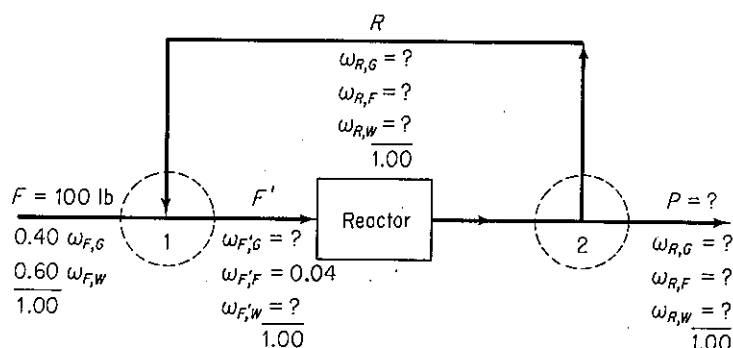


Figure E2.26b

Step 4 Pick as a basis $F = 100$ lb.

Steps 5 and 6 The sum of the mass fractions is one in each stream so that the unknown compositions can be picked to be $\omega_{F',G}$, $\omega_{R,G}$, $\omega_{R,W}$, and the unknown stream values are F' and P ($R = P/8.33$). Let f be the fraction conversion in the reactor. Three balances

each can be made about the mixing point 1, the reactor, and the separation point 2. Not all the balances will be independent, but sufficient independent balances should exist to solve this problem. We can check as we proceed with the calculations rather than going through an extensive analysis at the beginning.

Steps 7, 8, and 9 We will start with overall balances, as they are the easiest to write.

Overall:

$$\text{Total: } 100 = P$$

Consequently,

$$R = \frac{100}{8.33} = 12.0 \text{ lb}$$

No water is generated or consumed, hence

$$\begin{aligned} \text{Water: } 100(0.60) &= P(\omega_{R,W}) = 100\omega_{R,W} \\ \omega_{R,W} &= 0.60 \end{aligned}$$

We now have left three unknown values of the variables plus f .

Mixing Point 1:

$$\begin{aligned} \text{Total: } 100 + 12 &= F' = 112 \\ \text{Glucose: } 100(0.40) + 12(\omega_{R,G}) &= 112(\omega_{F',G}) \\ \text{Fructose: } 0 + 12(\omega_{R,F}) &= 112(0.04) \end{aligned}$$

or

$$\omega_{R,F} = 0.373$$

Also, because $\omega_{R,F} + \omega_{R,G} + \omega_{R,W} = 1$,

$$\omega_{R,G} = 1 - 0.373 - 0.600 = 0.027$$

and then from the glucose balance,

$$\omega_{F',G} = 0.360$$

Reactor Plus Separator 2:

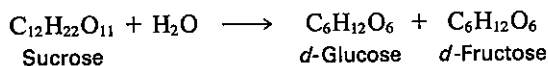
$$\text{Total: } F' = 12 + 100 = 112 \quad (\text{redundant equation})$$

Glucose:

In	Out	Consumed
$F' \omega_{F',G}$	$-(R + P) \omega_{R,G}$	$-f F' \omega_{F',G}$
$112(0.360)$	$- 112(0.027)$	$- f(112)(0.360)$
		$f = 0.93$

EXAMPLE 2.27 Recycle with a Reaction Occurring

Refined sugar (sucrose) can be converted to glucose and fructose by the inversion process.



The combined quantity glucose/fructose is called inversion sugar. If 90% conversion of sucrose occurs on one pass through the reactor, what would be the recycle stream flow per 100 lb of sucrose solution entering the process shown in Fig. E2.27a? What is the concentration of inversion sugar (*I*) in the recycle stream and in the product stream? The concentrations of components in of the recycle stream and product stream are the same.

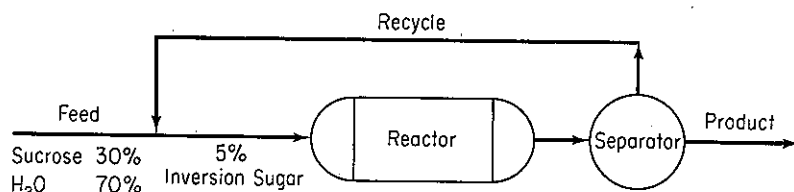


Figure E2.27a

Solution

Steps 1, 2, and 3 First we need to enter the concentrations and stream flows on the diagram. See Fig. E2.27b. (*W* stands for water, *S* for sucrose, and *I* for inversion sugar in the mass fraction subscripts.)

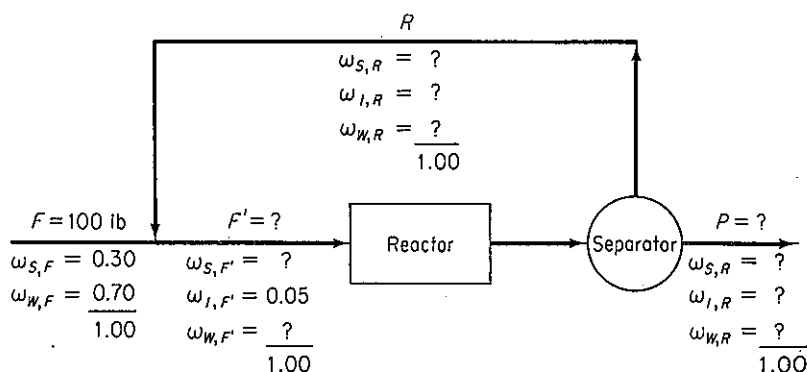


Figure E2.27b

Step 4

Basis: $F = 100$ lb

Steps 5 and 6 Keeping in mind that the sum of the mass fractions in each stream is unity, we have six unknown values of the variables, say $\omega_{S,F'}$, $\omega_{S,R}$, $\omega_{I,R}$, R , F' , and P .

Steps 7, 8, and 9 Let us start with the necessary balances. Only the total balance in the overall balance is directly useful because the *S* and *I* balances involve the generation and consumption terms in Eq. (2.12).

Overall:

$$\text{Total: } 100 = P$$

Mixing Point:

$$\text{Total: } 100 + R = F' \quad (\text{a})$$

$$\text{Sucrose: } 100(0.30) + R\omega_{S,R} = F'\omega_{S,F'} \quad (\text{b})$$

$$\text{Inversion: } 0 + R\omega_{I,R} = F'(0.05) \quad (\text{c})$$

Reactor plus Separator To avoid calculating the reactor output stream properties, we will make the system the reactor plus the separator. First we need to calculate the pounds of water consumed in the reaction per pound of sucrose consumed in the reaction.

1 mole sucrose uses 1 mole water

$$\frac{1 \text{ mol } W}{1 \text{ mol } S} \mid \frac{1 \text{ mol } S}{342.35 \text{ lb } S} \mid \frac{18 \text{ lb } W}{1 \text{ mol } W} = 0.0526 \frac{\text{lb } W}{\text{lb } S}$$

Total: $F' = R + P = R + 100$ [redundant equation to (a)]

Water: $\begin{array}{ccc} \text{In} & \text{Out} & \text{Consumed} \end{array}$ (d)

$$F'(1 - 0.05 - \omega_{S,F'}) - (R + 100)(1 - \omega_{S,R} - \omega_{I,R}) - (F'\omega_{S,F'})(0.90)(0.0526) = 0$$

Sucrose: $\begin{array}{ccc} \text{In} & \text{Out} & \text{Consumed} \end{array}$ (e)

$$F'\omega_{S,F'} - (R + 100)\omega_{S,R} - (F'\omega_{S,F'})(0.90) = 0$$

We have five independent equations (a)–(e) that can be solved for the five unknown values of the variables listed in steps 5 and 6; the value of P is given by the overall total materials balance. Either by successive substitution of Eqs. (a)–(e) into each other or by use of a computer program (see Sec. 2.7), you can find

$$R = 20.9 \text{ lb}$$

$$\omega_{I,R} = \omega_{I,P} = 0.279$$

2.6-3 Bypass and Purge

Two additional commonly encountered types of process streams are shown in Fig. 2.19.

- (a) A **bypass** stream—one that skips one or more stages of the process and goes directly to another stage
- (b) A **purge** stream—a stream bled off to remove an accumulation of inerts or unwanted material that might otherwise build up in the recycle stream

A bypass stream is used to control the composition of a final exit stream from a unit by mixing the bypass stream and the unit exit stream in suitable proportions to obtain the desired final composition.

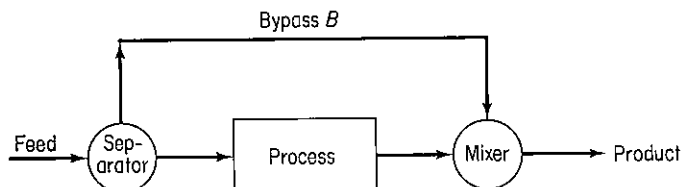


Figure 2.19a

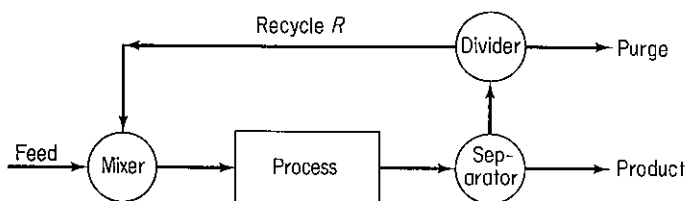
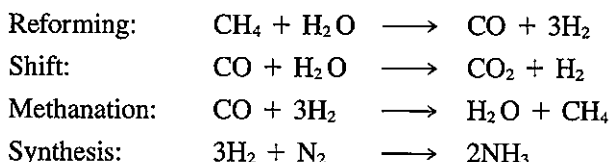


Figure 2.19b Recycle stream with purge.

As an example of the use of a purge stream, consider the production of NH_3 . Steam reforming, with feedstock natural gas, LPG, or naphtha, is the most widely accepted process for ammonia manufacture. The route includes four major chemical steps:



In the final stage, for the fourth reaction, the synthesis gas stream is approximately a 3:1 mixture of hydrogen to nitrogen, with the remainder about 0.9% methane and 0.3% argon.

Compressors step up the gas pressure from atmospheric to about 3000 psi—the high pressure that is needed to favor the synthesis equilibrium. Once pressurized and mixed with recycle gas, the stream enters the synthesis converter, where ammonia is catalytically formed at 400 to 500°C. The NH_3 is recovered as liquid via refrigeration, and the unreacted syngas is recycled.

In the synthesis step, however, some of the gas stream must be purged to prevent buildup of argon and methane. But purging causes a significant loss of hydrogen that could be used for additional ammonia manufacture, a loss that process designers seek to minimize.

Do you understand why the recycle process without a purge stream will cause an impurity to build up even though the recycle rate is constant? The purge rate is adjusted so that the amount of purged material remains below an acceptable specified economic level or so that the

$$\left\{ \begin{array}{c} \text{rate of} \\ \text{accumulation} \end{array} \right\} = 0 = \left\{ \begin{array}{c} \text{rate of entering material} \\ \text{and/or production} \end{array} \right\} - \left\{ \begin{array}{c} \text{rate of purge} \\ \text{and/or loss} \end{array} \right\}$$

Calculations for bypass and purge streams introduce no new principles or techniques beyond those presented so far. Two examples will make this clear.

EXAMPLE 2.28 Bypass Calculations

In the feedstock preparation section of a plant manufacturing natural gasoline, isopentane is removed from butane-free gasoline. Assume for purposes of simplification that the process and components are as shown in Fig. E2.28. What fraction of the butane-free gasoline is

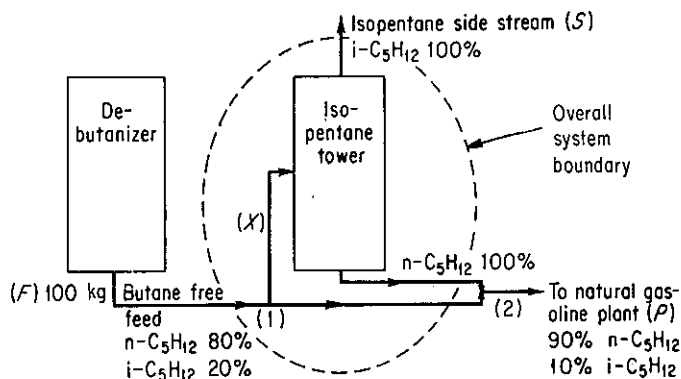


Figure E2.28

passed through the isopentane tower? Detailed steps will not be listed in the analysis and solution of this problem.

Solution

By examining the flow diagram you can see that part of the butane-free gasoline bypasses the isopentane tower and proceeds to the next stage in the natural gasoline plant. All the compositions are known. What kind of balances can we write for this process? We can write the following:

Basis: 100 kg feed

- (a) *Overall balances* (each stream is designated by the letter F , S , or P):

Total material balance:

$$\begin{array}{r} \text{In} \\ 100 \end{array} = \begin{array}{r} \text{Out} \\ S + P \end{array} \quad (a)$$

Component balance ($n\text{-C}_5$), tie component:

$$\begin{array}{r} \text{In} \\ 100(0.80) \end{array} = \begin{array}{r} \text{Out} \\ S(0) + P(0.90) \end{array} \quad (b)$$

Consequently,

$$P = 100 \left(\frac{0.80}{0.90} \right) = 88.9 \text{ kg}$$

$$S = 100 - 88.9 = 11.1 \text{ kg}$$

The overall balances will not tell us the fraction of the feed going to the isopentane tower; for this we need another balance.

- (b) *Balance around isopentane tower:* Let x = lb of butane-free gas going to isopentane tower and y be the $n\text{-C}_5\text{H}_{12}$ stream leaving the isopentane tower.

Total material balance:

$$\begin{array}{r} \text{In} \\ x \end{array} = \begin{array}{r} \text{Out} \\ 11.1 + y \end{array} \quad (c)$$

Component ($n\text{-C}_5$), a tie component:

$$x(0.80) = y \quad (d)$$

Consequently, combining (c) and (d),

$$x = 11.1 + 0.8x$$

$$x = 55.5 \text{ kg or the desired fraction is } 0.555$$

Another approach to this problem is to make a balance at mixing points (1) or (2). Although there are no pieces of equipment at those points, you can see that streams enter and leave the junction.

(c) Balance around mixing point (2):

$$\text{material into junction} = \text{material out}$$

Total material:

$$(100 - x) + y = 88.9 \quad (e)$$

Component (iso- C_5):

$$(100 - x)(0.20) + 0 = 88.9(0.10) \quad (f)$$

Equation (f) avoids the use of y . Solving yields

$$20 - 0.2x = 8.89$$

$$x = 55.5 \text{ kg as before}$$

EXAMPLE 2.29 Purge

Considerable interest exists in the conversion of coal into more convenient liquid products for subsequent combustion. Two of the main gases that can be generated under suitable conditions from insitu coal combustion in the presence of steam (as occurs naturally in the presence of groundwater) are H_2 and CO . After cleanup, these two gases can be combined to yield methanol according to the following equation

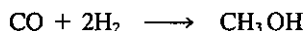


Figure E2.29 illustrates the process.

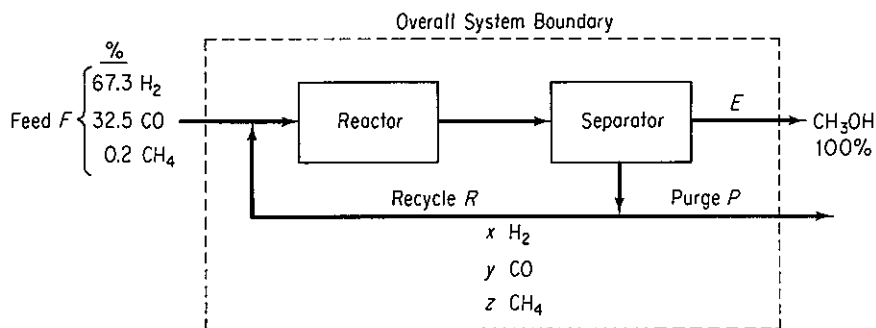


Figure E2.29

You will note in Fig. E2.29 that some CH_4 enters the process, but the CH_4 does not participate in the reaction. A purge stream is used to maintain the CH_4 concentration in the exit to the separator at no more than 3.2 mol%. The once-through conversion of the CO in the reactor is 18%.

Compute the moles of recycle, CH_3OH , and purge per mole of feed, and also compute the purge gas composition.

Solution

Steps 1, 2 and 3 Figure E2.29 is the sketch of the process. We will make balances about the whole process using the system designated by the dashed line, and also make a balance about the reactor plus separator to calculate the amount of recycle. Each of the stream flows has been labeled, and because the composition of the recycle and purge stream is not known, we have designated by x , y , and z , respectively, the mole fractions of H_2 , CO, and CH_4 . The ethanol stream is 100% CH_3OH ; the purge and recycle streams have the same compositions.

Step 4 Take a basis of 100 moles of feed.

Step 5 The unknown stream flows and mole fractions of the components are R , E , P , x , y , and z .

Step 6 We can make three independent elemental material balances for the overall process: H_2 , C, and O balances. A CO balance on the reactor plus separator will provide one additional balance. How can we obtain fifth and sixth balances so that the system of equations is determinate? One piece of information given in the problem statement that we have not used is the information about the upper limit on the CH_4 concentration in the purge stream. This limit can be expressed as $z \leq 0.032$. Let us make

$$z = 0.032 \quad (\text{a})$$

Another piece of information is the implicit balance,

$$x + y + z = 1 \quad (\text{b})$$

Steps 7 and 8 The overall balances are (in moles):

$$\text{H}_2: \quad 67.3 + 0.2(2) = E(2) + P(x + 2z) \quad (\text{c})$$

$$\text{C}: \quad 32.5 + 0.2 = E(1) + P(y + z) \quad (\text{d})$$

$$\text{O}: \quad 32.5 = E(1) + P(y) \quad (\text{e})$$

For a system composed of the reactor plus the separator (chosen to avoid calculating the unknown information about the reactor direct output), the CO balance is

$$\begin{array}{ccc} & \text{In} & \text{Out} & \text{Consumed} \\ \text{CO:} & 32.5 + Ry & - y(R + P) & = (32.5 + Ry)(0.18) \end{array} \quad (\text{f})$$

Would a H_2 balance on the reactor plus separator yield any additional information not given in Eq. (f)? The balance would be

$$\text{H}_2: \quad 67.3 + Rx - x(R + P) = (67.3 + Rx)(0.18) \quad (\text{g})$$

Experienced engineers would say that the hydrogen balance is redundant. You can verify this conclusion by looking at the coefficient matrix of Eqs. (f) and (g):

$$0.82(32.5 + Ry) = (R + P)y = Ry + Py$$

$$0.82(67.3 + Rx) = (R + P)x = Rx + Px$$

or

$$0.82(32.5) = (0.18y)R + (y)P$$

$$0.82(67.3) = (0.18x)R + (x)P$$

What is the rank of

$$M = \begin{bmatrix} 0.18y & y \\ 0.18x & x \end{bmatrix}$$

The $\det [M] = 0.18yx - 0.18yx = 0$, hence the rank of M is one, and Eqs. (f) and (g) are not independent.

Step 9 Equation (a) can be substituted into Eqs. (b)–(f) and the resulting five equations solved by successive substitution or by using one of the computer programs on the disk in the pocket in the back of this book. The resulting values obtained are (in moles)

E	CH_3OH	31.25
P	purge	6.25
R	recycle	705
x	H_2	0.768
y	CO	0.200
z	CH_4	0.032

Step 10 Check to see that each of the balances (b)–(f) is satisfied.

Up to now we have discussed material balances of a rather simple order of complexity. If you try to visualize all the calculations that might be involved in even a moderate-sized plant, as illustrated in Fig. 2.4, the stepwise or simultaneous solution of material balances for each phase of the entire plant may seem to be a staggering task, but is a task that can be eased considerably by the use of computer codes as discussed in Sec. 2.7. Keep in mind that a plant can be described by a number of individual, interlocking material balances each of which, however tedious they are to set up and solve, can be set down according to the principles and techniques discussed in this chapter. In application there is always the problem of collecting suitable information and evaluating its accuracy, but this matter calls for detailed familiarity with any specific process and is not a suitable topic for discussion here. We can merely remark that some of the problems you will encounter have such conflicting data or so little useful data that the ability to perceive what kind of data are needed is the most important attribute you can bring to bear in their solution.

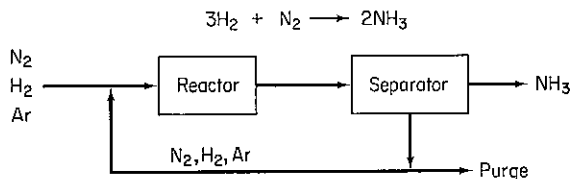
Self-Assessment Test

1. Explain what recycle and bypassing involve by means of words and also by a diagram.
2. Repeat for the term "purge."
3. If the components in the feed to a process appear in stoichiometric quantities and the subsequent separation process is complete so that all the unreacted reactants are recycled, what is the ratio of reactants in the recycle stream?

4. A material containing 75% water and 25% solid is fed to a granulator at a rate of 4000 kg/hr. The feed is premixed in the granulator with recycled product from a dryer which follows the granulator (to reduce the water concentration of the overall material in the granulator to 50% water, 50% solid). The product that leaves the dryer is 16.7% water. In the dryer, air is passed over the solid being dried. The air entering the dryer contains 3% water by weight (mass), and the air leaving the dryer contains 6% water by weight (mass).
- (a) What is the recycle rate to the granulator?
- (b) What is the rate of air flow to the dryer on a dry basis?
5. In the famous Haber process to manufacture ammonia, the reaction is carried out at pressures of 800 to 1000 atm and at 500 to 600°C using a suitable catalyst. Only a small fraction of the material entering the reactor reacts on one pass, so recycle is needed. Also, because the nitrogen is obtained from the air, it contains almost 1% rare gases (chiefly argon) that do not react. The rare gases would continue to build up in the recycle until their effect on the reaction equilibrium would become adverse so that a small purge stream is used.

The fresh feed composed of 75.16% H_2 , 24.57% N_2 , and 0.27% Ar is mixed with the recycled gas and enters the reactor with a composition of 79.52% H_2 . The gas leaving the ammonia separator contains 80.01% H_2 and no ammonia. The product ammonia contains no dissolved gases. Per 100 moles of fresh feed:

- (a) How many moles are recycled and purged?
- (b) What is the percent conversion of hydrogen per pass?

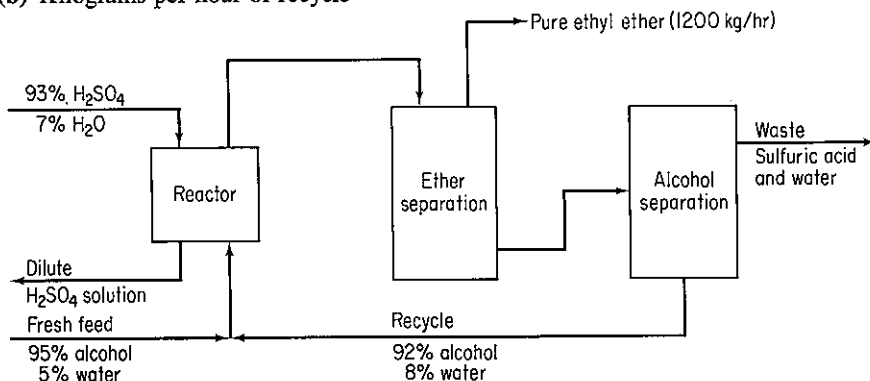


6. Ethyl ether is made by the dehydration of ethyl alcohol in the presence of sulfuric acid at 140°C:



A simplified process diagram is shown below. If 87% conversion of the alcohol fed to the reactor occurs per pass in the reactor, calculate:

- (a) Kilograms per hour of fresh feed
- (b) Kilograms per hour of recycle



2.7 SOLVING MATERIAL BALANCE PROBLEMS WITH THE AID OF A COMPUTER

Your objectives in studying this section are to be able to:

1. Locate computer programs that will solve coupled sets of linear and/or nonlinear equations simultaneously.
2. Use such programs for which you have access.

The great potential of computers is their capacity to do anything that can be described mathematically as a series of operations and logical decisions—theoretically. From a practical viewpoint, we need to ask ourselves not merely whether it is feasible for a task to be performed on a computer, but whether it is sensible. Two criteria need to be applied to reach a decision: (1) can the task be performed (or the problem solved) at all without the use of a computer; and (2) is it better or cheaper or faster to use a computer to solve a problem than a hand-held calculator (or no machine at all)? To decide whether it is better or cheaper or faster to use a computer, you must consider, among other factors, the necessary investment of effort, time, and money. Is the problem to be solved just one time or many times? Is the accuracy of the solution enhanced through the use of the computer? Do you have a computer handy with the right program stored to solve the problem, or do you have to code an algorithm and debug it first? Answers to these and other questions will guide your choice of tools.

In this section we examine briefly how equation solvers, generic codes, spreadsheets, and flowsheeting codes can be used to solve material (and energy) balances.

2.7-1 Equation-Solving Programs

You can use inexpensive *equation-solving programs* available for microcomputers to solve linear and nonlinear material balances. Such codes may actually run faster than spreadsheet programs but not as fast as BASIC or Fortran programs. They are faster to program than BASIC or Fortran programs. A typical example of an equation-solving package is TK Solver Plus,⁵ which contains the following sheets (modules):

1. The Variable Sheet, where you name and define variables to be used in your equations and models.
2. The Rule Sheet, where you define relationships among variables.
3. The Function Sheet, which contains a summary of all user-defined functions that are part of a model.
4. The Unit Sheet, where you enter rules (if needed) for the conversion of calculation and display units.

⁵ A trademark of Universal Technical Systems, Inc., 1220 Rock St., Rockford, IL 61101.

5. The List Sheet, which summarizes all lists that have been created in the model.
6. The Plot Sheet, where you define the type and structure of plots of the model variables and relationships.
7. The Table Sheet, which summarizes all tables created in the model.
8. The Numeric Format Sheet, which lets you set numeric conventions such as number of decimal places, monetary format, etc.
9. The Global Sheet, which controls settings and defaults in effect throughout the model. Figure 2.20 illustrates two sheets.

<3u> Unit:
25+/OK

VARIABLE SHEET

St	Input	Name	Output	Unit	Comment
3		X1		Mass fr.	
56		X2		Mass fr.	
33.3		F		lb	
		P		lb	

/ List options

List solve

Block solve

Put to element

Get from element

RULE SHEET

S Rule

Mass balances on unit 1

$F \cdot X1 + 200 \cdot 0.777 = P \cdot X2$

$F \cdot X3 + 200 \cdot 0.223 = P \cdot X4$

F1 Help F2 Cancel F5 Edit F9 Solve / Commands = Sheets ; Window switch

Figure 2.20 TK Solver Plus lets you view any two of its nine worksheets onscreen at once. Shown here are the Variable Sheet (top) and the Rule Sheet (bottom). After known values are entered into the Variable Sheet, the ! key is pressed to start the calculations. Once the solution is found, the values of the unknown variables appear in the output column spaces in the Variable Sheet.

Sheets generally have predefined row and column titles, guiding you as to what information should be placed in which cells.

All sheets but the Unit Sheet may have subsheets. The Plot Sheet, for example, may have a Line Chart, Bar Chart, or Pie Chart Subsheet, or all three at once. These subsheets in turn can be based on List Subsheets, on which the actual data values are entered. You can add text comments to any statement or expression by enclosing the comment in quotes.

Models (for our purposes material balances) are built on the Rules Sheet; everything else is support structure. A model's rules are just equations, which may be entered in any number and sequence. Equations are written in ordinary algebraic form, and the calculations can involve table lookup, user-defined functions, and so on.

TK Solver has two methods of solving equations. (1) The Direct Solver is just what it sounds like: values are substituted for variables, and both sides of an expres-

sion are calculated directly. (2) The Iterative Solver is used for nonlinear or simultaneous equations that cannot be evaluated directly. Iterative solving begins with a user-generated guess as to what a value might be. The software then uses this approximate value to generate a new guess that comes closer to the solution, and so on until a desired solution is reached. You do not have to choose the method of solution; the program makes the choice for you.

A number of iterative methods exist, as described in Appendix L. TK Solver uses a modified Newton–Raphson iterative procedure (see Sec. L.2), which is satisfactory for a wide variety of problems.

TK Solver Plus has 71 built-in functions, which can be used in rules, statements, and expressions. Built-in functions range from the mathematical constants π and e (natural log) through trigonometric, logarithmic, hyperbolic, mathematical, Boolean, and complex functions. Complex functions include such things as “Power,” which returns real and imaginary values of a complex number raised to some power, and “Ptord,” which returns the real and imaginary components of a complex number in rectangular form. The code does not let you use scientific notation such as the summation sign; instead, a command string has to be entered.

There are also several TK-specific functions, such as Element, which returns the value of a specified list element; Place, which puts a value into a specific place in a list; and Given, which returns one specific value if a particular variable is assigned an input value and another if the variable does not have an input value.

TK Solver Plus also provides a large set of library models (programs) that you can use as is or change for use in modified versions. Library models include: Roots of Equations, Differentiation and Integration, Differential Equations, Special Functions, Complex Variables, Optimization, Matrix Calculations, Arbitrary Length Integers, Interactive Tables, Graphics, Statistics and Curve Fitting, Engineering and Science (mostly chemistry), and Finance.

For example calculations, refer to Singh⁶ or the manual that comes with the calculation program.

2.7-2 Generic Computer Codes to Solve Sets of Equations

After you have transformed a word problem into a formal set of linear or nonlinear equations, there remains the question of how to solve the equations. Appendix L discusses several techniques to accomplish this task using general computer programs. From a classical viewpoint, two major categories of solution techniques exist: (1) direct or (2) interactive. For linear equations, the direct methods come for the most part from the well-known Gaussian elimination algorithm explained in Sec. L1. Gaussian methods are far more rapid than iterative methods, but computer time is not the only factor to take into account. Direct methods need considerable memory (in order to store the entire matrix and the right-hand side, both of which are constantly modified in the course of the calculations). Furthermore, no approximate solution can be obtained before the end of the calculations when you arrive at the (theoretically) exact solution.

⁶V. Singh, *Chem. Eng.*, p. 149 (February 17, 1986).

On the other hand, the situation is rather different in the case of iterative methods. If the system of equations is written in matrix form (see Sec. L.1) as $\mathbf{Ax} = \mathbf{b}$, the concept is to produce, starting with an initial guess x_0 , a sequence of vectors x_k converging toward the desired solution reasonably rapidly. Substitution methods (outlined in Sec. L.2 for nonlinear equations) are one way to implement the concept.

In the pocket in the back of this book are four simple Fortran generic codes that can be used to solve sets of linear and nonlinear equations on microcomputers or mainframes. Because these programs are simple, and thus may on occasion fail to solve your problem, you may want to use more polished computer codes that are available in your computer center software library. Such codes are more robust, but of course it takes more of your time to understand how to use them proficiently.

If you formulate a material balance problem that results in nonlinear equations, you should review Sec. L.2. There you will find an outline of techniques to solve sets of nonlinear equations, and also recommendations as to computer codes that are state of the art. In the pocket in the back of the text is a simple Fortran code to solve sets of nonlinear equations based on Newton's method and another one based on an optimization technique. You can use them to solve one or more nonlinear equations, and they will be effective for most of the problems in this book.

2.7-3 Spreadsheet Programs

Within the last few years, spreadsheet programs^{7,8} for microcomputers have achieved widespread acceptance. Spreadsheets were originally developed for business use as an electronic bookkeeping worksheet, hence they are a natural for solving material balance problems. They represent an easy-to-use, interactive visual program with user-friendly menus. A spreadsheet program can be applied to solving material balance problems without knowing a programming language. Furthermore, to enhance the capabilities of spreadsheet programs, numerous add-on products, such as word processors, three-dimensional graphics, general utilities, and communications links can be purchased.

A spreadsheet is nothing more than the storage and display on the monitor screen of a portion of a large two-dimensional matrix (array). It consists of numbered rows and columns identified by sequential letters. (Sometimes the rows have letters and the columns numbers.) Each cell or element in the array can be operated on by placing the cursor in it. The cursor can be moved from one cell to another by means of the arrow keys or a mouse. A cell can be used to store descriptive characters, a number (integer or real), or formulas containing standard mathematical functions and conditional relationships. The formulas may refer to data from any number of numeric or formula cells, and the results of calculations from applying such data to the data in the referenced cells appear in the formula cell. If the data in any cell are changed, the effect of the change is instantly propagated to the whole spreadsheet. This capability makes a spreadsheet well suited to study the effect of modifying one or more parameters in a material balance. Because process design is frequently broken down into problem definition, generation of alternative solutions,

⁷D. F. Cobb and G. T. LeBlond, *Using 1-2-3*, 2nd ed., Que., Indianapolis, Ind., 1985.

⁸D. Ewing, *1-2-3 Macro Library*, Que., Indianapolis, Ind., 1985.

evaluation of alternatives, selection of the recommended solution, and implementation of the selected alternative, the "what if" capabilities of spreadsheets are a powerful aid to decision making.

One advantage of spreadsheets is that the creation and execution of spreadsheets usually involves significantly less effort than writing and running a user-friendly program in BASIC or Fortran. Other advantages in engineering practice are low cost of execution, ease of execution, reasonable speed, and specific knowledge of the equations used in the calculations (in contrast to flowsheeting codes). Of course, the disadvantages are that spreadsheets cannot handle very complex material (and energy) balance problems more suitable for mainframe or minicomputers, and the speed is much less than Fortran programs executed on big computers. Furthermore, they are not particularly effective for the iterative solution of sets of simultaneous nonlinear equations or problems with many recycle streams.

Figure 2.21 illustrates a typical Lotus 1-2-3 template used to solve the problem in Example 2.21. Our purpose here is not to explain how to use your particular software package; refer to the documentation accompanying your disk. But to solve material balances without frustration, you need at the very least to be able to accomplish the following tasks:

- Open, close, save, and delete spreadsheet files
- Enter labels, data, and formulas that compute values for each cell
- Introduce into a formula in one cell values of a variable(s) or formulas from another cell(s)
- Copy the value of a variable or a formula in one cell into other cells
- Execute an iterative calculation using the spreadsheet function

Other features of spreadsheet codes you can use include:

1. *Macros*. A simple command language can be used to create a series of predetermined steps that the program will execute whenever ALT and an appropriate label key are pressed. Such a grouping of commands is called a macro. A macro can be used to extract data from a data base, to perform a series of mathematical operations (such as finding the roots of a nonlinear equation), or to prompt you for desired data changes.
2. *Window*. The window feature allows the screen to be doubled horizontally or vertically and enables you to view two otherwise widely separated areas of the spreadsheet at the same time. One window may be used as a spreadsheet work area and the other for displaying instructions.
3. *Graph*. Results of a problem can be displayed graphically on the monitor screen. Using the "Graph" feature, you can create XY plots, bar charts, pie charts, and histograms.
4. *Translate*. You can transfer data from the spreadsheet program to other application programs, and vice versa. For instance, ASCII text files from BASIC programs can be transferred to and from the spreadsheet program through appropriate commands.

A20:	Worksheet Range Copy Move File Print Graph Data Quit							MENU
	Global, Insert, Delete, Column-Width, Erase, Titles, Window, Status							
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		G	F	A	A*	W	RHS
H2		1.94	0.47	0	0	-1	0
N2		0	0	0.79	0.76	0	5269.91
O2		0.02	0	0.21	0.24	-0.5	935.09
S		0	0.03	0	0	0	6.205
C		1.02	0.5	0	0.04	0	672.62

G =	F =	A =	A* =	W =
0	0	0	0	0

CAPS

A20:	Worksheet Range Copy Move File Print Graph Data Quit							MENU
	Global, Insert, Delete, Column-Width, Erase, Titles, Window, Status							
	A	B	C	D	E	F	G	H
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		G	F	A	A*	W	RHS
H2		1.94	0.47	0	0	-1	0
N2		0	0	0.79	0.76	0	5269.91
O2		0.02	0	0.21	0.24	-0.5	935.09
S		0	0.03	0	0	0	6.205
C		1.02	0.5	0	0.04	0	672.62

G =	F =	A =	A* =	W =
499.15	206.83	5226.04	1501.76	1065.57

CAPS

Figure 2.21 Lotus 1 2 3 solution for Example 2.21.

Spreadsheet codes commonly use a method called the Gauss-Seidel iterative method to solve a set of linear equations. The procedure is roughly as follows. Take an arbitrary value (often zero) and use it to initialize one or several of the unknowns in order to break all of the recycle of information in the system of equations. Then compute the rest of the unknowns via simple substitution. What results is only an approximation to a solution because at least one unknown has been given an arbitrary value. Because of the recycle of information in the equations, however, you can compute the values of the unknown variables again, and in so doing, you will get a better guess to use in a second round of calculations, and so on. During each iteration, the values of the unknown variables tend to change less and less. After a certain number of iterations, they do not change any more. The process has converged,

and you have reached the solution. (Refer to Sec. L.2, under "Substitution," for the concept applied to nonlinear equations.)

To start out solving a material balance problem via a spreadsheet, you first must determine the size of the spreadsheet and its boundaries. Then you initialize all the points in that array (to zero, usually). Next, set aside a section of the spreadsheet with cells reserved for the known constants in the problem. You should label the constants across the page using the labeling features of your spreadsheet to avoid confusion. Then enter the constants.

You are now ready to enter the equations for the internal and boundary nodes. Some spreadsheets may start solving the equations as you enter them leading to all sorts of error messages. To avoid this outcome, define a constant in, say, cell A3 (first column, third row) to be zero, and multiply each equation by A3 as you enter it. Then, when you have completed entering all the equations, you need to solve the problem, change A3 from 0 to 1, and begin iteration. A formula is entered in a cell, but the formulas themselves are not displayed in the cell on the screen. What is displayed is the value given by the formula. For example, to show the product in cell D10 of the feed located in cell F24 and the concentration located in F25, you would enter into cell D10 the formula $F24 \cdot F25$. It is recommended that you set up the overall material balances about specific process units in addition to the equations entered on the main part of the spreadsheet. If you take this step, you can quickly check for errors in the setup of the balances. Also, as you enter equations and data, check interim calculations just as you would check out a computer code as it was written.

2.7-4 Flowsheeting Programs

In the 1960s, the chemical process industry initiated the use of large-scale programs for computer-aided process design. Such programs were frequently referred to as heat and material balancing programs, but now are generally called **flowsheeting programs**. Such programs accept information about a chemical process at the flowsheet level of detail, and make calculations that provide data about not only material and energy flows but also about costs, pipe layout, time effects, and other useful information for design and operation. Figure 2.22 illustrates the structure of a flowsheeting code. Such codes can simulate the steady-state (and unsteady-state in some cases) performance of large integrated chemical plants consisting of interconnected process units with recycle streams. Table 2.5 lists a number of flowsheeting codes and their sources.

Two extremes are encountered in flowsheeting software. At one extreme, the entire set of equations (and inequalities) representing the process is employed. This representation is known as the *equation-oriented* method of flowsheeting. The equations can be solved in a sequential fashion analogous to the modular representation described below or simultaneously by Newton's method, Broyden's method, or by employing sparse matrix techniques to reduce the extent of matrix manipulations. Refer to the review by Evans and Chapter 5.⁹

⁹L. B. Evans, "Process Flowsheeting," in Proc. *Chem comp.* 1982, G. F. Froment, ed., KVI, Amsterdam, 1982.

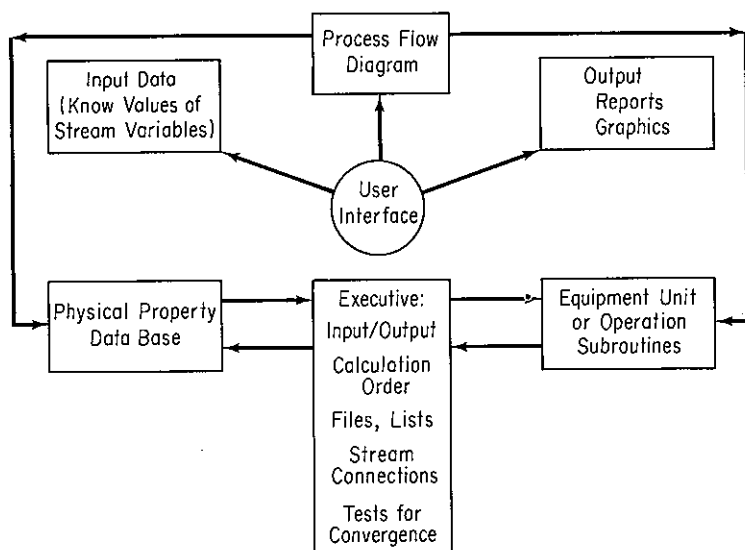


Figure 2.22 Structure of a Generic Flowsheeting Code.

At the other extreme, the process can be represented by a collection of modules (the *modular* method of flowsheeting) in which the equations (and other information) representing each subsystem or piece of equipment are collected together and coded so that the module may be used in isolation from the rest of the flowsheet and hence is portable from one flowsheet to another. Most of the flowsheeting codes of commercial importance are modular in character. Each module contains the equipment sizes, the material and energy balance relations, the rate equations, and allocated variables for component flow rates, and the temperatures, pressures, and phase conditions of each stream that enters and leaves the physical equipment represented by the module. Interconnections must be set up for the modules so that information can be transferred from module to module concerning the streams, compositions, flow rates, coefficients, and so on. In other words, the modules comprise a set of building blocks that can be arranged in general ways to represent any process. An executive routine calls the modules in the proper order, transmits information from a library of calculational subroutines, and picks out information on physical properties from an associated data base. Both sequential and simultaneous calculational sequences have been proposed for the modular approach as well as the equation-oriented approach (and intermediate mixtures of the two are possible as well). Either the program and/or the user must select the decision variables for recycle and provide estimates of certain stream values to make sure that convergence of the calculations occurs, especially in a process with many recycle streams.

Keep in mind that despite their portability, certain difficulties exist with modular codes relative to equation-based codes:

The output of one module is the input to another. The input and output variables in a computer module are fixed so that you cannot arbitrarily introduce an output and generate an input.

TABLE 2.5 Flowsheeting Codes

Name	Author	Reference	Number of components in data bank	Availability
ASPENPLUS	Aspen Technology	[H]	Large	Aspen Technology Corp., Cambridge, Mass.; original Aspen from National Technical Information Service, Springfield, Va.
CAPEX	Chiyoda Ltd.	[B]		Chiyoda Ltd., Yokohama, Japan
CHESS	R.L. Motard	[A]	98	Dept. Chem. Eng., Washington University, St. Louis, Mo.; microcomputer version from COADE, Houston, Tex.
CONCEPT	Computer Aided Design Center	[D]	105	Computer Aided Design Center, Cambridge, England.
DESIGN/2000	Chem Share	[I]		Chem Share, Houston, Tex.
FLOWTRAN	Various at Monsanto Co.	[C]	>200	Monsanto Co., St. Louis, Mo.
MPBII	Sood and Reklaitis	[J]		Dept. Chem Eng., Purdue University, West Lafayette, Ind.
PROCESS	Simulation Sciences	[E]	>600	Simulation Sciences, Fullerton, Calif.
SIMMOD	Chen and Stadtherr	[K]		Dept. Chem. Eng., University of Illinois at Urbana-Champaign, Urbana, Ill.
SPEEDUP	R. Sargent	[G]	100	British Technology Group, London
SYMBOL	Cambridge University	[F]	No 120	Computer Aided Design Center, Cambridge, England

TABLE 2.5 (cont.)

- [A] Motard, R.A., Dept. Chem. Eng., Washington University, St. Louis, Mo.
 [B] Chiyoda Ltd., Systems Eng., Section, *Outline of Capes*, Chiyoda Ltd., Yokohama, Japan, 1979.
 [C] Rosen, E.M., and A.C. Pauls, "Computer Aided Chemical Process Design The FLOWTRAN System." *Comput. & Chem. Eng.*, v.1, p. 11 (1977).
 [D] Tambynayagam, R.K.M., S.J. Banch, and P. Winter, "CONCEPT MK IV: A New Approach to Process Flowsheeting." *277th Event Eur. Fed. Chem. Eng. (CHEMPLANT '80)*, Sept. 3-5, 1980, Heviz, Hungary.
 [E] Brannock, N.F., V.S. Verneuil, and Y.L. Wang, "PROCESS Simulation Program—A Comprehensive Flowsheeting Tool for Chemical Engineers." *Comput. Chem. Eng.*, v.3, p. 329 (1979).
 [F] Computer Aided Design Center, *SYMBOL*, Cambridge, England.
 [G] Perkins, I.D., R. W.H. Sargent and S. Thomas in *Computer Aided Process Plant Design*, M.E. Leesley (Ed.), p. 566, Gulf Publishing Company, Houston (1982).
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 [J] Sood, M.K., and G.V. Reklaitis, "Material Balance Program—II," School of Chemical Engineering, Purdue University, West Lafayette, Ind., 1977.
 [K] Chen, H.S., and M.A. Stadtherr, "A Simultaneous-Modular Approach to Process Flowsheeting and Optimization: I Theory and Implementation," *AIChE J.* v.32, p. 184 (1984).

2. The modules may require a fixed precedence order of solution (i.e., the output of one module must become the input of another); hence, convergence may be slower than in an equation-solving code, and the computational costs may be high.
3. To specify a parameter in a module as a design variable, you have to place a control block around the module and adjust the parameter such that design specifications are met. This arrangement creates a loop. If the values of many design variables are to be determined, you might end up with several nested loops of calculation.

Consequently, the concept of **tearing** has evolved in connection with modular flowsheeting codes to solve material (and energy) balances. Tearing involves decoupling the interconnections between the modules so that sequential information flow takes place. Tearing is required because of loops of information created by recycle streams. What you do in tearing is to provide guesses for values of some of the unknowns (the **tear variables**), usually the recycle streams, and then calculate the values of the tear variables from the modular subroutines. These calculated values form new guesses, and so on, until the differences between the guessed values and the calculated values are sufficiently small.

We cannot go into all of the details of flowsheeting program construction and application here. Refer to Chapter 5 and the to supplementary references at the end of this chapter for further information. However, let us look at an example of a very simple process with recycle to compare the simultaneous equation and modular solution techniques just to illustrate the concepts.

EXAMPLE 2.30 Comparison of Simultaneous Equation and Modular Techniques of Solving Material Balances

Figure E2.30 illustrates a two-stage process with one recycle stream. The objective is to use the total material balances for the process to calculate the amount of the recycle stream R as a function of α , the fraction of A recycled. Use $\alpha = \frac{1}{3}$ and $\alpha = 0.90$ for the comparison.

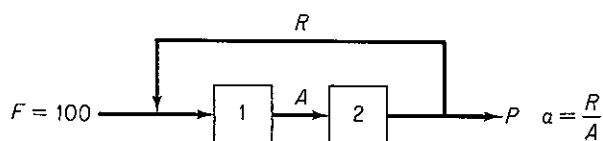


Figure E2.30

Solution

The material balances are

$$\text{Unit 1: } 100 + R = A \quad (a)$$

$$\text{Unit 2: } A = R + P \quad (b)$$

The fraction recycle is

$$A\alpha = R \quad (c)$$

We have three independent equations to be solved containing three variables whose values are unknown. Any of the methods mentioned in Appendix L might be used to solve the equations. Gaussian elimination starts with the format

$$A - R = 100 \quad (a')$$

$$A - R - P = 0 \quad (b')$$

$$\alpha A - R = 0 \quad (c')$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 100 \\ 1 & -1 & -1 & 0 \\ \alpha & -1 & 0 & 0 \end{array} \right]$$

and successive elementary operations result in the matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 100\left(\frac{1}{1-\alpha}\right) \\ 0 & 1 & 0 & 100\left(\frac{\alpha}{1-\alpha}\right) \\ 0 & 0 & 1 & 100 \end{array} \right]$$

leading to the solution

$$A = 100\left(\frac{1}{1-\alpha}\right)$$

$$R = 100\left(\frac{\alpha}{1-\alpha}\right)$$

$$P = 100$$

For the two given values of α ,

	$\alpha = \frac{1}{3}$	$\alpha = 0.9$
A	150	1000
R	50	900
P	100	100

Next, the modular approach to the solution of the problem would involve solving unit 1 for A first assuming a value for R , the tear variable. Then unit 2 would be solved, the value of R calculated, and the value calculated compared with the assumed value. If the error is not small enough, the new value of R from unit 2 would become the assumed value of R for unit 1, unit 1 solved again for A , and unit 2 solved again. The sequence of solutions would be repeated until the error in R became sufficiently small. Because R is initially not known, suppose that we start with $R = 0$ and $A = 100$ as the initial guesses. Thereafter

$$\text{Unit 1: } A_k = 100 + R_k \quad (d)$$

$$\text{Unit 2: } R_k = A_k \alpha \quad (e)$$

where k designates the stage in the iteration.

Successive calculations yield

k	$\alpha = \frac{1}{3}$			$\alpha = 0.9$		
	R	$A = 100 + R$	$R = A\alpha$	R	$A = 100 + R$	$R = A\alpha$
1	0	100	33.33	0	100	90
2	33.33	133.33	44.44	90	190	171
3	44.44	144.44	48.15	171	271	243.9
4	48.15	148.15	49.38	243.9	343.9	309.5
⋮	⋮	⋮	⋮	⋮	⋮	⋮
∞	50.00	150.00	50.00	900	1000	900

Insofar as this book is concerned, if you are not familiar with a flowsheeting code, you will find it will be more trouble to solve a material balance problem using a flowsheeting code than it will be to use a standard equation-solving code such as one of those in the disk pocket in the back of the book.

Self-Assessment Test

1. Solve one or two of the examples in Sec. 2.6 and 2.7 using
 - (a) A personal computer-based equation-solving code
 - (b) A code from the disk in the pocket in the back of this book
 - (c) A code taken from your computer center library
 - (d) A spreadsheet program
 - (e) A flowsheeting code

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PROBLEMS

An asterisk designates problems appropriate for solution using a computer. Refer also to the problem that requires writing a computer program at the end of the chapter.

Section 2.1

- 2.1. By use of an overall steady-state material balance determine whether or not the petrochemical process indicated in Fig. P2.1 has been properly formulated. The block diagrams represent the steam cracking of naphtha into various products, and all flows are on an annual basis (i.e., per year).
- 2.2. Examine Fig. P2.2. Is the material balance satisfactory?
- 2.3. Does the plant shown in Fig. P2.3 yield a material balance that balances? Explain a possible reason for the outcome of your calculations.
- 2.4. Water balances on river basins for a season or for a year can be used to check predicted groundwater infiltration, evaporation, or precipitation in the basin. Prepare a water balance, in symbols, for a large river basin and its reservoir, including the physical processes indicated in Fig. P2.4 (all symbols are for 1 year and S = storage or inventory).
- 2.5. Examine the processes in Fig. P2.5. Each box represents a system. For each, state whether:
 - (a) The process is in the (1) steady state, (2) unsteady state, or (3) unknown condition
 - (b) The system is (1) closed, (2) open, (3) neither, or (4) bothThe wavy line represents the initial fluid level when the flows begin. In case (c), the tank stays full.

Section 2.2

- 2.6. Make a set of all the independent material balances you can for the steady-state operation of the cooling tower in Fig. P2.6. Use F for the mass flow rate and ω for the

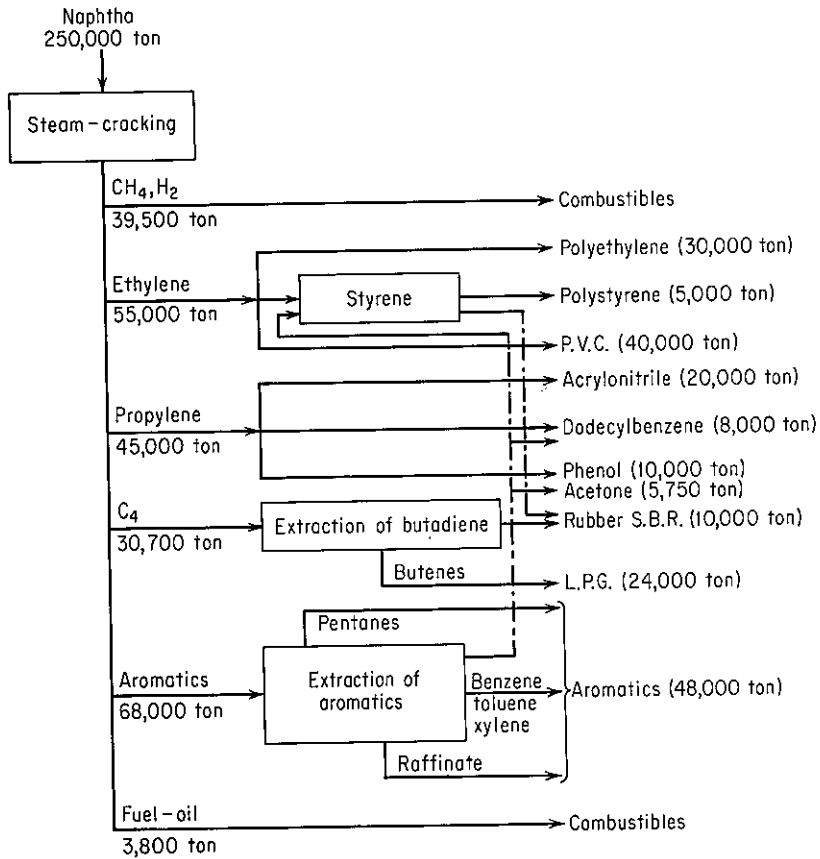


Figure P2.1

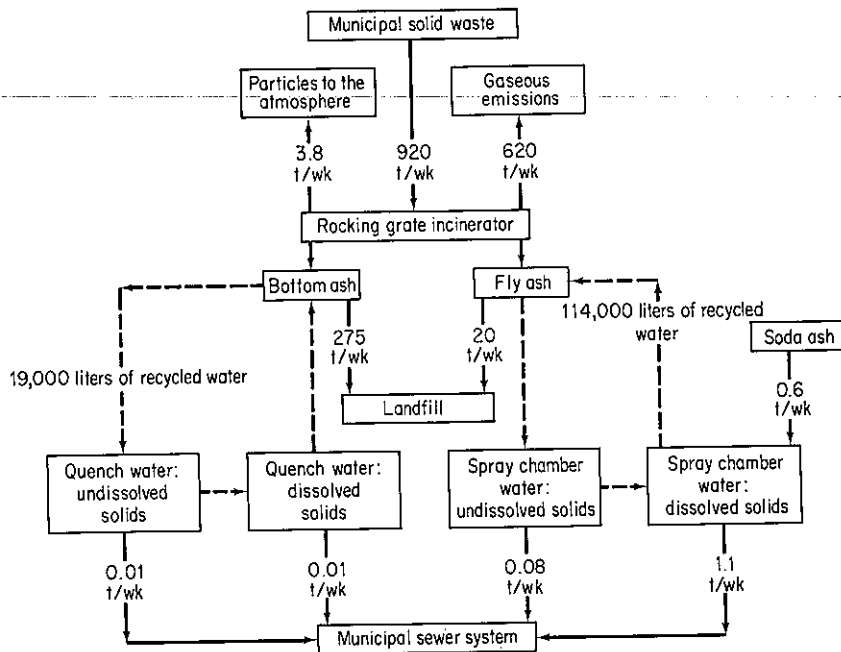


Figure P2.2

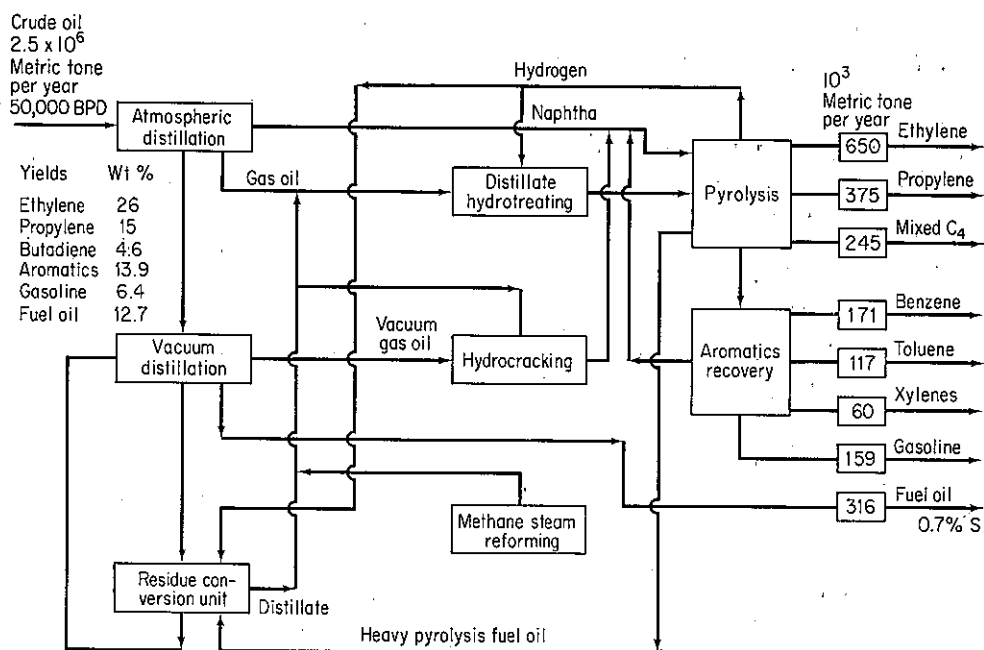


Figure P2.3

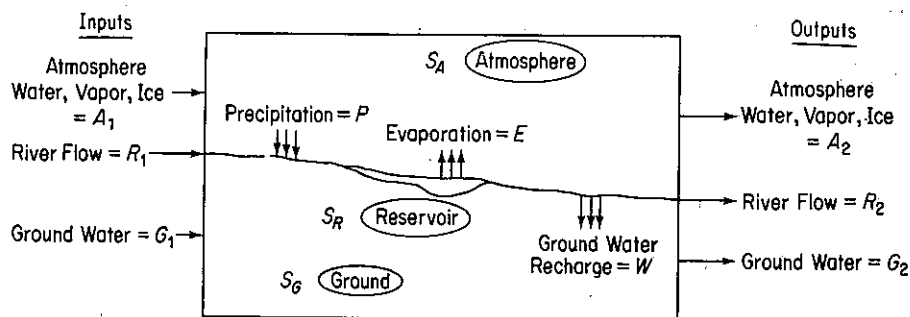


Figure P2.4

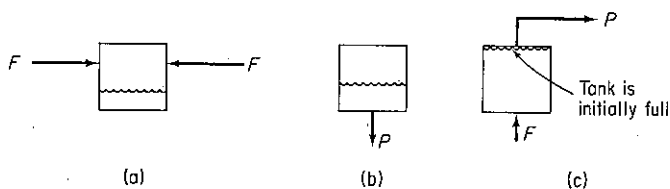


Figure P2.5

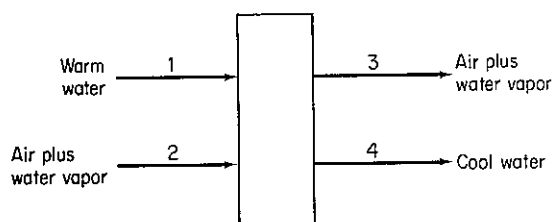


Figure P2.6

mass fraction, with the appropriate subscripts. Suppose that F was the molar flow rate. How would your equations change?

- 2.7. A cylinder containing CH_4 , C_2H_6 , and N_2 has to be prepared containing a mole ratio of CH_4 to C_2H_6 of 1.5 to 1. Available are (1) a cylinder of a N_2 - CH_4 mixture (80% N_2 , 20% CH_4), (2) a cylinder of 90% N_2 and 10% C_2H_6 , and (3) a cylinder of pure N_2 . List the total number and names (or symbols) of the unknown quantities and concentrations, and list by name the balances that you would use to solve this problem (do not make the balances). What is the residual number of degrees of freedom that have to be specified to make the problem completely determined (i.e., have a unique solution)?
- 2.8. For the process shown in Fig. P2.8, how many material balance equations can be written? Write them. How many independent material balance equations are there in the set?

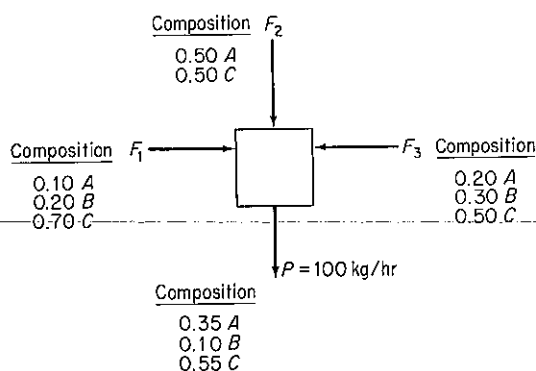


Figure P2.8

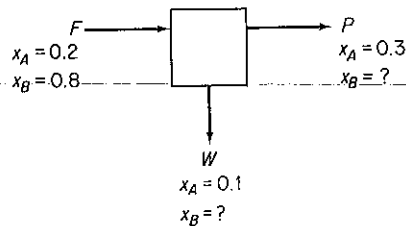


Figure P2.9

- 2.9. Examine the process in Fig. P2.9. No chemical reaction takes place, and x stands for mole fraction. How many stream values are unknown? How many concentrations? Can this problem be solved uniquely for the unknowns?
- 2.10. The separation of solids and liquids by washing is usually carried out in counter-current staged apparatus. One such stage is shown in Fig. P2.10. For this stage with the analyses as shown in the figure, (a) state how many unknown variables exist, (b) list them by symbol, (c) state how many independent material balance equations can be written for the stage, and (d) write each one down. (ω = mass fraction; the subscripts designate components.)
- 2.11. Repeat Problem 2.9 for the process in Fig. P2.11.

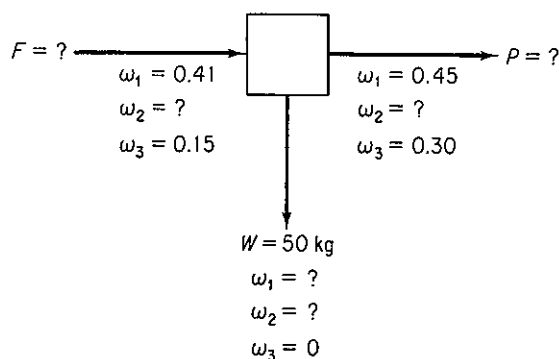


Figure P2.10

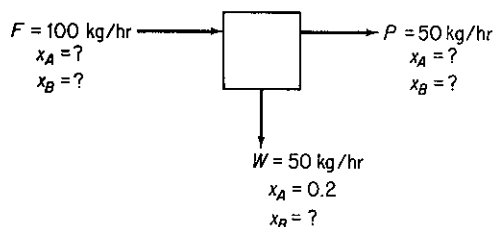


Figure P2.11

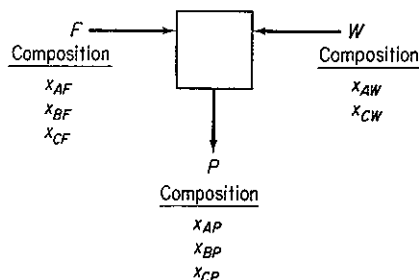


Figure P2.12

- 2.12. You have been asked to check out the process shown in Fig. P2.12. What will be the minimum number of measurements to make in order to compute the value of each of the stream flow rates and stream concentrations? Explain your answer. Can any arbitrary set of five be used; that is, can you measure just the three flow rates and two concentrations? Can you measure just three concentrations in stream F and two concentrations in stream W ? No chemical reaction takes place and x is the mole fraction of component A , B , or C .

Section 2.3

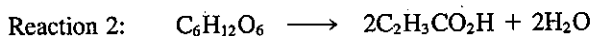
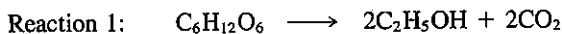
- 2.13. A synthesis gas analyzing 6.4% CO_2 , 0.2% O_2 , 40.0% CO , and 50.8% H_2 (the balance is N_2) is burned with 40% dry excess air. What is the composition of the flue gas?
- 2.14. Hydrogen-free carbon in the form of coke is burned:
- With complete combustion using theoretical air
 - With complete combustion using 50% excess air
 - Using 50% excess air but with 10% of the carbon burning to CO only
- In each case calculate the flue gas analysis and the Orsat analysis.
- 2.15. Thirty pounds of coal (analysis 80% C and 20% H ignoring the ash) are burned with 600 lb of air, yielding a gas having an Orsat analysis in which the ratio of CO_2 to CO is 3 to 2. What is the percent excess air?

- 2.16. An experimental solid rocket fuel consists of

H	5%
S	4%
C	65%
O	10%
Inert	<u>16%</u>
Total	100%

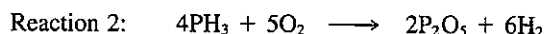
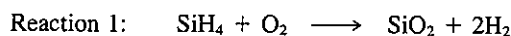
Assume on combustion that no C is left unburned and that the S is oxidized to SO_2 . The inert does not go into the gas phase.

- (a) If the fuel-air mixture calls for 20% excess air to oxidize the fuel, calculate the composition of the exit combustion gas.
- (b) If the same fuel were oxidized in a laboratory furnace with 20% excess air and only 80% of the C was oxidized to CO_2 , with 15% going to CO and 5% becoming soot (just C), what would the composition of the exit combustion gas be?
- 2.17. Pure methane is completely burned with air. The outlet gases from the burner, which contain no oxygen, are passed through a cooler, where some of the water is removed by condensation. The gases leaving the cooler have a nitrogen mole fraction of 0.8335. Calculate the following:
- (a) The analysis of the gases leaving the cooler (mol % CO_2 , H_2O , and N_2)
- (b) The pounds of H_2O condensed per mole of CH_4 burned
- (c) The average molecular weight of the gases leaving the cooler
- 2.18. The utilization of coal to produce gaseous fuels for heating and generating power as a supplement to the dwindling supplies of natural gas is a major research and development activity in the United States at the present time. Barriers to synthetic fuel production are more economic than technological, but there are still many technical problems to be resolved. In a test of pilot plant coal gasifier with a production rate of $25.8 \text{ m}^3/\text{hr}$, the output gas with a composition of 4.5% CO_2 , 26% CO, 13% H_2 , 0.5% CH_4 , and 56% N_2 is burned with 10% excess air. Calculate the Orsat analysis of the combustion products.
- 2.19. In the anaerobic fermentation of grain, the yeast *Saccharomyces cerevisiae* digests glucose from plants to form the products ethanol and propenoic acid by the following overall reactions:



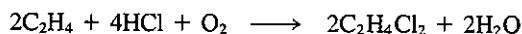
In a batch process, a tank is charged with 4000 kg of a 12% glucose/water solution. After fermentation 120 kg of carbon dioxide are produced together with 90 kg of unreacted glucose. What are the weight percents of ethyl alcohol and propenoic acid remaining in the broth? Assume that none of the glucose is assimilated into the bacteria.

- 2.20. Semiconductor microchip processing often involves chemical vapor deposition (CVD) of thin layers. The material being deposited needs to have certain desirable properties. For instance, to overlay on aluminum or other bases, a phosphorus pentoxide-doped silicon dioxide coating is deposited as passivation (protective) coating, by the simultaneous reactions



Determine the relative masses of SiH_4 and PH_3 required to deposit a film of 5% by weight of phosphorus (P) in the protective coating.

- 2.21. In many fermentations, the maximum amount of cell mass must be obtained. However, the amount of mass that can be made is ultimately limited by the cell volume. Cells occupy a finite volume and have a rigid shape so that they cannot be packed beyond a certain limit. There will always be some water remaining in the interstices between the adjacent cells, which represents the void volume that at best can be as low as 40% of the fermenter volume. Calculate the maximum cell mass on a dry basis that can be obtained if the wet cell density is 1.1 g/cm^3 . Assume the 40% void volume applies. Note that cells themselves consist of about 75% water and 25% solids, and cell mass is reported as dry weight in the fermentation industry.
- 2.22. Aviation gasoline is isooctane, C_8H_{18} . If it is burned with 20% excess air and 30% of the carbon forms carbon monoxide, what is the Orsat analysis?
- 2.23. Oxychlorination of hydrocarbons refers to a chemical reaction in which oxygen and hydrogen chloride react with a hydrocarbon in the vapor phase over a supported copper chloride catalyst to produce a chlorinated hydrocarbon and water. The oxychlorination of ethylene to produce 1, 2-dichloroethane (commonly, ethylene dichloride (EDC)) is of the greatest commercial importance. EDC is the precursor for vinyl chloride monomer, which when polymerized to polyvinyl chloride (PVC), becomes one of the most commonly used commercial plastics. The overall oxychlorination reaction of ethane is given by



Determine the mass, moles, and weight percent of the reactant and product streams if 200 kilogram moles of the limiting reactant, HCl , is fed with 10% excess air and 5% excess ethane. Ninety-five percent conversion of ethane occurs in the reactor.

- 2.24. Synthetic fibers are made in a wet spinning process by pumping a polymer solution (density ρ in g/cm^3 , solids fraction ω_s) at a volumetric rate of Q (cm^3/min) through a spinnerette with n holes. See Fig. P2.24. The filaments emerge into a bath containing both solvent and nonsolvent. Part of the solvent is lost from the filaments and an equivalent weight of nonsolvent is picked up; this solidifies the filaments. Next, the filaments are washed free of solvent as they are stretched by a factor S between two sets of rolls. Finally, the filaments are dried in a hot roll rotating at V_f feet per

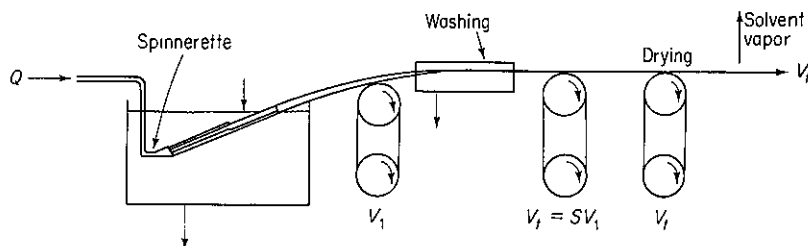


Figure P2.24

minute. Calculate the denier per filament (d.p.f.) (weight in grams of a filament 9000 m long) in terms of Q , n , ρ , ω_s , S (a constant), and V_i .

Section 2.4

- 2.25. To prepare a solution of 50.0% sulfuric acid, a dilute waste acid containing 28.0% H_2SO_4 is fortified with a purchased acid containing 96.0% H_2SO_4 . How many kilograms of the purchased acid must be bought for each 100 kg of dilute acid?
- 2.26. A synthetic fuel oil known to contain only H and C gives on combustion an Orsat analysis of CO_2 : 14.6%, CO : 2.0%, O_2 : 2.8%, and N_2 : 80.6%. Calculate the composition of the fuel oil.
- 2.27. A gas containing only CH_4 and N_2 is burned with air yielding a flue gas that has an Orsat analysis of CO_2 : 8.7%, CO : 1.0%, O_2 : 3.8%, and N_2 : 86.5%. Calculate the percent excess air used in combustion and the composition of the CH_4 - N_2 mixture.
- 2.28. A natural gas consisting entirely of methane (CH_4) is burned with an oxygen-enriched air of composition 40% O_2 and 60% N_2 . The Orsat analysis of the product gas as reported by the laboratory is CO_2 : 20.2%, O_2 : 4.1%, and N_2 : 75.7%. Can the reported analysis be correct? Show all calculations.
- 2.29. When pure carbon is burned in air, some of it oxidizes to CO_2 and some to CO . If the molar ratio of N_2 to O_2 is 7.18, and the ratio of CO to CO_2 is 2.0 in the product gas, what is the percent excess air used? The exit gases contain only N_2 , O_2 , CO , and CO_2 .
- 2.30. The products and by-products from coal combustion can create environmental problems if the combustion process is not carried out properly. A fuel analyzing 74% C, 14% H, and 12% ash is burned to yield a flue gas containing 12.4% CO_2 , 1.2% CO , 5.7% O_2 , and 80.7% N_2 on a dry basis. Your boss asks you to determine:
- The lb of coal fired per 100 lb mol of flue gas
 - The percent excess air used
 - The lb of air used per lb of coal
- Can you accomplish the calculations? Explain.
- 2.31. A fuel composed of ethane (C_2H_6) and methane (CH_4) in unknown proportions is burned in a furnace with oxygen-enriched air (50.0 mole % O_2). Your Orsat analysis is: 25% CO_2 , 60% N_2 , and 15% O_2 . Find:
- The composition of the fuel, that is, the mole percent methane in the methane-ethane mixture
 - The moles of oxygen-enriched air used per mole of fuel
- 2.32. A power company operates one of its boilers on natural gas and another on oil (for peak period operation). The analysis of the fuels is as follows:

Natural Gas	Oil
96% CH_4	$(\text{CH}_{1.8})_n$
4% CO_2	

When both boilers are on the line, the flue gas shows (Orsat analysis) 10.0% CO_2 , 4.5% O_2 , and the remainder N_2 . What percentage of the total carbon burned comes from the oil? (*Hint*: Do not forget the H_2O in the stack gas.)

- 2.33. A fuel oil and a sludge are burned together in a furnace with dry air. Assume that the fuel oil contains only C and H.

Fuel oil (%)	Sludge (%)		Flue gas (%)
	Wet	Dry	
C = ?	Water = 50	S = 32	SO ₂ = 1.52
	Solids = 50	C = 40	CO ₂ = 10.14
H = ?		H ₂ = 4	CO = 2.02
		O ₂ = 24	O ₂ = 4.65
			N ₂ = 81.67

- (a) Determine the weight percent composition of the fuel oil.
 (b) Determine the ratio of kilograms of sludge to kilograms of fuel oil.
- 2.34. A power company operates one of its boilers on natural gas and another on oil. The analyses of the fuels show 96% CH₄, 2% C₂H₂, and 2% CO₂ for the natural gas and C_nH_{1.8n} for the oil. The flue gases from both groups enter the same stack, and an Orsat analysis of this combined flue gas shows 10.0% CO₂, 0.63% CO, and 4.55% O₂. What percentage of the total carbon burned comes from the oil?
- 2.35. How much CaCl₂ · 6H₂O must be dissolved in 100 kg of water at 20°C to form a saturated solution? The solubility of CaCl₂ at 20°C is 6.70 g mol of anhydrous salt (CaCl₂) per kg of water.
- 2.36. One ton of a 30% (by weight) solution of Na₂CO₃ in water is cooled slowly to 20°C. During the cooling, 10% of the water originally present is evaporated. The carbonate precipitating out forms crystals of Na₂CO₃ · 10H₂O. If the solubility of anhydrous Na₂CO₃ at 20°C is 2.15 lb/100 lb H₂O, what weight of Na₂CO₃ · 10H₂O crystallizes out?
- 2.37. Crystals of Na₂CO₃ · 10H₂O are dropped into a saturated solution of Na₂CO₃ in water at 100°C. What percent of the Na₂CO₃ in the Na₂CO₃ · 10H₂O is recovered in the precipitated solid? The precipitated solid is Na₂CO₃ · H₂O. Data at 100°C: the saturated solution is 31.2% Na₂CO₃; the molecular weight of Na₂CO₃ is 106.
- 2.38. A water solution contains 60% Na₂S₂O₂ together with 1% soluble impurity. Upon cooling to 10°C, Na₂S₂O₂ · 5H₂O crystallizes out. The solubility of this hydrate is 1.4 lb Na₂S₂O₂ · 5H₂O/lb free water. The crystals removed carry as adhering solution 0.06 lb solution/lb crystals. These are dried to remove the remaining water (but not the water of hydration). The final dry Na₂S₂O₂ · 5H₂O crystals must not contain more than 0.1% impurity. To meet this specification, the original solution, before cooling, is further diluted with water. On the basis of 100 lb of the original solution, calculate:
 (a) The amount of water added before cooling
 (b) The percentage recovery of the Na₂S₂O₂ in the dried hydrated crystals
- 2.39. A mixed acid containing 65% (by weight) H₂SO₄, 20% HNO₃, and 15% H₂O is to be made by blending the following liquids:
 (1) A spent acid containing 10% HNO₃, 60% H₂SO₄, and 30% H₂O
 (2) A concentrated nitric acid containing 90% HNO₃ and 10% H₂O
 (3) Concentrated sulfuric acid containing 98% H₂SO₄ and 2% H₂O

How many pounds of each of the three must be used to obtain 1000 lb of the mixed acid?

- 2.40. An aqueous etching solution containing 8.8% KI is to be prepared to etch gold in printed circuit boards. The desired solution is to be formed by combining a strong solution (12% KI and 3% I_2 in H_2O) with a weak solution (2.5% KI and 0.625% I_2).
- (a) What should be the value of R , the ratio of the weights of the strong to the weak solution, to make up the desired etching solution? What will be the concentration of I_2 in the final solution?
 - (b) Note that you cannot independently vary the concentration of both KI and I_2 in the final mixture simply by varying the value of R . Derive a relationship between the weight fraction of KI and the weight fraction of I_2 in the mixture to illustrate this point.
- 2.41. The thermal destruction of hazardous wastes involves the controlled exposure of waste to high temperatures (usually 900°C or greater) in an oxidizing environment. Types of thermal destruction equipment include high-temperature boilers, cement kilns, and industrial furnaces in which hazardous waste is burned as fuel. In a properly designed system, primary fuel (100% combustible material) is mixed with waste to produce a feed for the boiler.
- (a) Sand containing 30% by weight of 4,4'-dichlorobiphenyl [an example of a polychlorinated biphenyl (PCB)] is to be cleaned by combustion with excess hexane to produce a feed that is 60% combustible by weight. To decontaminate 8 tons of such contaminated sand, how many pounds of hexane would be required?
 - (b) Write the two reactions that would take place under ideal conditions if the mixture of hexane and the contaminated sand were fed to the thermal oxidation process to produce the most environmentally satisfactory products. How would you suggest treating the exhaust from the burner? Explain.
 - (c) The incinerator is supplied with an oxygen-enriched airstream containing 40% O_2 and 60% N_2 to promote high-temperature operation. The exit gas is found to have a $x_{CO_2} = 0.1654$ and $x_{O_2} = 0.1220$. Use this information and the data about the feed composition above to find: (1) the complete exit gas concentrations and (2) the % excess O_2 used in the reaction.
- 2.42. A gas containing 20% N_2 and 80% CH_4 passes through a liquefaction process. Thirty weight percent of the entering gas is recovered and the composition of the recovered gas which is a liquid is 60% N_2 by weight. Find the composition of the non-recovered gas if 65 lb of entering gas is processed per minute. The temperature of the entering stream is 70°F; of the exit stream, 0°F. The pressures of all streams are about 1 atmosphere.
- 2.43. The organic fraction in the wastewater is measured in terms of the biological oxygen demand (BOD) material, namely the amount of dissolved oxygen required to biodegrade the organic contents. If the dissolved oxygen (DO) concentration in a body of water drops too low, the fish in the stream or lake may die. The Environmental Protection Agency has set the minimum summer levels for lakes at 5 mg/L DO.
- (a) If a stream is flowing at 0.3 m^3/s and has an initial BOD of 5 mg/L before reaching the discharge point of a sewage treatment plant, and the plant discharges 3.785 ML/day of wastewater, with a concentration of 0.15 g/L BOD, what will be the BOD concentration immediately below the discharge point of the plant?

- (b) The plant reports a discharge of 15.8 ML/day having a BOD of 72.09 mg/L. If the EPA measures the flow of the stream before the discharge point at 530 ML/day with 3 mg/L of BOD, and measures the downstream concentration of 5 mg/L BOD, is the report correct?
- 2.44. Printed circuit boards (PCBs) are used in the electronic industry to both connect and hold components in place. In production, 0.03 in. of copper foil is laminated to an insulating plastic board. A circuit pattern made of a chemically resistant polymer is then printed on the board. Next, the unwanted copper is chemically etched away by using selected reagents. If copper is treated with $\text{Cu}(\text{NH}_3)_4\text{Cl}_2$ (cupric ammonium chloride) and NH_4OH (ammonium hydroxide), the products are water and $\text{Cu}(\text{NH}_3)_4\text{Cl}$ (cuprous ammonium chloride). Once the copper is dissolved, the polymer is removed by solvents leaving the printed circuit ready for further processing. If a single-sided board 4 in. by 8 in. is to have 75% of the copper layer removed using the reagents above, how many grams of each reagent will be consumed?
- Data: The density of copper is 8.96 g/cm^3 .
- 2.45. A polymer blend is to be formed from the three compounds whose compositions and approximate formulas are listed in the table. Determine the percentages of each compound A, B, and C to be introduced into the mixture to achieve the desired composition.

Composition	Compound (%)			Desired mixture
	A	B	C	
$(\text{CH}_4)_x$	25	35	55	30
$(\text{C}_2\text{H}_6)_x$	35	20	40	30
$(\text{C}_3\text{H}_8)_x$	40	45	5	40
Total	100	100	100	100

How would you decide to blend compounds A, B, C, and D [$(\text{CH}_4)_x = 10\%$, $(\text{C}_2\text{H}_6)_x = 30\%$, $(\text{C}_3\text{H}_8)_x = 60\%$] to achieve the desired mixture?

- 2.46.* In a gas-separation plant, the feed-to-butane splitter has the following constituents:

Component	Mole %
C_3	1.9
<i>i</i> - C_4	51.6
<i>n</i> - C_4	46.0
C_5^+	0.6
Total	100.0

The flow rate is 5804 kg mol/day. If the overhead and bottoms streams from the butane splitter have the following compositions, what are the flow rates of the overhead and bottoms streams in kg mol/day?

Component	Mole %	
	Overhead	Bottoms
C ₃	3.4	—
<i>i</i> -C ₄	95.7	1.1
<i>n</i> -C ₄	0.9	97.6
C ₃ ⁺	—	1.3
Total	100.0	100.0

- 2.47. A low-grade pyrites containing 32% S is mixed with 10 lb of pure sulfur per 100 lb of pyrites so the mixture will burn readily, forming a burner gas that analyzes (Orsat) 13.4% SO₂, 2.7% O₂, and 83.9% N₂. No sulfur is left in the cinder. Calculate the percentage of the sulfur fired that burned to SO₃. (The SO₃ is not detected by the Orsat analysis.)
- 2.48. Methane is flowing in a pipeline at the rate of 20.7 m³/min at 30°C and 250 kPa. To check the flow rate, He is to be introduced at 25°C and 300 kPa. What is the minimum number of cm³ per minute of He must be introduced if the minimum detectable downstream concentration of the He in the methane is 10 ppm?
- 2.49. In the manufacture of vinyl acetate, there is some unreacted acetic acid and other compounds that are discharged to the sewer. Your company has been cited as discharging more than the specified limit of several pollutants. It is not possible to measure the discharge directly because no flow measuring devices are in place, but you can take samples of liquid at different places in the sewer line, and measure the concentration of potassium chloride. At one manhole, the concentration is 0.105%. You introduce a solution of 400 g of KCl in 1L at a steady rate of 1 L per minute over $\frac{1}{2}$ hr at a manhole 500 ft downstream, and at 1200 ft downstream measure the average steady-state concentration of KCl as 0.281%. What is the flow rate of fluid in the sewer in kg/min?
- 2.50. Trace components can be used to measure flow rates in pipelines, process equipment, and rivers that might otherwise be quite difficult to measure. Suppose that the water analysis in a flowing creek shows 180 ppm Na₂SO₄. If 10 lb of Na₂SO₄ is added to the stream uniformly over a 1-hr period and the analysis downstream where mixing is complete indicates 330 ppm Na₂SO₄, how many gallons of water are flowing per hour? See Fig. P2.50. Assume that no Na₂SO₄ is lost in the streambed or in the vegetation in the stream.

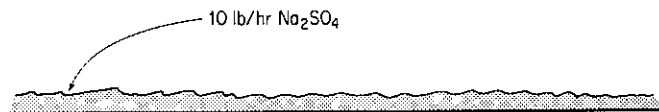


Figure P2.50

- 2.51. Your boss asks you to calculate the flow through a natural-gas pipeline. Since it is 26 in. in diameter, it is impossible to run the gas through any kind of meter or measuring device. You decide to add 100 lb of CO₂ per minute to the gas through a small $\frac{1}{2}$ -in. piece of pipe, collect samples of the gas downstream, and analyze them for CO₂. Several consecutive samples after 1 hr are:

Time	Percent CO ₂
1 hr, 0 min	2.0
10 min	2.2
20 min	1.9
30 min	2.1
40 min	2.0

- (a) Calculate the flow of gas in pounds per minute at the point of injection.
 (b) Unfortunately for you, the gas at the point of injection of CO₂ already contained 1.0% CO₂. How much was your original flow estimate in error (in percent)?
 [Note: In part (a) the natural gas is all methane, CH₄.]

- 2.52. The nose is an extremely sensitive detector of odors; it can be matched only by the most sensitive instrumental techniques. Many smells are identified by comparison with pure compounds diluted by odor-free air until they are near the odor threshold limit. However, with the introduction of flame photometric detectors it has been possible to measure the concentration of the sulfur compounds giving rise to odors in the ppb range with the aid of preconcentration (by a factor of 10^3 to 10^4).

The main offenders are known to be H₂S, mercaptans, and disulfides, which are present in crude-oil fractions. Because of their offensive smell and low olfactory levels, these compounds give rise to the majority of complaints. Typical detectable limits for the most important sulfur compounds are:

Compound	ppm
Hydrogen sulfide, H ₂ S	0.001–0.014
Methyl mercaptan, CH ₃ SH	0.001–0.0085
Ethyl mercaptan, C ₂ H ₅ SH	0.001–0.0026
Dimethyl sulfide, (CH ₃) ₂ S	0.002–0.0052

If the concentration of methyl mercaptan is measured as 0.035 ppm in a waste stream flowing at 1.7 m³/s (specific volume = 0.021 m³/kg mol), how many kilograms of air have to be bled into the waste stream per second to reduce the concentration of methyl mercaptan in the stream to below detectable levels?

- 2.53. The Clean Air Act requires automobile manufacturers to warrant their control systems as satisfying the emission standards for 50,000 mi. It requires owners to have their engine control systems serviced exactly according to manufacturers' specifications and to always use the correct gasoline. In testing an engine exhaust having a known Orsat analysis of 16.2% CO₂, 4.8% O₂, and 79% N₂ at the outlet, you find to your surprise that at the end of the muffler the Orsat analysis is 13.1% CO₂. Can this discrepancy be caused by an air leak into the muffler? (Assume that the analyses are satisfactory.) If so, compute the moles of air leaking in per mole of exhaust gas leaving the engine.
- 2.54. A dairy produces casein which when wet contains 23.7% moisture. They sell this for \$8.00/100 lb. They also dry this casein to produce a product containing 10% moisture. Their drying costs are \$0.80/100 lb water removed. What should be the selling price of the dried casein to maintain the same margin of profit?

Section 2.5

2.55. See Fig. P2.55

$$F_1 + F_2 - F_3 = 0 \quad (1)$$

$$F_3 - F_4 - F_5 = 0 \quad (2)$$

$$F_1 + F_2 - F_4 - F_5 = 0 \quad (3)$$

For Fig. P2.55 how many independent equations are obtained from the overall balance around the entire system [Eq. (3)], plus the balances on units A [Eq. (1)], and B [Eq. (2)]?

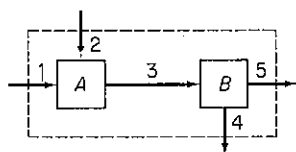
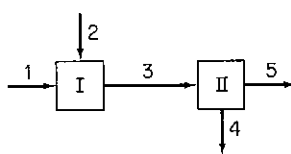


Figure P2.55



Steam composition (known)

- | | |
|---|----------------------|
| 1 | Pure A |
| 2 | Pure B |
| 3 | } Mixture of A and B |
| 4 | |
| 5 | |

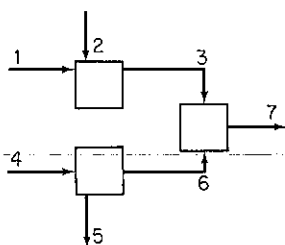
Steam flows (unknown)

Figure P2.56

2.56. What is the maximum number of independent material balances that can be written for the process in Fig. P2.56?

2.57. What is the maximum number of independent material balances that can be written for the process in Fig. P2.57? The stream flows are unknown.

Suppose you find out that A and B are always combined in each of the streams in the same ratio. How many independent equations could you write?



Steam composition (known)

- | | |
|---|----------------|
| 1 | A and B |
| 2 | Pure C |
| 3 | A, B, and C |
| 4 | D and E |
| 5 | Pure E |
| 6 | Pure D |
| 7 | A, B, C, and D |

Figure P2.57

2.58. The diagram in Fig. P2.58 represents a typical but simplified distillation column. Streams 3 and 6 consist of steam and water, and do not come in contact with the fluids in the column which contains two components. Write the total and component material balances for the three sections of the column. How many independent equations would these balances represent?

2.59. A distillation process is shown in Fig. P2.59. You are asked to solve for all the values of the stream flows and compositions. How many unknowns are there in the system? How many independent material balance equations can you write? Explain each answer and show all details whereby you reached your decision. For each stream, the only components that occur are shown below the stream.

2.60. Metallurgical-grade silicon is purified to electronic grade for use in the semiconductor industry by chemically separating it from its impurities. Si metal reacts in varying degrees with hydrogen chloride gas at 300°C to form several polychlorinated

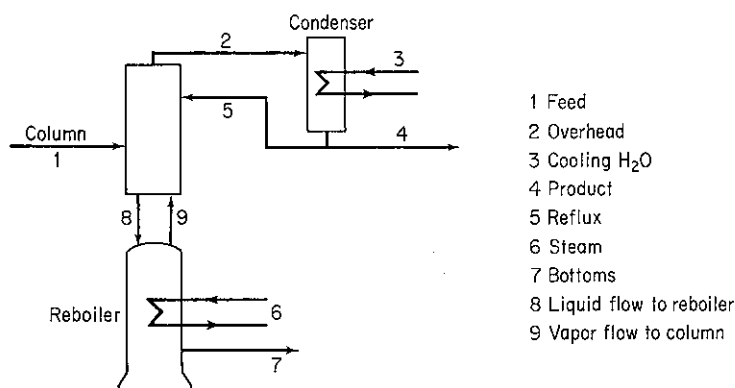


Figure P2.58

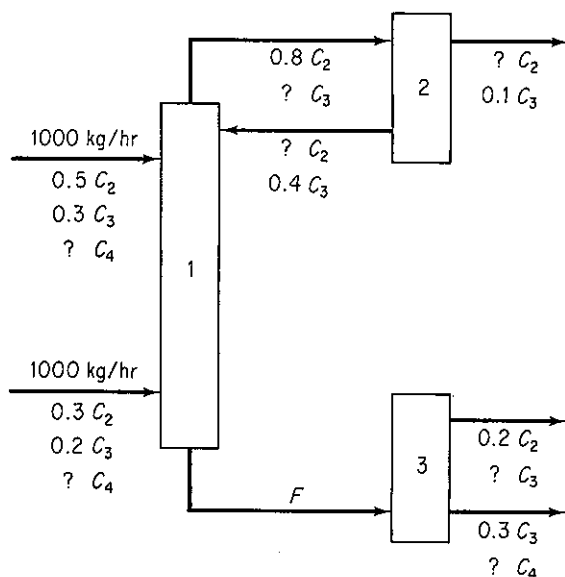


Figure P2.59

silanes. Trichlorosilane is liquid at room temperature and is easily separated by fractional distillation from the other gases. If 100 kg of silicon is reacted as shown in Fig. P2.60, how much trichlorosilane is produced?

- 2.61.*** Two major types of “gasifiers” for solid fossil fuels are available commercially at the present time. The occurrent or downdraft versions are used in small internal combustion engine applications for sawmills, remote communities, and to satisfy power needs for developing nations. A more recent advance is the fluidized-bed gasifier, which is appropriate for larger-scale conversion of biomass to “producer gas” or other products.

To take advantage of the most economical fuels, a company can burn producer gas (4.3% CO_2 , 27% CO , 10% H_2 , 1.0% CH_4 , and the residual N_2) generated from a fluidized-bed gasifier using coal waste in one boiler or No. 6 fuel oil (87% C,

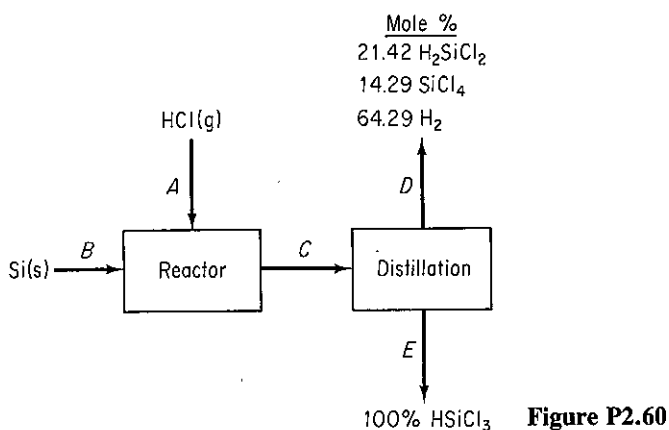


Figure P2.60

13% H_2) in a second boiler. One day both boilers are running and the Orsat analysis from the common stack is 8.2% CO_2 , 0.4% CO , and 10% O_2 and the residual N_2 .

- Calculate the fraction of the total carbon in the stack gas that comes from the fluidized-bed gasifier.
- How many moles of water would be produced in the gasifier per 100 moles of dry gas having the Orsat analysis above?
- How many total moles of air are fed to the process, and what is the overall effective percent excess air that was used?

- 2.62. In a distillation train a liquid hydrocarbon containing 20 mol % ethane, 40 mol % propane, and 40 mol % butane is to be fractionated into essentially pure components as shown in Fig. P2.62. On the basis of $F = 100$ moles, what is P (in moles) and the composition of stream A?

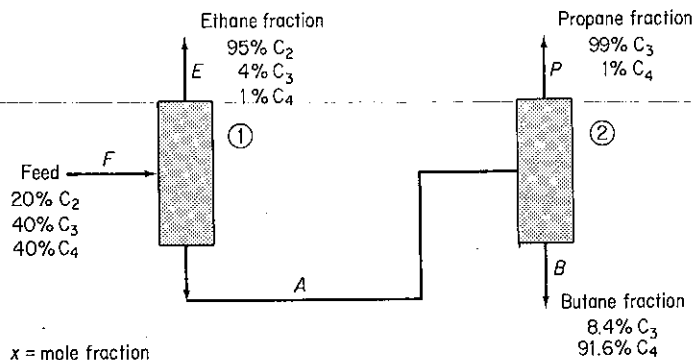


Figure P2.62

- 2.63.* A simplified flowsheet for the manufacture of sugar is shown in Fig. P2.63. Sugar-cane is fed to a mill where a syrup is squeezed out, and the resulting "bagasse" contains 80% pulp. The syrup (E) containing finely divided pieces of pulp is fed to a screen which removes all the pulp and produces a clear syrup (H) containing 15% sugar and 85% water. The evaporator makes a "heavy" syrup and the crystallizer produces 1000 lb/hr of sugar crystals.

- Find the water removed in the evaporator, lb/hr

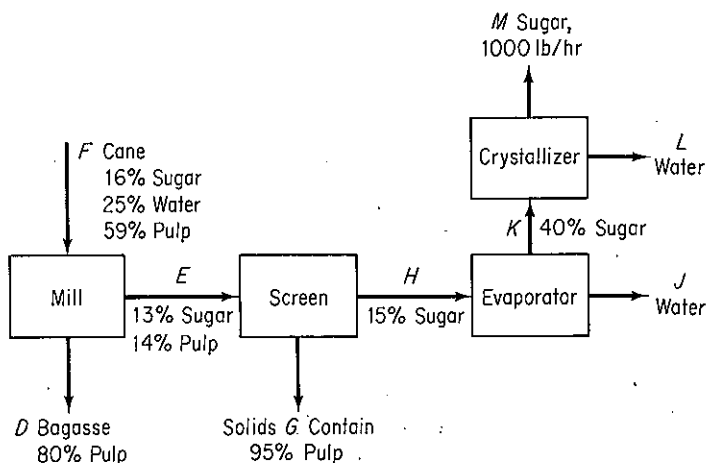


Figure P2.63

- (b) Find the composition of the waste stream G
 (c) Find the rate of feed of cane to the unit, lb/hr
 (d) Of the sugar fed in the cane, what percentage is lost with the bagasse?
 (e) Is this an efficient operation? Explain why or why not.
- 2.64.* Sodium hydroxide is usually produced from common salt by electrolysis. The essential elements of the system are shown in Fig. P2.64.
- (a) What is the percent conversion of salt to sodium hydroxide?
 (b) How much chlorine gas is produced per pound of product?
 (c) Per pound of product, how much water must be evaporated in the evaporator?

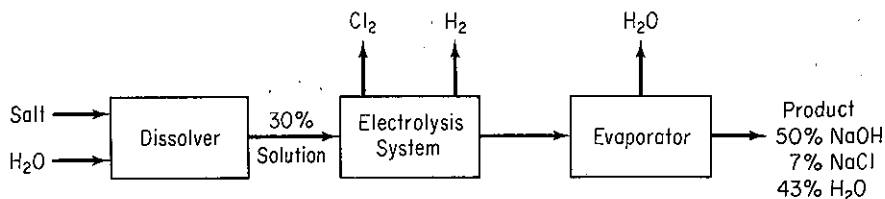


Figure P2.64

- 2.65.* The flowsheet shown in Fig. P2.65 represents the process for the production of titanium dioxide (TiO_2) used by Canadian Titanium Pigments at Varennes, Quebec. Sorel slag of the following analysis:

	Wt %
TiO_2	70
Fe	8
Inert silicates	22

is fed to a digester and reacted with H_2SO_4 , which enters as a 67% by weight

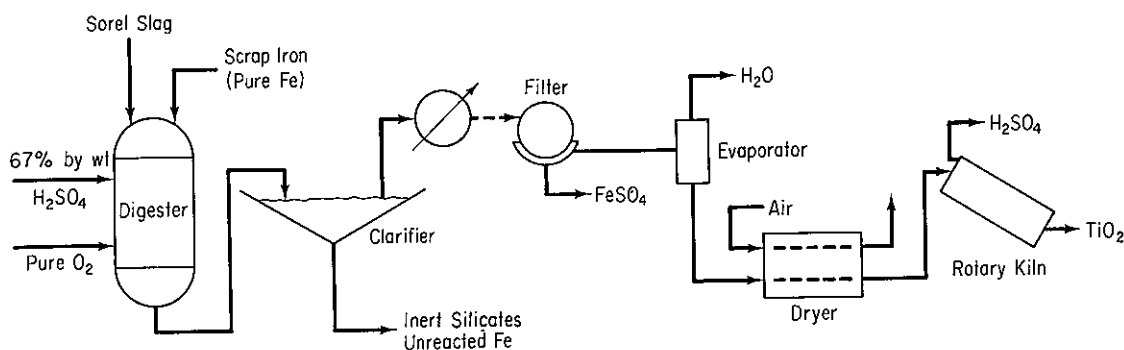
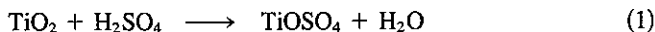


Figure P2.65

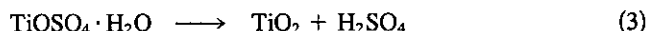
H_2SO_4 in a water solution. The reactions in the digester are as follows:



Both reactions are complete. The theoretically required amount of H_2SO_4 for the Sorel slag is fed. Pure oxygen is fed in the theoretical amount for all the Fe in the Sorel slag. Scrap iron (pure Fe) is added to the digester to reduce the formation of ferric sulfate to negligible amounts. Thirty-six pounds of scrap iron are added per pound of Sorel slag.

The products of the digester are sent to the clarifier, where all the inert silicates and unreacted Fe are removed. The solution of TiOSO_4 and FeSO_4 from the clarifier is cooled, crystallizing the FeSO_4 , which is completely removed by a filter. The product TiOSO_4 solution from the filter is evaporated down to a slurry that is 82% by weight TiOSO_4 .

The slurry is sent to a dryer from which a product of pure hydrate, $\text{TiOSO}_4 \cdot \text{H}_2\text{O}$, is obtained. The hydrate crystals are sent to a direct-fired rotary kiln, where the pure TiO_2 is produced according to the following reaction:

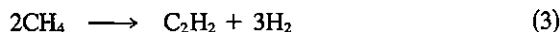
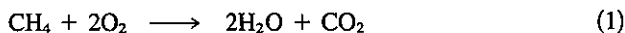


Reaction (3) is complete.

On the basis of 100 lb of Sorel slag feed, calculate:

- The pounds of water removed by the evaporator
- The exit lb of H_2O per lb dry air from the dryer if the air enters having 0.036 moles H_2O per mole dry air and the air rate is 18 lb mol of dry air per 100 lb of Sorel slag
- The pounds of product TiO_2 produced

2.66.* In the process for the production of pure acetylene, C_2H_2 , (see Fig. P2.66), pure methane (CH_4) and pure oxygen are combined in the burner, where the following reactions occur:



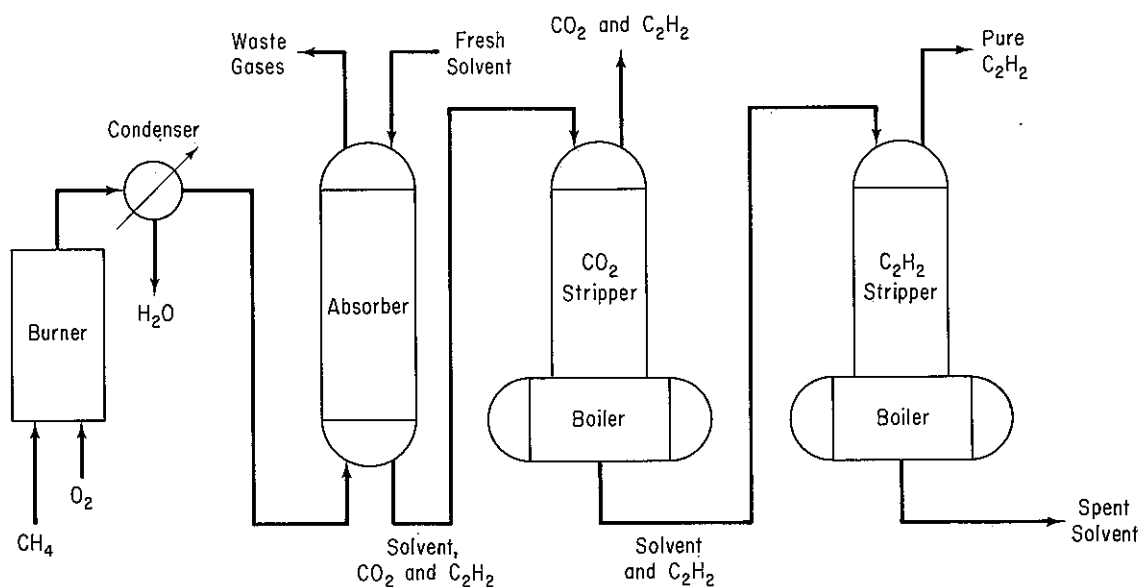


Figure P2.66

The gases from the burner are cooled in a condenser that removes all the water. The analysis of the gases leaving the condenser is as follows:

	Mol %
C_2H_2	8.5
H_2	25.5
CO	58.3
CO_2	3.7
CH_4	4.0
Total	100.0

These gases are sent to an absorber where 97% of the C_2H_2 and essentially all the CO_2 are removed with the solvent. The solvent from the absorber is sent to the CO_2 stripper, where all the CO_2 is removed. The analysis of the gas stream leaving the top of the CO_2 stripper is as follows:

	Mol %
C_2H_2	7.5
CO_2	92.5
Total	100.0

The solvent from the CO_2 stripper is pumped to the C_2H_2 stripper, which removes all the C_2H_2 as a pure product:

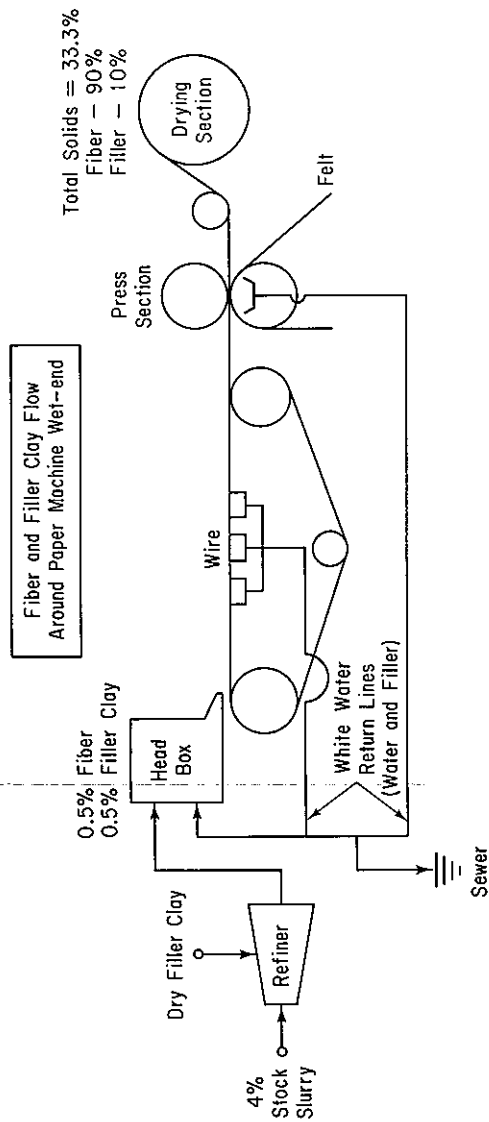
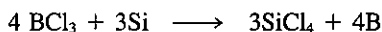


Figure P2.67

- (a) Calculate the ratio of the moles of O_2 to moles of CH_4 fed to the burner.
- (b) On the basis of 100 lb mol of gases leaving the condenser, calculate how many pounds of water are removed by the condenser.
- (c) What is the overall percentage yield of product (pure) C_2H_2 , based on the carbon in the natural gas entering the burner?
- 2.67. In a paper machine system (see Fig. P2.67), a stock suspension of cellulose fibers in water enters the continuous refining system at 4% consistency (4 lb of fiber per 100 lb of mixture). A dry filler clay is added in the refining system, and this slurry then goes to the paper machine headbox. The headbox distributes the suspension of cellulose fibers and filler clay uniformly over a wire belt in the form of a thin sheet. This wet sheet becomes the desired paper after it is dried. In the headbox, the slurry is diluted with a recycle stream of white water that passed through the wire belt during the formation of the sheet, to a fiber consistency of 0.5%. Analysis shows that the filler consistency is also 0.5% at this point. The damp sheet leaving the press section and entering the drying section, where water is removed by the application of heat, contains 33.3% solids consisting of 90% fiber and 10% filler. Any excess white water from the machine, not needed for headbox dilution, is discarded.
- Assuming no fiber loss through the wire and at the presses, calculate:
- (a) The overall percent filler retention on the wet sheet
- (b) The percent filler retention per pass (on the basis of the total filler that can be retained)

Section 2.6

- 2.68. Boron trichloride (BCl_3) gas can be fed into a gas stream and used for doping silicon. The simplest reaction (not the only one) is



If all the BCl_3 not reacted is recycled (see Fig. P2.68), what is the mole ratio of recycle to $SiCl_4$ exiting the separator? The conversion on one pass through the reactor is 87% and 1 mole per hour of BCl_3 is fed to the reactor.

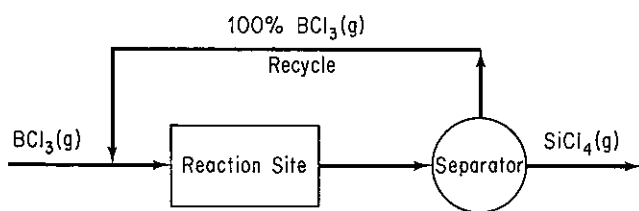


Figure P2.68

- 2.69. Examine Fig. P2.69. What is the quantity of the recycle stream in kg/hr?
- 2.70. The connections of computer components are often electroplated with relatively inert materials such as gold to prevent short circuiting by oxide buildup. Gold cyanide solutions have traditionally been used in batch processing. A process for continuous plating and replenishing of the electrolyte bath as shown in Fig. P2.70 has been patented (U.S. patent 4,340,451). If the throughput of components requires 30 g/hr of gold plating, how many cubic centimeters of water are consumed per hour?

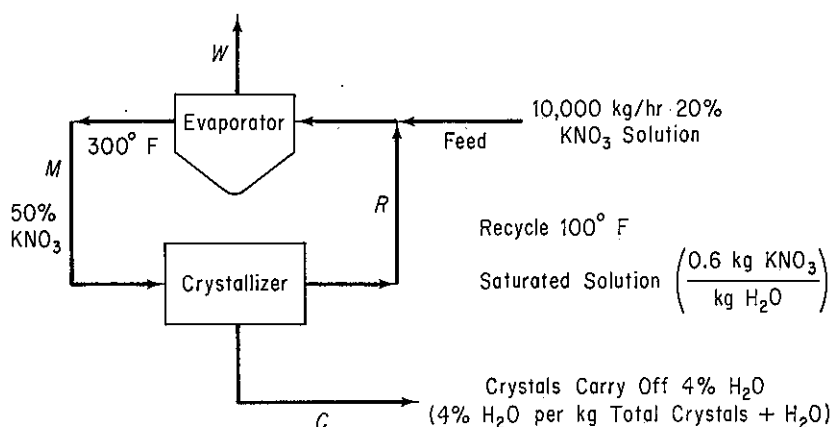


Figure P2.69

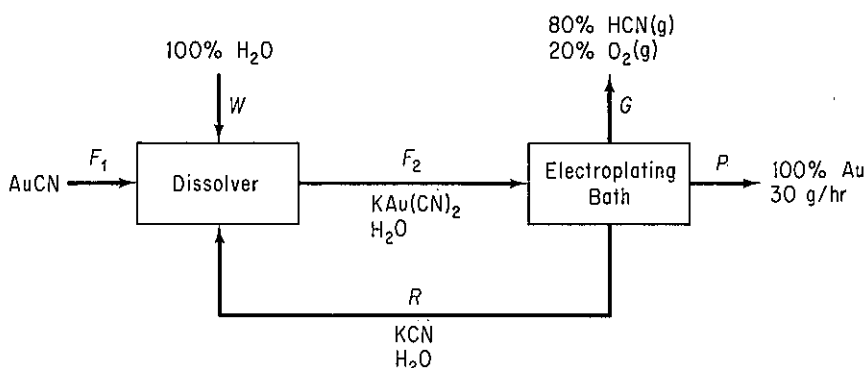


Figure P2.70

- 2.71.* The process shown in Fig. P2.71 is the catalytic dehydrogenation of propane to propylene. The composition and flow rate of the recycle stream are unknown. Certain data are known for the reactor, absorber, and distillation column as follows:

Reactor Effluent Composition Based on
100 kg mol of Propane Fed to Reactor

Component	Mol % in Product from reactor
Hydrogen	25.4
Methane	3.2
Ethylene	0.3
Ethane	5.3
Propylene	21.3
Propane	44.5
Total	100.0

Fractionator Performance

Component	Fraction of feed recovered in the overhead product
Methane	1.00
Ethane	1.00
Ethylene	1.00
Propylene	0.96
Propane	0.002

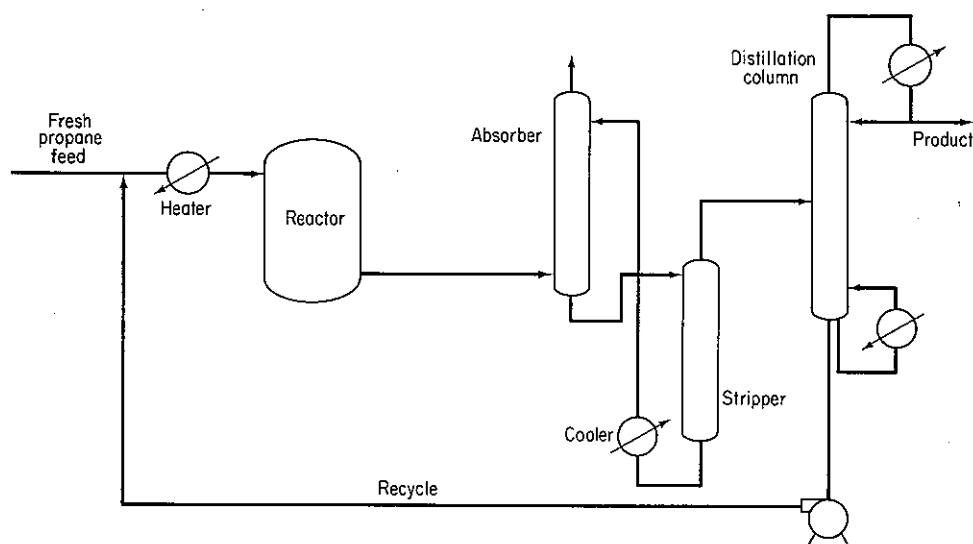


Figure P2.71

Absorber Performance Based
on 100 kg mol of Absorber Feed

Component	kg mol of absorber feed	Fraction of absorber feed not absorbed
Hydrogen	25.4	1.00
Methane	3.2	0.95
Ethylene	0.3	0.70
Ethane	5.3	0.60
Propylene	21.3	0.01
Propane	44.5	0.003
Total	100.0	

Assume that the source of all the hydrogen in the product from the reactor is the propane feed, and that any loss of mass in the reactor is due to carbon formation. Also assume that none of the hydrogen in the reactor product is absorbed in the absorber.

Calculate the flow rates in the reactor feed (after the heater), the reactor product stream, the absorber bottoms stream, the distillation column bottoms, and the recycle stream in kilograms per 100 kg of propane fed to the process. Can you calculate the composition of the recycle stream as well?

- 2.72. The rate of production of ethanol by fermentation is slowed by the increasing concentrations of ethanol formed in the process. A method is being investigated in which the ethanol is to be separated continuously in situ with solvent. The solvent is then pumped off and the ethanol recovered. How many moles/hr of ethanol are recycled in the solvent for the system shown in Fig. P2.72?

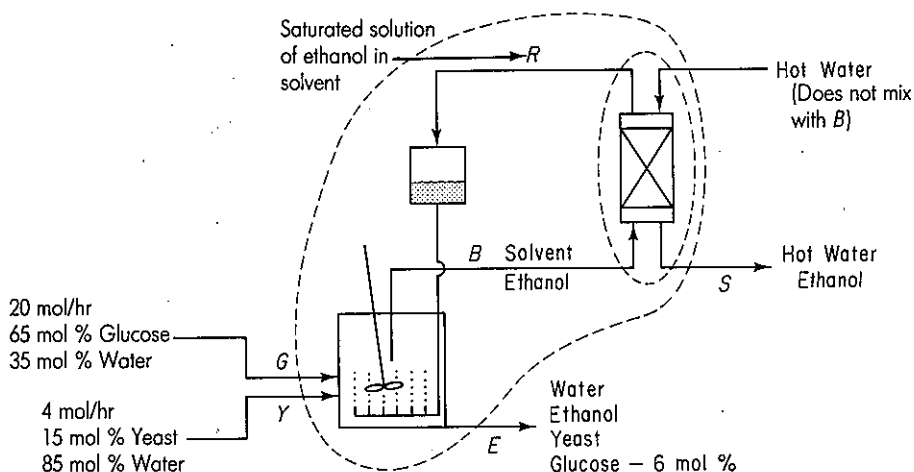


Figure P2.72

Data: glucose $\xrightarrow{\text{yeast}}$ 2 ethanol + 2CO₂

92% conversion of glucose

saturated solvent: 0.24 $\frac{\text{g EtOH}}{\text{g solvent}}$ in B

density of solvent: 0.84 g/cm³

saturated solvent flow: 4.5 L/hr

Assume the participation of the yeast in the reaction is negligible.

- 2.73. A waste treatment plant operating under steady-state conditions processes a waste liquor containing 500 parts per million (ppm) of a noxious impurity. The impurity is removed with practically no loss of the carrier fluid. The treating process can remove the impurity down to a level of 10 ppm. By local ordinance, a maximum of 100 ppm is allowed in the discharge of the processed effluent to a nearby river. What fraction of the waste liquor must be sent through the waste treatment unit, and what fraction may be bypassed?
- 2.74. Seawater is to be desalinated by reverse osmosis using the scheme indicated in Fig. P2.74. Use the data given in the figure to determine:
- The rate of waste brine removal (B)
 - The rate of desalinated water (called potable water) production (D)

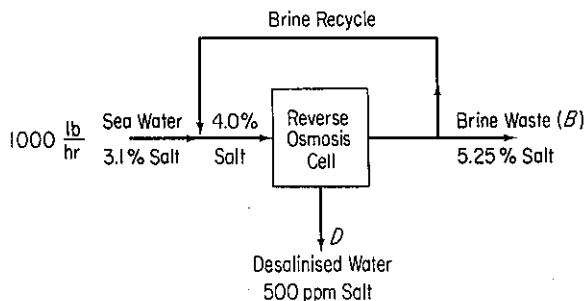


Figure P2.74

(c) The fraction of the brine leaving the osmosis cell (which acts in essence as a separator) that is recycled

(Note: ppm designates parts per million).

- 2.75. A solution containing 10% NaCl, 3% KCl, and 87% water is fed to the process shown in Fig. P2.75 at the rate of 18,400 kg/hr. The compositions of the streams are as follows in percent. Evaporator product, P: NaCl: 16.8, KCl: 21.6, H₂O: 61.6; Recycle, R: NaCl: 18.9 and the balance water. Calculate the kilograms per hour and complete compositions of every stream.

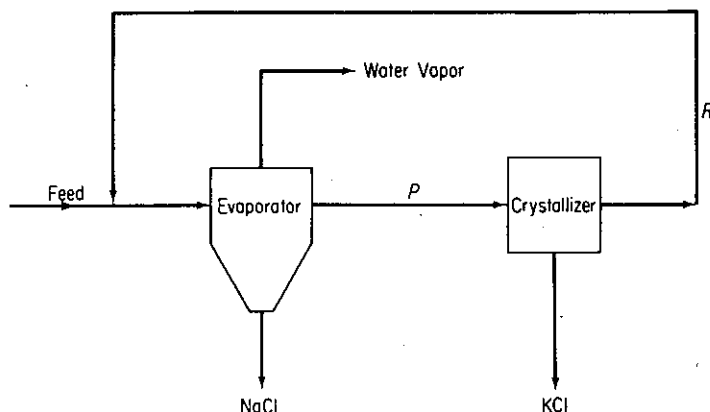


Figure P2.75

- 2.76. A 50% NaCl solution is to be concentrated in a triple-effect evaporator as shown in Fig. P2.76. (Each individual evaporator is termed an *effect*.) An equal amount of water is evaporated in each effect. Determine the composition of the outlet stream from effect 2 if the internal contents of effect 2 are uniformly mixed so that the outlet stream has the same composition as the internal contents of effect 2. The steam lines in each effect are completely separate from the evaporator contents so that no mixing of the steam with the contents occurs.

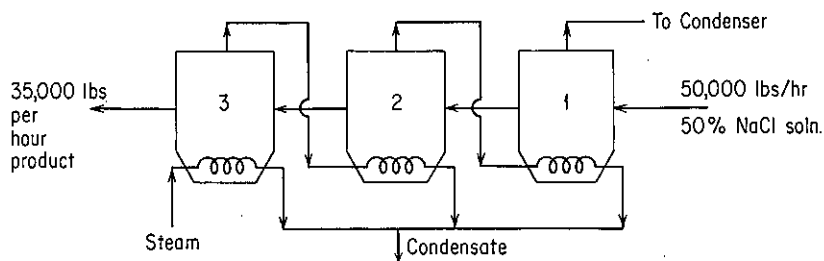


Figure P2.76

- 2.77.* A natural-gas plant at Short Junction, Oklahoma, produces gasoline by removing condensable vapors from the gas flowing out of gas wells. The gas from the well has the following composition:

Component	Mol %
CH ₄	77.3
C ₂ H ₆	14.9
C ₃ H ₈	3.6
iso- and <i>n</i> -C ₄ H ₁₀	1.6
C ₅ and heavier	0.5
N ₂	2.1
Total	100.0

This gas passes through an absorption column called a scrubber (see Fig. P2.77), where it is scrubbed with a heavy, nonvolatile oil. The gas leaving the scrubber has the following analysis:

Component	Mol %
CH ₄	92.0
C ₂ H ₆	5.5
N ₂	2.5

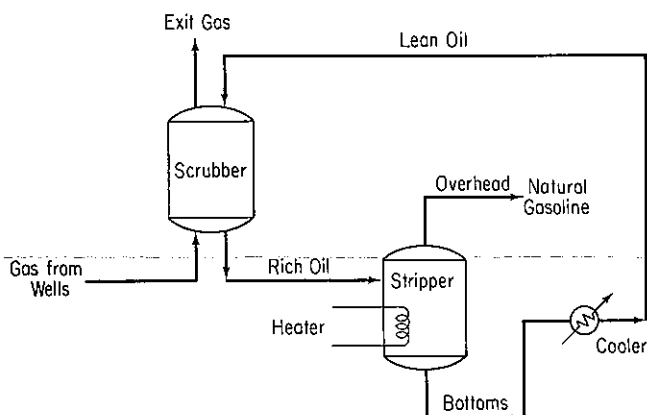


Figure P2.77

The scrubbing oil absorbs none of the CH₄ or N₂, much of the ethane, and all of the propane and higher hydrocarbons in the gas stream. The oil stream is then sent to a stripping column, which separates the oil from the absorbed hydrocarbons. The overhead from the stripper is termed *natural gasoline*, while the bottoms from the stripper is called *lean oil*. The lean-oil stream is cooled and returned to the absorption column. Assume that there are no leaks in the system. Gas from the wells is fed to the absorption column at the rate of 52,000 lb moles/day and the flow rate of the rich oil going to the stripper is 1230 lb/min (MW = 140). Calculate:

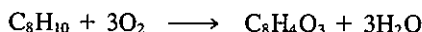
- The pounds of CH₄ passing through the absorber per day
- The pounds of C₂H₆ absorbed from the gas stream per day
- The weight percentage of propane in the rich oil stream leaving the scrubber

- 2.78. Given that the reaction for fresh feed A and B is



in a reactor with recycle, find the desired recycle ratio (moles recycle/moles fresh feed) if in the fresh feed A is 20% in excess, the once-through conversion of B is 60%, and the overall conversion of B to products for the overall process is 90%. After the products exit from the reactor, a stream of pure B is separated from the products and forms the recycle.

- 2.79. Phthalic anhydride (PA) is produced by the oxidation of o -xylene (C_8H_{10}) in a fixed-bed reactor (Fig. P2.79) at 350°C according to the reaction



A condenser is used to remove all of the PA ($C_8H_4O_3$) as a mixture of solid and liquid. How many kilograms of o -xylene are recycled per hour if the purge gas is 78% N_2 and 60% conversion of o -xylene takes place on one pass through the reactor?

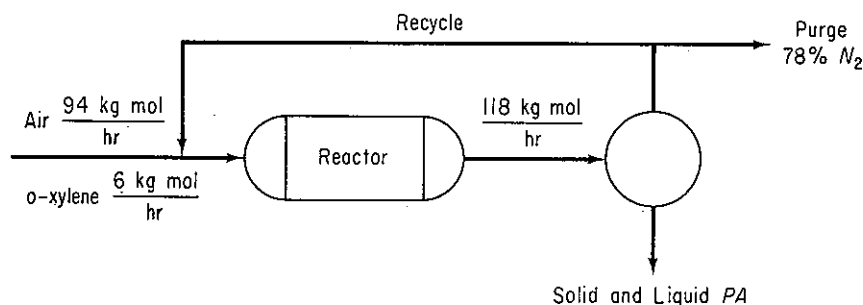
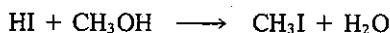


Figure P2.79

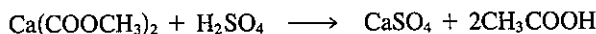
- 2.80. In the operation of a synthetic ammonia plant, an excess of hydrogen is burned with air so that the burner gas contains nitrogen and hydrogen in a 1:3 mole ratio and no oxygen. Argon is also present in the burner gas since it accounts for 0.94% of air. The burner gas is fed to a converter where a 25% conversion of the N_2 - H_2 mixture to ammonia is produced. The ammonia formed is separated by condensation and the unconverted gases are recycled to the converter. To prevent accumulation of argon in the system, some of the unconverted gases are vented before being recycled to the converter. The upper limit of argon in the converter is to be 4.5% of the entering gases. What percentage of the original hydrogen is converted into ammonia?
- 2.81. Methanol (CH_3OH) can be converted into formaldehyde ($HCHO$) either by oxidation to form formaldehyde and water or by direct decomposition to formaldehyde and hydrogen. Suppose that in the fresh feed to the process the ratio of methanol to oxygen is 4 moles to 1, the conversion of methanol per pass in the reactor is 50%, and all of the oxygen reacts in the reactor—none leaves with the hydrogen. After the reaction a separation process removes first all the formaldehyde and water, and then all the hydrogen is removed from the recycled methanol. Determine the flow rates of each species and the total flow rates at each point in the process.
- 2.82. In a proposed process for the preparation of methyl iodide, 2000 kg/day of hydroiodic acid is added to an excess of methanol:



If the product contains 81.6 wt % CH_3I along with the unreacted methanol, and the waste contains 82.6 wt % hydroiodic acid and 17.4% H_2O , calculate, assuming that the reaction is 40% complete in the process vessel:

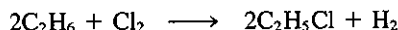
- The weight of methanol added per day
- The amount of HI recycled

- 2.83. Acetic acid is to be generated by the addition of 10% excess sulfuric acid (of strength 100%) to calcium acetate. The reaction in the reactor



goes to 90% completion. The unused calcium acetate and sulfuric acid are separated from the reaction products and are recycled back to the process. The acetic acid is separated from the reaction products and sold. The atomic weights are $\text{Ca} = 40.08$, $\text{S} = 32$.

- Draw a block diagram sketch for the process.
 - Calculate the total amount of the recycle stream per hour based on 1000 lb of calcium acetate fed per hour.
 - Calculate the pounds of acetic acid manufactured per hour.
- 2.84. Alkyl halides are used as an alkylating agent in various chemical transformations. The alkyl halide ethyl chloride can be prepared by the following chemical reaction:



In the reaction process shown in Fig. P2.84, fresh ethane and chlorine gas and recycled ethane are combined and fed into the reactor. A test shows that if 100% excess chlorine is mixed with ethane, a single-pass optimal conversion of 60% results, and of the ethane that reacts, all is converted to products and none goes into undesired products. Calculate:

- The fresh feed concentrations required for operation
- The moles of $\text{C}_2\text{H}_5\text{Cl}$ produced in P per mole of C_2H_6 in the fresh feed F_1

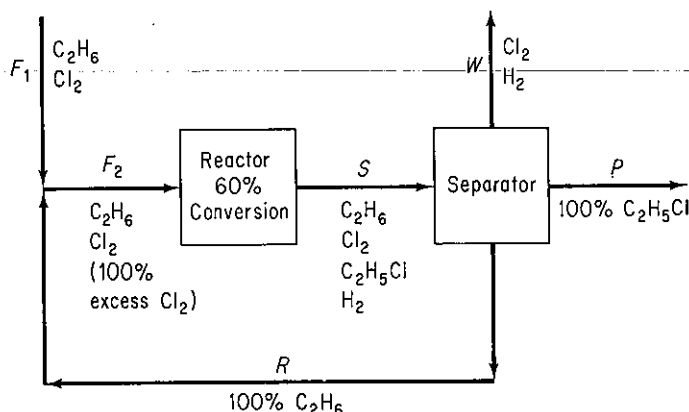


Figure P2.84

- 2.85.* A process for methanol synthesis is shown in Fig. P2.85. The pertinent chemical reactions involved are



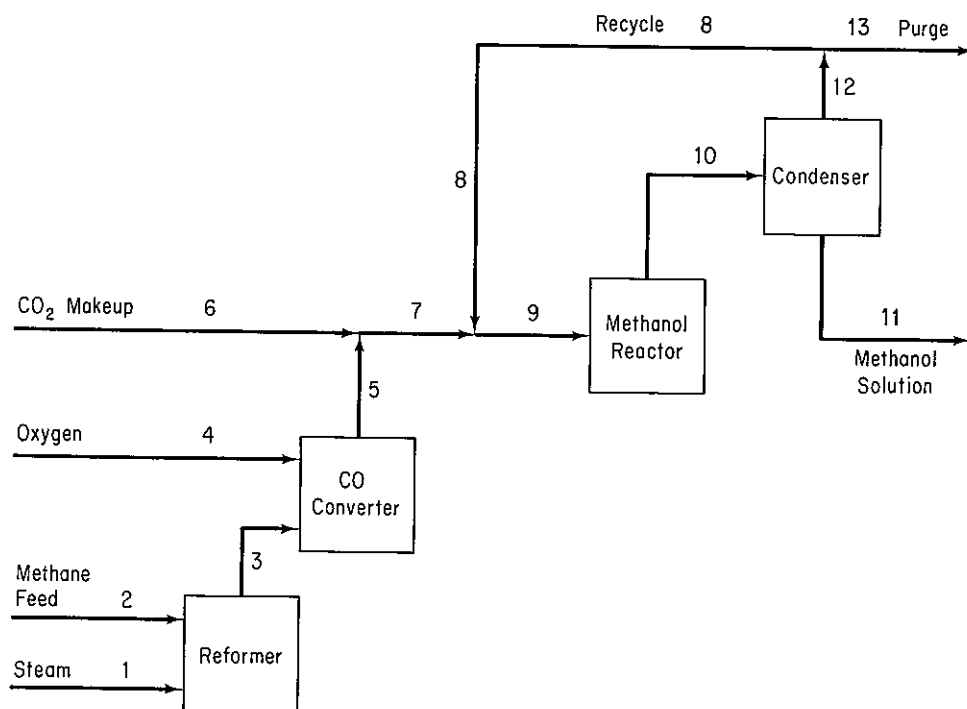
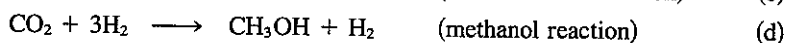


Figure P2.85



Ten percent excess steam, based on reaction (a), is fed to the reformer, and conversion of methane is 100%, with a 90% yield of CO_2 . Conversion in the methanol reactor is 55% on one pass through the reactor.

A stoichiometric quantity of oxygen is fed to the CO converter, and the CO is completely converted to CO_2 . Additional makeup CO_2 is then introduced to establish a 3-to-1 ratio of H_2 to CO_2 in the feed stream to the methanol reactor.

The methanol reactor effluent is cooled to condense all the methanol and water, with the noncondensable gases recycled to the methanol reactor feed. The H_2/CO_2 ratio in the recycle stream is also 3 to 1.

Because the methane feed contains 1% nitrogen as an impurity, a portion of the recycle stream must be purged as shown in the flow diagram to prevent the accumulation of nitrogen in the system. The purge stream analyzes 5% nitrogen.

On the basis of 100 mol of methane feed (including the N_2), calculate:

- How many moles of H_2 are lost in the purge
- How many moles of makeup CO_2 are required
- The recycle to purge ratio in mol/mol
- How much methanol solution (in kg) of what strength (weight percent) is produced

2.86.* An isomerizer is a catalytic reactor that simply tends to rearrange isomers. The number of moles entering an isomerizer is equal to the number of moles leaving. A

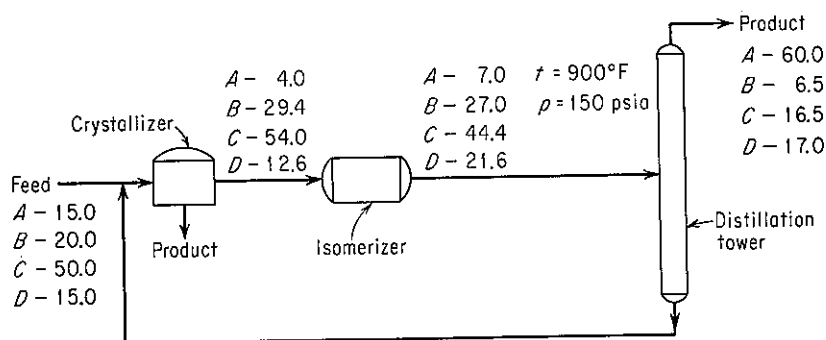


Figure P2.86

process, as shown in Fig. P2.86, has been designed to produce a *p*-xylene-rich product from an aromatic feed charge. All compositions on the flow sheet are in mole percent. The components are indicated as follows:

- A ethyl benzene
- B *o*-xylene
- C *m*-xylene
- D *p*-xylene

Eighty percent of the ethyl benzene entering the distillation tower is removed in the top stream from the tower. The ratio of the moles of fresh feed to the process as a whole to the moles of product from the crystallizer is 1.63. Find:

- (a) The reflux ratio (ratio of moles of stream from the bottom of the distillation tower per mole of feed to the tower)
- (b) The composition (in mole percent) of the product from the crystallizer
- (c) The moles leaving the isomerizer per mole of feed

PROBLEM THAT REQUIRES WRITING A COMPUTER PROGRAM

- 2.1. A 15-stage extraction column similar to the one in Example 2.22 is set up to separate acetone and ethanol using two solvents, pure water and pure chloroform. The only added features to Fig. E2.22 in Example 2.22 are (1) that you must consider two solutes in each solvent instead of one solute in the two solvents and (2) the acetone-ethanol feed is introduced into stage 6. You know that the feed composition is 50 mol % ethanol and 50 mol % acetone and that the feed rate is 20 lb mol/hr. Assume that the water and chloroform are not soluble in each other. You want to have 1.50 mol % ethanol in the exit stream from the bottom of the column, and 2×10^{-5} mol % acetone in the exit stream from the top of the column. Determine the rates of flow of the two solvents. In each stage you are given a relationship between X_i^A and X_i^B ($i = 1, 2, 3$) in the two phases in terms of *mole fractions*:

$$\frac{x_i^B}{x_i^A} = \frac{\gamma_i^A}{\gamma_i^B} \quad (a)$$

Each of the γ 's in phase *A* or phase *B* can be expressed in terms of some known constants and the mole fraction of the other two components. For component *i*,

$$\ln \gamma_i = 2x_i \sum_{j=1}^3 x_j a_{ji} + \sum_{j=1}^3 x_j^2 a_{ij} + \sum_{j=1}^3 \sum_{\substack{k=1 \\ j \neq i \\ k \neq i \\ j < k}}^3 x_j x_k a_{ijk}^* - 2 \sum_{i=1}^3 x_i^2 \sum_{j=1}^3 x_j a_{ij} - 2 \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 x_i x_j x_k a_{ijk}^* \quad (\text{b})$$

where $a_{ijk}^* = a_{ij} + a_{ji} + a_{ik} + a_{ki} + a_{jk} + a_{kj}$. The values of the coefficients are

- | | |
|------------|---------------|
| 1: acetone | 3: chloroform |
| 2: ethanol | 4: water |

$a_{11} = 0.0$	$a_{12} = 0.5446$	$a_{13} = 0.9417$	$a_{14} = 1.872$
$a_{21} = 0.599$	$a_{22} = 0.0$	$a_{23} = 1.61$	$a_{24} = 1.46$
$a_{31} = 0.674$	$a_{32} = 0.501$	$a_{33} = 0.0$	$a_{34} = 5.91$
$a_{41} = 1.338$	$a_{42} = 0.877$	$a_{43} = 4.76$	$a_{44} = 0.0$

The γ 's are obtained for each component, 1, 2, and 3, by permuting the subscripts cyclically $i \rightarrow j \rightarrow k \rightarrow i$. [Hints: Select some appropriate initial values for the flow rates of *A* and *B*. Divide the feed (stage 6) equally between both phases. The initial compositions can be provided by specifying both stage 1 and stage 15 and generating the remaining compositions by linear interpolation.] Be sure to include (1) one equation such as Eq. (a) for each component for each stage, (2) the correct number of component material balances for each stage (a total balance for the stage can be substituted for one component balance), and (3) $\sum x_i = 1$ for each stage for one phase (similar equations for the other phase are redundant).

GASES, VAPORS, LIQUIDS, 3 AND SOLIDS

3.1	Ideal Gas Laws	237
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In planning and decision making for our modern technology, engineers and scientists must know with reasonable accuracy the properties of the fluids and solids with which they deal. By **property** we shall mean any measurable characteristic of a substance, such as pressure, volume, or temperature, or a characteristic that can be calculated or deduced, such as internal energy, to be discussed in Chap. 4. If you are engaged in the design of equipment, say calculating the volume required for a process vessel, you need to know the specific volume or density of the gas or liquid that will be in the vessel as a function of temperature and pressure. If you are interested in predicting the possibility or extent of rainfall, you have to know something about the relation between the vapor pressure of water and the temperature. Whatever your current or future job, you must be aware of the character and be able to locate sources of information concerning the physical properties of fluids and solids.

Clearly, it is not possible to have reliable, detailed experimental data at hand on all the useful pure compounds and mixtures that exist in the world. Consequently, in the absence of experimental information, we estimate (predict) properties based on modifications of well-established principles, such as the *ideal* gas law, or based on empirical correlations. Thus the foundation of the estimation methods ranges from quite theoretical to completely empirical, and their reliability ranges from excellent to terrible.

Before starting our examination of physical properties, we need to mention two concepts: **state** and **equilibrium**. A system will *possess a unique set of properties*, such as temperature, pressure, density, and so on, at a given time, *and thus is said to be in a particular state*. A change in the state of a system results in a change in at least one of its properties.

By *equilibrium* we mean a state in which there is no tendency toward spontaneous change. Another way to say the same thing is to say that equilibrium is a state

in which all the rates of attaining and departing from the state are balanced. When a system is in equilibrium with another system, or the surroundings, it will not change its state unless the other system, or the surroundings, also changes.

What are solids, liquids, fluids, vapors, and gases? Rather than define the states of matter, a task that is not easy to accomplish with precision in the brief space we have here, let us instead characterize the states in terms of two quantities, flow and structure of the molecules:

State	Flow of collections of molecules	Structure of collections of molecules
Perfect gas	Extensive	None
Real gas	Extensive	Almost none
Liquid	Short distance	Related structure
Liquid crystal	Some	Some crystal structure
Amorphous solid	Little	Little
Real crystal	Almost none	Highly structured
Perfect crystal	None	Completely structured

At any temperature and pressure, a pure compound can exist as a gas, liquid, or solid, and at certain specific values of T and p , mixtures of phases exist, such as when water boils or freezes. Thus a compound (or a mixture of compounds) may consist of one or more phases. A **phase** is defined as a completely homogeneous and uniform state of matter. Liquid water would be a phase and ice would be another phase. Two immiscible liquids in the same container, such as mercury and water, would represent two different phases because the liquids have different properties.

In this chapter we first discuss ideal and real gas relationships, including some of the gas laws for pure components and mixtures of ideal gases. You will learn about methods of expressing the p - V - T properties of real gases by means of equations of state and, alternatively, by compressibility factors. Next we introduce the concepts of vaporization, condensation, and vapor pressure and illustrate how material balances are made for saturated and partially saturated gases. Finally, we examine the qualitative characteristics of gas-liquid-solid phases with the aid of diagrams. Figure 3.0 shows the interrelationships among the topics discussed in this chapter and how they relate to the making of material and energy balances.

3.1 IDEAL GAS LAWS

In 1787, Jacques Charles, a French chemist and physicist, published his conclusions about the relationship between the volume of gases and temperature. He demonstrated that the volume of a dry gas varies directly with temperature if the pressure remains constant. Charles, Boyle, Gay-Lussac, Dalton, and Amagat, the investigators who originally developed correlating relations among gas temperature, pressure, and volume, worked at temperatures and pressures such that the average distance be-

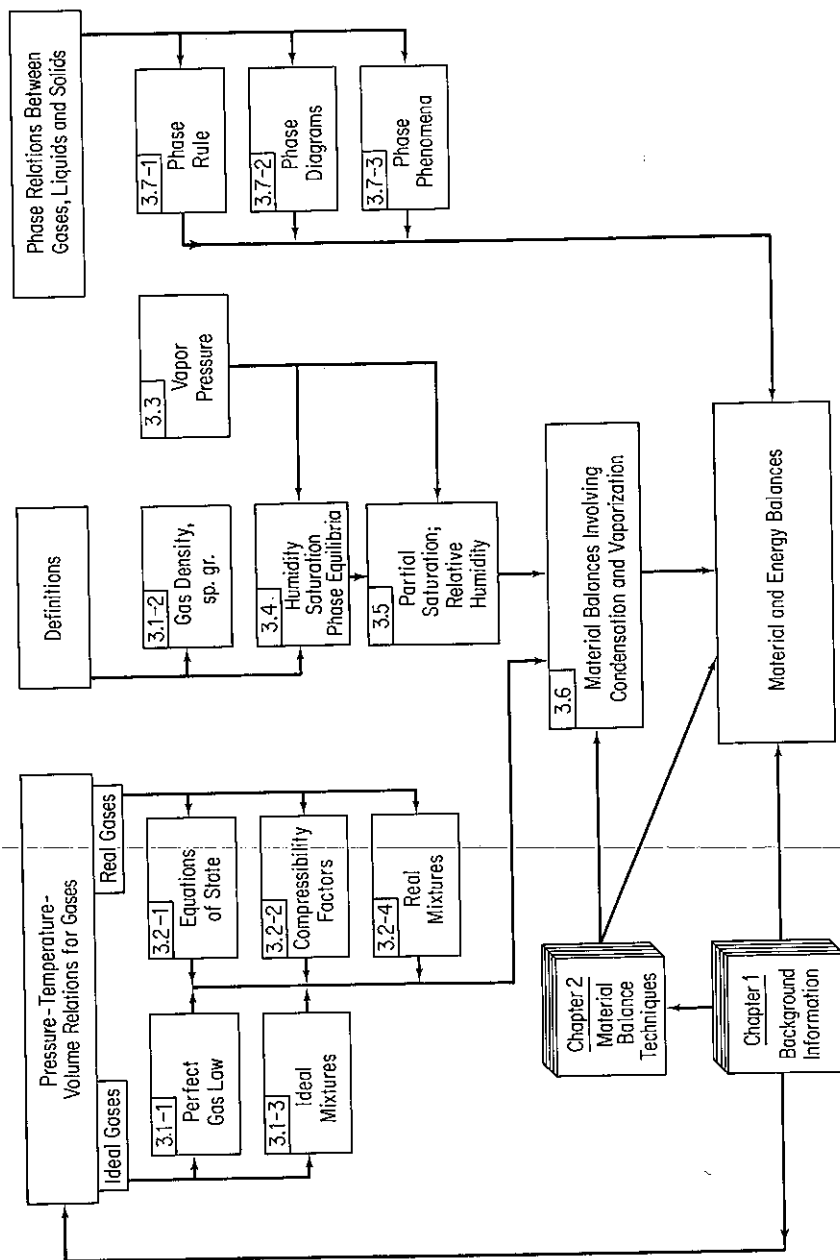


Figure 3.0 Hierarchy of topics to be studied in this chapter (section numbers are in the upper lefthand corner of the boxes).

tween the molecules was great enough to neglect the effect of the intermolecular forces and the volume of the molecules themselves. Under these conditions a gas came to be termed an **ideal gas**. More properly, an *ideal gas* is an imaginary gas which obeys exactly certain simple laws such as the laws of Boyle, Charles, Dalton, and Amagat. No real gas obeys these laws exactly over all ranges of temperature and pressure, although the "lighter" gases (hydrogen, oxygen, air, etc.) under ordinary circumstances obey the ideal gas laws with but negligible deviations. The properties of the "heavier" gases, such as sulfur dioxide and hydrocarbons, particularly at high pressures and low temperatures, deviate considerably from those predicted by the ideal gas laws. Vapors, under conditions near the boiling point, deviate markedly from the ideal gas laws. However, at low pressures and high temperatures, the behavior of a vapor approaches that of an ideal gas. Thus for many engineering purposes, the ideal gas laws, if properly applied, will give answers that are correct within a few percent or less. But for liquids and solids with the molecules compacted relatively close together, we do not have such general laws.

3.1-1 Calculations Using the Ideal Gas Law

Your objectives in studying this section are to be able to:

1. Write down the ideal gas law, and define all its variables and parameters and their associated dimensions.
2. Calculate the values and units of the ideal gas law constant in any set of units from the standard conditions.
3. Convert gas volumes to moles (and mass), and vice versa.
4. Use ratios of variables in the ideal gas law to calculate p , V , T , or n .

From the work of Boyle and Charles, scientists developed the relationship now called the **ideal gas law** (or sometimes the *perfect gas law*).

$$pV = nRT \quad (3.1)$$

where p = absolute pressure of the gas

V = total volume occupied by the gas

n = number of moles of the gas

R = ideal gas constant in appropriate units

T = absolute temperature of the gas

Sometimes the ideal gas law is written as

$$p\hat{V} = RT \quad (3.1a)$$

where $\hat{V}$ is the specific volume (volume per mole or mass) of the gas. Figure 3.1 illustrates the surface generated by Eq. (3.1a) in terms of the three properties p , $\hat{V}$, and T .

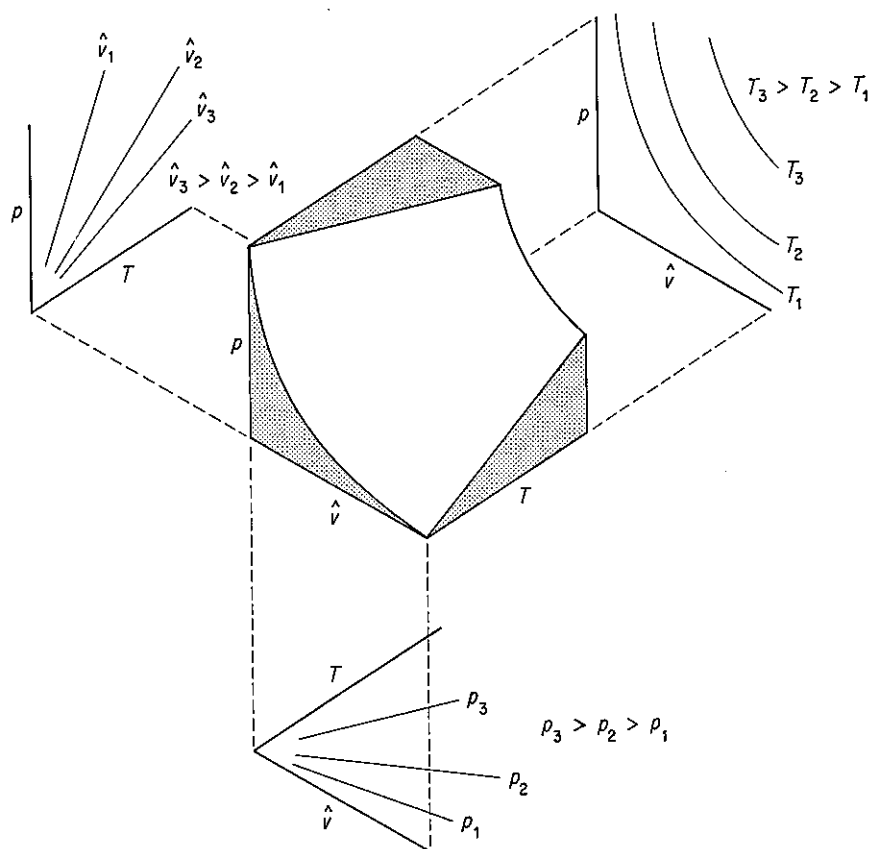


Figure 3.1 Representation of the ideal gas law in three dimensions as a surface.

Several arbitrarily specified standard states (usually known as *standard conditions*, or S.C.) of temperature and pressure have been selected by custom. See Table 3.1 for the most common ones. The fact that a substance cannot exist as a gas at 0°C and 1 atm is immaterial. Thus, as we see later, water vapor at 0°C cannot exist at a pressure greater than its saturation pressure of 0.61 kPa (0.18 in. Hg) without condensation occurring. However, the imaginary volume at standard conditions can be calculated and is just as useful a quantity in the calculation of volume-mole relation-

TABLE 3.1 Common Standard Conditions for the Ideal Gas

System	T	p	$\hat{v}$
SI	273.15K	101.325 kPa	22.415 m ³ /kg mol
Universal scientific	0.0°C	760 mm Hg	22.415 liters/g mol
Natural gas industry	60.0°F (15.0°C)	14.696 psia (101.325 kPa)	379.4 ft ³ /lb mol
American engineering	32°F	1 atm	359.05 ft ³ /lb mol

ships as though it could exist. In the following, the symbol V will stand for total volume and the symbol $\hat{V}$ for volume per mole, or per unit mass.

Because the SI, universal scientific, and American engineering standard conditions are identical, you can use the values in Table 3.1 with their units to change from one system of units to another. Knowing the standard conditions also makes it easy for you to work with mixtures of units from different systems.

The next example illustrates how the standard conditions can be employed to convert mass or moles to volume. Do you see how to convert volume to moles or mass?

EXAMPLE 3.1 Use of Standard Conditions

Calculate the volume, in cubic meters, occupied by 40 kg of CO_2 at standard conditions.

Solution

Basis: 40 kg of CO_2

$$\frac{40 \text{ kg CO}_2}{44 \text{ kg CO}_2} \times \frac{1 \text{ kg mol CO}_2}{1 \text{ kg mol CO}_2} \times \frac{22.42 \text{ m}^3 \text{ CO}_2}{1 \text{ kg mol CO}_2} = 20.4 \text{ m}^3 \text{ CO}_2 \text{ at S.C.}$$

Notice in this problem how the information that $22.42 \text{ m}^3 \text{ at S.C.} = 1 \text{ kg mol}$ is applied to transform a known number of moles into an equivalent number of cubic meters.

Incidentally, whenever you use cubic measure for volume, you must establish the conditions of temperature and pressure at which the cubic measure for volume exists, since the term " m^3 " or " ft^3 ," standing alone, is really not any particular *quantity* of material.

You can apply the ideal gas law, Eq. (3.1), directly by introducing values for three of the four quantities, n , p , T , and V , and solving for the fourth. To do so, you need to look up or calculate R in the proper units. Inside the front cover you will find selected values of R for different combinations of units. Example 3.2 illustrates how to calculate the value of R in any set of units you want from the values of p , T , and $\hat{V}$ at standard conditions.

In many processes going from an initial state to a final state, you can use the ratio of ideal gas law in the respective states and eliminate R as follows (the subscript 1 designates the initial state, and the subscript 2 designates the final state)

$$\frac{p_1 V_1}{p_2 V_2} = \frac{n_1 R T_1}{n_2 R T_2}$$

or

$$\left(\frac{p_1}{p_2}\right)\left(\frac{V_1}{V_2}\right) = \left(\frac{n_1}{n_2}\right)\left(\frac{T_1}{T_2}\right) \quad (3.2)$$

Note how Eq. (3.2) involves ratios of the same variable. This arrangement of the ideal gas law has the convenient feature that the pressures may be expressed in any system of units you choose, such as kPa, in. Hg, mm Hg, atm, and so on, as long as the same units are used for both conditions of pressure (do not forget that the pres-

sure must be *absolute* pressure in both cases). Similarly, the grouping together of the *absolute* temperature and the volume terms gives ratios that are dimensionless. Notice how the ideal gas constant R is eliminated in taking the ratio of the initial to the final state.

Let us see how we can apply the perfect gas law both in the form of Eq. (3.2) and Eq. (3.1) to problems.

EXAMPLE 3.2 Calculation of R

Find the value for the universal gas constant R for the following combinations of units:

(a) For 1 lb mol of ideal gas when the pressure is expressed in psia, the volume is in ft³/lb mol, and the temperature is in °R.

(b) For 1 g mol of ideal gas when the pressure is in atm, the volume in cm³, and the temperature in K.

(c) For 1 kg mol of ideal gas when the pressure is in kPa, the volume is in m³/kg mol, and the temperature is in K.

Solution

(a) At standard conditions we will use the approximate values

$$p = 14.7 \text{ psia}$$

$$\hat{V} = 359 \text{ ft}^3/\text{lb mol}$$

$$T = 492^\circ\text{R}$$

Then

$$R = \frac{p\hat{V}}{T} = \frac{14.7 \text{ psia}}{492^\circ\text{R}} \left| \frac{359 \text{ ft}^3}{1 \text{ lb mol}} \right| = 10.73 \frac{(\text{psia})(\text{ft}^3)}{(^{\circ}\text{R})(\text{lb mol})}$$

(b) Similarly, at standard conditions,

$$p = 1 \text{ atm}$$

$$\hat{V} = 22,415 \text{ cm}^3/\text{g mol}$$

$$T = 273.15 \text{ K}$$

$$R = \frac{p\hat{V}}{T} = \frac{1 \text{ atm}}{273.15 \text{ K}} \left| \frac{22,415 \text{ cm}^3}{1 \text{ g mol}} \right| = 82.06 \frac{(\text{cm}^3)(\text{atm})}{(\text{K})(\text{g mol})}$$

(c) In the SI system of units standard conditions are

$$p = 1.013 \times 10^5 \text{ Pa (or N/m}^2\text{)}$$

$$\hat{V} = 22.415 \text{ m}^3/\text{kg mol}$$

$$T = 273.15 \text{ K}$$

$$R = \frac{p\hat{V}}{T} = \frac{1.013 \times 10^5 \text{ Pa}}{273.15 \text{ K}} \left| \frac{22.415 \text{ m}^3}{1 \text{ kg mol}} \right| = 8.313 \times 10^3 \frac{(\text{Pa})(\text{m}^3)}{(\text{K})(\text{kg mol})} = 8.313 \frac{\text{kJ}}{(\text{K})(\text{kg mol})}$$

To summarize, we want to emphasize that R does not have a universal value even though it is sometimes called the *universal gas constant*. The value of R depends on the units of p , $\hat{V}$, and T .

EXAMPLE 3.3 Application of the Ideal Gas Law

Calculate the volume occupied by 88 lb of CO_2 at a pressure of 32.2 ft of water and at 15°C .

Solution

See Fig. E3.3.

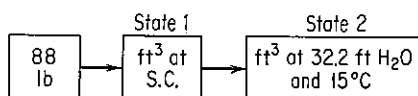


Figure E3.3

Solution 1 [Using Eq. (3.2)]:

At S.C. (state 1)	At state 2
$p = 33.91 \text{ ft H}_2\text{O}$	$p = 32.2 \text{ ft H}_2\text{O}$
$\hat{V} = 359 \frac{\text{ft}^3}{\text{lb mol}}$	$\hat{V} = ?$
$T = 273 \text{ K}$	$T = 273 + 15 = 288 \text{ K}$

Basis: 88 lb of CO_2

Assume that the pressure is absolute pressure. First, we convert the mass of gas to moles and get the equivalent volume at standard conditions. Finally, we use Eq. (3.2), in which both R and (n_1/n_2) cancel out:

$$V_2 = V_1 \left(\frac{p_1}{p_2} \right) \left(\frac{T_2}{T_1} \right)$$

You can say to yourself: The temperature goes up from 0°C at S.C. to 15°C at the final state, hence the volume must increase from S.C., hence the temperature ratio must be greater than unity. Similarly, you can say: The pressure goes down from S.C. to the final state, so that the volume must increase from S.C., hence the pressure ratio must be greater than unity. The calculations are

$$\begin{array}{c|c|c|c|c}
 88 \text{ lb CO}_2 & & 359 \text{ ft}^3 & 288 & 33.91 \\
 & \frac{44 \text{ lb CO}_2}{1 \text{ lb mol CO}_2} & 1 \text{ lb mol} & 273 & 32.2
 \end{array} = 798 \text{ ft}^3 \text{ CO}_2 \text{ at } 32.2 \text{ ft H}_2\text{O and } 15^\circ\text{C}$$

Solution 2 [using the gas constant R and the ideal gas law Eq. (3.1)].

First, the value of R must be obtained in the same units as the variables p , $\hat{V}$, and T . For 1 lb mol,

$$R = \frac{p\hat{V}}{T}$$

and at S.C.,

$$p = 33.91 \text{ ft H}_2\text{O}$$

$$\hat{V} = 359 \text{ ft}^3/\text{lb mol}$$

$$T = 273 \text{ K}$$

$$R = \frac{33.91}{359} \times \frac{359}{273} = 44.59 \frac{(\text{ft H}_2\text{O})(\text{ft}^3)}{(\text{lb mol})(\text{K})}$$

Now, using Eq. (3.1), insert the given values, and perform the necessary calculations.

Basis: 88 lb of CO₂

$$V = \frac{nRT}{p} = \frac{88 \text{ lb CO}_2}{44 \text{ lb CO}_2} \times \frac{44.59 (\text{ft H}_2\text{O})(\text{ft}^3)}{(\text{lb mol})(\text{K})} \times \frac{288 \text{ K}}{32.2 \text{ ft H}_2\text{O}}$$

$$= 798 \text{ ft}^3 \text{ CO}_2 \text{ at } 32.2 \text{ ft H}_2\text{O and } 15^\circ\text{C}$$

If you will inspect both solutions closely, you will observe that in both cases the same numbers appear and that both are identical except that in the second solution using R two steps are used to obtain the solution.

EXAMPLE 3.4 Ideal Gas Law

An oxygen cylinder used as a standby source of oxygen contains O₂ at 70°F. To calibrate the gauge on the O₂ cylinder which has volume of 1.01 ft³, all of the oxygen, initially at 70°F, is released into an evacuated tank of known volume (15.0 ft³). At equilibrium, the gas pressure was measured as 4 in. H₂O gauge and the gas temperature in both cylinders was 75°F. See Fig. E3.4. The barometer read 29.99 in. Hg.

What did the pressure gauge on the oxygen tank read in psig if it was a Bourdon gauge?

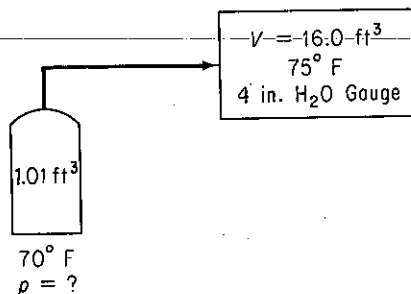


Figure E3.4

Solution

You must first convert the temperatures and pressures into absolute units:

$$460 + 70 = 530^\circ\text{R}$$

$$460 + 75 = 535^\circ\text{R}$$

$$\text{atmospheric pressure} = 29.99 \text{ in. Hg} = 14.73 \text{ psia}$$

$$\begin{aligned} \text{final pressure} &= 29.99 \text{ in. Hg} + \frac{4 \text{ in. H}_2\text{O}}{12 \text{ in. H}_2\text{O}} \left| \frac{29.92 \text{ in. Hg}}{33.91 \text{ ft H}_2\text{O}} \right. \\ &= 29.99 + 0.29 = 30.28 \text{ in. Hg absolute} \end{aligned}$$

The simplest way to proceed, now that the data are in good order, is to apply the ideal gas law, Eq. (3.2). Take as a basis, 16.011 ft³ (do not forget to include the volume of the O₂ tank in your system!) of O₂ at 75°F and 30.28 in. Hg. We want to determine the initial pressure in the O₂ tank alone.

$$\begin{aligned} p_1 &= p_2 \left(\frac{V_2}{V_1} \right) \left(\frac{n_1}{n_2} \right) \left(\frac{T_1}{T_2} \right) \\ p_1 &= 30.28 \text{ in. Hg} \left(\frac{16.01 \text{ ft}^3}{1 \text{ ft}^3} \right) \left(\frac{530^\circ\text{R}}{535^\circ\text{R}} \right) = 480 \text{ in. Hg absolute} \end{aligned}$$

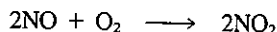
In gauge pressure,

$$p_1 = \frac{(480 - 29.99) \text{ in. Hg}}{29.92 \text{ in. Hg}} \left| \frac{14.696 \text{ psia}}{29.92 \text{ in. Hg}} \right. = 221 \text{ psig}$$

EXAMPLE 3.5 Application of the Ideal Gas Law

One important source of emissions from gasoline-powered automobile engines that causes smog is the nitrogen oxides NO and NO₂. They are formed whether combustion is complete or not as follows. At the high temperatures that occur in an internal combustion engine during the burning process, oxygen and nitrogen combine to form nitric oxide (NO). The higher the peak temperatures and the more oxygen available, the more NO is formed. There is insufficient time for the NO to decompose back to O₂ and N₂ because the burned gases cool too rapidly during the expansion and exhaust cycles in the engine. Although both NO and nitrogen dioxide (NO₂) are significant air pollutants (together termed NO_x), the NO₂ is formed in the atmosphere as NO is oxidized.

Suppose that you collect a sample of a NO-NO₂ mixture (after having removed the other combustion gas products by various separations procedures) in a 100-cm³ standard cell at 30°C. Certainly some of the NO will have been oxidized to NO₂,



during the collection, storage, and processing of the combustion gases, so that measurement of NO alone will be misleading. If the standard cell contains 0.291 g of NO₂ plus NO and the pressure measured in the cell is 170 kPa, what percent of the NO + NO₂ is in the form of NO? See Fig. E3.5.

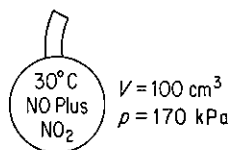


Figure E3.5

Solution

The gas in the cell is composed partly of NO and partly of NO₂. We can use the ideal gas law to calculate the total gram moles present in the cell.

Basis: 100 cm³ of gas at 170 kPa and 30°C

$$R = \frac{101.3 \text{ kPa}}{273 \text{ K}} \left| \frac{22.41 \text{ L}}{1 \text{ g mol}} \right| \left| \frac{1000 \text{ cm}^3}{1 \text{ L}} \right| = 8.316 \times 10^3 \frac{(\text{kPa})(\text{cm}^3)}{(\text{K})(\text{g mol})}$$

$$n = \frac{pV}{RT} = \frac{170 \text{ kPa}}{8.316 \times 10^3 \frac{(\text{kPa})(\text{cm}^3)}{(\text{K})(\text{g mol})}} \left| \frac{100 \text{ cm}^3}{303 \text{ K}} \right| = 0.00675 \text{ g mol}$$

If the mixture is composed of NO (MW = 30) and NO₂ (MW = 46), because we know the total mass in the cell we can compute the fraction of, say, NO. Let x = grams of NO; then $(0.291 - x)$ = g NO₂. In a table format the calculation is

Component	g	Mol. Wt.	g mol
NO	x	30	$\frac{x}{30}$
NO ₂	$0.291 - x$	46	$\frac{0.291 - x}{46}$
Total	0.291		0.00675

or

$$\frac{x}{30} + \frac{0.291 - x}{46} = 0.00675$$

$$0.0333x + (0.291 - x)(0.0217) = 0.00675$$

$$x = 0.0366 \text{ g}$$

The weight percent NO is

$$\frac{0.0366}{0.291}(100) = 12.5\%$$

and the mole percent NO is

$$\frac{0.0366 \text{ g NO}}{0.00675 \text{ g mol total}} \left| \frac{1 \text{ g mol NO}}{30 \text{ g NO}} \right| (100) = 18\%$$

Can you let x be the mole fraction NO and obtain the same result? (Answer: Yes.) Try the calculation.

Self-Assessment Test

1. Write down the ideal gas law.
2. What are the dimensions of T , p , V , n , and R ?
3. List the standard conditions for a gas in the SI, universal scientific, and American engineering systems of units.

4. Calculate the volume in ft^3 of 10 lb mol of an ideal gas at 68°F and 30 psia.
5. A steel cylinder of volume 2 m^3 contains methane gas (CH_4) at 50°C and 250 kPa absolute. How many kilograms of methane are in the cylinder?
6. What is the value of the ideal gas constant R to use if the pressure is to be expressed in atm, the temperature in kelvin, the volume in cubic feet, and the quantity of material in pound moles?
7. Twenty-two kilograms per hour of CH_4 are flowing in a gas pipeline at 30°C and 920 mm Hg. What is the volumetric flow rate of the CH_4 in m^3 per hour?

Thought Problems

1. A candle is placed vertically in a soup plate, and the soup plate filled with water. Then the candle is lit. An inverted water glass is carefully placed over the candle. The candle soon goes out, and the water rises inside the glass. It is often said that this shows how much oxygen in the air has been used up. Is this conclusion correct?
2. A scientific supply house markets aerosol-type cans containing compressed helium for filling balloons, doing demonstrations, and the like. On the label there appears the notice: "Because the can contains helium, it quite naturally feels empty. It is actually lighter full than empty."

Is this statement correct? If so, why? If not, why not?

3.1-2 Gas Density and Specific Gravity

Your objectives in studying this section are to be able to:

1. Define gas density and specific gravity.
2. Calculate the specific gravity of a gas even if the reference condition is not clearly specified.
3. Calculate the density of a gas given its specific gravity.

The **density of a gas** is defined as the mass per unit volume and can be expressed in kilograms per cubic meter, pounds per cubic foot, grams per liter, or other units. Inasmuch as the mass contained in a unit volume varies with the temperature and pressure, as we have previously mentioned, you should always be careful to specify these two conditions. If not otherwise specified, the densities are presumed to be at S.C. Density can be calculated by selecting a unit volume as the basis and calculating the mass of the contained gas.

EXAMPLE 3.6 Calculation of Gas Density

What is the density of N_2 at 27°C and 100 kPa expressed in:

- (a) SI units?
- (b) American engineering units?

Solution

(a)

Basis: 1 m³ of N₂ at 27°C and 100 kPa

$$\frac{1 \text{ m}^3}{300 \text{ K}} \left| \frac{273 \text{ K}}{101.3 \text{ kPa}} \right| \left| \frac{100 \text{ kPa}}{22.4 \text{ m}^3} \right| \left| \frac{1 \text{ kg mol}}{28 \text{ kg}} \right| = 1.123 \text{ kg}$$

density = 1.123 kg/m³ of N₂ at 27°C (300 K) and 100 kPa

(b)

Basis: 1 ft³ of N₂ at 27°C and 100 kPa

$$\frac{1 \text{ ft}^3}{300 \text{ K}} \left| \frac{273 \text{ K}}{101.3 \text{ kPa}} \right| \left| \frac{100 \text{ kPa}}{359 \text{ ft}^3} \right| \left| \frac{1 \text{ lb mol}}{28 \text{ lb}} \right| = 0.0701 \text{ lb}$$

density = 0.0701 lb/ft³ of N₂ at 27°C (80°F) and 100 kPa (14.5 psia)

The **specific gravity of a gas** is usually defined as the ratio of the density of the gas at a desired temperature and pressure to that of air (or any specified reference gas) at a certain temperature and pressure. The use of specific gravity occasionally may be confusing because of the manner in which the values of specific gravity are reported in the literature. You must be very careful in using literature values of specific gravity to ascertain that the conditions of temperature and pressure are known both for the gas in question and for the reference gas. Among the examples below, several represent inadequate methods of expressing specific gravity.

- (a) *What is the specific gravity of methane?* Actually, this question may have the same answer as the question: How many grapes are in a bunch? Unfortunately, occasionally one may see this question and the best possible answer is

$$\text{sp gr} = \frac{\text{density of methane at S.C.}}{\text{density of air at S.C.}}$$

- (b) *What is the specific gravity of methane (H₂ = 1.00)?* Again a poor question. The notation of (H₂ = 1.00) means that H₂ at S.C. is used as the reference gas, but the question does not tell what the conditions of temperature and pressure of the methane are. Therefore, the best interpretation is

$$\text{sp gr} = \frac{\text{density of methane at S.C.}}{\text{density of H}_2 \text{ at S.C.}}$$

- (c) *What is the specific gravity of ethane (air = 1.00)?* Same as question (b) except that in the petroleum industry the following is used:

$$\text{sp gr} = \frac{\text{density of ethane at } 60^\circ\text{F and } 760 \text{ mm Hg}}{\text{density of air at S.C. (} 60^\circ\text{F, } 760 \text{ mm Hg)}}$$

- (d) *What is the specific gravity of butane at 50°C and 90 kPa?* No reference gas nor state of reference gas is mentioned. However, when no reference gas is mentioned, it is taken for granted that air is the reference gas. In the case at hand

the best thing to do is to assume that the reference gas and the desired gas are under the same conditions of temperature and pressure:

$$\text{sp gr} = \frac{\text{density of butane at } 50^{\circ}\text{C and } 90 \text{ kPa}}{\text{density of air at } 50^{\circ}\text{C and } 90 \text{ kPa}}$$

(e) What is the specific gravity of CO_2 at 60°F and 740 mm Hg (air = 1.00)?

$$\text{sp gr} = \frac{\text{density of } \text{CO}_2 \text{ at } 60^{\circ}\text{F and } 740 \text{ mm Hg}}{\text{density of air at S.C.}}$$

(f) What is the specific gravity of CO_2 at 60°F and 740 mm Hg (ref. air at S.C.)?

sp gr = same as question (e)

EXAMPLE 3.7 Specific Gravity of a Gas

What is the specific gravity of N_2 at 80°F and 745 mm Hg compared to

(a) Air at S.C. (32°F and 760 mm Hg)?

(b) Air at 80°F and 745 mm Hg ?

Solution

First you must obtain the density of the N_2 and the air at their respective conditions of temperature and pressure, and then calculate the specific gravity by taking a ratio of their densities. Example 3.6 covers the calculation of the density of a gas, and therefore, to save space, no units will appear in the intermediate calculations:

(a)

Basis: 1 ft^3 of N_2 at 80°F and 745 mm Hg

$$\frac{1}{540} \left| \frac{492}{760} \right| \left| \frac{745}{359} \right| \left| \frac{28}{359} \right| = 0.0697 \text{ lb } \text{N}_2/\text{ft}^3 \text{ at } 80^{\circ}\text{F}, 745 \text{ mm Hg}$$

Basis: 1 ft^3 of air at 32°F and 760 mm Hg

$$\frac{1}{492} \left| \frac{492}{760} \right| \left| \frac{760}{359} \right| \left| \frac{29}{359} \right| = 0.0808 \text{ lb air}/\text{ft}^3 \text{ at } 32^{\circ}\text{F}, 760 \text{ mm Hg}$$

Therefore,

$$\text{sp.gr.} = \frac{0.0697}{0.0808} = 0.862 \frac{\text{lb } \text{N}_2/\text{ft}^3 \text{ N}_2 \text{ at } 80^{\circ}\text{F}, 745 \text{ mm Hg}}{\text{lb air}/\text{ft}^3 \text{ air at S.C.}}$$

Note: Specific gravity is not a dimensionless number.

(b)

Basis: 1 ft^3 of air at 80°F and 745 mm Hg

$$\frac{1}{540} \left| \frac{492}{760} \right| \left| \frac{745}{359} \right| \left| \frac{29}{359} \right| = 0.0721 \text{ lb}/\text{ft}^3 \text{ at } 80^{\circ}\text{F} \text{ and } 745 \text{ mm Hg}$$

$$\begin{aligned} (\text{sp. gr.})_{\text{N}_2} &= \frac{0.0697}{0.0721} = 0.967 \frac{\text{lb } \text{N}_2/\text{ft}^3 \text{ N}_2 \text{ at } 80^{\circ}\text{F}, 745 \text{ mm Hg}}{\text{lb air}/\text{ft}^3 \text{ air at } 80^{\circ}\text{F}, 745 \text{ mm Hg}} \\ &= 0.967 \text{ lb } \text{N}_2/\text{lb air} \end{aligned}$$

Note that you can also obtain the specific gravity of a gas directly by taking the ratio of the ideal gas law densities of each gas. Thus (the gases are denoted by the respective subscripts A and B)

$$pV = \frac{mRT}{\text{mol. wt.}}$$

where m is the mass of gas. The gas density is

$$\rho = \frac{m}{V} = \frac{p(\text{mol. wt.})}{RT}$$

and the ratio of gas densities is

$$\text{sp gr} = \frac{\rho_A}{\rho_B} = \left(\frac{p_A}{p_B}\right) \left(\frac{\text{mol. wt.}_A}{\text{mol. wt.}_B}\right) \left(\frac{T_B}{T_A}\right) \quad (3.3)$$

For gases at the *same* temperature and pressure, the specific gravity is just the ratio of the respective molecular weights.

Self-Assessment Test

1. What is the density of a gas that has a molecular weight of 0.123 kg/kg mol at 300 K and 1000 kPa?
2. What is the specific gravity of CH_4 at 70°F and 2 atm compared to air at S.C.?

Thought Problem

1. Sea breezes provide welcome relief from the summer heat for residents who live close to the shore. No matter what part of the world—the coast of California, Australia where sea breezes can be very strong, even along the shores of the Great Lakes—the daily pattern in summertime is the same. The sea breeze, a wind blowing from sea to land, begins to develop three or four hours after sunrise and reaches its peak intensity by mid-afternoon. It may penetrate inland as much as 60 or 70 km. The sea breeze dies out in the evening and three or four hours after sunset may be replaced by a land breeze, blowing from land to sea. The land breeze, much weaker than the sea breeze, reaches its peak intensity just before sunrise.

What causes these breezes?

2. A distillation column reboiler in a room, as shown in the figure, had been cleaned, but the

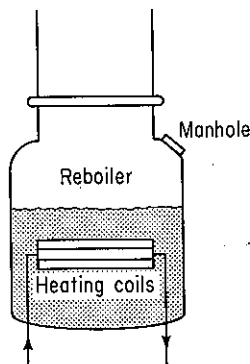


Figure TP3.2

manhole cover was not securely fastened on startup again. As a result benzene vapor escaped from the manhole and one operator died by asphyxiation. How could this occur?

3.1-3 Ideal Gas Mixtures

Your objectives in studying this section are to be able to:

1. Write down and apply Dalton's law and Amagat's law.
2. Define and use partial pressure in gas calculations.
3. Show that under certain assumptions the volume fraction equals the mole fraction in a gas.

Frequently, as an engineer, you will want to make calculations for mixtures of gases instead of individual gases. You can use the ideal gas law (under the proper assumptions of course) for a mixture of gases by interpreting p as the total pressure of the mixture, V as the volume occupied by the mixture, n as the total number of moles of all components in the mixture, and T as the temperature of the mixture. Is it possible to calculate a pressure or volume for the individual components? The answer is yes in a certain sense. Two classical definitions exist:

- (a) The **partial pressure** of Dalton, p_i , namely the pressure that would be exerted by a single component in a gaseous mixture if it existed by itself in the *same volume* as occupied by the mixture and at the *same temperature* of the mixture

$$p_i V_{\text{total}} = n_i RT_{\text{total}} \quad (3.4)$$

where p_i is the partial pressure of component i . If you divide Eq. (3.4) by Eq. (3.1), you find that

$$\frac{p_i V_{\text{total}}}{p_{\text{total}} V_{\text{total}}} = \frac{n_i RT_{\text{total}}}{n_{\text{total}} RT_{\text{total}}}$$

or

$$p_i = p_{\text{total}} \frac{n_i}{n_{\text{total}}} = p_{\text{total}} y_i \quad (3.5)$$

where y_i is the mole fraction of component i . Can you show that Dalton's law of the summation of partial pressures is true?

$$p_1 + p_2 + \cdots + p_n = p_{\text{total}} \quad (3.6)$$

- (b) The **partial volume** of Amagat, V_i , namely the volume that would be occupied by a single component of the mixture if the component were at the *same temperature and total pressure* as the mixture

$$p_{\text{total}} V_i = n_i RT_{\text{total}} \quad (3.7)$$

Division by Eq. (3.1) yields

$$\frac{p_i V_i}{p_i V_i} = \frac{n_i RT_i}{n_i RT_i}$$

or

$$V_i = V_i \left(\frac{n_i}{n_i} \right) = V_i y_i \quad (3.8)$$

Can you show that Amagat's law of the summation of partial volumes is true?

$$V_1 + V_2 + \cdots + V_n = V_i \quad (3.9)$$

To illustrate the significance of Eq. (3.4) and the meaning of partial pressure, suppose that you carried out the following experiment with ideal gases. Two tanks of 1.50 m³ volume, one containing gas *A* at 300 mm Hg and the other gas *B* at 400 mm of Hg (both gases being at the same temperature of 20°C), are connected together. All the gas in *B* is forced into tank *A* isothermally. Now you have a 1.50-m³ tank of *A* + *B* at 700 mm of Hg. For this mixture (in the 1.50-m³ tank at 20°C and a total pressure of 700 mm Hg) you could say that gas *A* exerts a partial pressure of 300 mm and gas *B* exerts a partial pressure of 400 mm. Of course you cannot put a pressure gauge on the tank and check this conclusion because the pressure gauge will read only the total pressure. These partial pressures are hypothetical pressures that the individual gases would exert and are equivalent to the pressures they actually would have if they were put into the same volume at the same temperature all by themselves. If the total pressure of the gaseous mixture is known as well as the mole fraction of a component, the partial pressure of the component can be calculated via Eq. (3.5):

$$p_A = 700 \left(\frac{3}{7} \right) = 300 \text{ mm Hg}$$

An analogous explanation can be made for the partial volume of *A* in the mixture using Eq. (3.8):

$$V_A = 1.50 \left(\frac{3}{7} \right) = 0.64 \text{ m}^3 \text{ at 700 mm Hg and 20}^\circ\text{C}$$

Finally, Eq. (3.8) leads to the principle, stated earlier in Chap. 1, that

$$\text{volume fraction} = \text{mole fraction} = y_i \quad (3.10)$$

for an ideal gas.

EXAMPLE 3.8 Partial Pressures and Volumes

A gas-tight room has a volume of 1000 m³. This room contains air (considered to be 21% O₂ and 79% N₂) at 20°C and a total pressure of 1 atm.

- What is the partial volume of O₂ in the room?
- What is the partial volume of N₂ in the room?

- (c) What is the partial pressure of O_2 in the room?
 (d) What is the partial pressure of N_2 in the room?
 (e) If all of the O_2 were removed from the room by some method, what would be the subsequent total pressure in the room?

Solution

See Fig. E3.8

Basis: 1000 m^3 of air at 20°C and 1 atm

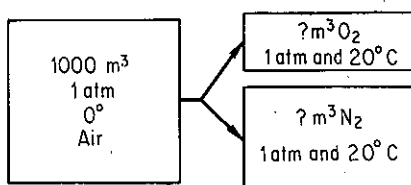


Figure E3.8

Partial volumes can be calculated by multiplying the total volume by the respective component mole fractions [Eq. (3.8)]:

(a) $V_{O_2} = (0.21)(1000) = 210\text{ m}^3\text{ }O_2\text{ at }20^\circ\text{C, 1 atm}$

(b) $V_{N_2} = (0.79)(1000) = 790\text{ m}^3\text{ }N_2\text{ at }20^\circ\text{C, 1 atm}$
 total volume = $1000\text{ m}^3\text{ air at }20^\circ\text{C, 1 atm}$

Note how the temperature and pressure have to be specified for the partial volumes to make them meaningful.

Partial pressures can be calculated by multiplying the total pressure by the respective component mole fractions [Eq. (3.5)]; the basis is still the same:

(c) $p_{O_2} = (0.21)(1\text{ atm}) = 0.21\text{ atm when }V = 1000\text{ m}^3\text{ at }20^\circ\text{C}$

(d) $p_{N_2} = (0.79)(1\text{ atm}) = 0.79\text{ atm when }V = 1000\text{ m}^3\text{ at }20^\circ\text{C}$
 total pressure = $1.00\text{ atm when }V = 1000\text{ m}^3\text{ at }20^\circ\text{C}$

(e) If a tight room held dry air at 1 atm and all the oxygen were removed from the air by a chemical reaction, the pressure reading would fall to 0.79 atm.

For use in our subsequent calculations you should clearly understand now that the original room contained:

- (1) 790 m^3 dry N_2 at 1 atm and 20°C
- (2) 210 m^3 dry O_2 at 1 atm and 20°C
- (3) 1000 m^3 dry air at 1 atm and 20°C (add 1 and 2)

or

- (1) 1000 m^3 dry O_2 at 0.21 atm and 20°C
- (2) 1000 m^3 dry N_2 at 0.79 atm and 20°C
- (3) 1000 m^3 dry air at 1.00 atm and 20°C (add 1 and 2)

EXAMPLE 3.9 Calculation of Partial Pressures from Gas Analysis

A flue gas analyzes 14.0% CO_2 , 6.0% O_2 , and 80.0% N_2 . It is at 400°F and 765.0 mm Hg pressure. Calculate the partial pressure of each component.

Solution

Use Eq. (3.5).

Basis: 1.00 kg (or lb) mol flue gas

<i>Component</i>	<i>kg (or lb) mol</i>	<i>p (mm Hg)</i>
CO ₂	0.140	107.1
O ₂	0.060	45.9
N ₂	0.800	612.0
Total	1.000	765.0

On the basis of 1.00 mole of flue gas, the mole fraction y of each component, when multiplied by the total pressure, gives the partial pressure of that component.

Self-Assessment Test

1. Write down Dalton's law and Amagat's law.
2. A gas has the following composition at 120°F and 13.8 psia.

<i>Component</i>	<i>Mol %</i>
N ₂	2
CH ₄	79
C ₂ H ₆	19

- (a) What is the partial pressure of each component?
 - (b) What is the partial volume of each component if the total volume of the container is 2 ft³?
 - (c) What is the volume fraction of each component?
3. (a) If the C₂H₆ were removed from the gas in problem 2, what would be the subsequent pressure in the vessel?
 (b) What would be the subsequent partial pressure of the N₂?

3.1-4 Material Balances Involving Gases

Now that you have had a chance to practice applying the ideal gas law to simple transformations, we turn our attention back to material balances. The only difference between the subject matter of Chapter 2 and this section is that here the amount of material flow can be specified in terms of p , V , and T rather than solely mass or moles. For example, the basis for a problem, or the quantity to be solved for, might be a volume of gas at a given temperature and pressure rather than a mass of gas. The next two examples illustrate two of the same types of the problems you have encountered before.

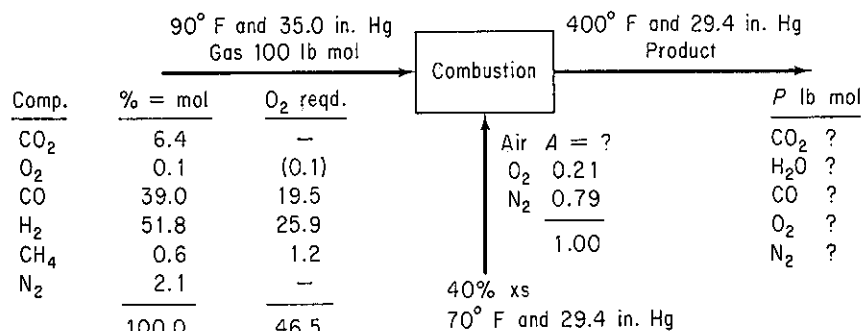
EXAMPLE 3.10 Material Balance with Combustion

A synthesis gas analyzes 6.4% CO₂, 0.1% O₂, 39% CO, 51.8% H₂, 0.6% CH₄, and 2.1% N₂. It enters the combustion chamber at 90°F and a pressure of 35.0 Hg, and is burned with 40% excess air (dry) which is at 70°F and the atmospheric pressure of 29.4 in. Hg; 10% of the CO remains unburned. How many cubic feet of air are supplied per cubic foot of entering gas? How many cubic feet of product are produced per cubic foot of entering gas if the exit gas is at 29.4 in. Hg and 400°F?

Solution

This is a steady-state problem with reaction in which you can directly add and subtract quantities; you do not have to solve simultaneous equations.

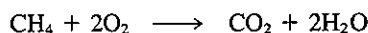
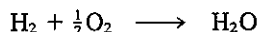
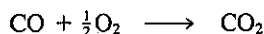
Step 1, 2, and 3 Figure E3.10 illustrates the process and notation. With 40% excess air, certainly all the CO, H₂, and CH₄ should burn to CO₂ and H₂O; apparently, for some unknown reason, not all of the CO burns to CO₂. The product gases are shown in the figure.

**Figure E3.10**

Step 4 You could take 1 ft³ at 90°F and 35.0 in. Hg as the basis, but it is just as easy to take 1 lb mol because then % = lb mol. At the end of the problem you can convert lb mol to ft³.

Basis: 100 lb mol synthesis gas

Step 3 (continued) The entering air can be calculated from the specified 40% excess air; the reactions for complete combustion are



the moles of oxygen required are listed in Fig. E3.10. The excess oxygen is

$$\text{Excess O}_2: 0.4(46.5) = 18.6$$

$$\text{Total O}_2 = 46.5 + 18.6 = 65.10$$

$$\text{N}_2 \text{ in } 65.10 \left(\frac{79}{21} \right) = 244.9$$

Total moles of air are $244.9 + 65.10 = 310$ lb mol.

Steps 5 and 6 Five unknowns exist, the five products. You can make four element balances plus know the fraction of the entering CO that exits in *P*. Hence the problem has a unique solution.

Steps 7, 8, and 9 We can use the chemical equations plus elemental balances to directly calculate the products.

<i>In with Fuel</i>		<i>Out</i>	
N ₂ :	2.1	N ₂ :	2.1 + 244.9 = 247.0
H ₂ :	51.8	H ₂ O:	51.8 + 0.6(2) = 53.0
CO:	39.0	CO:	39.0(0.10) = 3.9
CO ₂ :	6.4	CO ₂ :	6.4 + 39.0(0.9) + 0.6 = 42.1

The oxygen (O₂) balance is

$$6.4 + 0.1 + \frac{1}{2}(39.0) + 65.1 = \frac{1}{2}(53.0) + \frac{1}{2}(3.9) + 42.1 + \text{O}_{2\text{out}}$$

so that the exit O₂ is 20.55 lb mol. The total moles exiting sum up to be 366.55 lb mol.

Finally, we can convert the lb mol of air and product into the volumes requested:

$$T_{\text{gas}} = 90 + 460 = 550^{\circ}\text{R}$$

$$T_{\text{air}} = 70 + 460 = 530^{\circ}\text{R}$$

$$T_{\text{product}} = 400 + 460 = 860^{\circ}\text{R}$$

$$\text{ft}^3 \text{ of gas: } \frac{100 \text{ lb mol entering gas}}{1 \text{ lb mol}} \left| \frac{359 \text{ ft}^3 \text{ at SC}}{492^{\circ}\text{R}} \right| \frac{550^{\circ}\text{R}}{492^{\circ}\text{R}} \left| \frac{29.92 \text{ in. Hg}}{35.0 \text{ in. Hg}} \right| = 343 \times 10^2$$

$$\text{ft}^3 \text{ of air: } \frac{310 \text{ lb mol air}}{1 \text{ lb mol}} \left| \frac{359 \text{ ft}^3 \text{ at SC}}{492^{\circ}\text{R}} \right| \frac{530^{\circ}\text{R}}{492^{\circ}\text{R}} \left| \frac{29.92 \text{ in. Hg}}{29.4 \text{ in. Hg}} \right| = 1220 \times 10^2$$

$$\text{ft}^3 \text{ of product: } \frac{366.55 \text{ lb mol } P}{1 \text{ lb mol}} \left| \frac{359 \text{ ft}^3 \text{ at SC}}{492^{\circ}\text{R}} \right| \frac{860^{\circ}\text{R}}{492^{\circ}\text{R}} \left| \frac{29.92 \text{ in. Hg}}{29.4 \text{ in. Hg}} \right| = 2340.9 \times 10^2$$

The answers to the questions are

$$\frac{1220 \times 10^2}{343 \times 10^2} = 3.56 \frac{\text{ft}^3 \text{ air at } 530^{\circ}\text{R and } 29.4 \text{ in. Hg}}{\text{ft}^3 \text{ gas at } 550^{\circ}\text{R and } 35.0 \text{ in. Hg}}$$

$$\frac{2340.9 \times 10^2}{343 \times 10^2} = 6.82 \frac{\text{ft}^3 \text{ product at } 860^{\circ}\text{R and } 29.4 \text{ in. Hg}}{\text{ft}^3 \text{ gas at } 550^{\circ}\text{R and } 35.1 \text{ in. Hg}}$$

EXAMPLE 3.11 Material Balance without Reaction

Gas at 15°C and 105 kPa is flowing through an irregular duct. To determine the rate of flow of the gas, CO₂ from a tank was passed into the gas stream. The gas analyzed 1.2% CO₂ by volume before, and 3.4% CO₂ by volume after, addition. The CO₂ left the tank, was passed through a rotameter, and was found to flow at the rate of 0.0917 m³/min at 7°C and 131 kPa. What was the rate of flow of the entering gas in the duct in cubic meters per minute?

Solution

This is a steady-state problem without reaction.

Steps 1, 2, and 3

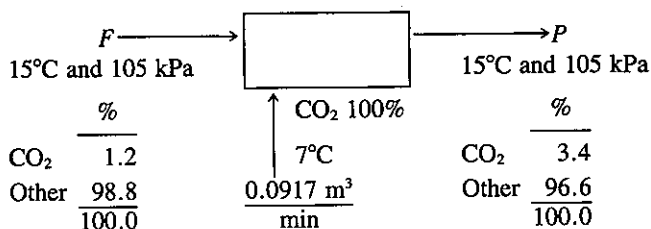


Figure E3.11

Assume that the entering and exit gases are at the same temperature and pressure.

Step 4 Take as a basis 1 min = 0.0917 m³ CO₂ at 7°C and 131 kPa. The gas analysis is in volume percent, which is the same as mole per cent. We could convert the 0.0917 m³ to moles and solve the problem in terms of moles, but there is no need to do so because we can calculate the flow rate of F in m³ at 7°C and 131 kPa, and then convert the calculated volume to 15°C and 105 kPa.

Steps 5 and 6 We do not know F and P , but can make two independent component balances, hence the problem has a unique solution.

Steps 7, 8, and 9

$$\text{"Other" balance (m}^3 \text{ at 7°C and 131 kPa): } F(0.988) = P(0.966) \quad (a)$$

$$\text{CO}_2 \text{ balance (m}^3 \text{ at 7°C and 131 kPa): } F(0.012) + 0.0917 = P(0.034) \quad (b)$$

$$\text{Total balance (m}^3 \text{ at 7°C and 131 kPa): } F + 0.0917 = P \quad (c)$$

Note that "other" is a tie component. Solution of Eqs. (a) and (c) gives

$$F + 0.0917 = F \left(\frac{0.988}{0.966} \right)$$

or $F = 4.0265 \text{ m}^3$ at 7°C and 131 kPa. Finally, we convert to a volumetric flow rate at 15°C and 105 kPa:

$$\frac{4.0265 \text{ m}^3 \text{ at 7°C and 131 kPa}}{\frac{288 \text{ K}}{280 \text{ K}} \cdot \frac{131 \text{ kPa}}{105 \text{ kPa}}} = 5.17 \text{ m}^3/\text{min at 15°C and 105 kPa}$$

Step 10 (Check)

$$\text{By Eq. (b): } 4.026(0.012) + 0.0917 \stackrel{?}{=} 4.026 \left(\frac{0.988}{0.966} \right) (0.034)$$

The equation checks out to a satisfactory degree of precision.

We have completed our review of the ideal gas law as applied to pure components and mixtures. In the next section we look at instances in which the ideal gas law does not apply.

Self-Assessment Test

1. A furnace is fired with 1000 ft³ per hour at 60°F and 1 atm of a natural gas having the following volumetric analysis: CH₄: 80%, C₂H₆: 16%, O₂: 2%, CO₂: 1%, N₂: 1%. The exit

flue gas temperature is 800°F and the pressure is 760 mm Hg absolute; 15% excess air is used and combustion is complete. Calculate the:

- (a) Volume of CO_2 produced per hour
 - (b) Volume of H_2O vapor produced per hour
 - (c) Volume of N_2 produced per hour
 - (d) Total volume of flue gas produced per hour
2. A flue gas contains 60% N_2 and is mixed with air. If the resulting mixture flows at a rate of 250,000 ft^3/hr and contains 70% N_2 , what is the flow rate of the flue gas? State all your assumptions concerning the temperatures and pressures of the gas streams.

3.2 REAL GAS RELATIONSHIPS

We have said that at room temperature and pressure many gases can be assumed to act as ideal gases. However, for some gases under normal conditions, and for most gases under conditions of high pressure, values of the gas properties that you might obtain using the ideal gas law would be at wide variance with the experimental evidence. You might wonder exactly how the behavior of **real gases** compares with that calculated from the ideal gas laws. In Fig. 3.2 you can see how the $p\bar{V}$ product of

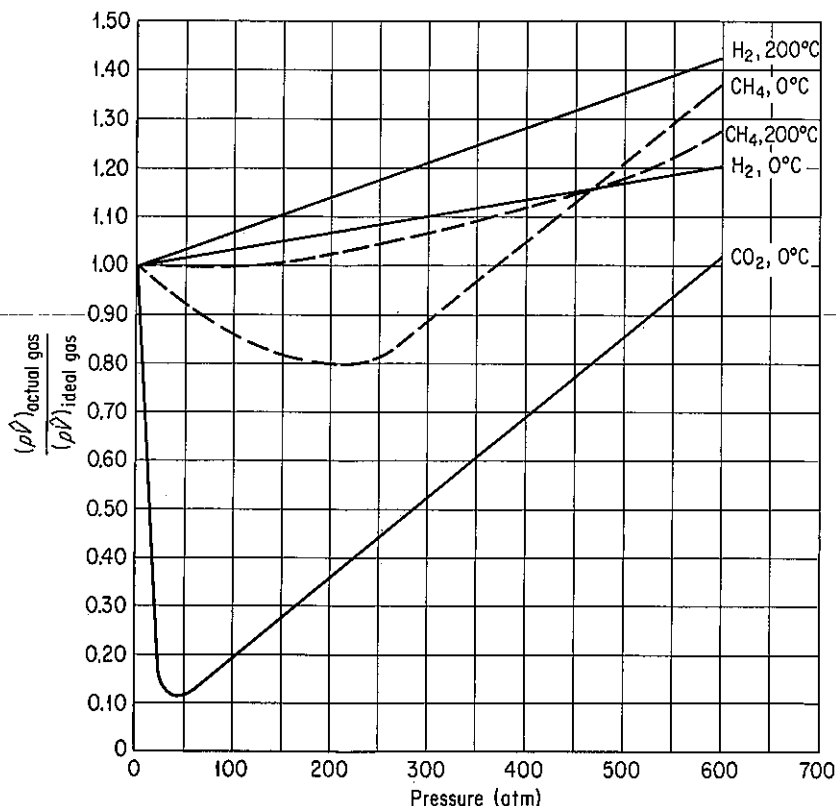


Figure 3.2 Deviation of real gases from the ideal gas law at high pressures.

several gases deviates from that predicted by the ideal gas laws as the pressure increases substantially. Thus it is clear that we need some way of computing the p - V - T properties of a gas that is not ideal, i.e. the real gas.

Essentially there are four methods of getting or predicting real gas properties:

- (a) Equations of state
- (b) Compressibility charts
- (c) Estimated properties
- (d) Actual experimental data

Even if experimental data are available, the other three techniques still may be quite useful for certain types of calculations. Of course, under conditions such that part of the gas liquefies, the gas laws apply only to the gas-phase portion of the system—you cannot extend these real gas laws into the liquid region any more than you can apply the ideal gas laws to a liquid.

Physical properties and/or the equations used to predict physical properties can be stored on spreadsheets. No programming experience is required, and simple calculations are easy to carry out using the spreadsheet software. Numerical data such as tables can be typed into cells, where they are easily seen. Equations can also be typed into cells and are introduced as needed. Labels and units as well as remarks can be added as needed.

3.2-1 Equations of State

Your objectives in studying this section are to be able to:

1. Cite two reasons for using equations of state to predict p , V , T properties of gases.
2. Write down the van der Waals equations.
3. Explain what the units are for the coefficients in van der Waals equation or other equations of state.
4. Solve the van der Waals equation or other equations of state for either p , V , n , or T given the values for the coefficients and the other variables.
5. Write down the names of five equations of state.

Equations of state relate the p - V - T properties of a pure substance (or mixtures) by theoretical or empirical relations. The simplest example of an equation of state is the ideal gas law itself. Table 3.2 lists a few of the commonly used equations of state (from among the hundreds that have been proposed) that involve two or more coefficients. In Secs. 1.7 and 4.3 and the supplementary references at the end of this chapter, you will find numerous sources of information about equations of state and

TABLE 3.2 Equations of State (for 1 Mole)*

Van der Waals:

$$\left(p + \frac{a}{\hat{V}^2}\right)(\hat{V} - b) = RT$$

$$a = \left(\frac{27}{64}\right) \frac{R^2 T_c^2}{P_c}$$

$$b = \left(\frac{1}{8}\right) \frac{RT_c}{P_c}$$

Dieterici:

$$p = \frac{RT}{\hat{V} - b} e^{-a/\hat{V}RT}$$

Berthelot:

$$p = \frac{RT}{\hat{V} - b} - \frac{a}{T \hat{V}^2}$$

Redlich-Kwong:

$$\left[p + \frac{a}{T^{1/2} \hat{V}(\hat{V} + b)}\right](\hat{V} - b) = RT$$

$$a = 0.4278 \frac{R^2 T_c^{2.5}}{P_c}$$

$$b = 0.0867 \frac{RT_c}{P_c}$$

Kammerlingh-Onnes:

$$p\hat{V} = RT \left(1 + \frac{B}{\hat{V}} + \frac{C}{\hat{V}^2} + \dots\right)$$

Holborn:

$$p\hat{V} = RT(1 + B'p + C'p^2 + \dots)$$

Beattie-Bridgeman:

$$p\hat{V} = RT + \frac{\beta}{\hat{V}} + \frac{\gamma}{\hat{V}^2} + \frac{\delta}{\hat{V}^3}$$

$$\beta = RTB_0 - A_0 - \frac{Rc}{T^2}$$

$$\gamma = -RTB_0b + aA_0 - \frac{RB_0c}{T^2}$$

$$\delta = \frac{RB_0bc}{T^2}$$

Benedict-Webb-Rubin:

$$p\hat{V} = RT + \frac{\beta}{\hat{V}} + \frac{\sigma}{\hat{V}^2} + \frac{\eta}{\hat{V}^4} + \frac{w}{\hat{V}^5}$$

$$\beta = RTB_0 - A_0 - \frac{C_0}{T^2}$$

$$\sigma = bRT - a + \frac{c}{T^2} \exp\left(-\frac{\gamma}{\hat{V}^2}\right)$$

$$\eta = cy \exp\left(-\frac{\gamma}{\hat{V}^2}\right)$$

$$w = a\alpha$$

Peng-Robinson:

$$p = \frac{RT}{\hat{V} - b} - \frac{a\alpha}{\hat{V}(\hat{V} + b) + b(\hat{V} - b)}$$

$$a = 0.45724 \left(\frac{R^2 T_c^2}{P_c}\right)$$

$$b = 0.07780 \left(\frac{RT_c}{P_c}\right)$$

$$\alpha = [1 + \kappa(1 - T_r^{1/2})]^2$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$$

$$\omega = \text{acentric factor (see p. 275)}$$

* T_c and p_c are explained in Sec. 3.2-2; $\hat{V}$ is the specific volume.

TABLE 3.3 Overall Average Deviation Between Predicted and Experimental Values Based on 4010 Experimental Points

Equation	No. of coefficients	Absolute deviation %	Reference
Adachi-Lu-Sugie	4	1.05	<i>Fluid Phase Equilibria</i> , v. 11, p. 29 (1983)
Harmens-Knapp	3	1.36	<i>Ind. Eng. Chem. Fundam.</i> , v. 19, p. 291 (1980)
Schmidt-Wenzel	3	1.39	<i>Chemical Eng. Sci.</i> , v. 35, p. 1503 (1980)
Kumar-Starling	4	1.39	<i>Ind. Eng. Chem. Fundam.</i> , v. 21, p. 255 (1982)
Peng-Robinson	2	1.51	<i>Ind. Eng. Chem. Fundam.</i> , v. 15, p. 59 (1976)
Redlich-Kwong	2	1.64	<i>Chem. Rev.</i> , v. 44, p. 233 (1949)
Soave-Redlich-Kwong	2	1.72	<i>Chem. Eng. Sci.</i> , v. 27, p. 1197 (1972)
Ishikawa-Chung-Lu	2	2.71	<i>AIChE J.</i> , v. 26, p. 372 (1980)
Van der Waals	2	3.68	Ph.D. dissertation, Leiden, the Netherlands (1873)
Clausius	3	4.18	<i>Ann. Phys. Chem.</i> , v. 9, 337 (1881)
Wilson-Redlich-Kwong	2	4.42	<i>Adv. Cryog. Eng.</i> , v. 9, p. 168 (1964)

SOURCE: Y. Adachi, H. Sugie, and B. C.-Y. Lu, *J. Chem. Eng. Jpn.*, v. 17, p. 624 (1984)

values of their respective coefficients for various compounds (and sometimes for mixture). Computer data bases¹ are also good sources of the same information in a convenient form. An equation of state to be effective must represent experimental data for p - V - T properties in the gas region with reasonable precision.

In an evaluation of 11 cubic equations of state, Adachi, Sugie, and Lu reported the overall ratings listed in Table 3.3 based on 21 substances for which detailed experimental data were available in the literature. You can see that some of the two parameter equations of state listed in Table 3.2 do as good a job in predicting p - V - T properties, as do some of the equations of state with more coefficients.

¹P. Benedek and F. Olti, *Computer Aided Chemical Thermodynamics of Gases and Liquids: Theory, Models and Programs*, Wiley, New York, 1985.

themselves by subtracting a term from the volume in the ideal gas law. These corrections led to the following equation:

$$\left(p + \frac{n^2 a}{V^2}\right)(V - nb) = nRT \quad (3.11)$$

where a and b are constants determined by fitting the van der Waals equation to experimental p - V - T data, particularly the values at the critical point. Values of the van der Waals constants for a few gases are given in Table 3.4.

Let us look at some of the computational problems that may arise when using equations of state to compute p - V - T properties by using van der Waals equation as an illustration. Van der Waals equation can easily be explicitly solved for p as follows:

$$p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} \quad (3.12)$$

However, if you want to solve for V (or n), you can see that the equation becomes cubic in V (or n):

$$V^3 - \left(nb + \frac{nRT}{p}\right)V^2 + \frac{n^2 a}{p}V - \frac{n^3 ab}{p} = 0 \quad (3.13)$$

Most of the equations of state are expressed explicitly in terms of p and sometimes T . To solve for $\hat{V}$, given p and T , you must extract a real root. In the disk in the back of the book you will find two Fortran computer programs that can solve

TABLE 3.4 Constants for the Van der Waals and Redlich-Kwong Equations Calculated From the Listed Values of the Critical Constants

	van der Waals		Redlich-Kwong	
	$\left[\text{atm} \left(\frac{\text{cm}^3}{\text{g mol}}\right)^2\right]$	$\left(\frac{\text{cm}^3}{\text{g mol}}\right)$	$\left[\text{(atm)(K)}^{1/2} \left(\frac{\text{cm}^3}{\text{g mol}}\right)\right]$	$\left(\frac{\text{cm}^3}{\text{g mol}}\right)$
Air	1.33×10^6	36.6	15.65×10^6	25.3
Ammonia	4.19×10^6	37.3	85.00×10^6	25.7
Carbon dioxide	3.60×10^6	42.8	63.81×10^6	29.7
Ethane	5.50×10^6	65.1	97.42×10^6	45.1
Ethylene	4.48×10^6	57.2	76.92×10^6	39.9
Hydrogen	0.246×10^6	26.6	1.439×10^6	18.5
Methane	2.25×10^6	42.8	31.59×10^6	29.6
Nitrogen	1.347×10^6	38.6	15.34×10^6	26.8
Oxygen	1.36×10^6	31.9	17.12×10^6	22.1
Propane	9.24×10^6	90.7	180.5×10^6	62.7
Water vapor	5.48×10^6	30.6	140.9×10^6	21.1

*To convert to psia (ft³/lb mol)², multiply table value by 3.776×10^{-3} .

†To convert to ft³/lb mol, multiply table value by 1.60×10^{-2} .

‡To convert to psia (°R)^{1/2}(ft³/lb mol)², multiply table value by 2.807×10^{-3} .

nonlinear equations and can be used to solve for $\hat{V}$ if you have a *reasonable initial guess* for $\hat{V}$, say from the ideal gas law. One code uses Newton's method and the other a minimization method; refer to Appendix L for details of these methods. An alternative technique for finding the root of a function of a single variable is to plot the nonlinear function obtained by substituting p and T into the equation of state, collecting all the terms on the left-hand side of the equation, and finding at what value of $\hat{V}$ the left-hand function crosses the zero axis as $\hat{V}$ changes. Tao² gives a basic program for plotting and discusses some of the difficulties. Quite a bit of inexpensive software is available to execute such plots on any type of computer hardware.

EXAMPLE 3.12 Application of Van der Waals' Equation

A 5.0-ft³ cylinder containing 50.0 lb of propane (C₃H₈) stands in the hot sun. A pressure gauge shows that the pressure is 665 psig. What is the temperature of the propane in the cylinder? Use van der Waals equation. See Fig. E3.12.

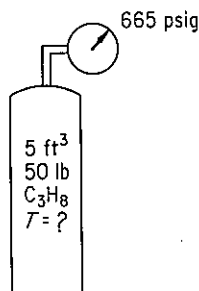


Figure E3.12

Solution

Basis: 50 lb of propane

The van der Waals constants obtained from any suitable handbook or Table 3.4 are

$$a = 3.49 \times 10^4 \text{ psia} \left(\frac{\text{ft}^3}{\text{lb mol}} \right)^2$$

$$b = 1.45 \frac{\text{ft}^3}{\text{lb mol}}$$

$$\left(p + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

All the additional information you need is as follows:

$$p = 665 \text{ psig} + 14.7 = 679.7 \text{ psia}$$

$$R \text{ in proper units is } = 10.73 \frac{(\text{psia})(\text{ft}^3)}{(\text{lb mol})(^\circ\text{R})}$$

²B. Y. Tao, *Chem. Eng.*, p. 145 (August 17, 1987).

$$n = \frac{50 \text{ lb}}{44 \text{ lb/lb mol}} = 1.136 \text{ lb mol propane}$$

$$\left[679.7 + \frac{(1.136)^2(3.49 \times 10^4)}{(5)^2} \right] [5 - 1.36(1.45)] = 1.136(10.73)T$$

$$T = 683^\circ\text{R} = 223^\circ\text{F}$$

Can you solve the Redlich-Kwong equation explicitly for T ?

EXAMPLE 3.13 Solution of the Van der Waals Equation for V

We will work Example 3.12 backward and solve for V given p , T , and n :

$$p = 679.7 \text{ psia} \quad a = 3.49 \times 10^4 \text{ psia} \left(\frac{\text{ft}^3}{\text{lb mol}} \right)^2$$

$$T = 683^\circ\text{R} \quad b = 1.45 \frac{\text{ft}^3}{\text{lb mol}}$$

We use Eq. (3.13), a cubic equation in V :

$$f(V) = V^3 - \frac{pnb + nRT}{p}V^2 + \frac{n^2a}{p}V - \frac{n^3ab}{p} = 0 \quad (\text{a})$$

Equation (a) might have one or three real roots, as indicated in Fig. E3.13, but we want a positive real root.

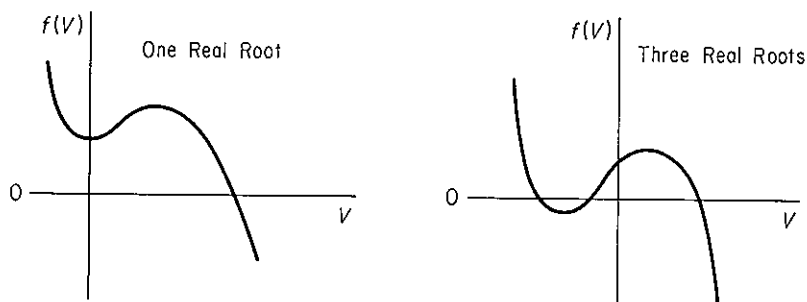


Figure E3.13

Solution

Let us apply Newton's method (refer to Appendix L) to obtain the desired root:

$$V_{k+1} = V_k - \frac{f(V_k)}{f'(V_k)} \quad (\text{b})$$

where $f'(V_k)$ is the derivative of $f(V)$ with respect to V evaluated at V_k :

$$f'(V) = 3V^2 - \frac{2(pnb + nRT)}{p}V + \frac{n^2a}{p} \quad (\text{c})$$

You can obtain a reasonably close approximation to V (or n) in many cases from the ideal gas law, useful at least for the first trial.

$$\begin{aligned}
 V_0 &= \frac{nRT}{p} \\
 &= \frac{1.136 \text{ lb mol}}{679.7 \text{ psia}} \left| \frac{10.73 \text{ (psia)(ft}^3\text{)}}{(\text{lb mol})(^\circ\text{R})} \right| \frac{683^\circ\text{R}}{679.7 \text{ psia}} \\
 &= 12.26 \text{ ft}^3 \text{ at } 679.7 \text{ psia and } 683^\circ\text{R}
 \end{aligned}$$

The second and subsequent estimates of V will be calculated using Eq. (b):

$$\begin{aligned}
 V_1 &= V_0 - \frac{f(V_0)}{f'(V_0)} \\
 f(V_0) &= (12.26)^3 - \frac{(679.7)(1.137)(1.45) + (1.137)(10.73)(683)}{679.7} (12.26)^2 \\
 &\quad + \frac{(1.137)^2(3.49 \times 10^4)}{679.7} (12.26) - \frac{(1.137)^3(3.49 \times 10^4)(1.45)}{679.7} = 738.3 \\
 f'(V_0) &= 3(12.26)^2 - \frac{2[(679.7)(1.137)(1.45) + (1.137)(10.73)(683)]}{679.7} (12.26) \\
 &\quad + \frac{(1.137)^2(3.49 \times 10^4)}{679.7} = 216.7 \\
 V_1 &= 12.26 - \frac{738.3}{216.7} = 8.85
 \end{aligned}$$

On the next iteration

$$V_2 = V_1 - \frac{f(V_1)}{f'(V_1)}$$

and so on until the change in V_k from one iteration to the next is sufficiently small. The Newton code on the disk in back of the book will execute this technique for you. The final solution is 5.09 ft^3 at 679.7 psia and 683°R .

Other equations of state. Table 3.2 lists a few of the many equations of state in addition to van der Waals equation. The form of these equations is of interest inasmuch as they are attempts to fit the experimental data with as few constants in the equation as possible.

The two-constant, Peng-Robinson and Redlich-Kwong equations appear to be quite good according to Table 3.3. Figure 3.4 compares the van der Waals and the Redlich-Kwong equations of state with experimental data.

One form of the equation known as the *virial* form is illustrated by the equations of Kammerlingh-Onnes and Holborn. These are essentially power series in $1/\hat{V}$ or in p , and the quantities B , C , D , and so on, are known as *virial coefficients*. These equations reduce to $p\hat{V} = RT$ at low pressures.

In spite of their complications, equations of state are important for several reasons. They permit a concise summary of a large mass of experimental data and also permit accurate interpolation between experimental data points. They provide a con-

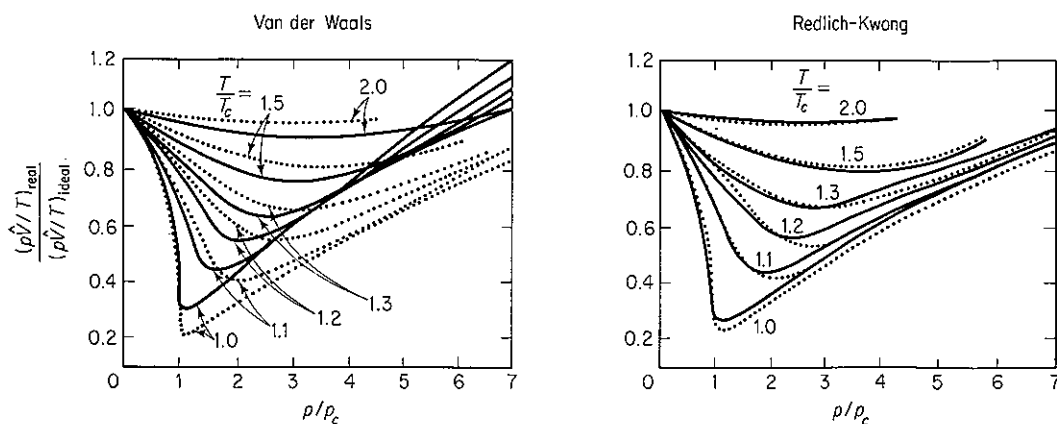


Figure 3.4 Comparison of experimental values (dots) with predicted values (solid lines) for two equations of state.

tinuous function to facilitate calculation of physical properties involving differentiation and integration. Finally, they provide a point of departure for the treatment of thermodynamic properties of mixtures. However, for instructional purposes, and for many engineering calculations, the techniques of predicting p - V - T values discussed in the next section are more convenient to use and usually have adequate accuracy.

Self-Assessment Test

- Equations of state for gases are often used to predict p - V - T properties of gas. Cite two reasons.
- The constants for an equation of state are usually evaluated in one of two ways.
 - What are these?
 - Which method would you expect to give an equation showing the least error?
- What are some of the tests that you might apply to see how an equation of state fits p - V - T data?
- What factors in a real gas cause the gas to behave in a nonideal manner? How are these factors taken into account in the van der Waals equation?
- What are the units of a and b in the SI system for the Redlich-Kwong equation?
- You measure that 0.00220 lb mol of a certain gas occupies a volume of 0.95 ft³ at 1 atm and 32°F. If the equation of state for this gas is $pV = nRT(1 + bp)$, where b is a constant, find the volume at 2 atm and 71°F.
- Calculate the temperature of 2 g mol of a gas using van der Waals equation with $a = 1.35 \times 10^{-6} \text{ (m}^6\text{)(atm)(g mol}^{-2}\text{)}$, $b = 0.0322 \times 10^{-3} \text{ (m}^3\text{)(g mol}^{-1}\text{)}$ if the pressure is 100 kPa and the volume is 0.0515 m³.
- Calculate the pressure of 10 kg mol of ethane in a 4.86-dm³ vessel at 300 K using two equations of state: (a) ideal gas and (b) Beattie-Bridgeman. For the latter, $A_0 = 5.880$, $a = 0.05861$, $B_0 = 0.09400$, $b = 0.01915$, and $c = 90.0 \times 10^4$. Compare with your answer the observed value of 34.0 atm.
- The van der Waals constants for a gas are $a = 2.31 \times 10^6 \text{ (atm)(cm}^3\text{/g mol)}^2$ and $b = 44.9 \text{ cm}^3\text{/g mol}$. Find the volume per kilogram mole if the gas is at 90 atm and 373 K.

Thought Problems

1. Municipal sludge is being converted to sterile ash and readily biodegradable liquid effluent in a mile-deep well at Longmont, Colo. The nation's first deep well wet-air oxidation sludge destruction process reduces chemical oxygen demand (COD) by up to 68% and destroys all living organisms, according to Howard C. Delaney, superintendent of Longmont's wastewater treatment plant.

The VerTech Treatment System is suspended in the conventionally drilled, concrete-encased well. Tubes of various diameters are concentrically fitted within the well to create annular spaces for the two-phase flow. Oxidation takes place at the bottom of the well.

Why is it advantageous to oxidize the sludge at the bottom of a well rather than in a pond at ground level?

2. From the *Oil and Gas Journal*, p. 55, December 2, 1985.

Gas trapped in a subsea blowout preventer (BOP) can be a serious problem in a deepwater drilling.

Amoco used nitrogen to confirm a safe method to remove gas trapped in a subsea BOP. The gas can be recovered up the choke line by displacing the kill mud with water and then allowing the gas to expand against the lower hydrostatic gradient of the water.

The test was conducted in the Gulf of Mexico from a semisubmersible working in 1,015 ft. of water.

What is the difficulty with a gas bubble trapped in the BOP?

3.2-2 Critical State, Reduced Parameters, and Compressibility

Your objectives in studying this section are to be able to:

1. State the law of corresponding states.
2. Define the critical state.
3. Calculate the reduced temperature, reduced pressure, and reduced volume, and to use any two of these parameters to obtain the compressibility factor, z , from the Nelson and Obert charts.
4. Use compressibility factors and appropriate charts to predict the p - V - T behavior of a gas, or given the required data, find compressibility factors.
5. List Newton's corrections for H_2 and He to T_c and p_c .
6. Calculate the ideal critical volume and ideal reduced volume, and be able to use V_{r_i} as a parameter in the Nelson and Obert charts.

In the attempt to devise some truly universal gas law for high pressures, the idea of corresponding states was developed. Early experimenters found that at the critical

point all substances are in approximately the same state of molecular dispersion. Consequently, it was felt that their thermodynamic and physical properties should be similar. The **law of corresponding states** expresses the idea that in the critical state all substances should behave alike.

Now that we have mentioned **critical state** several times, let us consider exactly what the term means. You can find many definitions, but the one most suitable for general use with pure component systems as well as with mixtures of gases is the following:

The critical state for the gas-liquid transition is the set of physical conditions at which the density and other properties of the liquid and vapor become identical.

This point, for a pure component (only), is the highest temperature at which liquid and vapor can exist in equilibrium.

Refer to Fig. 3.5. The vertical lines between the curves are lines of constant temperature connecting the vapor and liquid phases, which are in equilibrium. As the temperature increases, the specific volumes of the vapor and liquid approach each other until finally at 374.14°C , the values are the same. At the critical state water undergoes a transition from liquid to vapor without the appearance of a distinct two-phase region. If you watch a liquid held at or above the critical temperature and expand its volume, you cannot tell when the liquid becomes a vapor, because no interface is formed between the phases—no liquid surface can be seen. This phenomenon occurs at such a high pressure and temperature for water that it is outside your everyday experience.

You can find experimental values of the critical temperature (T_c) and the critical pressure (p_c) for various compounds in Appendix D. If you cannot find a desired critical value in this text or in a handbook, you can always consult Reid, Prausnitz,

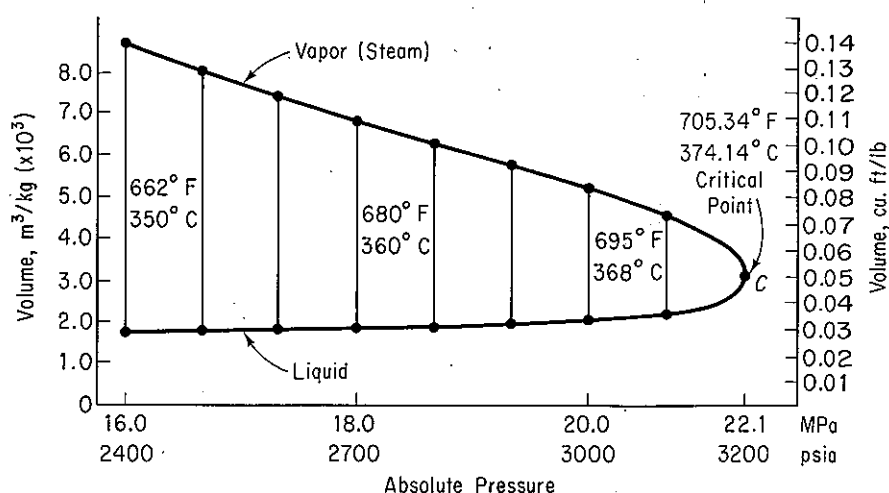


Figure 3.5 Critical point for a pure substance (water).

and Poling³ or Hakata and Hirata,⁴ which describe and evaluate methods of estimating critical constants for various compounds.

The gas-liquid transition described above is only one of several possible transitions exhibiting a critical point. Critical phenomena are observed in liquids and solids as well, as described in Sec. 3.7.

Another set of terms with which you should immediately become familiar are the **reduced conditions**. These are *corrected*, or *normalized*, conditions of temperature, pressure, and volume and are expressed mathematically as

$$T_r = \frac{T}{T_c} \quad (3.14)$$

$$p_r = \frac{p}{p_c} \quad (3.15)$$

$$V_r = \frac{V}{V_c} \quad (3.16)$$

The idea of using the reduced variables to correlate the p - V - T properties of gases, as suggested by van der Waals, is that all substances behave alike in their reduced (i.e., their corrected) states. In particular, any substance would have the same *reduced* volume at the same *reduced* temperature and pressure. If a relationship does exist involving the reduced variables that can be used to predict p_r , T_r , and V_r , what would the equation be? Certainly, the simplest form of such an equation would be to imitate the ideal gas law, or

$$p_r \hat{V}_r = \gamma RT_r \quad (3.17)$$

where γ is some universal constant. Although Eq. (3.17) works for many gases at high temperature and pressure, it fails to predict gas properties adequately in the region where the ideal gas law applies (at low pressure somewhat below $p_r = 1.0$).

A more convenient and accurate way that has been developed to tie together the concepts of the law of corresponding states and the ideal gas law is to revert to a correction of the ideal gas law, that is, a generalized equation of state expressed in the following manner:

$$pV = znRT \quad (3.18)$$

where the dimensionless quantity z is called the **compressibility factor** and is a function of the pressure and temperature:

$$z = \psi(p, T) \quad (3.19)$$

One way to look at z is to consider it to be a factor that makes Eq. (3.18) an equality. If the compressibility factor is plotted for a given temperature against the pressure for different gases, you obtain figures like Fig. 3.6(a). However, if the compressibility is plotted against the reduced pressure as a function of the reduced

³R. C. Reid, J. M. Prausnitz, and B. E. Poling, *The Properties of Gases and Liquids*, 4th ed., McGraw-Hill, New York, 1987.

⁴T. Hakata and M. Hirata, *J. Chem. Eng. Japan*, v. 3, p. 5(1970).

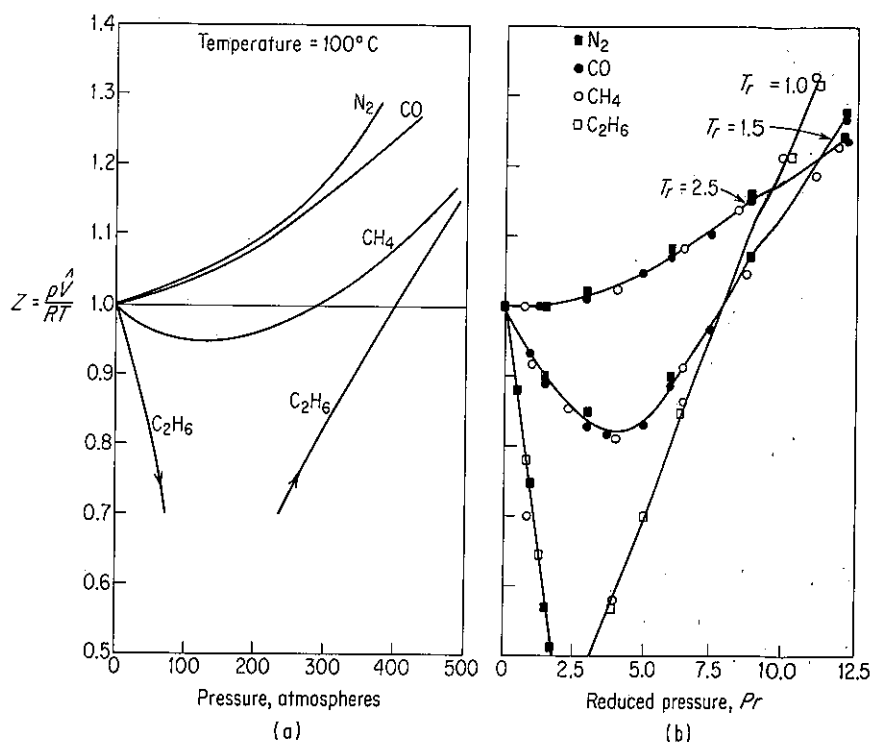


Figure 3.6 (a) Compressibility factor as a function of temperature and pressure; (b) compressibility as a function of reduced temperature and reduced pressure.

temperature,

$$z = \psi(p_r, T_r) \quad (3.20)$$

then for most gases the compressibility values at the same reduced temperature and reduced pressure fall at about the same point, as illustrated in Fig. 3.6(b).

This outcome permits the use of what is called a generalized compressibility factor, and Figs. 3.7 are the *generalized compressibility charts* or *z-factor charts* prepared by Nelson and Obert.⁵ These charts are based on 30 gases. Figures 3.7(b) and (c) represent z for 26 gases (excluding H_2 , He , NH_3 , H_2O) with a maximum deviation of 1%, and H_2 and H_2O within a deviation of 1.5%. Figure 3.7(d) is for 26 gases and is accurate to 2.5%, while Fig. 3.7(e) is for nine gases and errors can be as high as 5%. For H_2 and He only, Newton's corrections to the actual critical constants are used to give pseudocritical constants

$$T'_c = T_c + 8 \text{ K}$$

$$p'_c = p_c + 8 \text{ atm}$$

which enable you to use Figs. 3.7(a-e) for these two gases as well with minimum er-

⁵L. C. Nelson and E. F. Obert, *Chem. Eng.*, v. 61, No. 7, pp. 203-208(1954). Figures 3.7b and c include data reported by P. E. Liley, *Chem. Eng.*, p. 123 (July 20, 1987).

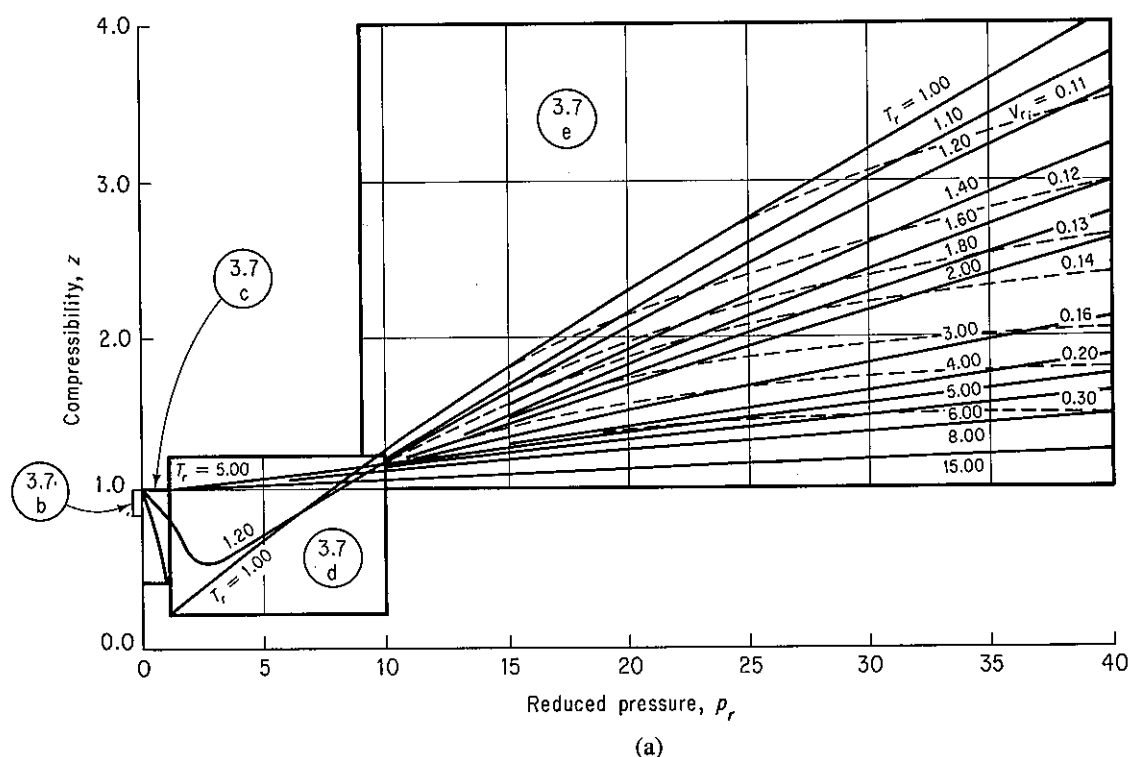


Figure 3.7 (a) Generalized compressibility chart showing the respective expanded portions. (b) Generalized compressibility chart, very low reduced pressure. (c) Generalized compressibility chart, low pressure (d) Generalized compressibility chart, medium pressures (e) Generalized compressibility chart, high pressures. (f) Generalized compressibility chart, with special scales.

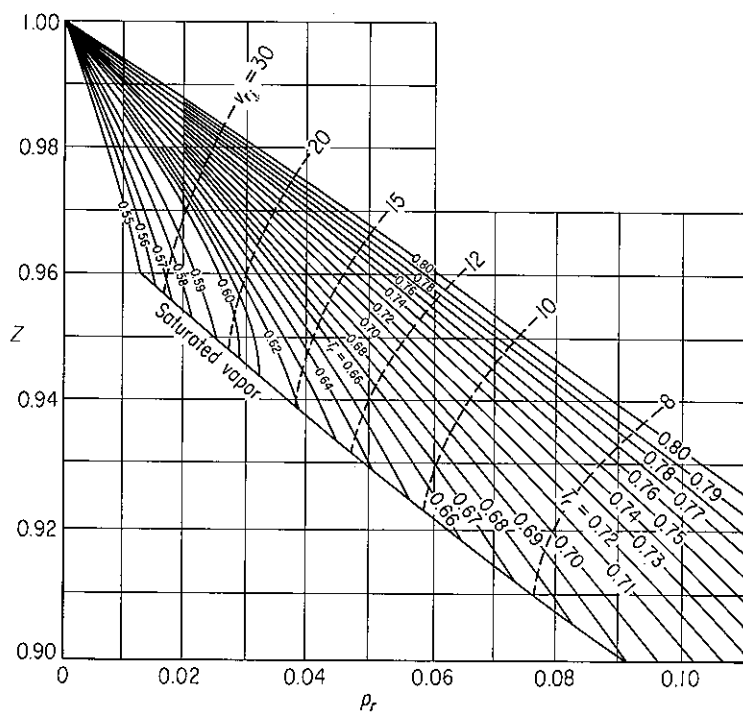
ror. Figure 3.7(f) is a unique chart which, by having several parameters plotted simultaneously on it, helps you avoid trial-and-error solutions or graphical solutions of real gas problems. One of these helpful parameters is the ideal reduced volume defined as

$$V_{ri} = \frac{\hat{V}}{\hat{V}_{ci}} \quad (3.21)$$

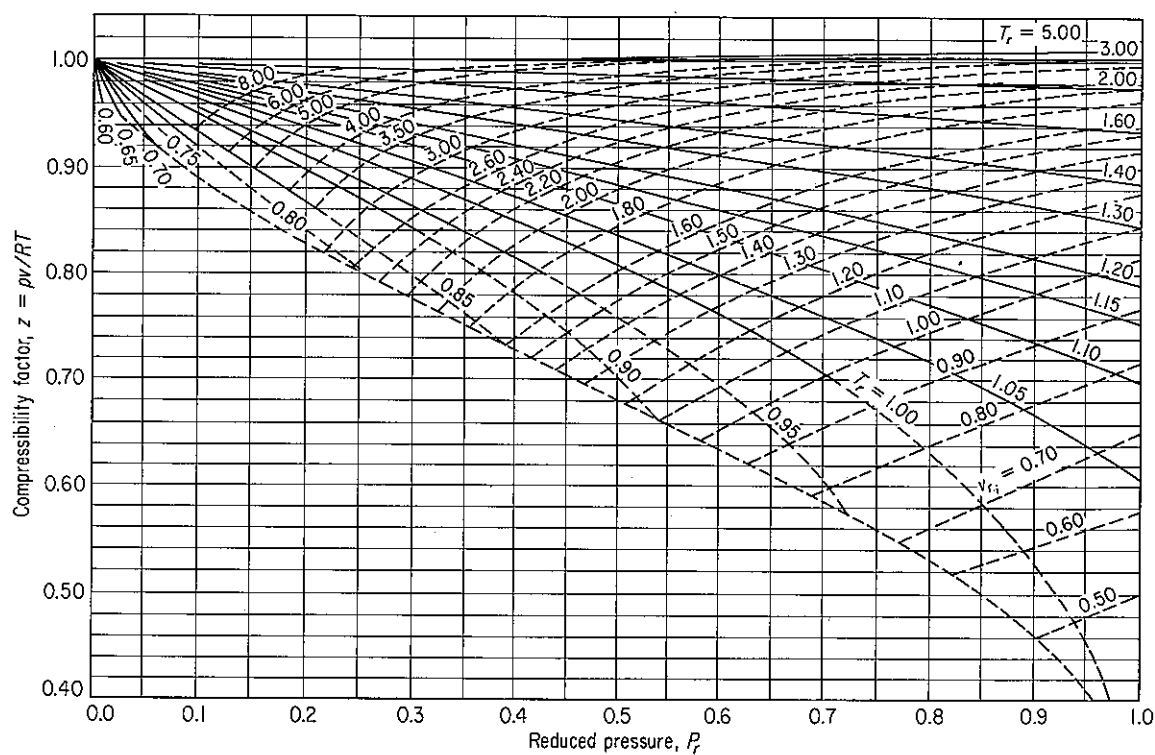
$\hat{V}_{ci}$ is the ideal critical volume, or

$$\hat{V}_{ci} = \frac{RT_c}{p_c} \quad (3.22)$$

Both V_{ri} and $\hat{V}_{ci}$ are easy to calculate since T_c and p_c are presumed known. The development of the generalized compressibility chart is of considerable practical as well as pedagogical value because it enables engineering calculations to be made with considerable ease and also permits the development of thermodynamic functions for gases for which no experimental data are available. All you need to know to use



(b)



(c)

Figure 3.7 (cont.)

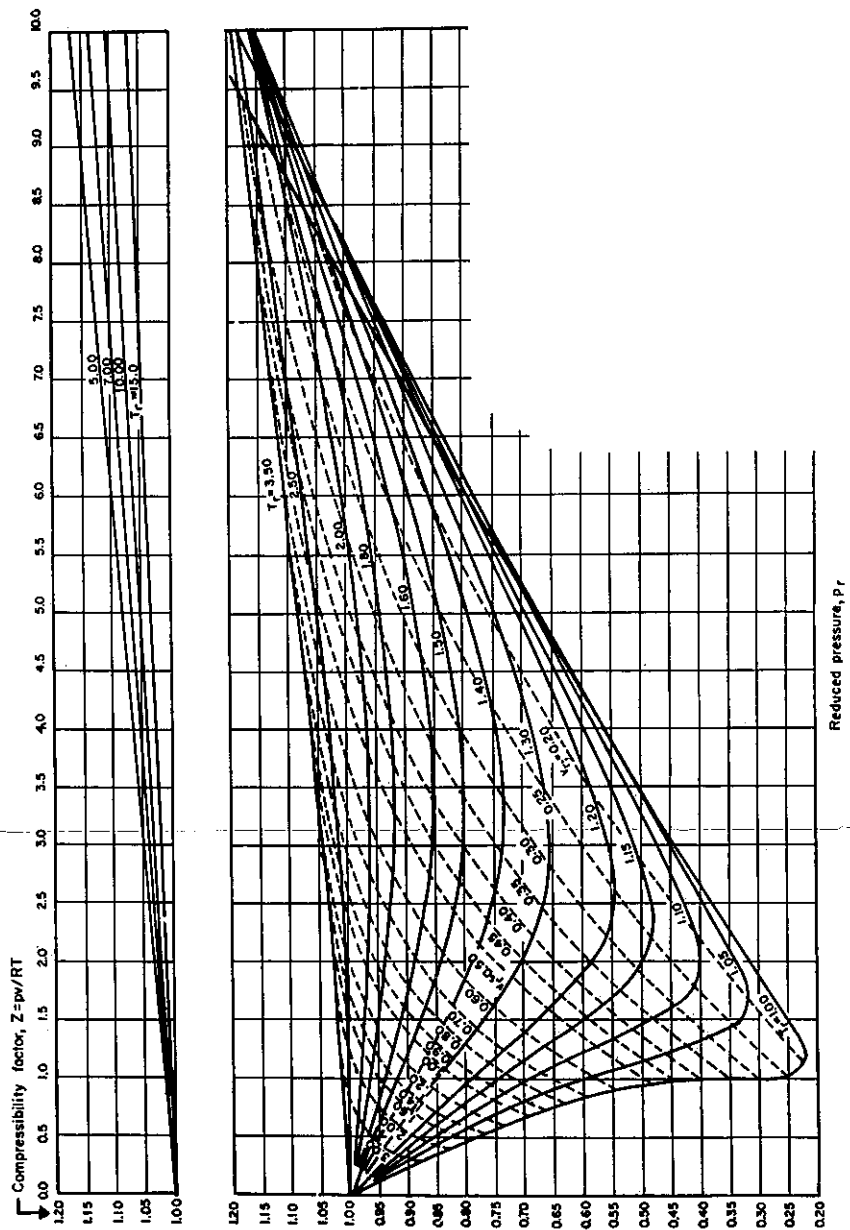


Figure 3.7 (cont.)

(d)

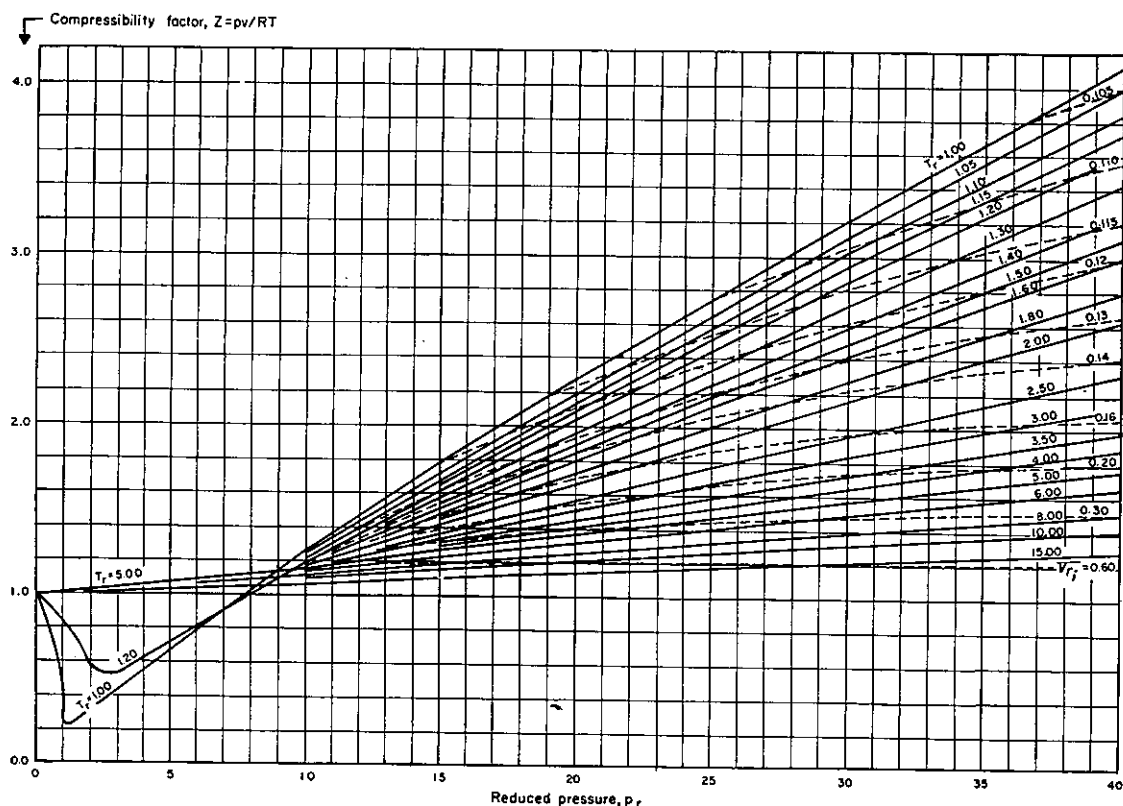


Figure 3.7 (cont.)

(e)

these charts are the critical temperature and the critical pressure for a pure substance (or the pseudovalues for a mixture, as we shall see later). The value $z = 1$ represents ideality, and the value $z = 0.27$ is the compressibility factor at the critical point.

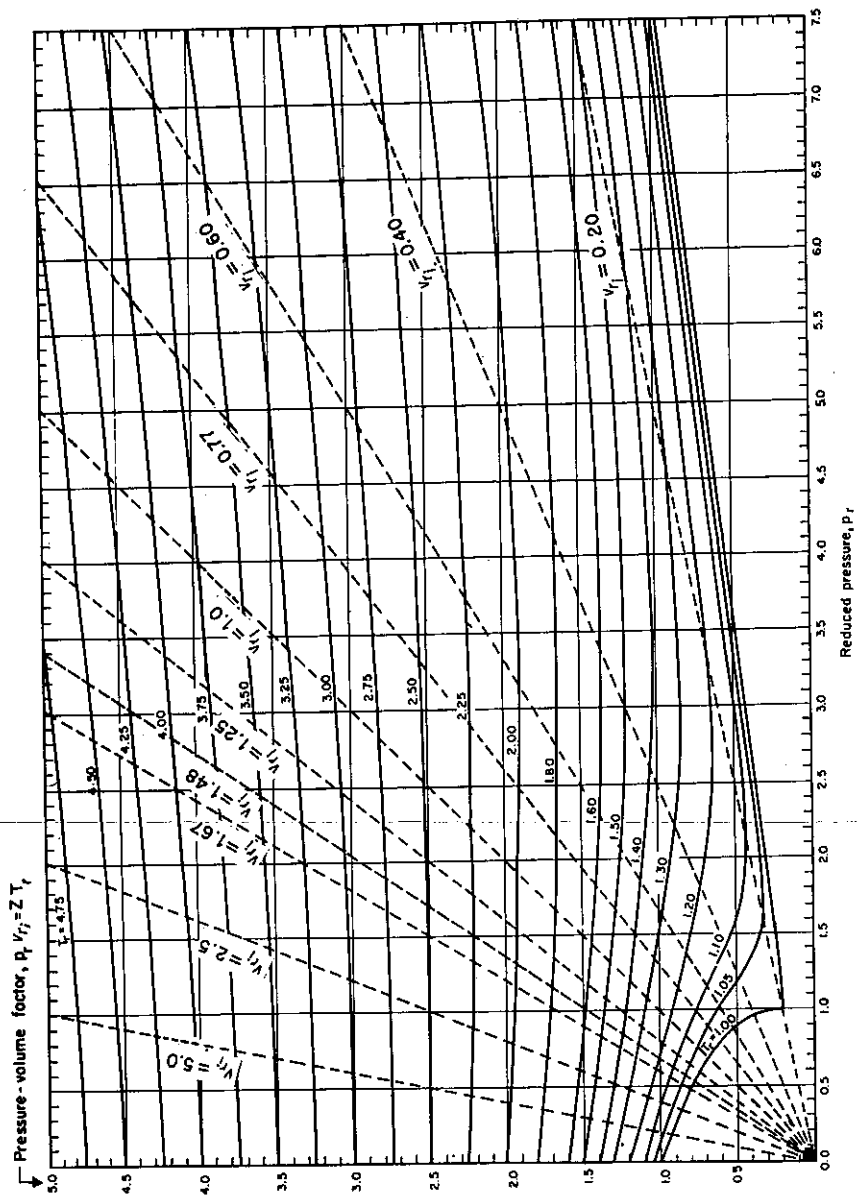
Quite a few authors have suggested that the generalized compressibility relation [Eq. (3.20)] could provide better accuracy if another parameter for the gas were included in the argument on the right-hand side (in addition to p_r and T_r). Clearly, if a third parameter is used, adding a third dimension to p_r and T_r , you have to employ a set of tables or charts rather than a single table or chart. Lydersen et al.⁶ developed tables of z that included as the third parameter the critical compressibility factor, $z_c = p_c \hat{V}_c / RT_c$. Pitzer⁷ used a different third parameter termed the *acentric* factor ω , defined as being equal to $-\ln p_{rs} - 1$, where p_{rs} is the value of the reduced vapor pressure at $T_r = 0.70$. Table 3.5 lists a few values of ω for common gases.

Viswanath and Su⁸ compared the z factors from the Nelson and Obert charts, the Lydersen–Greenkorn–Hougen charts, and the two-parameter (T_r and p_r)

⁶A. L. Lydersen, R. A. Greenkorn, and O. A. Hougen, *Univ. Wis. Eng. Exp. St. Rep. No. 4*, Madison, Wis., 1955.

⁷K. S. Pitzer, *J. Am. Chem. Soc.*, v. 77, p. 3427 (1955).

⁸D. S. Viswanath and G. J. Su, *AIChE J.*, v. 11, p. 202 (1965).



(f)

Figure 3.7 (cont.)

TABLE 3.5 Values Of The Pitzer Acentric Factor

Compound	Acentric factor
Ammonia	0.250
Argon	-0.002
Carbon dioxide	0.225
Carbon monoxide	0.049
Chlorine	0.073
Ethylene	0.085
Hydrogen	-0.220
Ethane	0.098
Hydrogen sulfide	0.100
Methane	0.008
Methanol	0.559
Nitrogen	0.040
Oxygen	0.021
Propane	0.152
Sulfur dioxide	0.251
Water vapor	0.344

Viswanath and Su chart with the experimental z 's for T_r 's of 1.00 to 15.00 and p_r 's of 0 to 40 for 19 gases with the following results:

	Viswanath-Su	Nelson-Obert	Lydersen-Greenkorn-Hougen
Number of points	415	355	339
Average deviation (%) in z	0.61	0.83	1.26

It is seen that all these charts yield quite reasonable values for engineering purposes. Because the increased precision that may accompany the use of a third parameter in calculating z is not necessary for our purposes, and because the presentation of z values so as to include the third parameter is considerably more cumbersome, we shall not show the three-parameter tables here.

EXAMPLE 3.14 Use of the Compressibility Factor

In spreading liquid ammonia fertilizer, the charges for the amount of NH_3 are based on the time involved plus the pounds of NH_3 injected into the soil. After the liquid has been spread, there is still some ammonia left in the source tank (volume = 120 ft³), but in the form of a gas. Suppose that your weight tally, which is obtained by difference, shows a net weight of 125 lb of NH_3 left in the tank as a gas at 292 psig. Because the tank is sitting in the sun, the temperature in the tank is 125°F.

Your boss complains that his calculations show that the specific volume of the gas is $1.20 \text{ ft}^3/\text{lb}$ and hence that there are only 100 lb of NH_3 in the tank. Could he be correct? See Fig. E3.14.

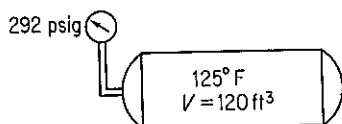


Figure E3.14

Solution

Basis: 1 lb of NH_3

Apparently, your boss used the ideal gas law in calculating his figure of $1.20 \text{ ft}^3/\text{lb}$ of NH_3 gas:

$$R = 10.73 \frac{(\text{psia})(\text{ft}^3)}{(\text{lb mol})(^\circ\text{R})}$$

$$T = 125^\circ\text{F} + 460 = 585^\circ\text{R}$$

$$p = 292 + 14.7 = 306.7 \text{ psia}$$

$$n = \frac{1 \text{ lb}}{17 \text{ lb/lb mol}}$$

$$\hat{V} = \frac{nRT}{p} = \frac{\frac{1}{17}(10.73)(585)}{306.7} = 1.20 \text{ ft}^3/\text{lb}$$

However, he should have used the compressibility factor because NH_3 does not behave as an ideal gas under the observed conditions of temperature and pressure. Let us again compute the mass of gas in the tank this time using

$$pV = znRT$$

What is known and unknown in the equation?

$$p = 306.7 \text{ psia}$$

$$V = 120 \text{ ft}^3$$

$$z = ?$$

$$n = \frac{1}{17} \text{ lb mol}$$

$$T = 585^\circ\text{R}$$

The additional information needed (taken from Appendix D) is

$$T_c = 405.5\text{K} \approx 729.9^\circ\text{R}$$

$$p_c = 111.3 \text{ atm} \approx 1636 \text{ psia}$$

Then, since z is a function of T_r and p_r ,

$$T_r = \frac{T}{T_c} = \frac{585^\circ\text{R}}{729.9^\circ\text{R}} = 0.801$$

$$p_r = \frac{p}{p_c} = \frac{306.7 \text{ psia}}{1636 \text{ psia}} = 0.187$$

From the Nelson and Obert chart, Fig. 3.7(c), you can read $z = 0.855$. Now $\hat{V}$ can be calculated as

$$\hat{V} = \frac{znRT}{p} = \hat{V}_{\text{ideal}} \left(\frac{z}{z_{\text{ideal}}} \right)$$

or

$$\hat{V} = \frac{1.20 \text{ ft}^3 \text{ ideal}}{\text{lb}} \left| \frac{0.855}{1} \right| = 1.03 \text{ ft}^3/\text{lb NH}_3$$

$$\frac{1 \text{ lb NH}_3}{1.03 \text{ ft}^3} \left| \frac{120 \text{ ft}^3}{1} \right| = 117 \text{ lb NH}_3$$

Certainly 117 lb is a more realistic figure than 100 lb, and it is easily possible to be in error by 8 lb if the residual weight of NH_3 in the tank is determined by difference. As a matter of interest you might look up the specific volume of NH_3 at the conditions in the tank in a handbook—you would find that $\hat{V} = 0.973 \text{ ft}^3/\text{lb}$, and hence the compressibility factor calculation yielded a volume with an error of only about 4%.

Example 3.15 Use of the Compressibility Factor

Suppose that 3.500 kg of liquid O_2 is vaporized into a tank of 0.0284-m^3 volume at -25°C . What will the pressure in the tank be; will it exceed the safety limit of the tank (100 atm)?

Solution

See Fig. E3.15.

Basis: 3.500 kg O_2

We know from Appendix D that

$$T_c = 154.4 \text{ K}$$

$$p_c = 49.7 \text{ atm}$$

However, this problem cannot be worked exactly the same way as the preceding problem because we do not know the pressure of the O_2 in the tank. Thus we need to use the other parameter, V_{ri} , that is available on the Nelson and Obert charts. We first calculate

$$\hat{V} (\text{molal volume}) = \frac{0.0284 \text{ m}^3}{3.500 \text{ kg}} \left| \frac{32 \text{ kg}}{1 \text{ kg mol}} \right| = 0.2597 \text{ m}^3/\text{kg mol}$$

Note that the *molal volume* must be used in calculating V_{ri} since $\hat{V}_{ci}$ is a volume per mole. Next,

$$\hat{V}_{ci} = \frac{RT_c}{p_c} = \frac{0.08206(\text{m}^3)(\text{atm})}{(\text{kg mol})(\text{K})} \left| \frac{154.4 \text{ K}}{49.7 \text{ atm}} \right| = 0.255 \frac{\text{m}^3}{\text{kg mol}}$$

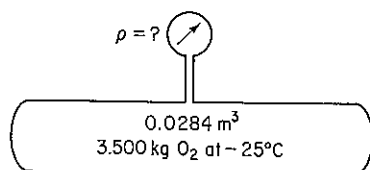


Figure E3.15

Then

$$V_{r_i} = \frac{\hat{V}}{\hat{V}_{c_i}} = \frac{0.260}{0.255} = 1.02$$

Now we know two parameters, V_{r_i} and,

$$T_r = \frac{T}{T_c} = \frac{248 \text{ K}}{154.4 \text{ K}} = 1.61$$

From the Nelson and Obert chart [Fig. 3.7],

$$p_r = 1.43$$

Then

$$\begin{aligned} p &= p_r p_c \\ &= 1.43(49.7) = 71.1 \text{ atm} \end{aligned}$$

The pressure of 100 atm is not exceeded.

Self-Assessment Test

1. What is the ideal critical volume? What is the advantage of using V_{c_i} ?
2. In a proposed low pollution vehicle burning $\text{H}_2\text{-O}_2$, the gases are to be stored in tanks at 2000 psia. The vehicle has to operate from -40 to 130°F .
 - (a) Is the ideal gas law a sufficiently good approximation for the design of these tanks?
 - (b) A practical operating range requires that 3 lb_m of hydrogen be stored. How large must the hydrogen tank be if the pressure is not to exceed 2000 psia?
 - (c) The H_2/O_2 ratio is 2 on a molar basis. How large must the oxygen tank be?
3. A carbon dioxide fire extinguisher has a volume of 40-L and is to be charged to a pressure of 20 atm at a storage temperature of 20°C . Determine the mass in kilograms of CO_2 within a charged extinguisher assuming it initially contains CO_2 at 1 atm.
4. Calculate the pressure of 4.00 g mol CO_2 contained in a $6.25 \times 10^{-3} \text{ m}^3$ fire extinguisher at 25°C .

3.2-3 Calculation of p - V - T Properties via Generalized Group Contribution Methods

The group contribution method has been successful in estimating p - V - T properties of pure components (as well as other thermodynamic properties). As indicated by the name, the idea is that compounds can be constituted from combinations of functional groups, the contribution of each group to a property can be tabulated, and the group contributions can be correlated and/or summed to give the desired property of the compound. The assumption is that a group such as $-\text{CH}_3$, or $-\text{OH}$, behaves identically no matter what the molecule may be in which it appears. This assumption is not quite true, so that any group contribution method yields approximate proper-

ties. Probably the most widely used group contribution method is UNIFAC,^{9,10} which forms a part of many data bases.

3.2-4 Gaseous Mixtures

Your objectives in studying this section are to be able to:

1. Use Dalton's law and an equation of state such as the van der Waals law to compute the total pressure of a gas mixture.
2. Use average constants given the desired weighting method in an equation of state.
3. Compute the mean compressibility factor.
4. Use Kay's method of pseudocritical values to calculate the pseudo-reduced values and predict p , V , T , and n via the compressibility factor.

So far we have discussed only pure compounds and their p - V - T relations. Most practical problems involve gaseous mixtures and there is very little experimental data available for gaseous mixtures. Thus the question is: How can we predict p - V - T properties with reasonable accuracy for gaseous mixtures? We treated ideal gas mixtures in Sec. 3.1.3, and saw, as we shall see later (in Chap. 4 for the thermodynamic properties), that for ideal mixtures the properties of the individual components can be added together to give the desired property of the mixture. But this technique does not prove to be satisfactory for real gases. The most desirable technique would be to develop methods of calculating p - V - T properties for mixtures based solely on the properties of the pure components. Possible ways of doing this for real gases are discussed below.

Averaged constants for equations of state. One way to employ equations of state is to use average values for the coefficients in the equation. Take, for example, the van der Waals equation [Eq. (3.12)]. How should the coefficients for the individual components be averaged to give the least error on the whole? For the van der Waals equation, you could proceed as follows. For b (use a linear sum of the mole fraction weights):

$$b_{\text{mixture}} = b_A y_A + b_B y_B + \cdots \quad (3.23)$$

⁹ A. Fredenslund, J. Gmehling, and P. Rasmussen, *Vapor-Liquid Equilibria Using UNIFAC*, Elsevier, Amsterdam, 1977.

¹⁰ D. Tiegs, J. Gmehling, P. Rasmussen, and A. Fredenslund, *Ind. Eng. Chem. Res.*, v. 26, p. 159 (1987).

For a (use a linear sum of the mole fraction weights of the square roots of a_i):

$$a_{\text{mixture}}^{1/2} = a_A^{1/2} y_A + a_B^{1/2} y_B + \dots \quad (3.24)$$

If you are puzzled as to the reasoning behind the use of the square-root weighting for a , remember that the term in the van der Waals equation involving a is

$$\frac{n^2 a}{V^2}$$

in which the number of moles is squared. The method of average constants can be successfully applied to most of the other equations of state listed in Table 3.2. The average constants can also be calculated from the pseudoreduced constants listed in Table 3.2. For example, for the van der Waals equation,

$$a = \frac{27}{64} \frac{R^2 T_c^2}{p_c} \quad \text{and} \quad b = \frac{RT_c}{8p_c}$$

you could average T_c and p_c and then compute a and b for the mixture.

Mixing rules used to estimate coefficients in gaseous mixtures via equations of state sometimes involve not only weighted sums of the coefficients of the pure components but interaction coefficients as well. Consequently, it is difficult to add new substances to a data base because of the large amount of data that has to be collected if the equation of state has more than two or three interaction coefficients.

Mean compressibility factor. Another approach toward the treatment of gaseous mixtures is to say that $pV = z_m nRT$, where z_m can be called the *mean compressibility factor*. With such a relationship the only problem is how to evaluate the mean compressibility factor satisfactorily. One obvious technique that might occur to you is to make z_m a mole average as follows:

$$z_m = z_A y_A + z_B y_B + \dots \quad (3.25)$$

Since z is a function of both the reduced temperature and the reduced pressure, it is necessary to decide what pressure will be used to evaluate p_r .

- (a) *Assume Dalton's law of partial pressures.* For each component z is evaluated at T_r and the reduced *partial pressure* for each gaseous component. The reduced partial pressure is defined as

$$p_{rA} = \frac{p_A}{p_{cA}} = \frac{(p_r) y_A}{p_{cA}} \quad (3.26)$$

- (b) *Assume Amagat's law of pure component volumes.* For each component z is evaluated at T_r and the reduced *total pressure* on the system.

Pseudocritical properties. Many weighting rules have been proposed to combine the critical properties of the components of a real gas mixture in order to get an effective set of critical properties (*pseudocritical*) that enable *pseudo-reduced*

properties of the mixture to be computed.¹¹ The pseudoreduced properties in turn are used in exactly the same way as the reduced properties for a pure compound, and in effect the mixture can be treated as a pure compound. In instances where you know nothing about the gas mixture, this technique is preferable to any of those discussed previously.

In Kay's method, pseudocritical values for mixtures of gases are calculated on the assumption that each component in the mixture contributes to the pseudocritical value in the same proportion as the number of moles of that component. Thus the pseudocritical values are computed as follows:

$$p'_c = p_{cA} y_A + p_{cB} y_B + \cdots \quad (3.27)$$

$$T'_c = T_{cA} y_A + T_{cB} y_B + \cdots \quad (3.28)$$

where p'_c = pseudocritical pressure and T'_c = pseudocritical temperature. (It has also been found convenient in some problems to calculate similarly a weighted pseudo-ideal-critical volume V'_{ci} .) You can see that these are linearly weighted mole average pseudocritical properties. (In Sec. 3.7 we compare the true critical point of gaseous mixtures with the pseudocritical point.) Then the respective pseudo-reduced values are

$$p'_r = \frac{p}{p'_c} \quad (3.29)$$

$$T'_r = \frac{T}{T'_c} \quad (3.30)$$

If you are faced with a complicated mixture of gases whose composition is not well known, you still can estimate the pseudocritical constants from charts¹² such as shown in Fig. 3.8 (a) and (b) if you know the gas specific gravity. Figure 3.8 is good only for natural gases composed mainly of methane that contain less than 5% impurities (CO_2 , N_2 , H_2S ; etc.).

Kay's method is known as a two-parameter rule since only p_c and T_c for each component are involved in the calculation of z . If a third parameter such as z_c , the Pitzer acentric factor, or $\hat{V}_{ci}$ is included in the determination of the mean compressibility factor, then we would have a three-parameter rule. All the pseudocritical methods do not provide equal accuracy in predicting p - V - T properties, but most suffice for engineering work. Stewart et al.¹³ reviewed 21 different methods of determining the pseudoreduced parameters by three-parameter rules (see Table 3.6). Although Kay's method was not the most accurate, it was easy to use and not consid-

¹¹ You should still add Newton's corrections for H_2 and He.

¹² Natural Gasoline Supply Men's Association, *Engineering Data Book*, Tulsa, Okla., issued periodically.

¹³ W. E. Stewart, S. F. Burkhart, and David Voo, paper given at the AIChE meeting in Kansas City, Mo., May 18, 1959. Also refer to A. Satter and J. M. Campbell, *Soc. Petrol. Engr. J.*, p. 333 (December 1963); and H. E. Barner and W. C. Quinlan, *I&EC Process Design and Development*, v. 8, p. 407 (1969).

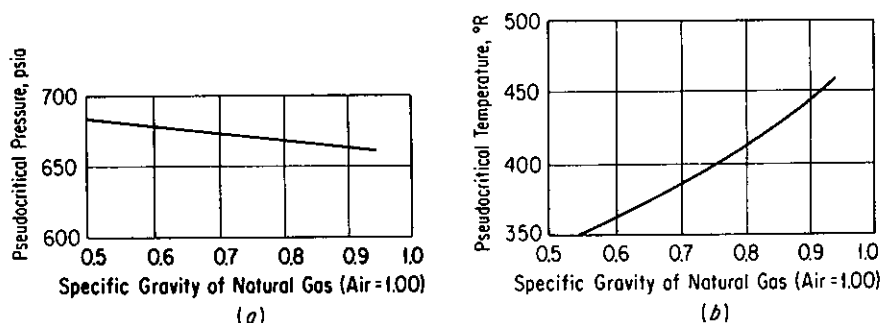


Figure 3.8 Estimation of critical properties of natural gases.

erably poorer than some of the more complex techniques of averaging critical properties.

In summary, we would have to state that no one method will consistently give the best results. Kay's pseudocritical method, on the average, will provide reasonable estimates, although other methods of greater accuracy (and of greater complexity) are available in the literature, as brought out in Table 3.6. All these methods begin to break down near the true critical point of the mixture and for highly polar compounds.

EXAMPLE 3.16 p - V - T Relations for Gas Mixture

A gaseous mixture has the following composition (in mole percent):

Methane, CH_4	20
Ethylene, C_2H_4	30
Nitrogen, N_2	50

at 90 atm pressure and 100°C . Compare the molal volume as computed by the methods of:

- the perfect gas law
- the van der Waals equation using averaged constants
- the mean compressibility factor and Dalton's law
- the mean compressibility factor and Amagat's law
- the pseudoreduced technique (Kay's method)

Solution

Basis: 1 g mol of gas mixture

Additional data needed are:

Component	$a(\text{atm})\left(\frac{\text{cm}^3}{\text{g mol}}\right)^2$	$b\left(\frac{\text{cm}^3}{\text{g mol}}\right)$	$T_c(\text{K})$	$p_c(\text{atm})$
CH_4	2.25×10^6	42.8	191	45.8
C_2H_4	4.48×10^6	57.2	283	50.9
N_2	1.35×10^6	38.6	126	33.5

$$R = 82.06 \frac{(\text{cm}^3)(\text{atm})}{(\text{g mol})(\text{K})}$$

(a) Perfect gas law:

$$\hat{V} = \frac{nRT}{p} = \frac{1(82.06)(373)}{90} = 340 \text{ cm}^3 \text{ at } 90 \text{ atm and } 373 \text{ K}$$

TABLE 3.6 Prediction Of p-V-T Values Of Gas Density By Pseudocritical Methods

Method	Percent root-mean-square deviation of density, 35 systems	Percent root-mean-square deviation of density, CO ₂ and H ₂ S free systems
(1) Three-parameter Kay's rule:		
$T'_c = \sum T_{ci} y_i$		
$p'_c = \sum p_{ci} y_i$	10.74	6.82
$z'_c = \sum z_{ci} y_i$		
(2) Empirical:		
$T'_c = \sum T_{ci} y_i$		
$\frac{T'_c}{p'_c} = \sum \frac{T_{ci} y_i}{p_{ci}}$	6.06	5.85
$\hat{V}'_c = \sum \hat{V}_{ci} y_i$		
(3) Virial approach (Joffe's method III):		
$\frac{T'_c}{p'_c} = \frac{1}{8} \sum_i \sum_j y_i y_j \left[\left(\frac{T_{ci}}{p_{ci}} \right)^{1/3} + \left(\frac{T_{cj}}{p_{cj}} \right)^{1/3} \right]$		
$\frac{T'_c}{\sqrt{p'_c}} = \sum y_i \frac{T_{ci}}{\sqrt{p_{ci}}}$	4.98	4.24
$\hat{V}'_c = \sum \hat{V}_{ci} y_i$		
(4) "Recommended" method		
$\frac{T'_c}{p'_c} = \frac{1}{3} \sum y_i \frac{T_{ci}}{p_{ci}} + \frac{2}{3} \left[\sum y_i \left(\frac{T_{ci}}{p_{ci}} \right)^{1/2} \right]^2$		
$\frac{T'_c}{\sqrt{p'_c}} = \sum y_i \frac{T_{ci}}{\sqrt{p_{ci}}}$	4.32	3.26
$z'_c = \sum z_{ci} y_i$		

(b) To use average constants in the van der Waals equation, write it in the following fashion:

$$V^3 - \left(\bar{b} + \frac{RT}{p}\right)nV^2 + \left(\frac{\bar{a}}{p}\right)n^2V - \frac{\bar{a}\bar{b}}{p}n^3 = 0$$

where $\bar{a}$ and $\bar{b}$ are the average constants.

$$(\bar{a})^{1/2} = 0.2a_{\text{CH}_4}^{1/2} + 0.3a_{\text{C}_2\text{H}_4}^{1/2} + 0.5a_{\text{N}_2}^{1/2}$$

$$\bar{a} = 2.30 \times 10^6 \text{ atm} \left(\frac{\text{cm}^3}{\text{g mol}} \right)^2$$

$$\bar{b} = 0.2b_{\text{CH}_4} + 0.3b_{\text{C}_2\text{H}_4} + 0.5b_{\text{N}_2} = 45.0 \frac{\text{cm}^3}{\text{g mol}}$$

$$n = 1 \text{ g mol (the basis)}$$

$$V^3 - \left[45.0 + \frac{(82.06)(373)}{90} \right] V^2 + \left[\frac{2.30 \times 10^6}{90} \right] V - \left[\frac{(2.30 \times 10^6)(45.0)}{90} \right] = 0$$

Let $V_0 = 340$ [based on (a) above]. You can solve the cubic equation by plotting it or by Newton's method (see the disk in the back of this book) to get $\hat{V} = 316 \text{ cm}^3$ at 90 atm and 373 K.

(c) The table below shows how Dalton's law can be used to estimate a mean compressibility factor.

Component	$p_c(\text{atm})$	$T_c(\text{K})$	y	$(90)y = p$	$p_r = \frac{p}{p_c}$	$T_r = \frac{373}{T_c}$
CH ₄	45.8	191	0.2	18	0.393	1.95
C ₂ H ₄	50.9	283	0.3	27	0.530	1.32
N ₂	33.5	126	0.5	45	1.343	2.96

Component	z	$(z)(y)$
CH ₄	0.99	0.198
C ₂ H ₄	0.93	0.279
N ₂	1.00	0.500
z_{mean}	$= 0.977$	

Then

$$V = \frac{z_{\text{mean}}RT}{p} = \frac{0.977(82.06)(373)}{90}$$

$$= 332 \text{ cm}^3 \text{ at 90 atm and 373 K}$$

(d) Combining Amagat's law and the z factor,

Component	$p_c(\text{atm})$	$T_c(\text{K})$	y	$p_r = \frac{90}{p_c}$	$T_r = \frac{373}{T_c}$
CH ₄	45.8	191	0.2	1.97	1.95
C ₂ H ₄	50.5	283	0.3	1.78	1.32
N ₂	33.5	126	0.5	2.69	2.96

Component	z	$(z)(y)$
CH ₄	0.97	0.194
C ₂ H ₄	0.75	0.225
N ₂	1.01	0.505
	$z_{\text{mean}} =$	0.924

$$V = \frac{0.924(82.06)(373)}{90} = 314 \text{ cm}^3 \text{ at } 90 \text{ atm and } 373 \text{ K}$$

(e) According to Kay's method, we first calculate the pseudocritical values for the mixture by Eqs. (3.27) and (3.28).

$$\begin{aligned}
 & \begin{array}{ccc} \text{CH}_4 & \text{C}_2\text{H}_4 & \text{N}_2 \end{array} \\
 p'_c &= p_{cA}y_A + p_{cB}y_B + p_{cC}y_C = (45.8)(0.2) + (50.9)(0.3) + (33.5)(0.5) \\
 &= 41.2 \text{ atm} \\
 T'_c &= T_{cA}y_A + T_{cB}y_B + T_{cC}y_C = (191)(0.2) + (283)(0.3) + (126)(0.5) \\
 &= 186 \text{ K}
 \end{aligned}$$

Then we calculate the pseudo-reduced values for the mixture by Eqs. (3.29) and (3.30):

$$p'_r = \frac{p}{p'_c} = \frac{90}{41.2} = 2.18, \quad T'_r = \frac{T}{T'_c} = \frac{373}{186} = 2.01$$

With the aid of these two parameters we can find from Fig. 3.7d that $z = 0.965$. Thus

$$V = \frac{zRT}{p} = \frac{0.965(82.06)(373)}{90} = 328 \text{ cm}^3 \text{ at } 90 \text{ atm and } 373 \text{ K}$$

EXAMPLE 3.17 Use of Pseudo-reduced Ideal Molal Volume

In instances where the temperature or pressure of a gas mixture is unknown, it is convenient, to avoid a trial-and-error solution using the generalized compressibility charts, to compute a pseudocritical ideal volume and a pseudoreduced ideal volume as illustrated below. Suppose we have given that the molal volume of the gas mixture in the preceding problem was 326 cm³ at 90.0 atm. What was the temperature?

Solution

Basis: 326 cm³ of gas at 90.0 atm and T K

Component	T_c (K)	p_c (atm)	y	yT_c	yp_c
CH ₄	190.7	45.8	0.20	38.14	9.16
C ₂ H ₄	283.1	50.5	0.30	84.93	15.15
N ₂	126.2	33.5	0.50	63.10	16.75
Total				186.2	41.06

Note we have used mole fractions as weighting factors to find an average $T'_c = 186.2$ and $p'_c = 41.2$ as in Example 3.16. Next we compute $\hat{V}'_{ci}$:

$$\hat{V}'_{ci} = \frac{RT'_c}{p'_c} = \frac{82.06(186.2)}{41.06} = 372.1 \text{ cm}^3/\text{g mol}$$

$$V'_{ti} = \frac{\hat{V}}{\hat{V}'_{ci}} = \frac{326}{372.1} = 0.876$$

$$p'_r = \frac{p}{p'_c} = \frac{90.0}{41.06} = 2.19$$

From Fig. 3.7d or Fig. 3.8, $T'_r = 1.98$. Then

$$T = T'_c T'_r = 186.2(1.98) = 369 \text{ K}$$

Self-Assessment Test

- One pound mole of a mixture containing 0.400 lb mol of N₂ and 0.600 lb mol C₂H₄ at 50°C occupies a volume of 1.44 ft³. What is the pressure in the vessel? Compute your answer by:
 - Averaged constants for the van der Waals equation
 - The mean compressibility factor (assuming Dalton's law)
 - Kay's method

3.3 VAPOR PRESSURE AND LIQUIDS

Your objectives in studying this section are to be able to:

- Define vapor pressure, triple point, equilibrium, dew point, bubble point, saturated, superheated, subcooled, and quality, and be able to locate the region or point in a p - T chart in which each term applies.
- Calculate the vapor pressure of a substance from a vapor-pressure equation given values for the coefficients in the equation, and look up the vapor pressure in reference books.
- Calculate the temperature of a substance from a vapor-pressure equation given the values for the coefficients in the equation, and the vapor pressure.

4. Calculate the properties of a wet mixture given the temperature, pressure, and volume of the mixture.
5. Prepare a Cox chart; calculate the scaling factors for each axis so that the chart can be used to predict the vapor pressure of a liquid.

The terms *vapor* and *gas* are used very loosely. A gas that exists below its critical temperature is usually called a vapor because it can condense. If you continually compress a pure gas at constant temperature, provided that the temperature is below the critical temperature, some pressure is eventually reached at which the gas starts to condense into a liquid. Further compression does not increase the pressure but merely increases the fraction of gas that condenses. A reversal of the procedure just described will cause the liquid to be transformed into the gaseous state again, (i.e., vaporize). From now on, the word **vapor** will be reserved to describe a gas below its critical point in a process in which the phase change is of primary interest, while the word **gas or noncondensable gas** will be used to describe a gas above the critical point or a gas in a process in which it cannot condense.

Vaporization and condensation at constant temperature and pressure are equilibrium processes, and the equilibrium pressure is called the vapor pressure. At a given temperature there is only one pressure at which the liquid and vapor phases of a pure substance may exist in equilibrium. Either phase alone may exist, of course, over a wide range of conditions.

You can understand vapor pressure, vaporization, and condensation more easily with the aid of Fig. 3.9. Figure 3.9 is an expanded p - T diagram for pure water. For each temperature you can read the corresponding pressure at which water vapor and water liquid exist in equilibrium. You have encountered this condition of equilibrium many times—for example, in boiling. Any substance has an infinite number of boiling points, but by custom we say the “**normal**” **boiling point** is the temperature at which boiling takes place under a pressure of 1 atm (101.3 kPa, 760 mm Hg). Unless another pressure is specified, 1 atm is assumed, and the term **boiling point** is taken to mean the “normal boiling point.” A piston exerting a force of 101.3 kPa could just as well take the place of the atmosphere, as shown in Fig. 3.10. For example, you know that at 100°C water will boil (vaporize) and the pressure will be 101.3 kPa or 1 atm (point *B*). Suppose that you heat water starting at 77°C (point *A*) in a container as in Fig. 3.10—what happens? We assume that the water vapor is at all times in equilibrium with the liquid water. This is a constant-pressure process. As the temperature rises and the confining pressure stays constant, nothing particularly noticeable occurs until 100°C is reached, at which time the water begins to boil (i.e., evaporate). It pushes back the piston and will completely change from liquid into vapor. If you heated the water in an enclosed cylinder, and if after it had all evaporated at point *B* you continued heating the water vapor formed at constant pressure, you could apply the gas laws in the temperature region *B*-*C* (and at higher temperatures). A reversal of this process from the temperature *C* would cause the vapor to condense at *B* to form a liquid. The temperature at point *B* would in these circumstances represent the **dew point**.

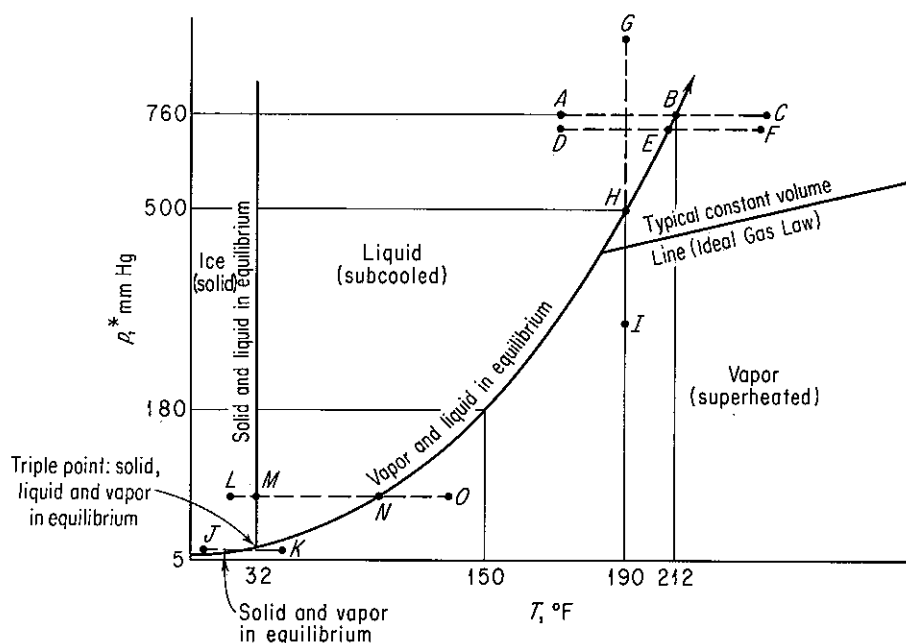


Figure 3.9 Vapor-pressure curve for water.

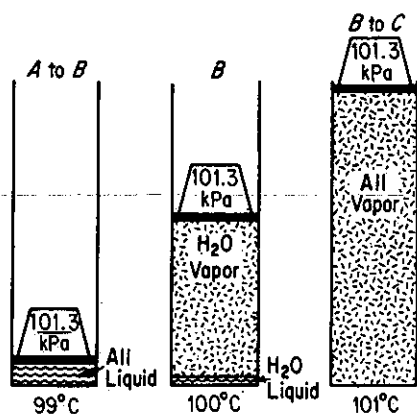


Figure 3.10 Transformation of liquid water into water vapor at constant pressure. The 101.3 kPa exerted by the piston includes the force of the atmosphere above the piston.

Suppose that you went to the top of Pikes Peak and repeated the experiment in the open air—what would happen then? Everything would be the same (points D-E-F) with the exception of the temperature at which the water would begin to boil, or condense. Since the pressure of the atmosphere at the top of Pikes Peak would presumably be lower than 101.3 kPa, the water would start to displace the air, or boil, at a lower temperature. You can see that (a) at any given temperature water exerts its vapor pressure (at equilibrium); (b) as the temperature goes up, the vapor pressure goes up; and (c) it makes no difference whether water vaporizes into air, into a cylinder closed by a piston, or into an evacuated cylinder—at any temperature it still exerts the same vapor pressure as long as the water is in equilibrium with its vapor.

A process of **vaporization or condensation at constant temperature** is illustrated by the lines $G-H-I$ or $I-H-G$, respectively, in Fig. 3.9. Water would vaporize or condense at constant temperature as the pressure reached point H on the vapor-pressure curve (also look at Fig. 3.11.)

Figure 3.9 also shows the p - T conditions at which ice (in its common form) and water vapor are in equilibrium. When the solid passes directly into the vapor phase without first melting to become a liquid (line $J-K$ as opposed to line $L-M-N-O$), it is said to **sublime**. Iodine crystals do this at room temperature; water sublimates only below 0°C , as when the frost disappears in the winter when the thermometer reads -6°C .

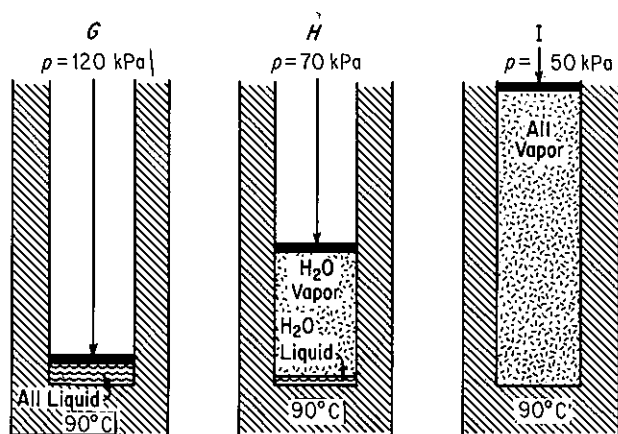


Figure 3.11 Transformation of liquid water into water vapor at constant temperature by changing the imposed pressure.

The vapor-pressure line extends well past the conditions shown in Fig. 3.9, all the way to the critical temperature and pressure (not shown). Why does it terminate at the critical temperature and pressure? Above the critical temperature, water can exist only as a gas.

A term commonly applied to the vapor-liquid portion of the vapor-pressure curve is the word **saturated**, meaning the same thing as **vapor and liquid in equilibrium** with each other. If a gas is just ready to start to condense its first drop of liquid, the gas is called a **saturated gas**; if a liquid is just about to vaporize, it is called a **saturated liquid**. These two conditions are also known as the **dew point** and **bubble point**, respectively.

The region to the right of the vapor-pressure curve in Fig. 3.9 is called the **superheated** region and the one to the left of the vapor-pressure curve is called the **sub-cooled** region. The temperatures in the superheated region, if measured as the difference ($O-N$) between the actual temperature of the superheated vapor and the saturation temperature for the same pressure, are called **degrees of superheat**. For example, steam at 500°F and 100 psia (the saturation temperature for 100 psia is 327.8°F) has $(500 - 327.8) = 172.2^\circ\text{F}$ of superheat. Another new term you will find used frequently is the word **quality**. A *wet* vapor consists of saturated vapor and saturated liquid in equilibrium. The mass fraction of vapor is known as the quality.

EXAMPLE 3.18 Vapor-Liquid Properties of Water

For each of the conditions of temperature and pressure listed below for water, state whether the water is a solid phase, liquid phase, vapor phase (superheated), or is a saturated mixture, and if the latter, calculate the quality. Use the steam tables (inside the back cover) to get your answers.

State	p (kPa)	T (K)	$\hat{V}$ (m ³ /kg)
1	2000	475	—
2	1000	500	0.2206
3	101.3	200	—
4	245.6	400	0.505

Solution

State 1: liquid

State 2: vapor

State 3: solid

State 4: saturated vapor and liquid

The properties of a mixture of vapor and liquid in equilibrium (for a single component) can be computed from the individual properties of the saturated vapor and saturated liquid. At 400 K and 245.6 kPa with a specific volume of a wet steam mixture being 0.505 m³/kg, what is the quality of the steam? From the steam tables the specific volumes of the saturated liquid and vapor are

$$\hat{V}_l = 0.001067 \text{ m}^3/\text{kg} \quad \hat{V}_g = 0.7308 \text{ m}^3/\text{kg}$$

Basis: 1 kg of wet steam mixture

Let x = mass fraction vapor.

$$\frac{0.001067 \text{ m}^3}{1 \text{ kg liquid}} \mid (1-x) \text{ kg liquid} + \frac{0.7308 \text{ m}^3}{1 \text{ kg vapor}} \mid x \text{ kg vapor} = 0.505 \text{ m}^3$$

$$0.001067 - 0.001067x + 0.7308x = 0.505$$

$$x = 0.69$$

Other properties of wet mixtures can be treated in the same manner.

3.3-1 Change of Vapor Pressure with Temperature

A large number of experiments on many substances have shown that a plot of the vapor pressure (p^*) of a compound against temperature does not yield a straight line but a curve, as you saw in Fig. 3.9. Many types of correlations have been proposed to transform this curve to a linear form ($y = mx + b$); a plot of $\ln(p^*)$ vs. $(1/T)$,

for moderate temperature intervals, is reasonably linear:

$$\underbrace{\ln(p^*)}_y = \underbrace{m\left(\frac{1}{T}\right) + b}_{mx + b} \quad (3.31)$$

Equation (3.31) is derived from the Clausius–Clapeyron equation (see Chap. 4). Empirical correlations of vapor pressure are frequently given in the form of the Antoine equation (refer to Appendix G for values of the constants):

$$\ln(p^*) = A - \frac{B}{T + C} \quad (3.32)$$

where A, B, C = constants different for each substance
 T = temperature, K

Other more complex equations can be found in the sources cited in Tables 1.10 and 1.11 in Sec. 1.7 and the supplementary references at the end of this chapter.

As explained in Appendix M, you can estimate the values of the coefficients in Eq. (3.32) by the method of least squares. We look at another way, a graphical method. Over very wide temperature intervals experimental data will not prove to be exactly linear as indicated by Eq. (3.31), but have a slight tendency to curve. This curvature can be straightened out by using a special plot known as a Cox chart.¹⁴ The $\ln$ or $\log_{10}$ of the vapor pressure of a compound is plotted against a special nonlinear temperature scale constructed from the vapor-pressure data for water (called a **reference substance**).

As illustrated in Fig. 3.12, the temperature scale is established as follows. First, you establish a logarithmic scale ($\log_{10}$ is easiest to use) as a horizontal axis to cover the desired range of vapor pressures. To get the special temperature scale with the desired labels for temperature in, say, common integers, such as 25, 50, 100, 200, and so on, you proceed as follows. For the first integer, say 100°F, you look up the vapor pressure of water at 100°F in the steam tables (Table C.1), namely 0.9487 psia. Locate this value on the vapor-pressure axis. Next, look up the vapor of water at, say, 200°F, namely 11.525 psia, and locate this value on the horizontal axis. You can compress or expand the temperature scale by adjusting the slope of the line you draw on the Cox chart for the vapor pressure of water. Draw a line with a slope so that it covers the desired range of temperatures. Proceed from 0.9487 psia on the horizontal axis vertically to the line you drew, and then proceed horizontally to the vertical (left-hand) axis. Make a tic mark labeled 100°F. Next, proceed from 11.525 psia on the vapor-pressure axis to the line you drew, and then go horizontally from the point so located to the vertical axis. Make a tic mark labeled 200°F. Continue as above to label the vertical axis with all the desired temperature labels. The vapor pressures of other substances plotted on this specially prepared set of coordinates will yield straight lines over extensive temperature ranges and thus facilitate the extrapolation and interpolation of vapor-pressure data. It has been found that lines so

¹⁴E. R. Cox, *Ind. Eng. Chem.*, v. 15, p. 592 (1923).

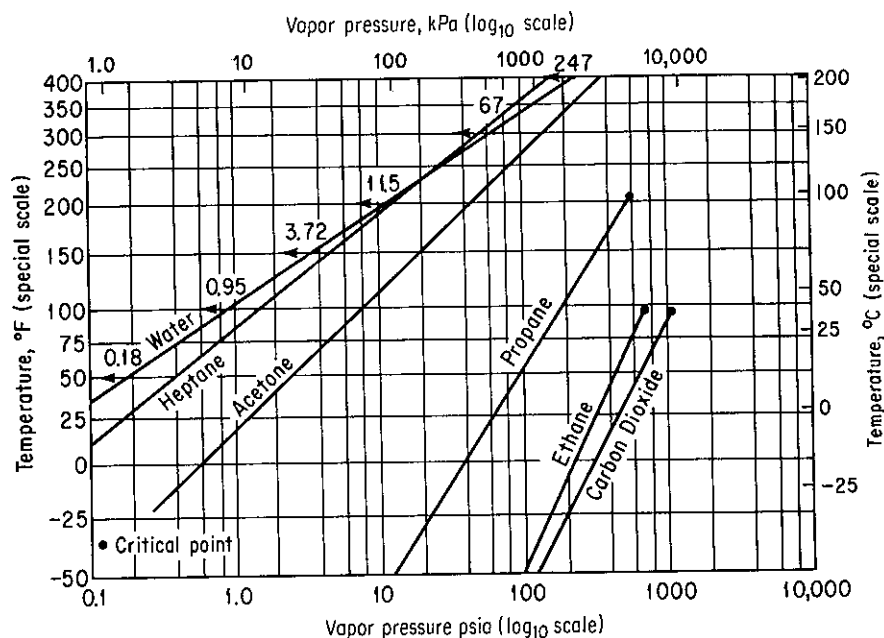


Figure 3.12 Cox chart.

constructed for closely related compounds, such as hydrocarbons, all meet at a common point. Since straight lines can be obtained on a Cox chart only two sets of vapor-pressure data are needed to provide complete information about the vapor pressure of a substance over a considerable temperature range. We discuss in Chap. 4 under the topic of the Clausius–Clapeyron equation other information that can be obtained from such vapor-pressure plots.

3.3-2 Change of Vapor Pressure with Pressure

The equation for the change of vapor pressure with total pressure at constant temperature in a system is

$$\left(\frac{\partial(p^*)}{\partial p_T} \right)_T = \frac{\hat{V}_l}{\hat{V}_g} \quad (3.33)$$

where $\hat{V}$ = molal volume of saturated liquid or gas
 p_T = total pressure on the system

Under normal conditions the effect is negligible.

EXAMPLE 3.19 Extrapolation of Vapor-Pressure Data

The control of existing solvents is described in the *Federal Register*, v. 36, no. 158, August 14, 1971, under Title 42, Chapter 4, Appendix B, Section 4.0, Control of Organic Compound Emissions. Section 4.6 indicates that reductions of at least 85% can be accomplished by (a) incineration or (b) carbon adsorption.

Chlorinated solvents and many other solvents used in industrial finishing and processing, dry-cleaning plants, metal degreasing, printing operations, and so forth, can be recycled and reused by the introduction of carbon adsorption equipment. To predict the size of the adsorber, you first need to know the vapor pressure of the compound being adsorbed at the process conditions.

The vapor pressure of chlorobenzene is 400 mm Hg at 110.0°C and 5 atm at 205°C. Estimate the vapor pressure at 245°C and at the critical point (359°C).

Solution

The vapor pressures will be estimated by use of a Cox chart. See Fig. E3.19. The temperature scale is constructed by using the following data from the steam tables:

$p^* \text{ H}_2\text{O}(\text{psia})$	$t(^{\circ}\text{F})$
0.95	100
3.72	150
11.5	200
29.8	250
67.0	300
247	400
680	500
1543	600
3094	700

Vapor pressures from 3.72 to 3094 psia are marked on the horizontal logarithmic scale as shown in Fig. E3.19. We selected the vapor pressures to correspond to even values of the temperature in $^{\circ}\text{F}$.

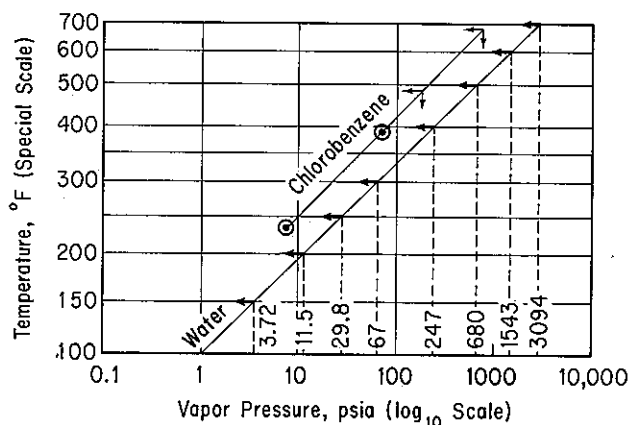


Figure E3.19

Next, we draw a line representing the vapor pressure of water at any suitable angle on the graph so as to stretch the desired temperature range from the bottom to the top of the vertical axis. For each vapor pressure, the temperature is marked and a horizontal line drawn to the ordinate. This establishes the temperature scale (which looks almost logarithmic in nature).

Next we convert the two vapor pressures of chlorobenzene into psia,

$$\frac{400 \text{ mm Hg}}{760 \text{ mm Hg}} \times 14.7 \text{ psia} = 7.74 \text{ psia} \quad 110^\circ\text{C} = 230^\circ\text{F}$$

$$\frac{5 \text{ atm Hg}}{1 \text{ atm Hg}} \times 14.7 \text{ psia} = 73.5 \text{ psia} \quad 205^\circ\text{C} = 401^\circ\text{F}$$

and plot these two points on the graph paper. Examine the encircled dots. Next we draw a straight line between the encircled points and extrapolate to 471°F (245°C) and 678°F (359°C). At these two temperatures read off the estimated vapor pressures:

	471°F(245°C)	678°F(359°C)
Estimated:	150 psia	700 psia
Experimental:	147 psia	666 psia

Experimental values are given for comparison.

An alternative way to extrapolate vapor pressure data would be to use the Othmer¹⁵ plot as explained in Sec. 4.4, or to fit a vapor pressure equation, $p^* = f(T)$, from experimental data.

3.3-3 Liquid Properties

Considerable experimental data are available for liquid densities of pure compounds as a function of temperature and pressure. Refer to the sources listed in Tables 1.10 and 1.11 or the supplementary reference to this chapter for data and correlation formulas. Wooley¹⁶ gives formulas (and a computer program) to estimate the liquid density given T_c , p , and ω , and/or any liquid density reference point.

As to liquid mixtures, it is even more difficult to predict the p - V - T properties of liquid mixtures than of real gas mixtures. Probably more experimental data (especially at low temperatures) are available than for gases, but less is known about the estimation of the p - V - T properties of liquid mixtures. For compounds with like molecular structures, such as hydrocarbons of similar molecular weight, called *ideal liquids*, the density of a liquid mixture can be approximated by assuming that the specific volumes are additive:

$$\hat{V}_{\text{average}} = \hat{V}_1 + \hat{V}_2 + \cdots + \hat{V}_n = \sum_{i=1}^n \hat{V}_i \quad (3.34)$$

where n is the number of components in the mixture. For *nonideal liquids* or solids dissolved in liquids, experimental data or estimation techniques described in many of the references listed in Sec. 1.7 and 4.3 must be employed, as the specific volumes are not additive. Corresponding state methods are usually used, but with less assurance than for gas mixtures.^{17,18}

¹⁵ D. F. Othmer and E. S. Yu, *Indus. Eng. Chem.*, v. 60, p. 20 (1968).

¹⁶ R. J. Wooley, *Chem. Eng.*, p. 109 (March 31, 1986).

¹⁷ N. Shah and C. L. Yaws, "Densities of Liquids," *Chem. Eng.*, p. 131 (October 25, 1976).

¹⁸ V. K. Mathur and R. N. Maddox, "Liquid Density Correlations," *Indian Chem. Eng.*, v. 18, no. 2, p. 29 (1976).

Self-Assessment Test

1. Draw a p - T chart for water. Label the following clearly: vapor-pressure curve, dew-point curve, saturated region, superheated region, subcooled region, and triple point. Show where evaporation, condensation, and sublimation take place by arrows.
2. Describe the state and pressure conditions of water initially at 20°F as the temperature is increased to 250°F in a fixed volume.
3. Look in Appendix J at the diagram for CO₂ (Fig. J.2).
 - (a) At what pressure is CO₂ solid in equilibrium with CO₂ liquid and vapor?
 - (b) If the solid is placed in the atmosphere, what happens?
4. Use the Antoine equation to calculate the vapor pressure of ethanol at 50°C, and compare with the experimental value.
5. Determine the normal boiling point for benzene from the Antoine equation.
6. Prepare a Cox chart from which the vapor pressure of toluene can be predicted over the temperature range -20 to 140°C.

Thought Problems

1. In the startup of a process, Dowtherm, an organic liquid with a very low vapor pressure, was being heated from room temperature to 335°F. The operator suddenly noticed that the gauge pressure was not the expected 15 psig but instead was 125 psig. Fortunately, a relief valve in the exit line ruptured into a vent (expansion) tank so that a serious accident was avoided.

Why was the pressure in the exit line so high?

2. A cylinder containing butadiene exploded in a research laboratory, killing one employee. The cylinder had been used to supply butadiene to a pilot plant. When butadiene gas was required, heat was supplied to the cylinder to raise the pressure of the butadiene in the tank. The maximum temperature that could be achieved in the tank on subsequent tests with a like tank was 160°C. At 152°C, the critical temperature for butadiene, the pressure is 628 lb/in.², less than one-half of the pressure required to rupture the tank by hydraulic test.

Why did the tank explode?

3. "Careless Campers Contaminate Mountain Water" was a recent headline in the newspaper. The article went on:

Beware! There are little monsters loose in those seemingly clean, pristine mountain streams. Their name: *Giardia*, and of specific interest to humans and pigs, *Giardia lamblia*. *Giardia* is a beet-shaped organism with no less than eight flagella. It is of concern to anyone who happens to slurp any down because it attaches itself by means of a sucking organism to the intestinal mucous membranes. The result is severe diarrhea, bordering on dysentery.

The incidence of giardia in the wilderness areas of New Mexico and Colorado has vastly increased over the past five years. The disease it causes, giardiasis, is contracted by drinking water containing the organism. Unfortunately for all backpackers, horse packers, and day hikers, many of the lakes and streams are already tainted. No problem, you say—just drop in a chemical purification tablet and let it do the job? While chemical purification such as Halazone, iodine, or chlorine may kill many bacteria, the hale and hearty giardia goes unscathed.

What steps would you take to avoid the problem other than carrying a portable water supply with you? You need at least 210°F to kill the organisms by boiling water.

3.4 SATURATION AND EQUILIBRIA

Your objectives in studying this section are to be able to:

1. Define saturated gas.
2. Calculate the partial pressure of the components of a saturated ideal gas given combinations of the temperature, pressure, volume and/or number of moles present; or calculate the partial volume; or calculate the number of moles of vapor.
3. Determine the condensation temperature of the vapor in a saturated gas given the pressure, volume, and/or number of moles.
4. Use the relationship $K_i = y_i/x_i$ to calculate any one of the parameters given the other two.

In this section we discuss two categories of phase phenomena for:

1. An equilibrium mixture of pure vapor and a noncondensable gas
2. An equilibrium mixture of two (or more) components both of which can condense and vaporize

3.4-1 Vapor and a Noncondensable Gas at Equilibrium

How can you predict the properties of a mixture of a *pure vapor* (which can condense) and a *noncondensable gas* at equilibrium? A mixture containing a vapor behaves somewhat differently than does a pure component by itself. A typical example with which you are quite familiar is that of water vapor in air.

Water vapor is a gas, and like all gases its molecules are free to migrate in any direction. They will do so as long as they are not stopped by the walls of a container. Furthermore, the molecules will distribute themselves evenly throughout the entire volume of the container.

When any pure gas (or a gaseous mixture) comes in contact with a liquid, the gas will acquire vapor from the liquid. If contact is maintained for a considerable length of time, vaporization continues until equilibrium is attained, at which time the *partial pressure of the vapor in the gas will equal the vapor pressure* of the liquid at the temperature of the system. Regardless of the duration of contact between the liquid and gas, after equilibrium is reached no more net liquid will vaporize into the gas phase. The gas is then said to be **saturated** with the particular vapor at the given

temperature. We can also say that the gas mixture is at its dew point. **Dew point for the mixture of pure vapor and noncondensable gas means the temperature at which the vapor just starts to condense when cooled at constant pressure.**

Now, what do these concepts mean with respect to the quantitative measurement of gas-vapor properties. Suppose that you inject liquid water at 65°C into a cylinder of air at the same temperature, and keep the system at a constant temperature of 65°C . The pressure at the top of the cylinder will be maintained at 101.3 kPa (1 atm). What happens to the volume of the cylinder as a function of time? Figure 3.13 shows that the volume of the air plus the water vapor increases until the air is saturated with water vapor, after which stage the volume remains constant. Figure 3.14(a) indicates how the partial pressure of the water vapor increases with time until it reaches its vapor pressure of 24.9 kPa (187 mm Hg). Why does the partial pressure of the air decrease?

Next, suppose that you carry out a similar experiment, but maintain the volume constant and let the total pressure vary in the cylinder. Will the pressure go up

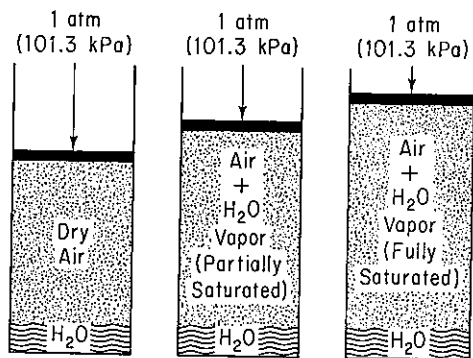


Figure 3.13 Evaporation of water at constant pressure and temperature of 65°C .

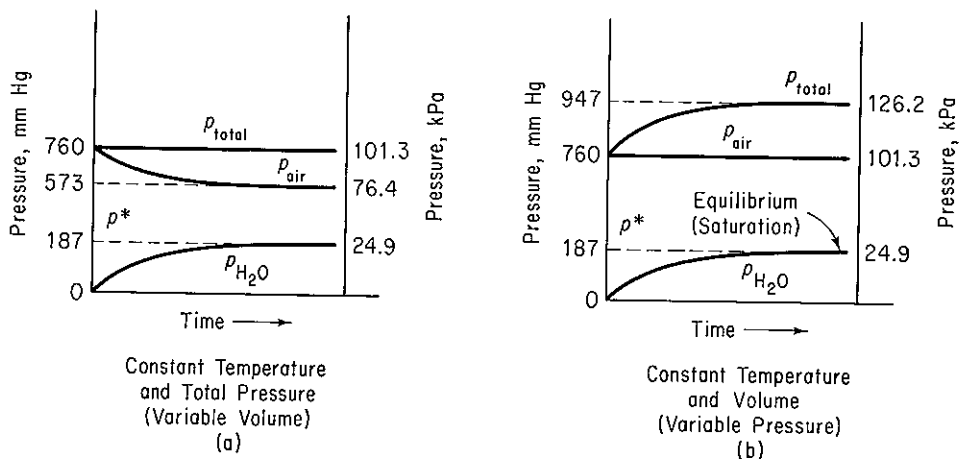


Figure 3.14 Change of partial and total pressures on vaporization of water into air at constant temperature: (a) constant temperature and pressure (variable volume); (b) constant temperature and volume (variable pressure).

or down with respect to time? What will be the asymptotic value of the partial pressure of the water vapor? The air? Look at Fig. 3.14(b) to see if your answers to these questions were correct.

Finally, is it possible to have the water evaporate into air and saturate the air, and yet maintain both a constant temperature, volume, and pressure in the cylinder? (Hint: What would happen if you let some of the gas-vapor mixture escape from the cylinder?)

Assuming that the ideal gas laws apply to both air and water vapor, as they do with excellent precision, we can say that the following relations hold *at saturation*:

$$\frac{p_{\text{air}} V}{p_{\text{H}_2\text{O}} V} = \frac{n_{\text{air}} RT}{n_{\text{H}_2\text{O}} RT}$$

or

$$\frac{p_{\text{air}}}{p_{\text{H}_2\text{O}}} = \frac{n_{\text{air}}}{n_{\text{H}_2\text{O}}} = \frac{p_{\text{air}}}{p_{\text{total}} - p_{\text{air}}} \quad (3.35)$$

in a volume V at temperature T .

EXAMPLE 3.20 Saturation

What is the minimum number of cubic meters of dry air at 20°C and 100 kPa that are necessary to evaporate 6 kg of ethyl alcohol if the total pressure remains constant at 100 kPa? Assume that the air is blown through the alcohol to evaporate it in such a way that the exit pressure of the air–alcohol mixture is at 100 kPa.

Solution

See Fig. E3.20. Assume that the process is isothermal. The additional data needed are

$$p_{\text{alcohol}}^* \text{ at } 20^\circ\text{C} = 5.93 \text{ kPa}$$

$$\text{mol. wt. ethyl alcohol} = 46.07$$

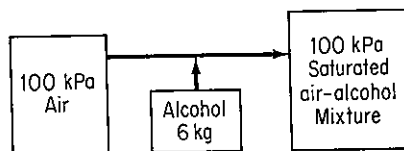


Figure E3.20

The minimum volume of air calls for a saturated mixture; any condition less than saturated would require more air.

Basis: 6 kg of alcohol

The ratio of moles of ethyl alcohol to moles of air in the final gaseous mixture is the same as the ratio of the partial pressures of these two substances. Since we know the moles of alcohol, we can find the number of moles of air:

$$\frac{p_{\text{alcohol}}^*}{p_{\text{air}}} = \frac{n_{\text{alcohol}}}{n_{\text{air}}}$$

From Dalton's law,

$$p_{\text{air}} = p_{\text{total}} - p_{\text{alcohol}}^*$$

$$p_{\text{alcohol}}^* = 5.93 \text{ kPa}$$

$$p_{\text{air}} = (100 - 5.93) \text{ kPa} = 94.07 \text{ kPa}$$

6 kg alcohol	1 kg mol alcohol	94.07 kg mol air
	46.07 kg alcohol	5.93 kg mol alcohol

22.4 m ³	293 K	101.3 kPa
1 kg mol	273 K	100 kPa

$$= 50.3 \text{ m}^3 \text{ air at } 20^\circ\text{C and } 100 \text{ kPa}$$

EXAMPLE 3.21 Saturation

A telescopic gas holder contains 10,000 ft³ of gas saturated with water vapor at 80°F and a pressure of 6.0 in. H₂O above atmospheric. The barometer reads 28.46 in. Hg. Calculate the weight of water vapor in the gas.

Solution

The total pressure of the saturated gas must be calculated first, say, in inches of mercury. See Fig. E3.21.

6.0 in. H ₂ O	1 ft.	29.92 in. Hg
	12 in.	33.91 ft H ₂ O

$$= 0.44 \text{ in. Hg}$$

$$\text{barometer} = 28.46 \text{ in. Hg}$$

$$p_{\text{total}} = 28.90 \text{ in. Hg}$$

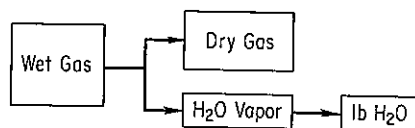


Figure E3.21

From the steam tables,

$$p_w^* = \text{vapor pressure of H}_2\text{O at } 80^\circ\text{F} = 1.032 \text{ in. Hg}$$

$$p_g = p_t - p_w^*$$

$$= 28.90 - 1.032 = 27.87$$

There exists

10,000 ft³ wet gas at 28.90 in. Hg and 80°F

10,000 ft³ dry gas at 27.87 in. Hg and 80°F

10,000 ft³ water vapor at 1.032 in. Hg and 80°F

$$^\circ\text{R} = 80 + 460 = 540^\circ\text{R}$$

From the volume of water vapor present at known conditions, you can calculate the volume at S.C. and hence the pound moles and finally the pounds of water.

Basis: 10,000 ft³ of water vapor at 1.032 in. Hg and 80°F

10,000 ft ³ H ₂ O vapor	492°R	1.032 in. Hg	1 lb mol	18 lb H ₂ O	= 15.8 lb H ₂ O
	540°R	29.92 in. Hg	359 ft ³	1 lb mol H ₂ O	

EXAMPLE 3.22 Smokestack Emission and Pollution

A local pollution-solutions group has reported the Simtron Co. boiler plant as being an air polluter and has provided as proof photographs of heavy smokestack emissions on 20 different days in January and February. As the chief engineer for the Simtron Co., you know that your plant is not a source of pollution because you burn natural gas (essentially methane) and your boiler plant is operating correctly. Your boss believes the pollution-solutions group has made an error in identifying the stack—it must belong to the company next door that burns coal. Is he correct? Is the pollution-solutions group correct? See Fig. E3.22a.



Figure E3.22a

Solution

Methane (CH₄) contains 2 kg mol of H₂ per kilogram mole of C, while coal (see references in Section 1.7) contains 71 kg of C per 5.6 kg of H₂ in 100 kg of coal. The coal analysis is equivalent to

$$\frac{71 \text{ kg C}}{12 \text{ kg C}} \times \frac{1 \text{ kg mol C}}{12 \text{ kg C}} = 5.92 \text{ kg mol C} \quad \frac{5.6 \text{ kg H}_2}{2.016 \text{ kg H}_2} \times \frac{1 \text{ kg mol H}_2}{2.016 \text{ kg H}_2} = 2.78 \text{ kg mol H}_2$$

or a ratio of 2.78/5.92 = 0.47 kg mol of H₂/kg mol of C. Suppose that each fuel burns with 40% excess air and that combustion is complete. We can compute the mole fraction of water vapor in each stack gas.

Steps 1, 2, and 3 The process is shown in Fig. E3.22b.

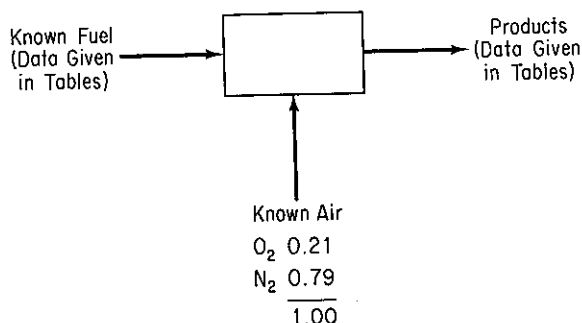


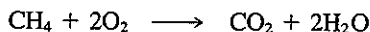
Figure E3.22b

Step 4

Basis: 1 kg mol C

Steps 5 and 6 The combustion problem is a standard type of problem having a unique solution in which both the fuel and air flows are given, and the product flows are calculated directly.

Steps 7, 8, and 9 Tables will make the analysis and calculations compact.

Natural Gas

Components	kg mol	Composition of combustion gases (kg mol)			
		CO ₂	H ₂ O	Excess O ₂	N ₂
C	1.0	1.0			
H ₂	2.0		2.0		
Air				0.80	10.5
Total		1.0	2.0	0.80	10.5

Required O₂: 2Excess O₂: 2(0.40) = 0.80N₂: (2.80)(79/21) = 10.5

The total kilogram moles of gas produced are 14.3 and the mole fraction H₂O is

$$\frac{2.0}{14.3} = 0.14$$

Coal

Components	kg mol	Composition of gas produced (kg mol)			
		CO ₂	H ₂ O	Excess O ₂	N ₂
C	1	1			
H ₂	0.47		0.47		
Air				0.49	6.5
Total		1	0.47	0.49	6.5

Required O₂: 1 + 0.47(1/2)

Excess O₂: $\frac{\text{C}}{1(0.40) = 0.40} \quad \frac{\text{H}_2}{(0.47)(1/2)(0.40) = 0.094}$

N₂: 1.40(79/21)[1 + 0.47(1/2)] = 6.50

The total kilogram moles of gas produced are 8.46 and the mole fraction H₂O is

$$\frac{0.47}{8.46} = 0.056$$

If the barometric pressure is, say, 100 kPa, the stack gas would become saturated and water vapor would start to condense at p^* :

	<i>Natural gas</i>	<i>Coal</i>
Pressure:	$100(0.14) = 14 \text{ kPa}$	$100(0.056) = 5.6 \text{ kPa}$
Equivalent temperature:	52.5°C	35°C

By mixing the stack gas with air, and by convective mixing above the stack, the mole fraction water vapor is reduced, and hence the condensation temperature is reduced. However, for equivalent dilution, the coal-burning plant will always have a lower condensation temperature. Thus, on cold winter days, the condensation of water vapor will occur more often and to a greater extent from power plants firing natural gas than from those using other fuels.

The public, unfortunately, sometimes concludes that all the emissions they perceive are pollution. Natural gas could appear to the public to be a greater pollutant than either oil or coal when, in fact, the emissions are just water vapor. The sulfur content of coal and oil can be released as sulfur dioxide to the atmosphere, and the polluting capacities of coal and oil are much greater than natural gas when all three are being burned properly. The sulfur contents as delivered to the consumers are as follows: natural gas, $4 \times 10^{-4}\%$ (as added mercaptans); number 6 fuel oil, up to 2.6%; and coal, from 0.5 to 5%. In addition, coal may release particulate matter into the stack plume.

What additional steps would you take to resolve the questions that were originally posed?

3.4-2 Vapor-Liquid Equilibria for Multicomponent Systems

In a two-phase vapor-liquid mixture at equilibrium, if all the components can vaporize and condense, a component in one phase is in equilibrium with the same component in the other phase. The equilibrium relationship depends on the temperature and pressure, and perhaps composition, of the mixture. Figure 3.15 illustrates two cases, one at constant pressure and the other at constant temperature. At the pairs of points *A* and *B*, and *C* and *D*, the respective pure components exert their respective vapor pressures at the equilibrium temperature. In between the pairs of points, as the overall composition of the mixture changes, two phases exist, each having a different composition for the same component as indicated by the dashed lines. Two useful linear ("ideal") equations exist to relate the mole fraction of one component in the vapor phase to the mole fraction of the same component in the liquid phase.

Raoult's law. Used primarily for a component whose mole fraction approaches unity or for solutions of components quite similar in chemical nature, such as straightchain hydrocarbons. Let the subscript *i* denote component, p_i be the partial pressure of component *i* in the gas phases, y_i be the gas-phase mole fraction, and x_i be the liquid-phase mole fraction. Then:

$$p_i = p_i^* x_i \quad (3.36)$$

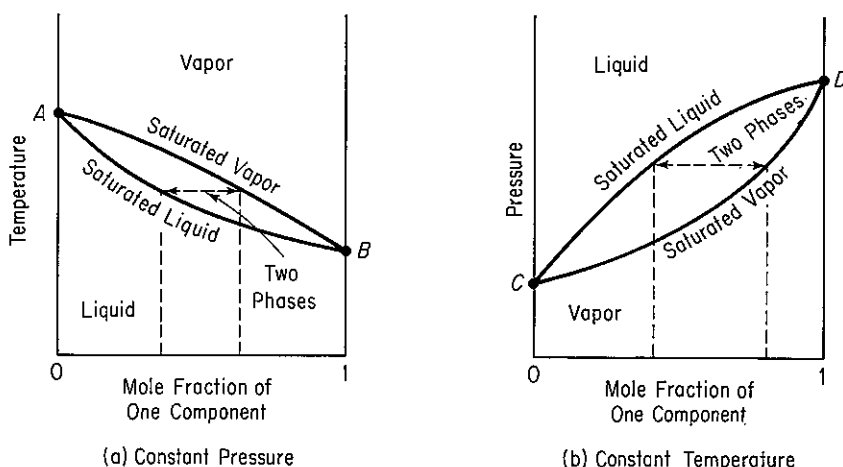


Figure 3.15 Vapor-liquid equilibria for a binary mixture. The dashed lines show the equilibrium compositions (a) when the total pressure is constant and (b) when the temperature is constant over the composition range.

Note that in the limit where $x_i \equiv 1$, $p_i \equiv p_i^*$. Often an equilibrium constant K_i is defined from Eq. (3.36) as follows by assuming that Dalton's law applies to the gas phase ($p_i = p_i y_i$):

$$K_i = \frac{y_i}{x_i} = \frac{p_i^*}{p_i} \quad (3.37)$$

Henry's law. Used primarily for a component whose mole fraction approaches zero, such as a dilute gas dissolved in a liquid:

$$p_i = H_i x_i \quad (3.38)$$

where H_i is the *Henry's law constant*. Note that in the limit where $x_i \equiv 0$, $p_i \equiv 0$. What would be the equilibrium constant K_i based on Henry's law if Dalton's law applied to the gas phase? Values of H_i can be found in Perry and many other handbooks.

Equation (3.37) gives reasonable estimates of K_i values at low pressures for components well below their critical temperatures, but yields values too large for components above their critical temperatures, at high pressures, and/or for polar compounds. Nevertheless, Eq. (3.37) can be adapted to nonideal mixtures if K_i is made a function of temperature, pressure, and composition so that relations for K_i can be fit by experimental data and used directly, or in the form of charts, for design calculations. In the absence of experimental data, as with compressibility, three different techniques are used to estimate K_i values for nonideal vapor-liquid mixtures:

1. Equations of state
2. Activity coefficient methods
3. Group contribution methods

Refer to Prausnitz et al.¹⁹ or the references in Sec. 1.7 for data to calculate K_i . If $T_{c,i}/T > 1.2$, Sandler²⁰ recommends the following empirical correlation:

$$K_i = \frac{p_{c,i}^{[7.224 - 7.534/T_{r,i} - 2.598 \ln T_{r,i}]}}{p_i} \quad (3.39)$$

Typical problems you may be asked to solve that involve the use of the equilibrium coefficient K_i are:

- Calculate the bubble point temperature of a liquid mixture given the total pressure and liquid composition.
- Calculate the dew point temperature of a vapor mixture given the total pressure and vapor composition.
- Calculate the related equilibrium vapor-liquid compositions over the range of mole fractions from 0 to 1 as a function of temperature given the total pressure.

Analogous problems occur with respect to calculating the total pressure given a fixed temperature. Also:

- Calculate the composition of the vapor and liquid streams, and their respective quantities, when a liquid of given composition is partially vaporized (flashed) at a given temperature and pressure (the temperature must lie between the bubble and dewpoint temperatures of the feed).

To calculate the **bubble point temperature** (given the total pressure and liquid composition), you can write Eq. (3.37) as $y_i = K_i x_i$ and you know that $\sum y_i = 1$. Consequently, you want to solve the equation

$$1 = \sum_{i=1}^n K_i x_i \quad (3.40)$$

in which the K_i 's are functions of solely the temperature. Because each of the K_i 's increases with temperature, Eq. (3.40) has only one positive root. You can employ Newton's method to get the root (see Appendix L) if you can express each K_i as an explicit function of temperature. For an ideal solution, Eq. (3.40) becomes

$$p_i = \sum_{i=1}^n p_i^* x_i \quad (3.41)$$

and you might use Antoine's equation for p_i^* . Once the bubble point temperature is determined, the vapor composition can be calculated from

$$y_i = \frac{p_i^* x_i}{p_i}$$

¹⁹ J. M. Prausnitz, *Computer Calculations for Multicomponent Vapor-Liquid and Liquid-Liquid Equilibria*, Prentice-Hall, Englewood Cliffs, N.J., 1980.

²⁰ S. I. Sandler, in *Foundations of Computer Aided Design*, Vol. 2, R. H. S. Mah and W. D. Seider, eds., p. 83, American Institute of Chemical Engineers, New York, 1981.

To calculate the **dew point temperature** (given the total pressure and vapor composition), you can write Eq. (3.37) as $x_i = y_i/K_i$, and you know $\sum x_i = 1$. Consequently, you want to solve the equation

$$1 = \sum_{i=1}^n \frac{y_i}{K_i} \quad (3.42)$$

in which the K 's are functions of temperature as explained for the bubble point temperature calculation. For an ideal solution,

$$1 = p_t \sum_{i=1}^n \frac{y_i}{p_i^*} \quad (3.43)$$

To calculate the amount of the respective vapor and liquid phases that evolve at equilibrium when a liquid of known composition flashes (**flash vaporization**) at a known temperature and pressure, you must use Eq. (3.37) together with a material balance. Figure 3.16 illustrates the steady-state process. A mole balance for component i gives

$$Fx_{Fi} = Lx_i + Vy_i \quad (3.44)$$

where F is the moles of liquid to be flashed, L is the moles of liquid at equilibrium, and V is the moles of vapor at equilibrium. Introduction of $y_i = K_ix_i$ into Eq. (3.44) gives

$$Fx_{Fi} = L\left(\frac{y_i}{K_i}\right) + Vy_i$$

so that

$$y_i = \frac{Fx_{Fi}}{\frac{L}{K_i} + (F - L)} = \frac{x_{Fi}}{1 - \frac{L}{F}\left(1 - \frac{1}{K_i}\right)} \quad (3.45)$$

where L/F is the fraction of liquid formed on vaporization. Consequently, after summing the y_i 's you want to solve the following equation

$$1 = \sum_{i=1}^n \frac{x_{Fi}}{1 - \frac{L}{F}\left(1 - \frac{1}{K_i}\right)} \quad (3.46)$$

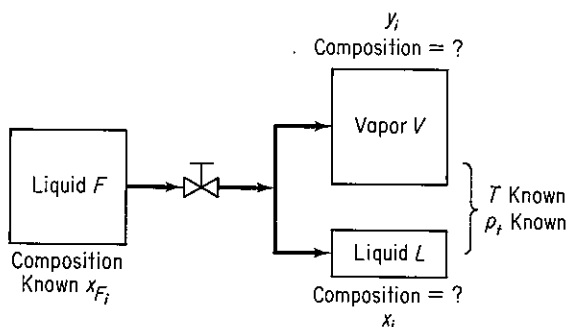


Figure 3.16 Flash vaporization with V and L in equilibrium.

for $L/F (>0)$. Numerous computer programs have been reported in the literature to solve the flash vaporization problem.

Using the above equations, you can prepare figures such as Fig. 3.15 for binary mixtures.

The equations above are not the only formulations in which vapor-liquid calculations are conducted. For example, for the bubble point calculation, an equation equivalent to Eq. (3.40) would be

$$\ln \sum_{i=1}^n K_i x_i = 0$$

and an equation equivalent to Eq. (3.46) would be

$$\sum_{i=1}^n \frac{x_{F_i}(K_i - 1)}{(K_i - 1)(V/F) + 1} = 0$$

In selecting a particular form of the equation to be used for phase equilibria calculations, you must select a method of solving the equation that has desirable convergence characteristics. Convergence to the solution should

- (a) Lead to the desired root if the equation has multiple roots
- (b) Be stable, that is, approach the desired root asymptotically rather than by oscillating
- (c) Be rapid, and not become slower as the solution is approached

3.4-3 Liquid-Liquid and Liquid-Solid Equilibria for Multicomponent Systems

Liquid-liquid equilibria are treated essentially in the same way as nonideal vapor-liquid equilibria with activity-coefficient corrections in both phases. For liquid-solid equilibria, engineers rely almost entirely on recorded experimental data, as prediction of the liquid-phase composition is quite difficult. Refer to Prausnitz et al. for more information.

EXAMPLE 3.23 Vapor-Liquid Equilibrium Calculation

Suppose that a liquid mixture of 4.0% *n*-hexane and *n*-octane is vaporized. What is the composition of the vapor formed if the total pressure is 1 atm?

Solution

What the problem asks for is: What is the composition of the first vapor formed at equilibrium with $x_{C_6} = 0.040$ and a pressure of 1 atm? Examine Fig. 3.16a. The mixture can be treated as an ideal mixture because the components are quite similar. As an intermediate step, you must calculate the bubble point temperature by Eq. (3.41). You have to look up the coefficients of the Antoine equation to obtain the vapor pressures of the two components:

$$\ln(p^*) = A - \frac{B}{C + T}$$

where p^* is in mm Hg and T is in K:

	A	B	C
n -hexane (C_6):	15.8737	2697.55	-48.784
n -octane (C_8):	15.9798	3127.60	-63.633

Basis: 1 kg mol of liquid

We need to solve the following equation to get the bubble point temperature using one of the techniques described in Sec. L.2:

$$760 = \exp\left(15.8737 - \frac{2697.55}{-48.784 + T}\right)0.040 + \exp\left(15.9787 - \frac{3127.60}{-63.633 + T}\right)0.960$$

The solution is $T = 393.3$ K, where the vapor pressure of a hexane is 3114 mm Hg and the vapor pressure of octane is 661 mm Hg.

$$y_{C_6} = \frac{p_{C_6}^*}{p_t} x_{C_6} = \frac{3114}{760}(0.040) = 0.164$$

$$y_{C_8} = 1 - 0.164 = 0.836$$

EXAMPLE 3.24 Flash Calculation

Calculate the fraction of liquid that exists at equilibrium at 150°F and 50 psia when the liquid concentrations of the solution to be vaporized are as follows:

Component	Initial liquid mole fraction	K
C_2	0.0079	16.20
C_3	0.1321	5.2
i - C_4	0.0849	2.6
n - C_4	0.2690	1.98
i - C_5	0.0589	0.91
n - C_5	0.1321	0.72
C_6	0.3151	0.28
Total	1.0000	

The K values come from the *Engineering Data Book* of the Gas Processors Supply Association (1980).

Solution

We want to solve Eq. (3.46) for L/F . Start with an initial guess of $L/F = 1.0$. Successive

stages of the iteration by Newton's method give

Stage	L/F
1	1.0
2	0.8565
3	0.6567
4	0.5102
5	0.4573
6	0.4511 which is sufficiently accurate

Self-Assessment Test

- What does the term "saturated gas" mean?
- If a container with a volumetric ratio of air to liquid water of 5 is heated to 60°C and equilibrium is reached, will there still be liquid water present? at 125°C?
- A mixture of air and benzene contains 10 mole % benzene at 43°C and 105 kPa pressure. At what temperature does the first liquid form? What is the liquid?
- The dew point of water in atmospheric air is 82°F. What is the mole fraction of water vapor in the air if the barometric pressure is 750 mm Hg?
- Ten pounds of KClO_3 is completely decomposed and the oxygen evolved collected over water at 80°F. The barometer reads 29.7 in. Hg. What weight of saturated oxygen is obtained?
- If a gas is saturated with water vapor, describe the state of the water vapor and the air if it is:
 - Heated at constant pressure
 - Cooled at constant pressure
 - Expanded at constant temperature
 - Compressed at constant temperature
- Calculate (a) the dew point pressure for the following mixture at 100°F and (b) the liquid composition.

Component	Mole fraction	K values at psia of		
		190	200	210
C_2H_6	0.2180	3.22	3.07	2.92
C_3H_8	0.6650	1.005	0.973	0.920
$i\text{-C}_4\text{H}_{10}$	0.1073	0.45	0.43	0.41
$n\text{-C}_4\text{H}_{10}$	0.0097	0.315	0.305	0.295
Total	1.0000			

Thought Problems

- Water was drained from the bottom of a gasoline tank into a sewer, and shortly thereafter a flash fire occurred in the sewer. The operator took special care to make sure that none of the gasoline entered the sewer.
What caused the sewer fire?

2. The fluid in a large tank caught on fire 40 minutes after the start of a blending operation in which one grade of naphtha was being added to another. The fire was soon put out and the naphtha was moved to another tank. The next day blending was resumed in the second tank; 40 minutes later another fire started.

Can you explain the reason for this sequence of events? What might be done to prevent such incidents?

3.5 PARTIAL SATURATION AND HUMIDITY

Your objectives in studying this section are to be able to:

1. Define relative saturation (humidity), molal saturation (humidity), absolute saturation (humidity), and humidity by formulas involving partial pressures of the gas components.
2. Given the partial saturation in one form, calculate the corresponding values in the other three forms as well as the dew point.

In Sec. 3.4 we dealt with mixtures of noncondensable gas and vapor in which the gas was saturated with the vapor. Often, the contact time required between the gas and liquid for equilibrium (or saturation) to be attained is too long, and the gas is not completely saturated with the vapor. Then the vapor is not in equilibrium with a liquid phase, and the partial pressure of the vapor is less than the vapor pressure of the liquid at the given temperature. This condition is called **partial saturation**. What we have is simply a mixture of two or more gases that obey the real gas laws. What distinguishes this case from the previous examples for gas mixtures is that under suitable conditions it is possible to condense part of one of the gaseous components. In Fig. 3.17 you can see how the partial pressure of the water vapor in a gaseous mixture at constant volume obeys the ideal gas laws as the temperature drops until saturation is reached, at which time the water vapor starts to condense. Until these conditions are achieved, you can confidently apply the gas laws to the mixture.

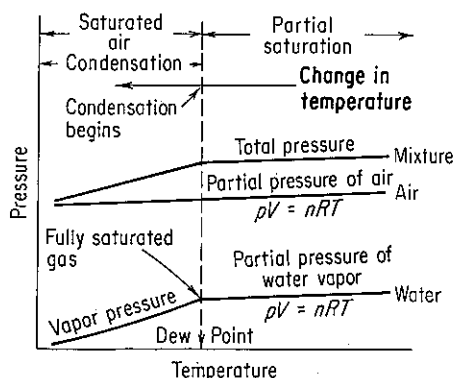


Figure 3.17 Transformation of a partially saturated water vapor-air mixture into a saturated mixture as the temperature is lowered (volume = constant).

Several ways exist to express the concentration of a vapor in a mixture with a noncondensable gas. You sometimes encounter mass or mole fraction (or percent), but more frequently one of the following:

- (a) Relative saturation (relative humidity); Sec. 3.5.1
- (b) Molal saturation (molal humidity); Sec. 3.5.2
- (c) "Absolute" saturation ("absolute" humidity) or percent saturation (percent humidity); Secs. 3.5.3 and 4.9
- (d) Humidity; Secs. 3.5.2 and 4.9

When the vapor is water vapor and the gas is air, the special term **humidity** applies. For other gases or vapors, the term **saturation** is used.

3.5-1 Relative Saturation

Relative saturation is defined as

$$\mathcal{R}\mathcal{S} = \frac{p_{\text{vapor}}}{p_{\text{satd}}} = \text{relative saturation} \quad (3.47)$$

where p_{vapor} = partial pressure of the vapor in the gas mixture

p_{satd} = partial pressure of the vapor in the gas mixture if the gas were saturated at the given temperature of the mixture (i.e., the vapor pressure of the vapor component)

Then, for brevity, if the subscript 1 denotes vapor,

$$\mathcal{R}\mathcal{S} = \frac{p_1}{p_1^*} = \frac{p_1/p_t}{p_1^*/p_t} = \frac{V_1/V_t}{V_{\text{satd}}/V_t} = \frac{n_1}{n_{\text{satd}}} = \frac{\text{mass}_1}{\text{mass}_{\text{satd}}} \quad (3.48)$$

You can see that relative saturation, in effect, represents the fractional approach to total saturation, as shown in Fig. 3.18. If you listen to the radio or TV and hear the announcer say that the temperature is 25°C (77°F) and the *relative humidity* is 60%, he or she implies that

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2\text{O}}^*}(100) = \% \mathcal{R}\mathcal{H} = 60 \quad (3.49)$$

with both the $p_{\text{H}_2\text{O}}$ and the $p_{\text{H}_2\text{O}}^*$ being measured at 25°C. Zero percent relative saturation means no vapor in the gas. What does 100% relative saturation mean? It

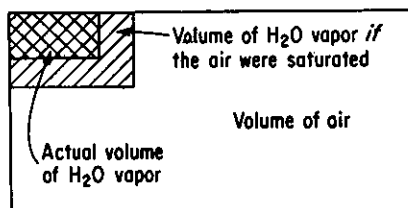


Figure 3.18 Partially saturated gas with the water and air separated conceptually.

means that the partial pressure of the vapor is the same as the vapor pressure of the substance that is the vapor.

EXAMPLE 3.25 Application of Relative Humidity

The weather reported on the radio this morning was that the temperature this afternoon would reach 94°F, the relative humidity would be 43%, the barometer 29.67 in. Hg, partly cloudy to clear, with the wind from SSE at 8 mi/hr. How many pounds of water vapor would be in 1 mi³ of afternoon air? What would be the dew point of this air?

Solution

The vapor pressure of water at 94°F is 1.61 in. Hg. We can calculate the partial pressure of the water vapor in the air from the given percent relative humidity; from this point forward, the problem is the same as the examples in Section 3.4.

$$p_w = (1.61 \text{ in. Hg})(0.43) = 0.692 \text{ in. Hg}$$

$$(p_{\text{air}} = p_t - p_w = 29.67 - 0.692 = 28.98 \text{ in. Hg})$$

Basis: 1 mi³ water vapor at 94°F and 0.692 in. Hg

$$\frac{1 \text{ mi}^3}{1 \text{ mi}} \left| \frac{(5280 \text{ ft})^3}{1 \text{ mi}} \right| \left| \frac{492^\circ\text{R}}{554^\circ\text{R}} \right| \left| \frac{0.692 \text{ in. Hg}}{29.92 \text{ in. Hg}} \right| \left| \frac{1 \text{ lb mol}}{359 \text{ ft}^3} \right| \left| \frac{18 \text{ lb H}_2\text{O}}{1 \text{ lb mol}} \right| = 1.52 \times 10^8 \text{ lb H}_2\text{O}$$

Now the dew point is the temperature at which the water vapor in the air will first condense on cooling at *constant total pressure and composition*. As the gas is cooled you can see from Eq. (3.49) that the percent relative humidity increases since the partial pressure of the water vapor is constant while the vapor pressure of water decreases with temperature. When the percent relative humidity reaches 100%,

$$100 \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2\text{O}}^*} = 100\% \quad \text{or} \quad p_{\text{H}_2\text{O}} = p_{\text{H}_2\text{O}}^*$$

the water vapor will start to condense. This means that at the dew point the vapor pressure of water will be 0.692 in. Hg. From the steam tables you can see that this corresponds to a temperature of about 68–69°F.

3.5-2 Molal Saturation

Another way to express vapor concentration in a gas is to use the ratio of the moles of vapor to the moles of vapor-free gas:

$$\frac{n_{\text{vapor}}}{n_{\text{vapor-free gas}}} = \text{molal saturation} \quad (3.50)$$

If subscripts 1 and 2 represent the vapor and the dry gas, respectively, then for a binary system,

$$p_1 + p_2 = p_t$$

$$n_1 + n_2 = n_t$$

$$\frac{n_1}{n_2} = \frac{p_1}{p_2} = \frac{V_1}{V_2} = \frac{n_1}{n_t - n_1} = \frac{p_1}{p_t - p_1} = \frac{V_1}{V_t - V_1} \quad (3.51)$$

By multiplying by the appropriate molecular weights, you can find the mass of vapor per mass of dry gas:

$$\frac{(n_{\text{vapor}})(\text{mol. wt. vapor})}{(n_{\text{dry gas}})(\text{mol. wt. dry gas})} = \frac{\text{mass}_{\text{vapor}}}{\text{mass}_{\text{dry gas}}} \quad (3.52)$$

The special term **humidity** ($\mathcal{H}$) refers to the mass of water vapor per mass of bone-dry air and is used in connection with the humidity charts in Sec. 4.9.

3.5-3 "Absolute" Saturation (Humidity); Percentage Saturation (Humidity)

"Absolute" saturation is defined as the ratio of the moles of vapor per mole of *vapor-free* gas to the moles of vapor *that would be present* per mole of *vapor-free* gas if the mixture were completely saturated at the existing temperature and total pressure:

$$\mathcal{AS} = \text{"absolute saturation"} = \frac{\left(\frac{\text{moles vapor}}{\text{moles vapor-free gas}} \right)_{\text{actual}}}{\left(\frac{\text{moles vapor}}{\text{moles vapor-free gas}} \right)_{\text{saturated}}} \quad (3.53)$$

Using the subscripts 1 for vapor and 2 for vapor-free gas,

$$\text{percent absolute saturation} = \frac{\left(\frac{n_1}{n_2} \right)_{\text{actual}}}{\left(\frac{n_1}{n_2} \right)_{\text{saturated}}} (100) = \frac{\left(\frac{p_1}{p_2} \right)_{\text{actual}}}{\left(\frac{p_1}{p_2} \right)_{\text{saturated}}} (100) \quad (3.54)$$

Since $p_1 \text{ saturated} = p_1^*$ and $p_t = p_1 + p_2$,

$$\text{percent absolute saturation} = 100 \frac{\frac{p_1}{p_t - p_1}}{\frac{p_1^*}{p_t - p_1^*}} = \frac{p_1}{p_1^*} \left(\frac{p_t - p_1^*}{p_t - p_1} \right) 100 \quad (3.55)$$

Now you will recall that $p_1/p_1^* = \text{relative saturation}$. Therefore,

$$\text{percent absolute saturation} = (\text{relative saturation}) \left(\frac{p_t - p_1^*}{p_t - p_1} \right) 100 \quad (3.56)$$

Percent absolute saturation is always less than relative saturation except at saturated conditions (or at zero percent saturation) when percent absolute saturation = percent relative saturation.

EXAMPLE 3.26 Partial Saturation

Helium contains 12% by volume of ethyl acetate. Calculate (a) the percent relative saturation and (b) the percent absolute saturation of the mixture at a temperature of 30°C and a pressure of 98 kPa.

Solution

The additional data needed are

$$p_{\text{EtAc}}^* \text{ at } 30^\circ\text{C} = 15.9 \text{ kPa} \quad (\text{from any suitable handbook})$$

Using Dalton's laws, we obtain

$$\begin{aligned} p_{\text{EtAc}} &= p_t y_{\text{EtAc}} = p_t \left(\frac{n_{\text{EtAc}}}{n_t} \right) = p_t \left(\frac{V_{\text{EtAc}}}{V_t} \right) \\ &= (98)(0.12) = 11.76 \text{ kPa} \\ p_{\text{He}} &= p_t - p_{\text{EtAc}} \\ &= 98 - 11.76 = 86.24 \text{ kPa} \end{aligned}$$

At 30°C :

(a) Percent relative saturation =

$$100 \frac{p_{\text{EtAc}}}{p_{\text{EtAc}}^*} = 100 \frac{11.76}{15.9} = 74.0\%$$

(b) Percent absolute saturation =

$$\begin{aligned} 100 \frac{\frac{p_{\text{EtAc}}}{p_t - p_{\text{EtAc}}}}{\frac{p_{\text{EtAc}}^*}{p_t - p_{\text{EtAc}}^*}} &= \frac{\frac{11.76}{98 - 11.76}}{\frac{15.9}{98 - 15.9}} 100 = \frac{\frac{11.76}{86.24}}{\frac{15.9}{82.1}} 100 \\ &= 70.4\% \end{aligned}$$

EXAMPLE 3.27 Partial Saturation

The percent absolute humidity of air at 86°F and a total pressure of 750 mm Hg is 20%. Calculate (a) the percent relative humidity and (b) the partial pressure of the water vapor in the air. (c) What is the dew point of the air?

Solution

Data from the steam tables are

$$p_{\text{H}_2\text{O}}^* \text{ at } 86^\circ\text{F} = 1.253 \text{ in. Hg} = 31.8 \text{ mm Hg}$$

To get the relative humidity, $p_{\text{H}_2\text{O}}/p_{\text{H}_2\text{O}}^*$, we need to find the partial pressure of the water vapor in the air. This may be obtained from

$$\mathcal{AH} = 20 = \frac{\frac{p_{\text{H}_2\text{O}}}{p_t - p_{\text{H}_2\text{O}}}}{\frac{p_{\text{H}_2\text{O}}^*}{p_t - p_{\text{H}_2\text{O}}^*}} 100 = \frac{\frac{p_{\text{H}_2\text{O}}}{750 - p_{\text{H}_2\text{O}}}}{\frac{31.8}{750 - 31.8}} 100$$

This equation can be solved for $p_{\text{H}_2\text{O}}$:

$$0.00886 = \frac{p_{\text{H}_2\text{O}}}{750 - p_{\text{H}_2\text{O}}}$$

$$(b) \quad p_{\text{H}_2\text{O}} = 6.58 \text{ mm Hg}$$

$$(a) \quad \% \mathcal{RH} = 100 \frac{6.58}{31.8} = 20.7\%$$

(c) The dew point is the temperature at which the water vapor in the air would commence to condense, when cooled at constant total pressure. This would be at the vapor pressure of 6.58 mm Hg, or about 42°F.

Self-Assessment Test

1. A mixture of air and benzene is found to have a 50% relative saturation at 27°C and an absolute pressure of 110 kPa. What is the mole fraction of benzene in the air?
2. A TV announcer says that the dew point is 92°F. If you compress the air to 110°F and 2 psig, what is the percent absolute humidity?
3. Nine hundred forty-seven cubic feet of wet air at 70°F and 29.2 in. Hg are dehydrated. If the water removed contains 0.94 lb of H₂O, what was the relative humidity of the wet air?

Thought Problems

1. A stirred tank containing liquid CS₂ solvent had to be cleaned because of solid residue that had accumulated in the stirrer. To avoid a possible fire and explosion, the CS₂ was pumped out and the tank blanketed with nitrogen. Then the manhole cover was removed from the top of the tank, and a worker started to remove the solid from the stirrer rod with a scraper. At this point the maintenance worker left for lunch. When he returned to complete the job, a spark caused by the scraper striking the stirrer rod started a flash fire. How could the fire have occurred despite the preventive measure of using a N₂ blanket?

2. From: Marine Board of Investigation

To: Commandant (MVI)

Subj.: SS V. A. FOGG, O. N. 244971; sinking with loss of life in Gulf of Mexico on 1 February 1972

Findings of Fact

At 1240 on February 1, 1972, the tankship V. A. FOGG departed Freeport, Texas, en route to the Gulf of Mexico to clean cargo tanks that carried benzene residue. The vessel was due to arrive in Galveston, Texas, at 0200, on February 2. At approximately 1545, February 1, the V. A. FOGG suffered multiple explosions and sank. All 39 persons aboard died as a result of this casualty. Three bodies were recovered; two of the bodies were identified and one remained unidentified. The other persons were missing and presumed dead.

You are asked your opinion of the most probable cause of the incident. What is your explanation?

3. On two different days the temperature and barometric pressure are the same. On day 1 the humidity is high, on day 2 the humidity is low. On which day is the air the most dense? Justify your answer with arguments using Dalton's law and the ideal gas law. Why does the air feel "heavier" on day 1 than on day 2?

3.6 MATERIAL BALANCES INVOLVING CONDENSATION AND VAPORIZATION

Your objective in studying this section is to be able to:

1. Solve material balance problems involving vaporization and condensation.

The solution of material balance problems involving partial saturation, condensation, and vaporization will now be illustrated. Remember the drying problems in Chap. 2? They included water and some bone-dry material, as shown at the top of Fig. 3.19. To complete the diagram, we add the air that is used to remove the water from the material being dried.

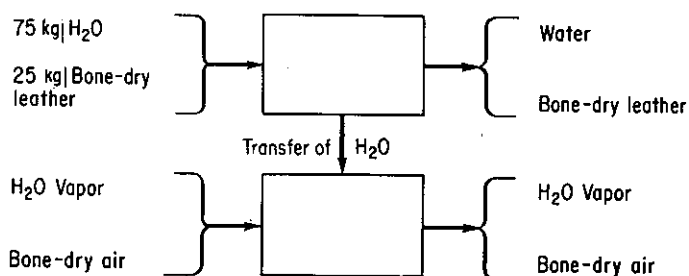


Figure 3.19 Complete drying schematic.

You can analyze material balance problems involving water vapor in air in exactly the same fashion as you analyzed the material balance problems for the drying of leather (or paper, etc.), depending on the information provided and sought. (Humidity and saturation problems that include the use of energy balances and humidity charts are discussed in Chap. 4.)

In connection with the examples that follow, we again stress that if you know the dew point of water in a gas, you automatically know the partial pressure of the water vapor in the gas. When a partially saturated gas is cooled at constant pressure, as in the cooling of air containing some water vapor at atmospheric pressure, the volume of the mixture may change slightly, but the mole fractions of the gas and water vapor remain constant until the dew point is reached. At this point water begins to condense; the gas remains saturated as the temperature is lowered. All that happens is that more water goes from the vapor into the liquid phase. **At the time the cooling is stopped, the gas is still saturated, and at its dew point.**

The material balance problems that follow are solved using the 10-step strategy presented in Table 2.4.

EXAMPLE 3.28 Material Balance with Condensation

If the atmosphere in the afternoon during a humid period is at 90°F and 80 RH (barometer reads 738 mm Hg) while at night it is at 68°F (barometer reads 745 mm Hg), what percent of the water in the afternoon air is deposited as dew at night?

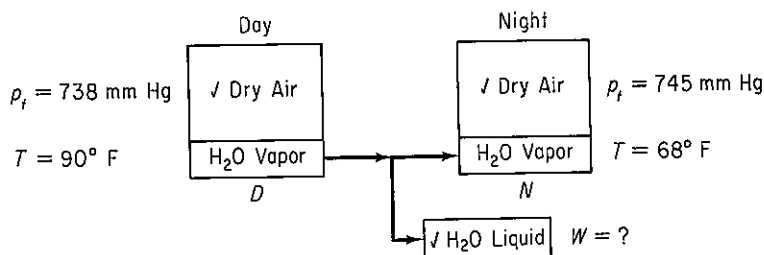
Solution

Assume a closed system that reaches equilibrium.

Steps 1, 2, and 3: The data are shown in Fig. E3.28. The check marks indicate that the compositions of the streams can be calculated from the information given. Two pieces of information needed are:

$$\text{vapor pressure H}_2\text{O at } 90^\circ\text{F} = 36.1 \text{ mm Hg}$$

$$\text{vapor pressure H}_2\text{O at } 68^\circ\text{F} = 17.5 \text{ mm Hg}$$

**Figure E3.28**

You can check to see if the dew point of the day air is below 68°F in which case no dew is deposited.

The partial pressure of the water pressure in the day air is

$$(36.1 \text{ mm Hg})(0.80) = 28.9 \text{ mm Hg} \approx 83^\circ\text{F dew point}$$

Consequently, water does condense from the air as it cools. Now we need to get the compositions of the day and night air so we can carry out a material balance.

We calculate all the partial pressures in the *D* and *N* streams, the results of which in effect give us the compositions of the two streams.

$$p_t = p_{\text{air}} + p_w \quad \text{so that}$$

$$\text{day: } p_{\text{air}} = 738 - 28.9 = 709.1 \text{ mm Hg}$$

$$\text{night: } p_{\text{air}} = 745 - 17.5 = 727.5 \text{ mm Hg}$$

Steps 4, 5, and 6: If we take a basis, we have only two stream variables unknown; two component material balances can be made, hence the problem has a unique solution.

As a basis we could select 1 ft³ of wet air, 1 lb mol wet (or dry) air, or many other suitable bases. The simplest basis to take is

Basis: 738 lb mol of moist day air

because then the moles = the partial pressures.

Steps 7, 8, 9: On the basis of 738 lb mol of moist day air, we have the following water in the saturated night air:

$$\frac{17.5 \text{ lb mol H}_2\text{O in night air}}{727.5 \text{ lb mol dry air}} \bigg| \frac{709.1 \text{ lb mol dry air}}{1} = 17.06 \text{ lb mol H}_2\text{O}$$

Then

$$\text{Water balance: } 28.9 - 17.06 = 11.84 \text{ lb mol H}_2\text{O deposited as dew}$$

$$100 \frac{11.84}{28.9} = 41\% \text{ of water in day air deposited as dew}$$

Step 10: Check your answer by selecting another basis on which to work the problem.

EXAMPLE 3.29 Dehydration

By absorption in silica gel you are able to remove all (0.93 kg) of the H_2O from moist air at 15°C and 98.6 kPa. The same air measures 1000 m^3 at 20°C and 108.0 kPa when dry. What was the relative humidity of the moist air?

Solution

Steps 1, 2, and 3: Figure E3.29 contains the known data. The check marks indicate that the moist air and other concentrations can be calculated from the data given.

Step 4: Either the W or the A stream can serve as the basis.

Basis: 1000 m^3 bone-dry air (BDA) at 20°C and 108.0 kPa

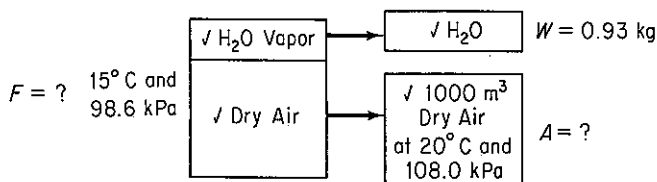


Figure E3.29

We first need to calculate the amounts (in kg mol) of water vapor and dry air in the original air. $W = 0.93 \text{ kg}$ or

$$\frac{0.93 \text{ kg H}_2\text{O}}{1} \bigg| \frac{1 \text{ kg mol H}_2\text{O}}{18 \text{ kg H}_2\text{O}} = 0.0517 \text{ kg mol H}_2\text{O}$$

As for the dry air

$$\frac{1000 \text{ m}^3 \text{ BDA}}{1} \bigg| \frac{273 \text{ K}}{293 \text{ K}} \bigg| \frac{108.0 \text{ kPa}}{101.3 \text{ kPa}} \bigg| \frac{1 \text{ kg mol}}{22.4 \text{ m}^3} = 44.35 \text{ kg mol BDA}$$

Steps 5, 6, 7, 8 and 9: All of the water and all of the air were in the original moist air so that balances are trivial, and the problem has a unique solution. The partial pressure of the water vapor in the original air was

$$p_{\text{H}_2\text{O}} = p_t y_{\text{H}_2\text{O}} = 98.6 \left(\frac{0.0517}{44.35 + 0.0517} \right) = 0.1147 \text{ kPa}$$

The vapor pressure at 15°C for water is 1.70 kPa, hence the fractional relative humidity of the original air was

$$\frac{0.1147}{1.70} = 0.067$$

EXAMPLE 3.30 Humidification

One thousand cubic meters of moist air at 101 kPa and 22°C and with a dew point of 11°C enters a process. The air leaves the process at 98 kPa with a dew point of 58°C. How many kilograms of water vapor are added to each kilogram of wet air entering the process?

Solution

Steps 1, 2, and 3 The known data appear in Fig. E3.30. The check marks indicate that the concentrations can be calculated from the given data. Additional data needed are

Dew point temp. (°C)	$p_{\text{H}_2\text{O}}^*$ (mm Hg)	$p_{\text{H}_2\text{O}}^*$ (kPa)
11	9.84	1.31 [†]
58	136.1	18.14 [†]

[†] These values give the partial pressures of the water vapor in the initial and final gas mixtures.

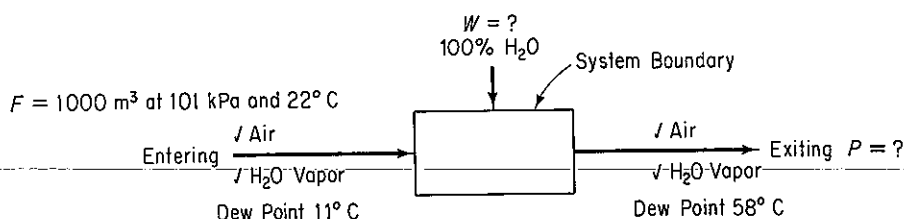


Figure E3.30

The partial pressures of the water vapor are the pressures at the dew point in each case, and the dry air has a partial pressure which is the difference between the total pressure and the partial pressure of the water vapor.

Let the subscript W stand for water vapor and DA stand for dry air:

$$\text{In: } p_{DA} = p_t - p_w = 101 - 1.31 = 99.69 \text{ kPa}$$

$$\text{Out: } p_{DA} = 98 - 18.14 = 79.86 \text{ kPa}$$

Step 4 Basis is 1000 m³ at 101 kPa and 22°C. Other bases could be selected such as 101 kg mol of moist entering air or 98 kg mol of moist exit air.

Steps 5 and 6 We have two unknowns, W and P , and can make both an air and water balance, so that the problem has a unique solution.

Steps 7, 8, and 9

$$\frac{1000 \text{ m}^3}{101.3 \text{ kPa}} \times \frac{273 \text{ K}}{295 \text{ K}} \times \frac{1 \text{ kg mol}}{22.4 \text{ m}^3} = 41.19 \text{ kg mol wet air}$$

Dry air is a tie component.

$$\text{DA balance: } 41.19 \left(\frac{99.69}{101} \right) = P \left(\frac{79.86}{98} \right)$$

$$P = 49.87 \text{ kg mol}$$

$$\text{Total balance: } F + W = P$$

$$W = 49.87 - 41.19 = 8.68 \text{ kg mol H}_2\text{O}$$

Step 10. Check using the water balance.

$$\text{Water balance: } 41.19 \left(\frac{1.31}{101} \right) + W = 49.87 \left(\frac{18.1}{98} \right)$$

$$W = 8.68 \text{ kg mol H}_2\text{O}$$

Step 9 (Continued) To calculate the kg of wet air entering, we proceed as in Chap. 1.

Component	kg mol	Mol. wt.	kg
Dry air	$41.19 \left(\frac{99.69}{101} \right)$	29	1179.0
H ₂ O	$41.19 \left(\frac{1.31}{101} \right)$	18	9.6
Total	41.19		1188.6

$$\text{Water added} = \frac{(8.68)(18)}{1188.6} = 0.131 \frac{\text{kg water}}{\text{kg wet air in}}$$

EXAMPLE 3.31 Vaporization of the Products of Ammonia Synthesis

After ammonia is produced from N₂ and H₂, the products are cooled and separated in a partial condenser operated at -28°F and 2000 psia (the reaction pressure). Figure E3.31 illus-

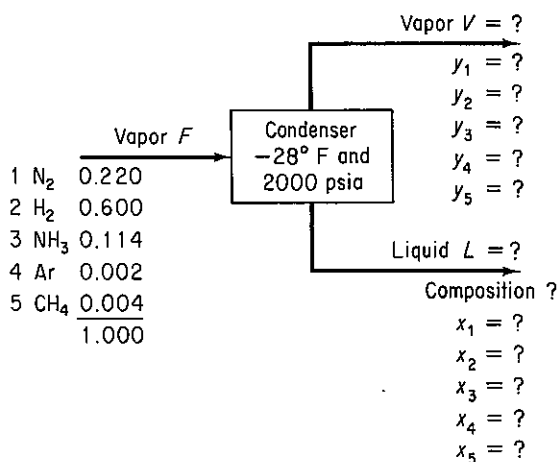


Figure E3.31

trates the separation process and gives the known data. Calculate the composition of the vapor and liquid streams. The equilibrium constants are

N ₂ :	66.67	Ar:	100
H ₂ :	50	CH ₄ :	33.33
NH ₃ :	0.015		

Solution

Steps 1, 2, and 3 Figure E3.31 contains the known data, and symbols for the unknown quantities.

Step 4

Basis: 100 lb mol = F

Steps 5 and 6 There are 12 variables whose values are unknown. You can make five component balances, invoke two sum of mole fraction relations for streams L and V , and write five equilibrium relations $y_i = K_i x_i$, hence the problem has a unique solution.

Step 7 Component material balances:

$$\text{N}_2: 100(0.220) = Vy_1 + Lx_1 \quad (\text{a})$$

$$\text{H}_2: 100(0.660) = Vy_2 + Lx_2 \quad (\text{b})$$

$$\text{NH}_3: 100(0.114) = Vy_3 + Lx_3 \quad (\text{c})$$

$$\text{Ar}: 100(0.002) = Vy_4 + Lx_4 \quad (\text{d})$$

$$\text{CH}_4: 100(0.004) = Vy_5 + Lx_5 \quad (\text{e})$$

Summation of mole fractions:

$$V: y_1 + y_2 + y_3 + y_4 + y_5 = 1 \quad (\text{f})$$

$$L: x_1 + x_2 + x_3 + x_4 + x_5 = 1 \quad (\text{g})$$

Equilibrium relations:

$$\text{N}_2: y_1 = 66.67x_1 \quad (\text{h})$$

$$\text{H}_2: y_2 = 50x_2 \quad (\text{i})$$

$$\text{NH}_3: y_3 = 0.015x_3 \quad (\text{j})$$

$$\text{Ar}: y_4 = 100x_4 \quad (\text{k})$$

$$\text{CH}_4: y_5 = 33.33x_5 \quad (\text{l})$$

Step 8 One method of solution is to solve all 12 equations simultaneously (the component material balances are nonlinear). By the following algebraic manipulations, the number of variables and equations can be reduced. Rearrange Eqs. (a)–(e). For example, for Eq. (a),

$$x_1 = \frac{100(0.220) - Vy_1}{L} \quad (\text{m})$$

and

$$y_1 = \frac{100(0.220) - Lx_1}{V} \quad (\text{n})$$

Substitute $y_i = K_i x_i$ into the set of equations corresponding to Eq. (m) and solve for x_i ; for example,

$$x_1 = \frac{100}{V} \frac{0.220}{(L/V) + 66.67} \quad (o)$$

Introduce all the equations corresponding to Eq. (o) into Eq. (g) to get a single equation in which the unknowns are L and V .

$$\sum_{i=1}^5 x_i(L, V) = 1 \quad (p)$$

A similar procedure applied with respect to Eqs. (n) and (f) gives another equation in which L and V are known; for example,

$$y_1 = \frac{100}{V} \frac{66.7(0.220)}{66.7 + (L/V)}$$

Insert all the y_i 's into Eq. (f) to get

$$\sum_{i=1}^5 y_i(L, V) = 1 \quad (q)$$

The overall material balance $100 = V + L$ can be combined with Eq. (p) and/or (q) to get a single equation in one unknown variable, V or L .

The solution is

Stream	lb mol	Concentration, mole fraction				
		N ₂	H ₂	NH ₃	Ar	CH ₄
V	89.73	0.2448	0.7339	0.0147	0.0022	0.0044
L	10.27	0.0037	0.0147	0.9815	0.0000	0.0001

You could use Eq. (3.46) to solve this problem without the detailed analysis above.

We have gone over a number of examples of condensation and vaporization, and you have seen how a given amount of air at atmospheric pressure can hold only a certain maximum amount of water vapor. This amount depends on the temperature of the air, and any decrease in the temperature will lower the water-bearing capacity of the air.

An increase in pressure also will accomplish the same effect. If a pound of saturated air at 75°F is isothermally compressed (with a reduction in volume, of course), liquid water will be deposited out of the air just like water being squeezed out of a wet sponge (Fig. 3.20). The vapor pressure of water is 0.43 psia at 75°F.

For example, if a pound of saturated air at 75°F and 1 atm is compressed isothermally to 4 atm (58.8 psia), almost three-fourths of the original content of water vapor now will be in the form of liquid, and the air still has a dew point of 75°F. Remove the liquid water, expand the air isothermally back to 1 atm, and you will find that the dew point has been lowered to about 36°F. Mathematically (1 = state at 1 atm, 4 = state at 4 atm) with $z = 1.00$ for both components:

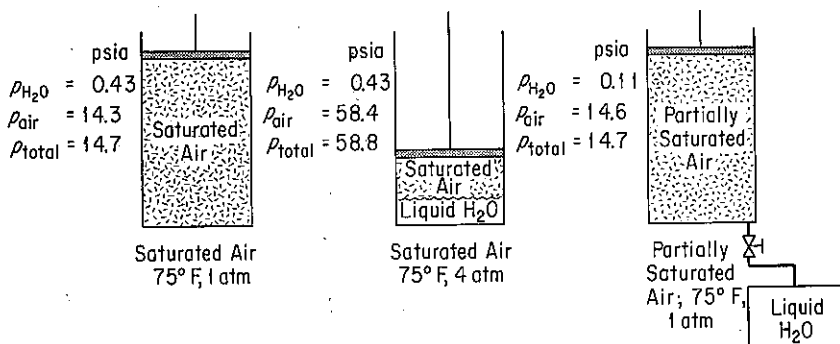


Figure 3.20 Effect of an increase of pressure on saturated air and a return to the initial pressure.

For saturated air at 75°F and 4 atm:

$$\left(\frac{n_{\text{H}_2\text{O}}}{n_{\text{air}}}\right)_4 = \left(\frac{p_{\text{H}_2\text{O}}^*}{p_{\text{air}}}\right)_4 = \frac{0.43}{58.4}$$

For the same air saturated at 75°F and 1 atm:

$$\left(\frac{n_{\text{H}_2\text{O}}}{n_{\text{air}}}\right)_1 = \left(\frac{p_{\text{H}_2\text{O}}^*}{p_{\text{air}}}\right)_1 = \frac{0.43}{14.3}$$

Since the air is the tie element in the process,

$$\left(\frac{n_4}{n_1}\right)_{\text{H}_2\text{O}} = \frac{\frac{0.43}{58.4}}{\frac{0.43}{14.3}} = \frac{14.3}{58.4} = 0.245$$

24.5% of the original water will remain as vapor after compression.

After the air-water vapor mixture is returned to a total pressure of 1 atm, the following two familiar equations now apply:

$$p_{\text{H}_2\text{O}} + p_{\text{air}} = 14.7$$

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{air}}} = \frac{n_{\text{H}_2\text{O}}}{n_{\text{air}}} = \frac{0.43}{58.4} = 0.00736$$

From these two relations you can find that

$$p_{\text{H}_2\text{O}} = 0.108 \text{ psia}$$

$$p_{\text{air}} = 14.6$$

$$p_{\text{total}} = 14.7 \text{ psia}$$

The pressure of the water vapor represents a dew point of about 36°F, and a relative humidity of

$$(100) \frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2\text{O}}^*} = \frac{0.108}{0.43} (100) = 25\%$$

Self-Assessment Test

1. Gas from a synthetic gas plant analyzes (on a dry basis) 4.8% CO_2 , 0.2% O_2 , 25.4% CO , 0.4% C_2H_4 , 12.2% H_2 , 3.6% CH_4 , and 53.4% N_2 . The coal used analyzes 70.0% C, 6.5% H, 16.0% O, and 7.5% ash. The entering air for combustion has a partial pressure of water equal to 2.67 kPa. The barometer reads 101 kPa. Records show that 465 kg of steam is supplied to the combustion vessel per metric ton of coal fired. Calculate the dew point of the exit gas.
2. A liquid solution of pharmaceutical material to be dried is sprayed into a stream of hot gas. The water evaporated from the solution leaves with the exit gases. The solid is recovered by means of cyclone separators. Operating data are:

Inlet air: 100,000 ft^3/hr , 600°F, 780 mm Hg, humidity of 0.00505 lb $\text{H}_2\text{O}/\text{lb}$ dry air
Inlet Solution: 300 lb/hr, 15% solids, 70°F
Outlet air: 200°F, 760 mm Hg, dew point 100°F
Outlet solid: 130°F

Calculate the composition of the outlet solid—it is not entirely dry.

3. A gas leaves a solvent recovery system saturated with benzene at 50°C and 750 mm Hg. The gas analyzes, on a benzene-free basis, 15% CO , 4% O_2 , and the remainder nitrogen. This mixture is compressed to 3 atm and is subsequently cooled to 20°C. Calculate the percent benzene condensed in the process. What is the relative saturation of the final gas?

Thought Problem

1. A large fermentation tank fitted with a 2-in. open vent was sterilized for 30 minutes by blowing in live steam at 35 psig. After the steam supply was shut off, a cold liquid substrate was quickly added to the tank at which point the tank collapsed inward.
What happened to cause the tank to collapse?

3.7 PHASE PHENOMENA***Your objectives in studying this section are to be able to:***

1. Write down the phase rule, define each parameter in the phase rule, and be able to apply the phase rule to determine the degrees of freedom, or the number of components, or the phases that exist in a system.
2. Sketch (roughly) the p - V - T surface and the two-dimensional projections thereof for water.
3. Predict the p - V - T behavior of a pure component or a mixture given the p - V - T diagram, and identify a specified state as residing in the solid, liquid, or vapor single-phase region or the solid plus liquid, solid plus vapor, or liquid plus vapor two-phase region.
4. Completely describe, from a phase diagram, all changes that occur on

cooling or heating a given composition of a two-component system; give all phases present at each temperature and approximate amounts of each phase; also, give the degrees of freedom at any point in the cooling or heating process.

5. Locate a reference containing the properties of a pure substance in your library.

We now consider briefly some of the qualitative characteristics of vapors, liquids, and solids. You need to know in any problem what materials are gases, liquids and solids so that you can apply the proper p - V - T relationships.

3.7-1 The Phase Rule

You will find the Gibb's phase rule a useful guide in establishing how many properties, such as pressure and temperature, have to be specified to definitely fix all the remaining properties and number of phases that can coexist for any physical system. **The rule can be applied only to systems in equilibrium.** It says that

$$F = C - \mathcal{P} + 2 \quad (3.57)$$

where F = number of degrees of freedom (i.e., the number of independent properties that have to be specified to determine all the intensive properties of each phase of the system of interest)

C = number of components in the system; for circumstances involving chemical reactions, C is *not* identical to the number of chemical compounds in the system but is equal to the number of chemical compounds less the number of independent reaction and other equilibrium relationships among these compounds as discussed below.

$\mathcal{P}$ = number of phases that can exist in the system; a phase is a homogeneous quantity of material such as a gas, a pure liquid, a solution, or a homogeneous solid

Variables of the kind with which the phase rule is concerned are called *phase-rule variables*, and they are **intensive** properties of the system. **By this we mean properties that do not depend on the quantity of material present.** If you think about the properties we have employed so far in this book, you have the feeling that pressure and temperature are independent of the amount of material present. So is concentration, but what about volume? The total volume of a system is called an **extensive property because it does depend on how much material you have; the specific volume, on the other hand, the cubic meter per kilogram, for example, is an intensive property** because it is independent of the amount of material present. In Chap. 4 we take up additional intensive properties, such as internal energy and enthalpy. You should remember that the specific (per unit mass) values of these quantities are intensive properties; the total quantities are extensive properties.

An example will clarify the use of these terms in the phase rule. You will remember for a pure gas that we had to specify three of the four variables in the ideal gas equation $pV = nRT$ in order to be able to determine the remaining one unknown. You might conclude that $F = 3$. If we apply the phase rule, for a single phase $\mathcal{P} = 1$, and for a pure gas $C = 1$, so that

$$F = C - \mathcal{P} + 2 = 1 - 1 + 2 = 2 \quad \text{variables to be specified}$$

How can we reconcile this apparent paradox with our previous statement? Since the phase rule is concerned with intensive properties only, the following are phase-rule variables in the ideal gas law:

$$\left. \begin{array}{l} P \\ \hat{V} \text{ (specific molar volume)} \\ T \end{array} \right\} \quad 3 \text{ intensive properties}$$

Thus the ideal gas law would be written

$$p\hat{V} = RT$$

and once two of the intensive variables are fixed (F equals 2), the third is also automatically fixed.

The intensive property must be uniform throughout the phase. If a system at equilibrium consists of two nonreacting components such as NH_3 and H_2O , and two phases, liquid and vapor, intensive variables would be temperature (the same in both phases), pressure (the same in both phases), concentration in a phase (different in each phase), specific volume (different in each phase), and so on. The overall concentration (including both phases) for the system would not be an intensive property because it is not a value equal to the actual value of the concentration in either phase.

An **invariant** system is one in which no variation of conditions is possible without one phase disappearing. An example with which you may be familiar is the ice-water-vapor system, which exists at only one temperature (0.01°C) and pressure (610 Pa):

$$F = C - \mathcal{P} + 2 = 1 - 3 + 2 = 0$$

With all three phases present, none of the physical conditions can be varied without the loss of one phase. As a corollary, if the three phases are present, the temperature, the specific volume, and so on, must always be fixed at the same values. This phenomenon is useful in calibrating thermometers and other instruments.

A complete discussion of the significance of the term C in the phase rule is beyond the scope of this book; however, a brief explanation will demonstrate the issues involved. The quantity C is calculated as

$$C = c^* - h$$

where c^* is the rank of the atom matrix and h is the number of stoichiometric constraints that exist in a reacting system. The atom matrix is just the matrix in which the rows represent the atomic species and the columns represent the chemical com-

pounds in the system. Each element in the matrix is the number of atoms in the particular compound. Thus (for water)

	H ₂ O	H ₂	O ₂
H:	2	2	0
O:	1	0	2

for which the rank is two. Refer to Sec. L.1 to ascertain how the rank of the atom matrix can be evaluated. In any case C is not necessarily the number of compounds present in the system, n , and in fact the following inequality holds:

$$C \leq c^* \leq n$$

We do not have the space here to explain completely how h is calculated, but an example will make the idea clear. Suppose that a portion of 100g of $\text{NH}_4\text{Cl(s)}$ decomposes in an evacuated tank to $\text{NH}_3\text{(g)}$, HCl(g) , $\text{N}_2\text{(g)}$, $\text{H}_2\text{(g)}$, and $\text{Cl}_2\text{(g)}$. Thus $n = 5$, but the rank of the atomic matrix is $c^* = 3$. Because all the N, H, and Cl in the gas phase came from the $\text{NH}_4\text{Cl(s)}$, in the gas phase you know the following additional ratios must exist (called *stoichiometric constraints*) among all the compounds

$$\frac{\text{N}}{\text{H}} = \frac{1}{4} \quad \frac{\text{Cl}}{\text{H}} = \frac{1}{4}$$

Consequently, $h = 2$, and the degrees of freedom for the system are

$$F = (3 - 2) - 2 + 2 = 1$$

in agreement with experimental observations. For more complicated reactions, consult Rao²¹ (or one of the supplementary references at the end of this chapter) as to how to calculate h . Without a reaction, occurring, $h = 0$.

EXAMPLE 3.32 Application of the Phase Rule

Calculate the number of degrees of freedom (how many additional intensive variables must be specified to fix the system) from the phase rule for the following materials at equilibrium:

- Pure liquid benzene
- A mixture of ice and water only
- A mixture of liquid benzene, benzene vapor, and helium gas
- A mixture of salt and water designed to achieve a specific vapor pressure

What variables might be specified in each case?

Solution

$$F = C - \mathcal{P} + 2$$

(a) $C = 1$, $\mathcal{P} = 1$, hence $F = 1 - 1 + 2 = 2$. The temperature and pressure might be specified in the range in which benzene remains liquid.

²¹ Y. K. Rao, *Stoichiometry and Thermodynamics of Metallurgical Processes*, Cambridge University Press, New York, 1984.

(b) $C = 1$, $\mathcal{P} = 2$, hence $F = 1 - 2 + 2 = 1$. Once either the temperature or the pressure is specified, the other intensive variables are fixed.

(c) $C = 2$, $\mathcal{P} = 2$, hence $F = 2 - 2 + 2 = 2$. A pair from temperature, pressure, or mole fraction can be specified.

(d) $C = 2$, $\mathcal{P} = 2$, hence $F = 2 - 2 + 2 = 2$. Since a particular pressure is to be achieved, you would adjust the salt concentration and the temperature of the solution.

We are now going to consider how phase phenomena can be illustrated by means of diagrams.

3.7-2 Phase Phenomena of Pure Components

If we are to clearly understand phase phenomena, the properties of a substance should be shown in at least three dimensions, particularly if there can be more than one phase in the region for which the p - $\hat{V}$ - T properties are to be presented. We previously overcame the handicap of using two dimensions to present three-dimensional data by showing, on the two-dimensionnal graph, lines of constant value for the other properties of the system (see Fig 3.3, which has p and $\hat{V}$ as axes, and lines of constant temperature as the third parameter). In this section we present a more elaborate picture of these p - $\hat{V}$ - T properties. Figure 3.21 illustrates the relationships between the three-dimensional presentations of p - $\hat{V}$ - T data for water and the conventional two-dimensional diagrams you usually encounter. Be certain that you view the central figure in Fig. 3.21 as a *surface* and not as solid figures; only the p - $\hat{V}$ - T points on the surface exist, and no point exists above or below the surface.

You can see that the two-dimensional diagrams are really projections of the three-dimensional surfaces onto suitable axes. On the p - $\hat{V}$ diagram, you will observe lines of constant temperature called **isothermal lines**, and on the p - T diagram you will see lines of constant volume, called **isometric lines** or **isochores**. On the T - $\hat{V}$ diagram, we might have put lines of constant pressure known as **isobaric lines**, but these are not shown since they would obscure the more important features which are shown. The two-phase vapor-liquid region is represented by the heavy envelope in the p - $\hat{V}$ and the T - $\hat{V}$ diagrams. On the p - T diagram this two-phase region appears only as a single curve because you are looking at the three-dimensional diagram from the side. The p - T diagram shows what we have so far called the *vapor-pressure curve*, but examination of the associated diagrams show that the "curve" is really a surface on which two phases coexist, each having the properties found at the appropriate bounding line for the surface. A point on the surface shows merely the overall composition of one phase in equilibrium with another. For example, at point *A* we have liquid and at point *B* we have vapor, and in between them no single phase exists. If we added the liquid at *A* to the vapor at *B*, we would have a two-phase mixture with the gross properties shown by point *M*. You will notice that in the p - $\hat{V}$ diagram the lines of constant temperature and pressure become horizontal in the two-phase region, because if you apply the phase rule

$$F = C - \mathcal{P} + 2 = 1 - 2 + 2 = 1$$

only one variable can change—the specific volume. In the two-phase region, where

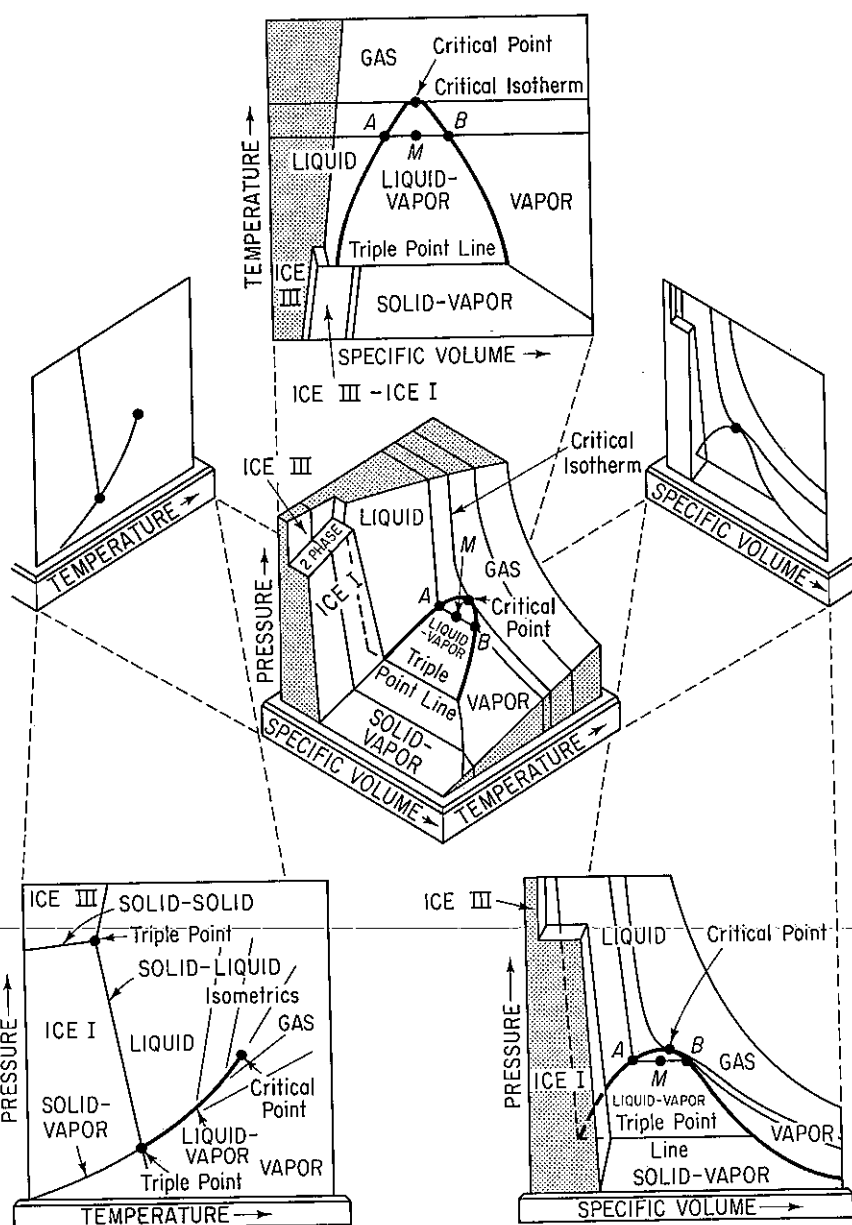


Figure 3.21 p - $\hat{V}$ - T surface and projections for H_2O .

condensation or evaporation takes place, the pressure and temperature remain constant. Do you recall our discussion of boiling water?

Other features illustrated on the diagrams are the critical point, the **critical isotherm**, the **triple point** in the p - T diagram, which actually becomes a **triple-point line** in the p - $\hat{V}$ and T - $\hat{V}$ diagrams; and the existence of two solid phases and a liquid phase in equilibrium, such as solid ice I, solid ice III, and liquid. The temper-

ature at the freezing point for water (i.e., the point where ice, water, air, and water vapor are in equilibrium) is not quite the same temperature as that at the triple point, which is the point where ice, water, and water vapor are in equilibrium. The presence of air at 1 atm lowers the freezing temperature by about 0.01°C , so that the triple point is at 0.00602 atm (0.610 Pa) and 0.01°C . Another triple point for the water system exists at 2200 atm and 20°C between ice III, ice I, and liquid water.

3.7-3 Phase Phenomena of Mixtures

Presentation of phase phenomena for mixtures that are completely miscible in the liquid state involves some rather complex reductions of three-, four-, and higher-dimensional diagrams into two dimensions. We shall restrict ourselves to two-component systems because, although the ideas discussed here are applicable to any number of components, the graphical presentation of more complex systems in an elementary text such as this is probably more confusing than helpful.

It would be convenient if the critical temperature of a mixture were the mole weighted average of the critical temperatures of its pure components, and the critical pressure of a mixture were simply a mole weighted average of the critical pressures of the pure components (the concept used in Kay's rule), but these maxims simply are not true, as shown in Fig. 3.22. The pseudocritical temperature falls on the dashed line between the critical temperatures of CO_2 and SO_2 , whereas the actual critical point for the mixture lies somewhere else. The solid line in Fig. 3.22 illustrates how the locus of the actual critical points diverges from the locus of the pseudo critical points.

Figure 3.23 is a p - T diagram for a 25% methane/75% butane mixture. You can compare this figure with the one for a pure substance, water, as in Fig. 3.9; the p - T curve for pure butane or pure methane would look just like the diagram for water

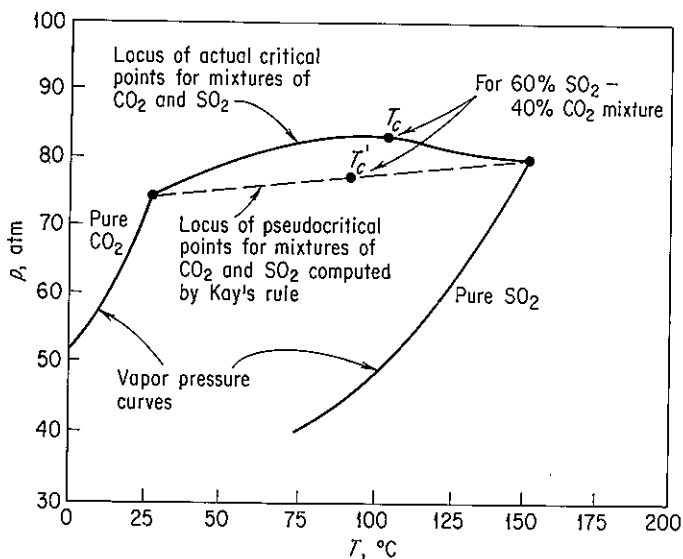


Figure 3.22 Critical and pseudocritical points for a mixture.

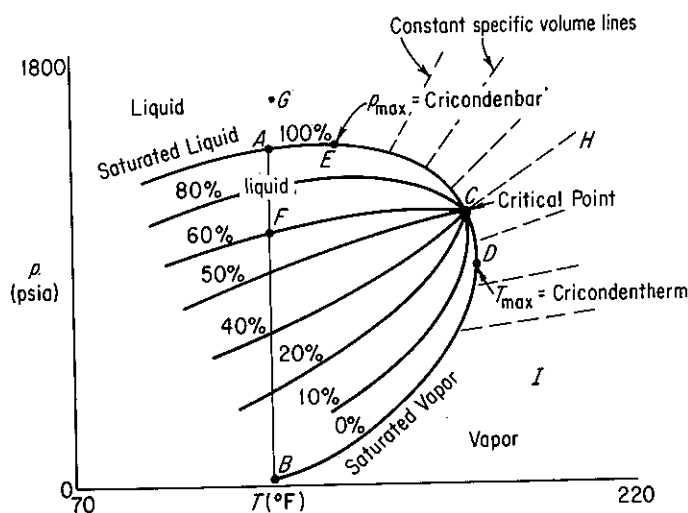


Figure 3.23 p - T diagram for binary system: 25% CH_4 —75% C_4H_{10} .

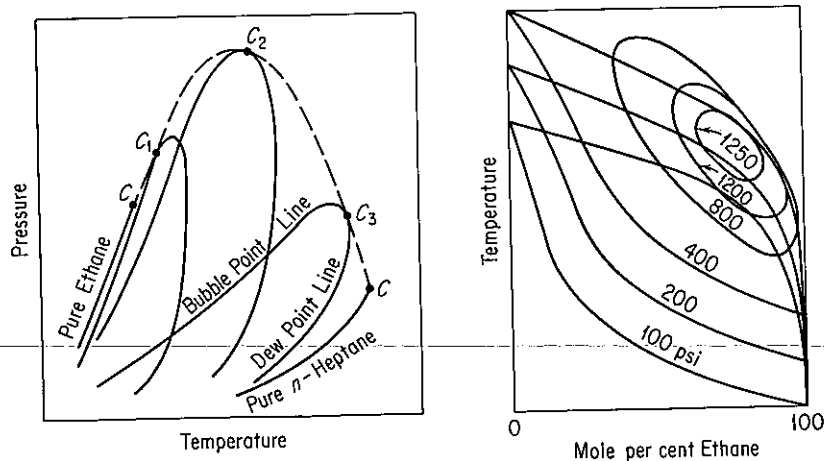


Figure 3.24 p - T and T - x diagrams for ethane-heptane mixtures of different compositions.

(excluding the solid phases). In Fig. 3.23, lines of constant specific volume are shown outside the envelope of the two-phase region. Inside the envelope are shown lines of constant fractions of liquid, starting at the high pressures, where 100% liquid exists (zero percent vapor), and ranging down to 0% liquid and 100% vapor at the low pressures. The 100% saturated liquid line is the bubble-point line and the 100% saturated vapor line is the dew-point line.

The critical point is shown as point C. You will note that this is not the point of maximum temperature at which vapor and liquid can exist in equilibrium; the latter is the **maximum cricondentherm**, point D. Neither is it the point of maximum pressure at which vapor and liquid can exist in equilibrium because that point is the

maximum cricondenbar (point *E*). The maximum temperature at which liquid and vapor can exist in equilibrium for this mixture is about 30° higher than the critical temperature, and the maximum pressure is 60 or 70 psia greater than the critical pressure. You can clearly see, therefore, that the best definition of the critical point was the one which stated that the density and other properties of the liquid and vapor become identical at the critical point. This definition holds for single components as well as mixtures.

Figure 3.23 represents a two-component system with a fixed overall composition. You might wonder what a diagram would look like if we were to try to show systems of several compositions on one page. This has been done in Fig. 3.24. Here we have a composite p - T diagram, which is somewhat awkward to visualize, but represents the bubble-point and dew-point curves for various mixtures of ethane and heptane. These curves in essence are intersections of surfaces in the composition coordinate sliced out of a three-dimensional system and are stacked one in front of the other, although in two dimensions it appears that they intersect one another. The vapor-pressure curves for the two pure components are at the extreme sides of the diagrams as single curves (as you might expect). Each of the loops represents the two-phase area for a system of a specific composition. An infinite number of these surfaces are possible, of course. The dashed line indicates the envelope of the critical points for each possible composition. Although this line appears to be two-dimensional in Fig. 3.24, it actually is a three-dimensional line of which only the projection is shown in the figure.

Another way to illustrate the phase phenomena for the two-component systems we have been discussing is to use pressure-composition diagrams at constant temperature or, alternatively, to use temperature-composition diagrams at constant pressure. A temperature-composition diagram with pressure as the third parameter is illustrated in Fig. 3.24 for the ethane-heptane system.

More complicated phase diagrams for multicomponent systems (ternary and higher) can be found in the references at the end of the chapter.

Self-Assessment Test

1. Is the critical point a single phase? If not, what phases are present? Repeat for the triple point (for water).
2. What does a p - V - T value lying above the surface in Fig. 3.23 mean? Below the surface?
3. Can a liquid be changed to a vapor without passing through a distinct discontinuity in properties? Can a solid be transformed similarly to vapor?
4. Sketch the p - V , p - T , and T - V diagrams for water.
5. A vessel contains air: $N_2(g)$, $O_2(g)$, and $Ar(g)$.
 - (a) How many phases, components, and degrees of freedom are there according to the phase rule?
 - (b) Repeat for a vessel one-third filled with liquid ethanol and two-thirds filled with N_2 plus ethanol vapor.
6. Can the following system exist at equilibrium: $H_2O(s)$, $H_2O(l)$, $H_2O(g)$, decane(s), decane(l), decane(g)? (*Hint*: Decane is insoluble in water.)
7. How many independent properties are required to fix the equilibrium state of a pure compound?

8. Consider the system Mn(s) , $\text{AlCl}_3(\text{g})$, $\text{MnCl}_2(\text{l})$, Al(l) . What are C , $\mathcal{P}$, and F in the Gibb's phase rule? Al(l) is not miscible with $\text{MnCl}_2(\text{l})$.

Thought Problems

1. The advertisement reads "Solid dry ice blocks in 60 seconds right in your own lab! Now you can have dry ice available to you at any time, day or night, with this small, safe, efficient machine and readily available CO_2 cylinders. No batteries or electrical energy are required."

How is it possible to make dry ice in 60 seconds without a compressor?

2. An inventor is trying to sell a machine that transforms water vapor into liquid water without ever condensing the water vapor. You are asked to explain if such a process is technically possible. What is your answer?

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PROBLEMS

An asterisk designates problems appropriate for solution using a computer. Refer also to the problems that require writing computer programs at the end of the chapter.

Section 3.1

- 3.1. One liter of a gas is under a pressure of 780 mm Hg. What will be its volume at standard pressure, the temperature remaining constant?
- 3.2. A gas occupying a volume of 1 m³ under standard pressure is expanded to 1.200 m³, the temperature remaining constant. What is the new pressure?
- 3.3. At standard conditions a gas that behaves as an ideal gas is placed in a 4.13-L container. By using a piston the pressure is increased to 31.2 psia, and the temperature is increased to 212°F. What is the final volume occupied by the gas?
- 3.4. An oxygen cylinder used as a standby source of oxygen contains 1.000 ft³ of O₂ at 70°F and 200 psig. What will be the volume of this O₂ in a dry-gas holder at 90°F and 4.00 in. H₂O above atmospheric? The barometer reads 29.92 in. Hg.
- 3.5. Probably the most important constituent of the atmosphere that fluctuates is the water. Rainfall, evaporation, fog, and even lightning are associated with water as a vapor or a liquid in air. To obtain some feeling for how little water vapor there is at higher altitudes, calculate the mass of 1.00 m³ of water vapor at 2.00 kPa and 23°C. Assume that water vapor is an ideal gas under these conditions.
- 3.6. You have 10 lb of CO₂ in a 20-ft³ fire extinguisher tank at 30°C. Assuming that the ideal gas law holds, what will the pressure gauge on the tank read in a test to see if the extinguisher is full?
- 3.7. Divers work as far as 500 ft below the water surface. Assume that the water temperature is 45°F. What is the molar specific volume (ft³/lb mol) for an ideal gas under these conditions?
- 3.8. A 25-L glass vessel is to contain 1.1 g moles of nitrogen. The vessel can withstand a pressure of only 20 kPa above atmospheric pressure (taking into account a suitable safety factor). What is the maximum temperature to which the N₂ can be raised in the vessel?

- 3.9. A gas in a storage container for plasma etching is at 90 mm Hg gauge pressure. What will be the (a) absolute pressure and (b) the gauge pressure if the volume of the container is doubled at the same temperature? The external atmospheric pressure is 14.7 psia.
- 3.10. *Columbia*, one of the Goodyear blimps, requires 5210 m³ of helium to fill the balloon at 20°C and 101.3 kPa. Helium may be purchased in standard gas cylinders of 23 cm diameter and 132 cm height pressurized to 15,200 kPa. If Goodyear were to fill the blimp with these cylinders at \$32 per cylinder, how much would it cost? Helium behaves ideally at these conditions.
- 3.11. Compressed air from a tank maintained at 125 psig is used to inflate to 30 psig a tire that has a volume of 0.50 ft³, the temperature remaining constant. What volume of air is taken from the tank? What volume is this at standard pressure?
- 3.12. A recent newspaper report states: "Home meters for fuel gas measure the volume of gas usage based on a standard temperature, usually 60 degrees. But gas contracts when it's cold and expands when warm. East Ohio Gas Co. figures that in chilly Cleveland, the homeowner with an outdoor meter gets more gas than the meter says he does, so that's built into the company's gas rates. The guy who loses is the one with an indoor meter: If his home stays at 60 degrees or over, he'll pay for more gas than he gets. (Several companies make temperature-compensating meters, but they cost more and aren't widely used. Not surprisingly, they are sold mainly to utilities in the North.)"

Suppose that the outside temperature drops from 60°F to 10°F. What is the percentage increase in the mass of the gas passed by a noncompensated outdoor meter that operates at constant pressure? Assume that the gas is CH₄.

- 3.13. In Bhopal, India, a Union Carbide storage tank containing methyl isocyanate (CH₃NCO, a component used for the production of insecticide) leaked, resulting in injury and death to thousands of people. The American Occupational Safety rules specify that workplace conditions are to be limited to concentrations less than 0.02 ppm of this compound. Assume ideal gas behavior. What is this concentration in mg/m³ at 20°C and atmospheric pressure?
- 3.14. One of the experiments in the fuel-testing laboratory has been giving some trouble because a particular barometer gives erroneous readings owing to the presence of a small amount of air above the mercury column. At a pressure of 755 mm Hg the barometer reads 748 mm Hg, and at 740 the reading is 736. What will the barometer read when the actual pressure is 760 mm Hg?
- 3.15. From the known standard conditions, calculate the value of the gas law constant *R* in the following sets of units:
(a) cal/(g mol)(K)
(b) Btu/(lb mol)(°R)
(c) (psia)(ft³)/(lb mol)(°R)
(d) J/(g mol)(K)
(e) (cm³)(atm)/(g mol)(K)
(f) (ft³)(atm)/(lb mol)(°R)
- 3.16. Two tanks are initially sealed off from one another by means of valve A. Tank I initially contains 1.00 ft³ of air at 100 psia and 150°F. Tank II initially contains a nitrogen-oxygen mixture containing 95 mole % nitrogen at 200 psia and 200°F. Valve A is then opened allowing the contents of the two tanks to mix. After complete mixing had been effected, the gas was found to contain 85 mole % nitrogen. Calculate the volume of tank II. See Fig. P3.16.

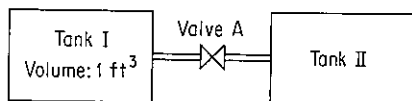
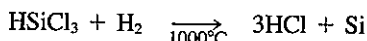


Figure P3.16

- 3.17. The majority of semiconductor chips used in the microelectronics industry are made of silicon doped with trace amounts of materials to enhance conductivity. The silicon initially must contain less than 20 parts per million (ppm) of impurities. Silicon rods are grown by the following chemical deposition reaction of trichlorosilane with hydrogen:



Assuming that the ideal gas law applies, what volume of hydrogen at 1000°C and 1 atm must be reacted to increase the diameter of a rod 1 m long from 1 cm to 10 cm? The density of solid silicon is 2.33 g/cm^3 .

- 3.18. The accident at the Three Mile Island nuclear plant began in March 1979 when a pressure relief valve of a water purifier stuck open. A series of operator errors and equipment failures stopped the flow of cooling water to the reactor core. As a result, the water began to boil and the core reached the temperature at which the steam reacted with the zirconium cladding of the fuel rods producing zirconium oxide (ZrO_2) and hydrogen gas (H_2). The hydrogen bubble that formed was estimated to be 28,000 L in volume and further impeded the flow of cooling water, which was at 250°C and 6900 kPa. From the measurements taken, determine the number of kilograms of zirconium reacted.
- 3.19. When a mixture of liquid hydrocarbons (C and H only) are burned in a test engine, the exhaust gas is found to contain 10.0% CO_2 on a dry basis. It is also found that the exhaust gas contains no oxygen or hydrogen (on the dry basis). Careful measurement indicates 173 ft^3 of air of 80°F and 740 mm Hg absolute enter the engine for every pound of fuel used.

Calculate the mass ratio of H to C in the fuel.

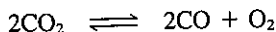
- 3.20. A medium-Btu gas analyzes 6.4% CO_2 , 0.1% O_2 , 39% CO, 51.8% H_2 , 0.6% CH_4 , and 2.1% N_2 . It enters the combustion chamber at 90°F and a pressure of 35.0 in. Hg. It is burned with 40% excess air (dry) at 70°F and the atmospheric pressure of 29.4 in. Hg, but 10% of the CO remains unburned. How many cubic feet of air are supplied per cubic foot of entering gas?
- 3.21. How many lb of CS_2 must be burned with the stoichiometric quantity of O_2 ($\text{CS}_2 + 3\text{O}_2 \rightarrow \text{CO}_2 + 2\text{SO}_2$) to yield 200 ft^3 of gas at 300°C and 1 atm?
- 3.22. Pine wood has the following composition:

C	50.31%
H_2	6.20%
O_2	43.08%
Ash	0.41%

If you burn this wood in an approved stove using 60% excess air, what is

- (a) The m^3 of air at 15°C and 98 kPa needed per kg of wood?
 (b) The analysis of the flue gas?
 (c) The m^3 of flue gas at 110°C and 98 kPa per m^3 air used (at 15°C and 98 kPa)?

- 3.23. Carbon dioxide dissociates at high temperatures according to the following equation:



At 3000°C and 1 atm CO_2 is 40% dissociated. If 5 L of CO_2 at 20°C and 1 atm is heated to 3000°C at constant pressure, what volume of the gas will result? State your assumptions.

- 3.24. An industrial fuel gas consists of 40% CO and 60% CH_4 ; 121 ft³/min of the gas, measured at 500 psia and 90°F, is completely burned with 25% excess air that enters at 70°F and 770 mm Hg. The resultant gases leave the furnace at 570°F and 770 mm Hg. Compute the composition and number of cubic feet per minute of the hot gases leaving the furnace.
- 3.25. Turbojet aircraft operating under most conditions produce lower concentrations of pollutants than do motor vehicles. At idle, CO and unburned hydrocarbons are higher than in motor vehicles at idle, but in the operating mode the emission index (grams of pollutant per kilogram of fuel) is:

	CO	Hydrocarbons	NO_x
Turbojet	8.7	0.16	2.7
Automotive piston	300	55	27

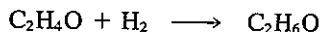
Given that the Orsat analysis from a turbojet shows 12.2% CO_2 , 0.4% CO, and 6.2% O_2 , compute the net hydrogen/carbon ratio in the fuel (the fuel contains negligible sulfur and nitrogen). Also compute the cubic meters of air used at 27°C and 101.4 kPa/kg of fuel burned.

- 3.26. In the first stage of the manufacture of sulfuric acid by the contact process, iron pyrites (FeS_2) is burned in dry air, the iron being oxidized to Fe_2O_3 . Sufficient air is supplied so that it is 40% in excess of that required if all the sulfur were oxidized to sulfur trioxide. Of the pyrites charges, 15% is lost by falling through the grate with the cinders and is not burned.

Calculate the cubic meters of air at 30°C and 150 kPa to be used per 100 kg of pyrites charged.

- 3.27. In a test on an oil-fired boiler, it is not possible to measure the amount of oil burned, but the air used is determined by inserting a venturi meter in the air line. It is found that 5000 ft³/min of air at 80°F and 10 psig is used. The dry gas analyzes 10.7% CO_2 , 0.55% CO, 4.75% O_2 , and 84.0% N_2 . If the oil is assumed to be all hydrocarbon, calculate the gallons per hour of oil burned. The specific gravity of the oil is 0.94.
- 3.28. Gas at 60°F and 42.1 in. Hg absolute is flowing through an irregular duct. To determine the rate of flow of the gas, He is passed into the gas stream. The gas analyzes 1.0% He by volume before, and 1.4% He after addition. The He tank is placed on a scale and is observed to lose 10 lb in 30 min. What is the rate of flow of the gas in the duct in cubic feet per minute?
- 3.29. A gas containing only C and H on complete combustion yields 0.302 kg mol of CO_2 and 0.309 kg mol of H_2O . What is the C/H ratio in the gas? The specific gravity of the gas at 20°C and 97.3 kPa compared to dry air at 80°C and 30.78 in. Hg is 1.631. Calculate the molecular formula of the gas.

- 3.30. A gaseous mixture consisting of 50 mol % hydrogen and 50 mol % acetaldehyde ($\text{C}_2\text{H}_4\text{O}$) is initially contained in a rigid vessel at a total pressure of 760 mm Hg abs. The formation of ethanol ($\text{C}_2\text{H}_6\text{O}$) occurs according to



After a time it was noted that the total pressure in the rigid vessel had dropped to 700 mm Hg abs. Calculate the degree of completion of the reaction using the following assumptions: (1) all reactants and products are in the gaseous state; and (2) the vessel and its contents were at the same temperature when the two pressures were measured.

- 3.31. Pure ethylene (C_2H_4) and oxygen are fed to a process for the manufacture of ethylene oxide ($\text{C}_2\text{H}_4\text{O}$):

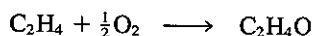


Figure P3.31 is the flow diagram for the process. The catalytic reactor operates at 300°C and 1.2 atm. At these conditions, single-pass measurements on the reactor show that 50% of the ethylene entering the reactor is consumed per pass, and of this, 70% is converted to ethylene oxide. The remainder of the ethylene consumed decomposes to form CO_2 and water.

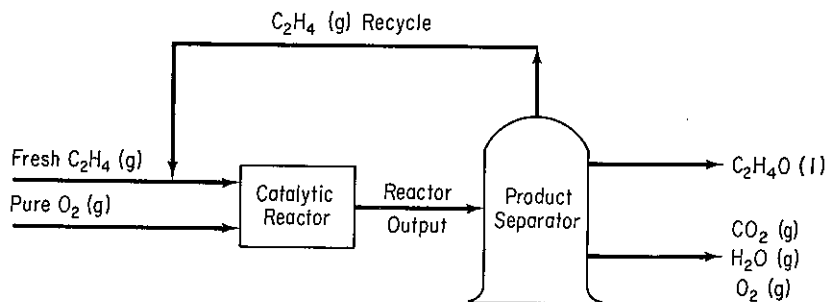
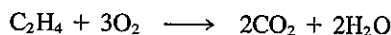


Figure P3.31

For a daily production of 10,000 kg of ethylene oxide:

- Calculate the m^3/hr of total gas entering the reactor at SC if the ratio of the O_2 (g) fed to fresh C_2H_4 (g) is 3 to 2.
 - Calculate the recycle ratio, m^3 at 10°C and 100 kPa of C_2H_4 recycled, per m^3 at SC of fresh C_2H_4 fed;
 - Calculate the m^3 of the mixture of O_2 , CO_2 and H_2O leaving the separator per day at 80°C and 100 kPa.
- 3.32. What is the density of propane gas (C_3H_8) in kg per cubic meter at 200 kPa and 40°C ? What is the specific gravity of propane?
- 3.33. What is the specific gravity of propane gas (C_3H_8) at 100°F and 800 mm Hg relative to air at 60°F and 760 mm Hg?
- 3.34. What is the mass of 1 m^3 of H_2 at 5°C and 110 kPa? What is the specific gravity of this H_2 compared to air at 5°C and 110 kPa?

- 3.35. Gas from the Padna Field, Louisiana, is reported to have the following components and volume percent composition. What is:
- The mole percent of each component in the gas?
 - The weight percent of each component in the gas?
 - The apparent molecular weight of the gas?
 - Its specific gravity?

Component	Percent	Component	Percent
Methane	87.09	Pentanes	0.46
Ethane	4.42	Hexanes	0.29
Propane	1.60	Heptanes	0.06
Isobutane	0.40	Nitrogen	4.76
Normal butane	0.5	Carbon dioxide	0.40
		Total	100.00

- 3.36. A gas used to extinguish fires is composed of 80% CO_2 and 20% N_2 . It is stored in a 2-m³ tank at 200 kPa and 25°C. What is the partial pressure of the CO_2 in the tank in kPa? What is the partial volume of the N_2 in the tank in m³?
- 3.37. Methane is completely burned with 20% excess air, with 30% of the carbon going to CO. What is the partial pressure of the CO in the stack gas if the barometer reads 740 mm Hg, the temperature of the stack gas is 300°F, and the gas leaves the stack at 250 ft above the ground level?
- 3.38. A mixture of 15 lb N_2 and 20 lb H_2 is at a pressure of 50 psig and a temperature of 60°F. Determine the following (assuming ideality for the gas mixture):
- The partial pressure of each component
 - The partial (or pure component) volumes
 - The specific volume of the mixture
 - The density of the mixture
- 3.39. The composition of mycelium from the production of antibiotics roughly resembles that of sewage sludge. In contrast to sewage sludge, however, the composition of mycelium remains largely constant. Therefore, mycelium is particularly suitable as model substrate for long-running experiments. In one experiment, 100 kg of product gas at 105 kPa and 460°C contains 60 kg of N_2 , 15 kg of O_2 , 20 kg of CO_2 , and 5 kg of CH_4 .
- Determine the following:
 - The mole fraction of each component
 - volume fraction of each component
 - partial pressure of each component
 - Determine the volume occupied by the mixture in m³.
 - What is the density of the gas?
 - What is the specific gravity of the gas?
- 3.40. Three thousand cubic meters per day of a gas mixture containing methane and *n*-butane at 21°C enters an absorber tower. The partial pressures at these conditions are 103 kPa for methane and 586 kPa for *n*-butane. In the absorber, 80% of the butane is removed and the remaining gas leaves the tower at 38°C and a total pressure of 550 kPa. What is the volumetric flow rate of gas at the exit? How many moles per day of butane are removed from the gas in this process? Assume ideal behavior.

- 3.41. Pure methane is completely burned with air. The outlet gases from the burner, which contain no oxygen, are passed through a cooler where some of the water is removed by condensation. The gases leaving the cooler have a nitrogen mole fraction of 0.8335. Calculate the following:
- The analysis of the gases leaving the cooler (mol % CO_2 , H_2O , and N_2)
 - The pounds of H_2O condensed per mole of CH_4 burned
 - The average molecular weight of the gases leaving the cooler
- If the exit gases from the cooler are at 130°F and 20 psia:
- What is the partial pressure of H_2O ?
 - What is the partial volume of H_2O per mole of CH_4 ?

Section 3.2

(Note: The use of a computer is recommended if equations of state are to be employed in solving a problem in this section.)

- 3.42.* Find the molar volume of methane at 350 K and 200 kPa abs (in $\text{cm}^3/\text{g mol}$). Use the van der Waals, Redlich-Kwong, and Benedict-Webb-Rubin equations. The nonlinear equation solving codes on the disk in the pocket at the back of this book will help you avoid trial-and-error solutions. The Benedict-Webb-Rubin constants are:

$$A_0 = 182277.6$$

$$B_0 = 0.4546 \times 10^{-1} \quad \text{Use } p \text{ in Pa}$$

$$C_0 = 0.3226 \times 10^{10} \quad T \text{ in K}$$

$$a = 4409.7 \quad R \text{ in } (\text{Pa})(\text{m}^3)/(\text{kg mol})(\text{K})$$

$$b = 0.252 \times 10^{-2} \quad \text{Then } V \text{ will be in } \text{m}^3/\text{kg mol}$$

$$c = 0.3635 \times 10^{-9}$$

$$\alpha = 0.33 \times 10^{-3}$$

$$\gamma = 0.105 \times 10^{-1}$$

- 3.43.* Find the molar volume (in $\text{cm}^3/\text{g mol}$) of propane at 375 K and 21 atm. Use the Redlich-Kwong and Peng-Robinson equations, and solve for the molar volume using the nonlinear equation solver NEWTON on the disk in the pocket at the back of this book. The acentric factor for propane to use in the Peng-Robinson equation is 0.1487.
- 3.44.* A 5-L tank of H_2 is left out overnight in Antarctica. You are asked to determine how many g moles of H_2 are in the tank. The pressure gauge reads 39 atm gauge and the temperature is -50°C . How many g moles of H_2 are in the tank? Use the van der Waals and Redlich-Kwong equations of state to solve this problem. (Hint: The nonlinear-equation-solving program on the disk in the pocket at the back of this book will make the execution of the calculations quite easy.)
- 3.45.* 4.00 g-mol of CO_2 is contained in a 6250- cm^3 vessel at 298.15 K and 14.5 atm. Use the nonlinear equation solver NEWTON on the disk in the back of the book to solve the Redlich-Kwong equation for the molar volume. Compare the calculated molar volume of the CO_2 in the vessel with the known molar volume.
- 3.46.* Calculate the volume (in ft^3/lb) occupied by 1 g mol of CO_2 at 100 atm abs. and 50°C . Solve the Dieterici, van der Waals, Beattie-Bridgeman, and Redlich-Kwong

equations of state using the nonlinear equation solver NEWTON on the disk in the back of the book. Which of the values from among the four equations would you use? Would the volume be more accurate if taken from the CO₂ chart (Fig. J.2)? Data for the coefficients are:

Dieterici		Van der Waals	
$a = 4.627 \times 10^6 \text{ (atm)(cm}^6\text{)/g mol}^2$		$a = 3.6 \times 10^6 \text{ (atm)(cm}^6\text{)/g mol}^2$	
$b = 46.3 \text{ cm}^3\text{/g mol}$		$b = 42.8 \text{ cm}^3\text{/g mol}$	
Beattie-Bridgeman		Redlich-Kwong	
$A_0 = 5.0065 \times 10^6$	$b = 72.35$	Find p_c and T_c from Table D.1 in the Appendix.	
$B_0 = 104.76$	$c = 66 \times 10^7$		
$a = 71.32$			

- 3.47. Calculate the pressure of 176 g of CO₂ in a 6250-cm³ tank at 298.15 K using the Redlich-Kwong equation of state.
- 3.48.* The fire department is inspecting the fire extinguishers in the chemical engineering building. A No. 2 gas cylinder, weighing 52.27 lb when completely evacuated, is placed on an accurate scale and filled with compressed carbon dioxide gas. When the gas in the cylinder has reached room temperature (54.5°F), the pressure in the cylinder is measured and found to be 338 psig. The capacity of the cylinder is 2.04 ft³. Use the van der Waals equation of state to estimate what the *scale* will read in lb.
- 3.49. It is desired to market oxygen in small cylinders having volumes of 0.5 ft³ and each containing 1.0 lb of oxygen. If the cylinders could be subjected to a maximum temperature of 120°F, calculate the minimum pressure for which they must be designed:
- Assuming the applicability of the perfect gas law
 - Using the compressibility factor
 - Using the van der Waals equation
- 3.50.* One of the biggest impediments to the realization of ambitious plans to build coal slurry pipelines has been lack of water in the right places. Much of the desirable low-sulfur coal in the United States comes from arid or semiarid regions in the West and in the northern Great Plains. Natives of these regions get upset and also litigious when coal companies talk of using thousands of acre-feet of scarce and valuable water every year just to carry powdered coal to the Midwest or East.

One way to get around the problem is to use liquid CO₂ instead of water as the transport medium. One advantage of liquid CO₂ is that it is less viscous than water. Friction in the pipeline would be lower, so less energy would be needed to transport a given amount of coal. Also there is little if any interaction between powdered coal and liquid carbon dioxide. Because of the lower viscosity and nonreactivity of liquid carbon dioxide (compared to water), slurries can carry more coal. That means additional energy savings and also means that a smaller pipelines could provide the same coal throughput.

About 8% of the coal shipped would be needed to produce the carbon dioxide if the CO₂ is discarded at the terminal. To have liquid CO₂ at, say 100°F, would not be possible because the critical temperature of CO₂ is 304.2 K (87.6°F) while the critical pressure is 72.9 atm. However, the fluid that exists at 100°F would serve satisfactorily as a transport medium.

Calculate by the following two ways the ft^3 of CO_2 at 120°F and 1200 psia that would be needed to transport 1 lb of coal (containing 74% carbon). Assume that 0.08 lb of (additional) coal per lb of transported coal is used to make the CO_2 .

(a) Compressibility factor

(b) Van der Waals equation

- 3.51. A 100- ft^3 tank contains 95.1 lb moles of a nonideal gas at 1250 atm and 440°F . The critical pressure is known to be 50 atm. What is the critical temperature? Use a generalized compressibility chart to obtain your answer.
- 3.52. A gas is flowing at a rate of 100,000 scfh (std. cond. = 1 atm, 32°F). What is the actual volumetric gas flow rate if the pressure is 50 atm and the temperature is 600°R ? The critical temperature is 40.0°F and the critical pressure is 14.3 atm.
- 3.53. Methyl chloride (mol. wt. = 50.49) is sold in small cylinders for medical purposes and as a refrigerant. A typical internal volume of a cylinder is 0.15 ft^3 . If the weight of the CH_3Cl in the cylinder is 0.0997 lb, what is the pressure in the cylinder at 60°C ?
- 3.54. Ethylene at 500 atm pressure and a temperature of 100°C is contained in a cylinder of internal volume of 1 ft^3 . How many pounds of C_2H_4 are in the cylinder?
- 3.55. Vapor degreasing removes soluble contamination such as oils and grease from solid surfaces that are to be subjected to electrical, chemical, or electrochemical treatment. To store the solvent Freon-12 (CCl_2F_2) ($T_c = 112^\circ\text{C}$, $P_c = 39.6$ atm), you have to design a cylinder to hold 750 kg of Freon-12 at a maximum temperature of 40°C and a pressure of 220 kPa absolute. What should the volume of the tank be?
- 3.56. Two identical cylinders contain different gases. One contains ethylene (C_2H_4) and the other contains nitrogen. Both have a molecular weight of 28, and both are at 25°C and 1000 psig pressure.
- (a) Which cylinder contains the larger volume of gas measured at standard conditions? Verify your answer by giving the ratio of the volumes at standard conditions.
- (b) Which cylinder weighs more? Verify your answer by giving the ratio of the weights of the gases in the two cylinders.
- (Note: On the next four problems check your calculations using the CO_2 chart in Appendix J, Fig. J.2.)**
- 3.57. One pound of carbon dioxide has a volume of 0.15 ft^3 at a pressure of 100 atm. What is the temperature of the gas?
- 3.58. A block of dry ice weighing 50 lb is dropped into an empty steel bomb, the volume of which is 5.0 ft^3 . The bomb is closed and heated until the pressure gauge reads 1600 psig. What was the temperature of the gas in the bomb?
- 3.59. Calculate the specific volume of CO_2 at 600 atm and 40°C .
- 3.60. One-half of a cubic meter of CO_2 gas is held at a constant pressure of 3690 kPa and is heated from 36°C to 77°C . Compare the volumes calculated for the gas after heating by considering it (a) as a nonideal gas and (b) as an ideal gas.
- 3.61. A steel cylinder contains ethylene (C_2H_4) at 200 psig. The cylinder and gas weigh 222 lb. The supplier refills the cylinder with ethylene until the pressure reaches 1000 psig, at which time the cylinder and gas weigh 250 lb. The temperature is constant at 25°C . Calculate the charge to be made for the ethylene if the ethylene is sold at \$0.41 per pound, and what the weight of the cylinder is for use in billing the freight charges. Also find the volume of the empty cylinder in cubic feet.
- 3.62. A steel cylinder contains ethylene (C_2H_4) at 10^4 kPa gauge. The weight of the cylin-

der and gas is 70 kg. Ethylene is removed from the cylinder until the gauge pressure measured falls to one-third of the original reading. The cylinder and gas now weigh 52 kg. The temperature is constant at 25°C. Calculate:

- (a) The fraction of the original gas (i.e., at 10⁴ kPa) that remains in the cylinder at the lower pressure
 (b) The volume of the cylinder in cubic meters

- 3.63. When a scuba diver goes to the dive shop to have his or her scuba tanks filled, the tank is connected to a compressor and filled to about 2100 psia while immersed in a tank of water. (Why immerse the tank in water? So that the compression of air into the tank will be approximately isothermal.)

Suppose that the tank is filled without inserting it into a water bath, and air at 27°C is compressed very rapidly from 1 atm absolute to the same final pressure. The final temperature would be about 700°C. Compute the fractional increase or decrease in the final quantity of air in the tank relative to the isothermal case:

- (a) Assuming that the air behaves as an ideal gas
 (b) Assuming that the air behaves as a real gas
 Treat air as a single component with $p_c = 37.2$ atm and $T_c = 132.5$ K.

- 3.64. What is the density in g/L of a 50 mol % propane (C₃H₈)–50 mol % isobutane (i-C₄H₁₀) gaseous mixture at 23.4 atm and 195°C?

- 3.65. A gas has the following composition:

CO ₂	10%
CH ₄	40%
C ₂ H ₄	50%

It is desired to distribute 33.6 lb of this gas per cylinder. Cylinders are to be designed so that the maximum pressure will not exceed 2400 psig when the temperature is 180°F. Calculate the volume of the cylinder required by Kay's method.

- 3.66. A gas composed of 20% ethanol and 80% carbon dioxide is at 500 K. What is its pressure if the volume per g mol is 180 cm³/g mol?

- 3.67. A sample of natural gas taken at 3500 kPa absolute and 120°C is separated by chromatography at standard conditions. It was found by calculation that the grams of each component in the gas were:

Component	g
Methane (CH ₄)	100
Ethane (C ₂ H ₆)	240
Propane (C ₃ H ₈)	150
Nitrogen (N ₂)	50
Total	540

What was the density of the original gas sample?

- 3.68.* A gaseous mixture has the following composition (in mol %):

C ₂ H ₄	57
Ar	40
He	3

at 120 atm pressure and 25°C. Compare the experimental volume of 0.14 L/g mol with that computed by one or more of the equations of state.

- 3.69. You are in charge of a pilot plant using an inert atmosphere composed of 60% ethylene (C_2H_4) and 40% argon (Ar). How big a cylinder (or how many) must be purchased if you are to use 300 ft³ of gas measured at the pilot-plant conditions of 100 atm and 300°F?

Cylinder type	Cost	Pressure (psig)	Gas (lb)
1A	\$40.25	2000	62
2	32.60	1500	47
3	25.50	1500	35

State any additional assumptions. You can buy only one type of cylinder.

Section 3.3

- 3.70. Prepare a Cox chart for:
 (a) Acetic acid vapor (c) Ammonia
 (b) Heptane (d) Ethanol
 from 0°C to the critical point (for each substance). Compare the estimated vapor pressure at the critical point with the critical pressure.
- 3.71. Use the Antoine equation (Appendix G) to estimate the vapor pressure of chloroform at 25°C and compare your calculated value with the experimental value obtained from a handbook. Do they agree? Then compute the temperature at which chloroform has a vapor pressure of 400 mm Hg absolute.
- 3.72. The vapor pressure of a highly volatile liquid (Q) was measured at 0°C and at 204°C. The results were 103 kPa and 998 kPa, respectively. If dry air in a container initially at 24°C and standard atmospheric pressure is saturated with Q at constant temperature and volume:
 (a) What is the final pressure of the mixture?
 (b) What is the composition of the final mixture?
- 3.73.* Estimate the vapor pressure of ethyl bromide at 125°C from vapor-pressure data taken from a handbook or a journal.
- 3.74.* From the following data, estimate the vapor pressure of sulfur dioxide at 100°C; the actual vapor pressure is about 29 atm.

t (°C)	-10	6.3	32.1	55.5
p^* (atm)	1	2	5	10

- 3.75. Use the Antoine equation to estimate the vapor pressure of sulfur dioxide at -10°C and compare with the experimental value (taken from a handbook).

Section 3.4

- 3.76. A large chamber contains dry N_2 at 27°C and 101.3 kPa. Water is injected into the chamber. After saturation of the N_2 with water vapor, the temperature in the chamber is 27°C.

- (a) What is the pressure inside the chamber after saturation?
 (b) How many moles of H_2O per mole of N_2 are present in the saturated mixture?
- 3.77. The vapor pressure of hexane (C_6H_{14}) at -20°C is 14.1 mm Hg absolute. Dry air at this temperature is saturated with the vapor under a total pressure of 760 mm Hg. What is the percent excess air for combustion?
- 3.78. In a search for new fumigants, chloropicrin (CCl_3NO_2) has been proposed. To be effective, the concentration of chloropicrin vapor must be 2.0% in air. The easiest way to get this concentration is to saturate air with chloropicrin from a container of liquid. Assume that the pressure on the container is 100 kPa. What temperature should be used to achieve the 2.0% concentration? From a handbook, the vapor pressure data are (T , $^\circ\text{C}$; vapor pressure, mm Hg): 0, 5.7; 10, 10.4; 15, 13.8; 20, 18.3; 25, 23.8; 30, 31.1.
 At this temperature and pressure, how many kg of chloropicrin are needed to saturate 100 m^3 of air?
- 3.79. Fifty cubic feet of air saturated with water 90.0°F and 29.80 in. Hg are dehydrated. Compute the volume of the dry air and the pounds of moisture removed.
- 3.80. Suppose that a vessel of dry nitrogen at 70.0°F and 29.90 in. Hg is saturated thoroughly with water. What will be the pressure in the vessel after saturation if the temperature is still 70.0°F ?
- 3.81. 100 lb mol per hour of hexane (C_6H_{14}) vapor are burned with 10% excess air in an efficient furnace. The barometric pressure is 760 mm Hg and the dew point of the ambient air is 80°F .
 (a) What is the flow rate of the entering air (do not neglect the ambient H_2O), in standard cubic feet per minute? Assume the air is saturated at 80°F .
 (b) What is the dew point of the exhaust gas if the exhaust pressure is 750 mm Hg?
 (c) What is the flow rate of the exhaust gas in actual cubic feet per minute if the exhaust stack pressure is 750 mm Hg and the temperature is 350°F ?
- 3.82. CH_4 is completely burned with air. The outlet gases from the burner, which contain no oxygen, are passed through an absorber where some of the water is removed by condensation. The gases leaving the absorber have a nitrogen mole fraction of 0.8335. If the exit gases from the absorber are at 130°F and 20 psia, calculate:
 (a) To what temperature must this gas be cooled at constant pressure in order to start condensing more water?
 (b) To what pressure must this gas be compressed at constant temperature before more condensation will occur?
- 3.83. A mixture of acetylene (C_2H_2) with an excess of oxygen measured 350 ft^3 at 25°C and 745 mm Hg pressure. After explosion, the volume of the dry gaseous product was 300 ft^3 at 60°C and the partial pressure of the dry product was 745 mm Hg. Calculate the partial volumes of acetylene and of oxygen in the original mixture. Assume that the final gas is saturated and that only enough water is formed to saturate the gas.
- 3.84. A mixture of air and benzene contains 10 mole % benzene at 38°C and 790 mm Hg pressure absolute. The vapor pressure of benzene is given as

$$\log_{10} p^* = 6.906 - \frac{1211}{220.8 + t}$$

where p^* is the vapor pressure in mm Hg and t is in $^\circ\text{C}$. What is the dew point of the mixture?

- 3.85. It is desired to represent the vapor pressure of an organic substance by an equation of the form

$$p^* = p_0^* e^{m/T}$$

where p^* is the vapor pressure, T the absolute temperature, and m and p_0^* are unknown constants. If 100 m³ of dry air (measured at standard conditions) is required to vaporize 0.5 kg mol of the organic material at a temperature of 290 K and a pressure of 1 atm and only 50 m³ are required for the 0.5 kg mol at 340 K and 1 atm, calculate the constants p_0^* and m .

- 3.86. The vapor pressure of hexane (C₆H₁₄) at -20°C is 14.1 mm Hg. Dry air at this temperature is saturated with the vapor under a total pressure of 760 mm Hg.
 (a) What is the percent excess air for combustion?
 (b) What is the flue-gas analysis if complete combustion occurs?
- 3.87. Liquid hexane is insoluble in water; hence each substance exerts its vapor pressure independently of the other. Compute the partial pressure of (a) the hexane and (b) the water if the gas mixture, containing 150,000 kg of hexane (C₆H₁₄) and 4200 kg of water, is initially at 100°C and 200 kPa absolute. To what temperature must the mixture be lowered before the water starts to condense out, if the pressure remains constant? Has hexane started to condense at this temperature? Support your answers with the necessary data and calculations.
- 3.88. A solution containing 12 wt % of dissolved nonvolatile solid is fed to a flash distillation unit. The molecular weight of the solid is 123.0. The effective vapor pressure of the solution is equal to the mole fraction of water,

$$p = p^*x$$

where p = effective vapor pressure of the solution

p^* = vapor pressure of pure water

x = mole fraction of water

The pressure in the flash distillation unit is 1.121 psia and the temperature is 110°F. Calculate the pounds of pure water obtained in the vapor stream per 100 lb of feed solution and the weight percent of the dissolved nonvolatile solid leaving in the liquid stream.

- 3.89. A mixture of water (15%), dimethyl acetamide (70%), and some inert liquid that has negligible vapor pressure (15%) is to be flashed at 194°F and 1.94 psia. The K values are water: 5.26 and dimethylacetamide: 0.64. How many moles of vapor and liquid are formed at equilibrium, and what are the mole fractions of the vapor and liquid components?
- 3.90.* A mixture of hydrocarbons consisting of an equal number of moles each of propane, normal butane and normal pentane is to be fractionated in a well insulated column. The figure shows the compositions and the overall material balance. The overhead product is completely condensed to a saturated liquid at 100°F. What is the temperature in the top tray of the column where the liquid vaporizes at equilibrium to form the overhead? Hint: Assume that the pressure in the condenser is the same as that in the top tray.
- 3.91.* A mixture of 50% benzene and 50% toluene is contained in a cylinder at 19.34 in. Hg absolute. Calculate the temperature range in which a two phase system can exist.

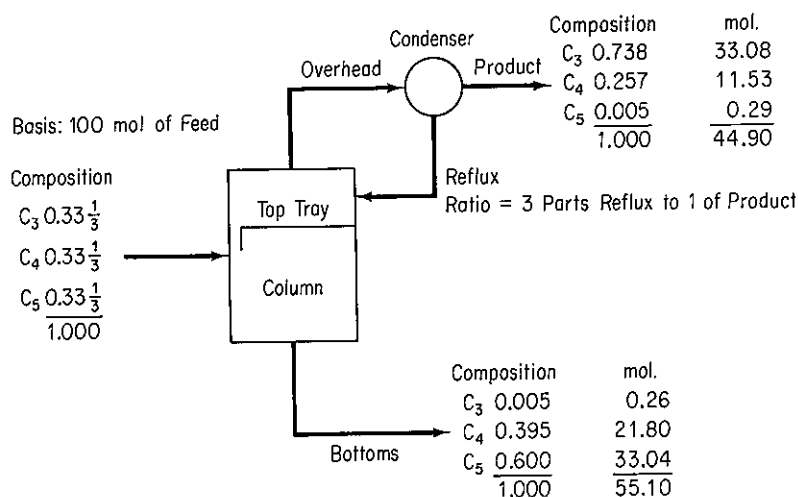


Figure P3.90

- 3.92. For an ideal binary system (of components designated by subscripts 1 and 2) with the vapor and liquid phases in equilibrium, show that if the temperature and pressure are specified

$$x_2 = \frac{K_1 - 1}{K_1 - K_2}$$

where x_i is the mole fraction of component i in the liquid phase, and K_i are the equilibrium constants $y_i = K_i x_i$.

- 3.93.* Calculate the liquid (L) and vapor (V) compositions and the temperature (in °F) for the flash vaporization of the following feed stream (F) at 300 psia for a value of V/F of (a) 0.3 and (b) 0.6.

Component	Mole fraction	b_i values		
		b_1	b_2	b_3
<i>n</i> -Hexane	0.05	0.2420	0.2713×10^{-2}	0.0876×10^{-4}
<i>n</i> -Pentane	0.20	0.5087	0.4625×10^{-2}	0.1051×10^{-4}
<i>i</i> -Pentane	0.20	0.5926	0.5159×10^{-2}	0.1094×10^{-4}
<i>n</i> -Butane	0.20	0.2126	0.7618×10^{-2}	0.1064×10^{-4}
<i>i</i> -Butane	0.35	0.2606	0.8650×10^{-2}	0.1017×10^{-4}

The equilibrium constants can be calculated from the relation

$$K_i = b_{1i} + b_{2i}(T - 250) + b_{3i}(T - 250)^2$$

where T is in °F.

- 3.94. For what temperature range can the following mixture be part liquid and part vapor at 100 psia?

	Mol %
Propane	10
<i>i</i> -Butane	10
<i>n</i> -Butane	40
<i>i</i> -Pentane	10
<i>n</i> -Pentane	30
Total	100

3.95. Assume that Raoult's law holds for the following mixture.

	Mol %
<i>n</i> -Butane	50
<i>i</i> -Pentane	30
<i>n</i> -Pentane	20
Total	100

- (a) What is the dew point pressure of the mixture at 100°F?
- (b) What is the dew point temperature at 50 psig? The barometer reads 775 mm Hg.
- (c) What is the bubble point temperature at 70 psia?
- (d) What is the bubble point pressure at 100°F?
- 3.96. Determine the moles of vapor and liquid that exist if 1128 moles of the following mixture is maintained at 325 psia and 100°F.

	Mole %	K value
Methane	13.7	8.8
Ethane	20.6	1.82
Propylene	15.9	0.78
Propane	33.2	0.71
Isobutane	7.2	0.35
<i>n</i> -Butane	9.4	0.27
Total	100.0	

3.97. A mixture of hydrocarbons has the following composition in mol %:

	Mol %
CH ₄	1.0
C ₂ H ₆	15.0
C ₃ H ₈	25.0
<i>i</i> -C ₄ H ₁₀	14.0
<i>n</i> -C ₄ H ₁₀	25.0
<i>n</i> -C ₅ H ₁₂	20.0
Total	100.0

The mixture exists at a pressure of 100 psia and is all in the vapor phase.

- (a) What is the minimum temperature at which it could be a vapor?
- (b) What mole % of the vapor is condensed if the mixture is cooled to 90°F?
- (c) Calculate the liquid and vapor compositions in part (b).
- (d) For the conditions in part (b), what percent of the total methane present is in the vapor phase? The pentane?
- (e) What temperature is required to condense 40 mol % of the total (still at 100 psia)?
- (f) What temperature is required to condense 100% of the vapor?
- (g) If 100,000 ft³ at S.C. of the original gas is introduced into a separator at 100 psia and 90°F, how many gallons per minute of liquid have to be handled by the pump removing liquid from this separator?

Assume that the liquid specific gravities are the same at 60°F and 90°F. Note the important effect of the 1% CH₄ on the bubble point and the insignificant effect of the pentane. Note also how the opposite is true for the dew point.

Section 3.5

- 3.98. The percent relative humidity of air at 90°F is 60% (where the vapor pressure of water is 0.698 psia). What is:
 - (a) The humidity of the air?
 - (b) The absolute humidity of the air?
- 3.99. Oxygen at 21°C and 1 atm with 80% relative humidity is supplied to a biotech reactor. Calculate:
 - (a) The molal humidity
 - (b) The humidity
 - (c) The percent saturation
 - (d) The relative humidity
 - (e) The dew point of the supplied gas.
- 3.100. If a gas at 140°F and 30 in. Hg. abs. has a molal humidity of 0.03 mole of H₂O per mole of dry air, calculate:
 - (a) The percentage humidity
 - (b) The relative humidity (%)
 - (c) The dew point of the gas (°F)
- 3.101. The Weather Bureau reports a temperature of 90°F, a relative humidity of 85% and a barometric pressure of 14.696 psia.
 - (a) What is the molar humidity?
 - (b) What is the humidity (weight basis)?
 - (c) What is the percentage "absolute" humidity?
 - (d) What is the saturation temperature or the dew point?
 - (e) What is the number of degrees of superheat of the water vapor?
 - (f) Determine the molar humidity and dew point if the air is heated to 105°F, pressure remaining steady.
 - (g) Determine the molar humidity and dew point if the air is cooled to 60°F, pressure remaining steady.
 - (h) What fraction of the original water is condensed at 60°F?
- 3.102. On a warm day the barometric pressure is 29.92 in. Hg, the temperature is 90°F, and the relative humidity is 70%.

- (a) What is the dew point of the air?
 - (b) At night the temperature drops to 60°F. What is the barometric pressure of the air? (State all assumptions but do not assume a constant pressure process.)
- 3.103. What is the mass of 3.00 m³ of air at 27°C and a total pressure of 115 kPa absolute if the relative humidity of the air is 65%?
- 3.104. The Environmental Protection Agency has promulgated a national ambient air quality standard for hydrocarbons: 160 µg/m³ is the maximum 3-hr concentration not to be exceeded more than once a year. It was arrived at by considering the role of hydrocarbons in the formation of photochemical smog. Suppose that in an exhaust gas benzene vapor is mixed with air at 25°C such that the partial pressure of the benzene vapor is 2.20 mm Hg. The total pressure is 800 mm Hg. Calculate:
- (a) The moles of benzene vapor per mole of gas (total)
 - (b) The moles of benzene per mole of benzene free gas
 - (c) The weight of benzene per unit weight of benzene-free gas
 - (d) The relative saturation
 - (e) The percent saturation
 - (f) The micrograms of benzene per cubic meter
 - (g) The grams of benzene per cubic foot
- Does the exhaust gas concentration exceed the national quality standard?
- 3.105. Sufficient oxygen at 21°C and 100 kPa absolute with 80% absolute saturation of water vapor is supplied to react with sewage to reduce the concentration of biological material to below the quality control limits. Calculate the following for the supplied gas:
- (a) The molal saturation
 - (b) kg water/kg bone-dry oxygen
 - (c) The relative saturation
 - (d) The dew point
- 3.106. Thermal pollution is the introduction of waste heat into the environment in such a way as to adversely affect environmental quality. Most thermal pollution results from the discharge of cooling water into the surroundings. It has been suggested that power plants use cooling towers and recycle water rather than dump water into the streams and rivers. In a proposed cooling tower, air enters and passes through baffles over which warm water from the heat exchanger falls. The air enters at a temperature of 80°F and leaves at a temperature of 70°F. The partial pressure of the water vapor in the air entering is 5 mm Hg and the partial pressure of the water vapor in the air leaving the tower is 18 mm Hg. The total pressure is 740 mm Hg. Calculate:
- (a) The relative humidity of the air-water vapor mixture entering and of the mixture leaving the tower
 - (b) The percentage composition by volume of the moist air entering and of that leaving
 - (c) The percentage composition by weight of the moist air entering and of that leaving
 - (d) The percent absolute humidity of the moist air entering and leaving
 - (e) The pounds of water vapor per 1000 ft³ of mixture both entering and leaving
 - (f) The pounds of water vapor per 1000 ft³ of vapor-free air both entering and leaving
 - (g) The weight of water evaporated if 800,000 ft³ of air (at 740 mm and 80°F) enters the cooling tower per day

Section 3.6

- 3.107. A drier must evaporate 200 lb/hr of H_2O . Air at $70^\circ F$ and 50% RH enters the drier, leaving at $140^\circ F$ and 80% RH. What volume of dry air is necessary per hour?
- 3.108. 1000 ft³ of air, saturated with H_2O , at $30^\circ C$ and 740 mm Hg, are cooled to a lower temperature and one-half of the H_2O is condensed out. Calculate:
(a) How many pounds of H_2O are condensed out
(b) The volume of dry air at $30^\circ C$ and 740 mm Hg
- 3.109. Moist air at $25^\circ C$ and 100 kPa with a dew point of $19.5^\circ C$ is to be dehydrated so that during its passage through a large cold room used for food storage excess ice formation can be avoided on the chilling coils in the room. Two suggestions have been offered: (1) cool the moist air to below the saturation temperature at 100 kPa, or (2) compress the moist air above the saturation pressure at $25^\circ C$. Calculate the saturation temperature for (1) and the total pressure at saturation for (2).
(a) If 60% of the initial water in the entering moist air has to be removed before the air enters the cold room, to what temperature should the air in process (1) be cooled?
(b) What pressure should the moist air in process (2) reach?
(c) Which process appears to be the most satisfactory? Explain.
- 3.110. A mixture of air and water at a temperature of $30^\circ C$ and a pressure of 750 mm Hg has a relative humidity of 75%. The wet gas is compressed to 40 psia and cooled to $20^\circ C$. How many cubic feet of original gas were compressed if 0.75 lb of condensate (water) was removed from a separator placed adjacent to the cooler?
- 3.111.* One hundred cubic meters of saturated gas at $25^\circ C$ and 100 kPa (saturated with ethyl alcohol) are cooled so that one-half of the ethyl alcohol originally present condenses out. To what temperature was the gas cooled if the cooling occurs at constant pressure? List by number all the equations and other information you need to solve this problem. Include the known values of the variables and constants in the equations. List the unknowns by symbol. State in order the specific steps you would take to solve for the final temperature. Solve for it.
- 3.112. To ensure a slow rate of drying and thereby prevent checking of the dried product, the inlet relative humidity to a drier is specified as 70% at $75^\circ F$. The air leaving the drier has a relative humidity of 90% at $70^\circ F$. If the outside air has a dew point of $40^\circ F$, what fraction of the air leaving the drier must be mixed and recycled with the outside air to provide the desired moisture content in the air fed to the drier?
- 3.113. A dryer used to season wood evaporates 27 kg/hr of water from freshly cut greenwood. Air enters the dryer at 40% relative humidity and $47^\circ C$ leaves the dryer at 80% relative humidity and $37^\circ C$. If the entire process operates at 100 kPa, how many m³/hr of moist air enter the dryer?
- 3.114. A hydrocarbon fuel is burned with bone-dry air in a furnace. The stack gas is at 116 kPa and has a dew point of $47^\circ C$. The Orsat analysis of the gas shows 10 mol % carbon dioxide; the balance consists of oxygen and nitrogen. What is the ratio of hydrogen to carbon in the hydrocarbon fuel?
- 3.115. A low-energy gas from in situ combustion of shale oil has the following composition CO_2 : 10.0%, CO : 20.0%, H_2 : 20.0%, CH_4 : 2.0%, and N_2 : 48.0%. On combustion the Orsat analysis of the flue gas gives CO_2 : 14.3%, CO : 1.0%, O_2 : 4.3%, and N_2 : 80.4%. Calculate the dew point of the flue gas if burned with air entering at $25^\circ C$ and 30% relative humidity. The barometer reads 100 kPa.

- 3.116.** Hot air that is used to dry pharmaceuticals is recycled in a closed loop to prevent the contamination of the moist material from atmospheric impurities. In the first conditioning step for the air, 5000 k mol/hr at 105 kPa and 42°C with a 90% relative humidity are fed to a condenser to remove some of the water picked up previously in the dryer. The air exists the condenser at 17°C and 100 kPa containing 91 k mol/hr of water vapor. Next, the air is heated in a heat exchanger to 90°C, and then goes to the dryer. By the time the air enters the dryer, the pressure of the stream has dropped to 95 kPa and the temperature is 82°C.
- How many moles of water per hour enter the condenser?
 - What is the flow rate of the condensate water in kg/hr?
 - What is the dew point of the air in the stream exiting the condenser?
 - What is the dew point of the air in the stream entering the dryer?
- 3.117.** Around airports jet aircraft can become major contributors to pollution, and as aircraft activity increases and public concern brings other sources under control, the relative contribution to aircraft to pollution could go up. Recently, federal-, state-, and local-government pressure has speeded the introduction of new combustors in aircraft. In a test for a supersonic aircraft fuel with the average composition $C_{1.20}H_{4.40}$, the fuel is completely burned with the exact stoichiometric amount of air required. The air is supplied at 25°C and 101 kPa, with an absolute humidity of 80%. The combustion products leave at 480°C and 106 kPa pressure and are passed through a heat exchanger from which they emerge at 57°C and 100 kPa pressure.
- For the entering air, compute:
 - The dew point
 - The molal humidity
 - The relative humidity
 - How much water is condensed in the heat exchanger per kilogram of gas burned, and hence must be removed as liquid water?
- 3.118.*** A certain gas contains moisture, and you have to remove the moisture by compression and cooling so that the gas will finally contain not more than 1% moisture (by volume). You decide to cool the final gas down to 21°C.
- Determine the minimum final pressure needed.
 - If the cost of the compression equipment is

$$\text{cost in \$} = (\text{pressure in psia})^{1.40}$$
 and the cost of the cooling equipment is

$$\text{cost in \$} = (350 - \text{temp. K})^{1.9}$$
 is 21°C the best temperature to use? Hint: Look at the list of computer programs on the disk in the back of this book.
- 3.119.** Oxalic acid ($H_2C_2O_4$) is burned with 248% excess air, 65% of the carbon burning to CO. Calculate:
- The Orsat gas analysis
 - The volume of air at 90°F and 785 mm Hg used per pound of oxalic acid burned
 - The volume of stack gases at 725°F and 785 mm Hg per pound of oxalic acid burned
 - The dew point of the stack gas
- 3.120.** A synthesis gas is made by partial oxidation of butane (C_4H_{10}) in the presence of steam and air. The product synthesis gas has the percent composition 3.5% CO_2 ,

2.3% C_2H_4 , 23.2% CO , 39.3% H_2 , 11.6% CH_4 , 1.7% C_2H_6 , and 18.4% N_2 . How many cubic meters of air (at 40% relative humidity, $30^\circ C$, 104 kPa) are used per cubic meter of butane (at $25^\circ C$, 120 kPa, dry)?

- 3.121. A flue gas from a furnace leaves at $315^\circ C$ and has an Orsat analysis of 16.7% CO_2 , 4.1% O_2 , and 79.2% N_2 . It is cooled in a spray cooler and passes under slight suction through a duct to an absorption system at $32.0^\circ C$ to remove CO_2 for the manufacture of dry ice. The gas at the entrance to the absorber analyzes 14.6% CO_2 , 6.2% O_2 , and 79.2% N_2 , due to air leaking into the system. Calculate the cubic meters of air leaked in per cubic meter of gas to the absorber, both measured at the same temperature and pressure.
- 3.122. Soybean flakes from an extraction process are reduced from 0.96 lb of C_2HCl_3 per pound of dry flakes to 0.05 lb of C_2HCl_3 per pound of dry flakes in a desolventizer by a stream of N_2 which vaporizes the C_2HCl_3 . The entering N_2 contains C_2HCl_3 such that its dew point is $30^\circ C$. The N_2 leaves at $90^\circ C$ with a relative saturation of 60%. The pressure in the desolventizer is 760 mmHg, and 1000 lb/hr of dry flakes pass through the drier.
- Compute the volume of N_2 plus C_2HCl_3 leaving the desolventizer at $90^\circ C$ and 760 mm Hg in cubic feet per minute.
 - The N_2 leaving the desolventizer is compressed and cooled to $40^\circ C$, thus condensing out the C_2HCl_3 picked up in the desolventizer. What must the pressure in the condenser be if the gas is to have a dew point of $30^\circ C$ at the pressure of the desolventizer?
- 3.123. One thousand pounds of a slurry containing 10% by weight of $CaCO_3$ is to be filtered on a rotary vacuum filter. The filter cake from the filter contains 60% water. This cake is then placed into a drier and dried to a moisture content of 9.09 lb of H_2O /100 lb of $CaCO_3$. If the humidity of the air entering the drier is 0.005 lb of water per pound of dry air and the humidity of the air leaving the drier is 0.015 lb of water per pound of dry air, calculate:
- Pounds of water removed by the filter
 - Pounds of dry air needed in the drier
- 3.124. Air is humidified in the spray chamber shown in Fig. P3.124. Calculate how much water must be added per hour to the tower to process 10,000 ft^3/hr of air metered at the entrance conditions.

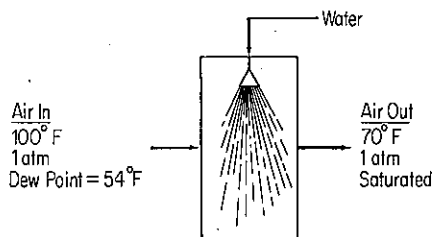


Figure P3.124

- 3.125. A wet sewage sludge contains 50% by weight of water. A centrifuging step removes water at a rate of 100 lb/hr. The sludge is dried further by air. Use the data in Fig. P3.125 to determine how much moist air (in cubic feet per hour) is required for the process shown.
- 3.126. Refer to the process flow diagram (Fig. P3.126) for a process that produces maleic anhydride by the partial oxidation of benzene. The moles of O_2 fed to the reactor

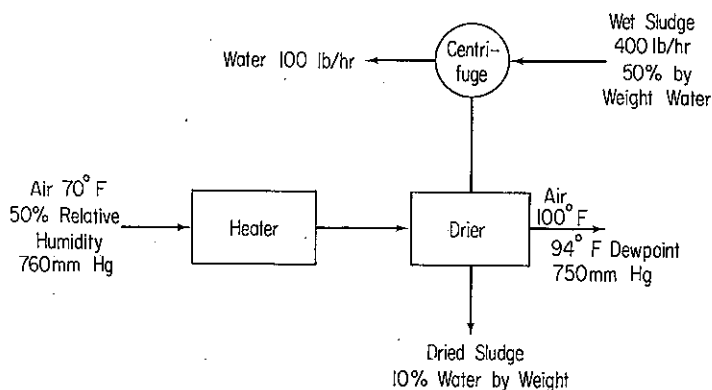


Figure P3.125

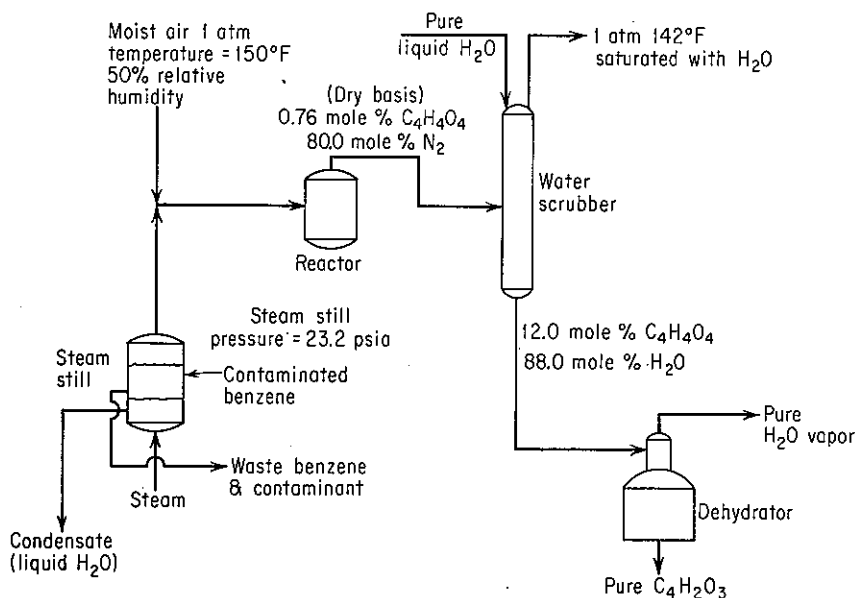
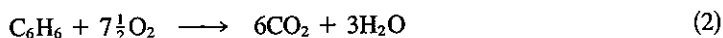
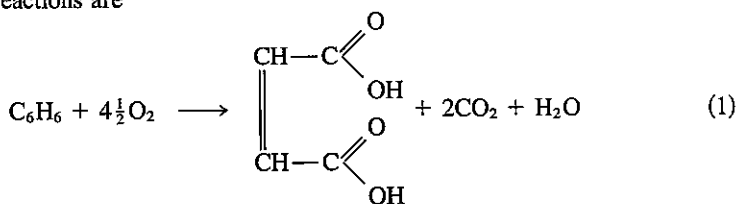


Figure P3.126

per mole of pure benzene fed to the reactor is 18.0. All the maleic acid produced in the reactor is removed with water in the bottom stream from the water scrubber. All the C₆H₆, O₂, CO₂, and N₂ leaving the reactor leave in the stream from the top of the water scrubber, saturated with H₂O. Originally, the benzene contains trace amounts of a nonvolatile contaminant that would inhibit the reaction. This contaminant is removed by steam distillation in the steam still. The steam still contains liquid phases of both benzene and water (benzene is completely insoluble in water). The benzene phase is 80% by weight, and the water phase is 20% by weight of the total of the two liquid phases in the still. Other process conditions are given in the flow sheet. Use the following vapor-pressure data:

Temperature (°F)	Benzene (psia)	Water (psia)
110	4.045	1.275
120	5.028	1.692
130	6.195	2.223
140	7.570	2.889
150	9.178	3.718
160	11.047	4.741
170	13.205	5.992
180	15.681	7.510
190	18.508	9.339
200	21.715	11.526

The reactions are



Calculate:

- The moles of benzene undergoing reaction (2) per mole of benzene feed to the reactor
- The pounds of H_2O removed in the top stream from the dehydrator per pound mole of benzene feed to the reactor
- The composition (mole percent, wet basis) of the gases leaving the top of the water scrubber
- The pounds of pure liquid H_2O added to the top of the water scrubber per pound mole of benzene feed to the reactor

Section 3.7

- When production is taken from a gas reservoir, is it possible to decrease the pressure at the well and recover liquids as well as gas? Explain using the phase diagram in Fig. P3.127 showing the p - T properties of a two-component gas condensate.
- At the critical point for Fig. P3.127, how many degrees of freedom exist according to the phase rule? How many degrees of freedom exist at each of the points in Fig. P3.127 marked A , B , C , D and E ? Show your computations.
- A vessel contains liquid ethanol, ethanol vapor, and N_2 gas. How many phases, components, and degrees of freedom are there according to the phase rule.
- The triple point and the ice point for water differ by 0.0095°C . Why?
- What is the number of degrees of freedom according to the phase rule for each of the following systems:
 - Solid iodine in equilibrium with its vapor
 - A mixture of liquid water and liquid octane (which is immiscible in water) both in equilibrium with their vapors

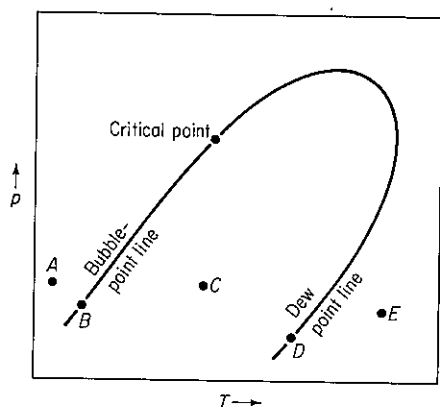
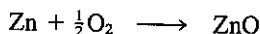
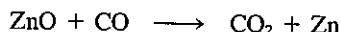
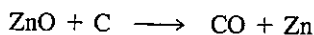


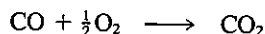
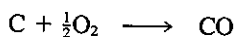
Figure P3.127

(c) CO , CO_2 , H_2 , H_2O , CH_3OH , and CH_4 in chemical equilibrium at high temperature

- 3.132. A mixture of zinc oxide (ZnO) and carbon reacts with the following reactions (the first listed reaction is the main reaction):



Also, the carbon reacts as follows:



If zinc produced by the reaction is in the liquid phase (and has a significant vapor pressure), at equilibrium how many degrees of freedom exist for the system?

- 3.133. In the decomposition of CaCO_3 in a sealed container from which the air was initially pumped out, you generate CO_2 and CaO . If not all of the CaCO_3 decomposes at equilibrium, how many degrees of freedom exist for the system according to the Gibbs phase rule?
- 3.134. Calculate C , $\mathcal{P}$, and F in the Gibbs phase rule for the following system in a closed vessel: $\text{AgCl}(\text{s})$, $\text{H}_2\text{O}(\text{s})$, $\text{H}_2\text{O}(\text{l})$, $\text{H}_2\text{O}(\text{g})$, $\text{H}^+(\text{aq})$, $\text{Cl}^-(\text{aq})$, $\text{HCl}(\text{g})$, $\text{CO}_2(\text{g})$, $\text{CO}_2(\text{aq})$, $\text{HCl}(\text{aq})$.
- 3.135. Water is pumped into a pipe at the rate of 250 kg/hr at 950 kPa and 175°C. Because of friction in the pipe, pressure losses due to pipe fittings and valves, the pressure at the exit of the pipe drops to 760 kPa. Is the entering water liquid, vapor, or a mixture of both? Determine the fraction of each. Is the exit water liquid, vapor, or both? Determine the fraction of each. From the steam tables or similar source, determine the density of the overall mixture in the pipe at both locations. How many degrees of freedom exist according to the phase rule at each location?
- 3.136. Examine the volume-pressure chart for toluene in Appendix J, Fig. J.1. If a cylinder contains toluene with a volume of 2.1 ft³/lb mol at a pressure of 30 psia, is the toluene all liquid, all vapor, or a mixture of the two phases?